

Notes on Classical Mechanics

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These notes develop classical mechanics as a self-contained sequence of models, structures, and examples. The first layer establishes the common language: configuration space, action principles, constraints, symmetries, momenta, the Legendre transform, Hamiltonian flow, and symplectic phase space. The second layer uses Kepler motion and rigid bodies to make that language concrete, then builds integrability through action-angle variables and invariant tori. The third layer studies perturbation: canonical transformations, resonances, KAM mechanisms, breakdown of tori, and restricted three-body dynamics. The final layer carries the same structural ideas into continuum mechanics: elastic media emphasize material deformation, shape-changing bodies emphasize connections and holonomy, and fluids emphasize Eulerian fields, Navier-Stokes dynamics, exact flows, stability criteria, and vorticity. Figures, exercises, Python demonstrations, and Wolfram Language notebooks are placed next to the derivations so that the architecture of the theory, its computations, and its physical examples reinforce one another.

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Preface

Mechanics is developed here from the structures that give its equations meaning. The starting point is the configuration space and the action. From there the notes pass to phase space, symplectic geometry, canonical transformations, integrability, perturbation, chaos, elasticity, gauge kinematics of deformation, and fluids. Kepler motion, the rigid body, the standard map, the restricted Sun-Jupiter-asteroid model, deforming bodies, laminar flows, and point vortices are the main laboratories.

The intended reading style is active. At each derivation, identify the space, the variables, the assumptions, the conserved quantities, and the geometric object being preserved or varied. At each figure, ask which equation or reduction the picture is meant to make visible. At each numerical demo, ask which continuous structure the discretization respects and which quantities must be monitored.

The exposition uses a repeated rhythm. First state the model in words. Then name the mathematical space on which the model lives. Then derive the equation from a principle or balance law. Coordinates, phase portraits, and simulations come after those choices have been made. In this order, coordinate formulas remain attached to the geometric or physical structures they represent.

The notes are also written for two passes. On a first pass, follow the definitions, boxed equations, examples, and the “what to take forward” sections. On a second pass, return to the derivations and ask why each assumption was needed. A formula may be a definition in one section, a constraint in another, and an equation of motion only after a principle has been applied. The text names these roles because mechanics becomes clearer when definitions, identities, constraints, and dynamical laws are kept separate.

How the chapters fit together. The notes are meant to be read linearly, but each chapter also has a precise role in the overall architecture:

reading rules	→	well-defined models introduced in context,
conventions and assumptions	→	
Lagrangian mechanics	→	equations from variation and symmetry,
Hamiltonian mechanics	→	phase space and generating functions,
rigid body, tori, and actions	→	integrability and action variables,
perturbation and chaos	→	canonical perturbation, KAM, and transport,
three-body problem	→	celestial reduction, restricted dynamics, and ejection,
elastic bodies	→	stress, strain, and boundary-value problems,
deforming bodies	→	shape space, connection, curvature, holonomy,
fluids	→	Eulerian fields, Navier-Stokes, exact flows, stability.

This dependency map records the conceptual order of the main constructions. KAM theory uses action-angle variables and near-identity generating functions: the Liouville-torus construction explains the invariant tori, and canonical perturbation theory tests what survives when the system is no longer exactly integrable. The gauge theory of deforming bodies uses the same separation of configuration, symmetry, momentum, and connection that first appears in the rigid-body analysis, while elasticity supplies the material-field language needed for continuum bodies.

How to use the computational material. The scripts are controlled experiments attached to the derivations. A useful run tells the same story as the equations: which invariant is monitored, which approximation is being made, which parameter controls the effect, and which observed feature might be a numerical artifact. When a demo reports drift in energy, Jacobi constant, symplectic area, or orthogonality, that diagnostic is part of the mechanics.

Chapter 1

Reading Rules, Conventions, And Standing Assumptions

1.1 Purpose Of These Notes

Each chapter states its assumptions, defines its symbols before use, and derives the central equations from a named structure: an action principle, a symplectic form, a symmetry, or a balance law. The standard for every major derivation is the same:

- model : what physical idealization is being made?
- space : where do the variables live?
- principle : what variational, Hamiltonian, or balance statement is used?
- calculation : which algebraic or analytic step produces the equation?
- interpretation : what does the result say invariantly?

These five questions are the local checklist. They connect particles, rigid bodies, celestial mechanics, elastic and deforming continua, and fluids.

Mechanics overloads its symbols. The letter ω can mean a symplectic form, an angular velocity, a vorticity, or a frequency vector, depending on the chapter. The word “action” can mean a path functional, an action variable, or a group action. The notes keep the conventional language but define it locally: each important symbol is tied to a space, a convention, and a physical interpretation.

The notes also distinguish three kinds of equality. A definition introduces a symbol, as in $p_i = \partial L / \partial \dot{q}^i$. A convention fixes a choice that could have been made differently: the sign convention for the canonical symplectic form on T^*Q is fixed by writing $\omega = -d\Theta$. After that choice has been made, the same displayed relation is used as an identity inside that convention. An equation of motion holds only along physical trajectories, as in $\dot{p}_i = -\partial H / \partial q^i$. Many confusions in mechanics come from silently moving an equation from one category to another.

There is a fourth reading habit of the same importance: keep track of the stage of a calculation. Before a variational principle is imposed, $q(t)$ is an arbitrary admissible curve. After the first variation is set to zero, it is a physical trajectory. Before a constraint is solved, a multiplier is an unknown reaction force. After a reduction is performed, the same multiplier may have disappeared from the reduced equation while still existing in the original force balance. The text calls attention to these stage changes because many formulas in mechanics change their logical role as a derivation proceeds.

Notation is introduced where it first does work. If a standard symbol is reused, the local definition governs the current section or chapter. Upper/lower indices in finite-dimensional mechanics label coordinates on a manifold and its cotangent bundle; in continuum chapters, capital indices such as A, B label material coordinates and lower-case indices such as i, j label spatial coordinates.

1.2 Notation And Conventions

A convention fixes how formulas are written or oriented. It is not a modelling assumption and does not restrict which motions, forces, or materials are being studied. The standing assumptions in the next section state genuine restrictions on regularity, endpoints, or invertibility.

Convention 1.1 (Einstein summation). *Repeated upper/lower coordinate indices are summed unless stated otherwise:*

$$p_i \dot{q}^i = \sum_{i=1}^n p_i \dot{q}^i.$$

This is a notational convention. It abbreviates sums and keeps covariance and contravariance visible; it does not add a physical hypothesis. In Euclidean vector formulas, dots and cross products use the standard metric and orientation unless a different metric or orientation is specified locally.

1.3 Standing Mathematical Assumptions

The assumptions below are different in kind from the conventions above. They identify the regularity and nondegeneracy conditions under which the first derivations are valid. Later chapters replace or refine them when a model requires weaker smoothness, boundary work, constraints, singular Legendre transforms, or continuum fields.

Assumption 1.1 (Smoothness). *Unless explicitly stated otherwise, all functions, curves, vector fields, and maps are assumed smooth enough for the differentiations and integrations being performed. For most finite-dimensional chapters, C^2 regularity is enough. For continuum chapters, additional differentiability is assumed locally.*

Assumption 1.2 (Finite-dimensional mechanics). *When the configuration space is a finite-dimensional manifold Q , a motion is a curve $q : I \rightarrow Q$ on a time interval $I = [t_0, t_1]$. The velocity $\dot{q}(t)$ lies in the tangent space $T_{q(t)}Q$. Local coordinates are written $q^1, \dots, q^n$.*

Assumption 1.3 (Endpoint variations). *For Hamilton's principle in its basic form, variations of the path satisfy $\delta q(t_0) = \delta q(t_1) = 0$. Free-endpoint and boundary-force variants are introduced when boundary work or transversality conditions are part of the model.*

Assumption 1.4 (Regular Lagrangians). *The Legendre transform from velocities to momenta is assumed locally invertible whenever we pass from $L(q, \dot{q}, t)$ to $H(q, p, t)$. In coordinates this means the matrix*

$$\frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$$

is nonsingular. This rank condition is intrinsic. Under a point change of coordinates $q = q(Q)$, velocities transform by $\dot{q}^i = (\partial q^i / \partial Q^a) \dot{Q}^a$, and the velocity Hessian transforms by congruence,

$$W'_{ab} = \frac{\partial q^i}{\partial Q^a} W_{ij} \frac{\partial q^j}{\partial Q^b}, \quad W_{ij} = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}.$$

Since the Jacobian is invertible, W is nonsingular if and only if W' is nonsingular. Singular Lagrangians lead to constrained Hamiltonian systems and gauge theories. The systematic theory of constrained Hamiltonian systems was developed by Bergmann (1949) and Dirac (1950). These notes name this boundary of the regular theory but do not develop the Dirac-Bergmann algorithm; the gauge and continuum chapters use constrained ideas only in settings where the relevant reduction is stated explicitly.

1.4 A Map Of The Course

The course moves through four enlargements of the same idea. The diagram below shows the loop revisited throughout the notes: a variational principle produces equations, Hamiltonian geometry organizes their phase space, action-angle variables describe integrable tori, perturbation theory tests the stability of those tori, and continuum mechanics returns to the same questions with fields instead of finitely many coordinates.

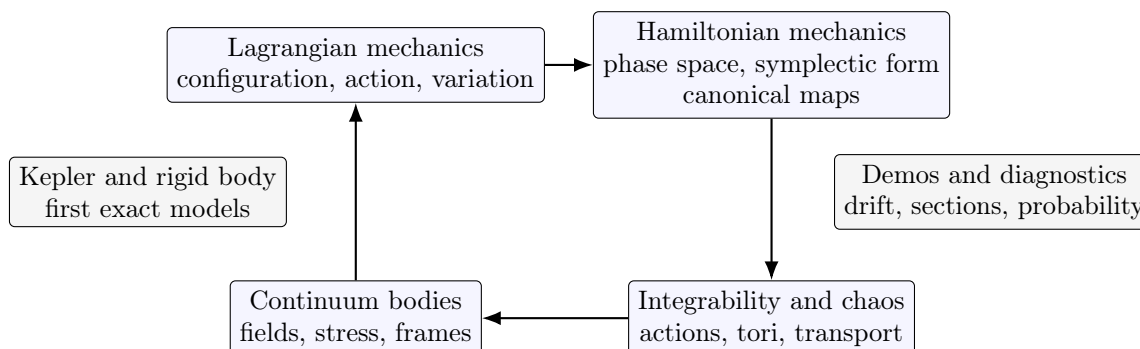


Figure 1.1: The course loop. The central theories are connected by examples and computational diagnostics: exact models teach the geometry, while numerical demos test which structures survive discretization or perturbation.

The arrows are only a first tour. Terms such as phase space, invariant torus, transport, and stress acquire their precise meanings in the chapters where they are used; here they mark the order in which the reader will meet them.

1.5 Units And Dimensions

Dimensional consistency is a constraint, not a convention. In celestial mechanics demos we often use astronomical units, years, and solar masses, so

$$G = 4\pi^2$$

when the central mass is measured in solar masses. In rigid body mechanics, moments of inertia have dimensions ML^2 , angular velocity has dimension T^{-1} , angular momentum has dimension

ML^2T^{-1} , and kinetic energy has dimension ML^2T^{-2} . Outside the celestial-mechanics simulations the notes keep dimensional constants symbolic unless a nondimensional scale choice is stated locally. This convention makes every displayed equation checkable by units before it is interpreted dynamically.

1.6 How To Read Derivations, Figures, And Laboratories

Figures

Figures in the notes are schematic unless explicitly generated from a numerical demo. Their purpose is to identify variables, spaces, and geometric relations. For quantitative plots, the corresponding Python or Mathematica script should be treated as part of the figure caption: the model assumptions and numerical parameters matter.

Derivations

Read long derivations in mechanics on two levels. The first level is formal: differentiate an action, apply a balance law, invert a Legendre transform, or change coordinates. The second level is structural: identify the conserved quantity, the reduced phase space, the constraint force, or the connection hidden in the algebra. The notes spell out both layers whenever the structural level is easy to miss.

To read each derivation, keep a small ledger:

input	output
model assumptions	equation of motion or constraint
symmetry	conserved quantity or reduced variable
constitutive law	closed continuum PDE
choice of frame	connection and curvature

Many mistakes in mechanics are ledger mistakes: using an equation outside the assumptions that produced it, changing a frame without transforming the components, or interpreting a reduced equation as if no reduction had occurred.

When a derivation contains a local coordinate calculation, the conclusion should be read as coordinate-independent only after the geometric object has been identified. For example, the symbols Γ_{ij}^ℓ , A_A , and ω^a_b are coordinate or frame components of, respectively, a connection, a principal connection, and torsion/curvature forms. Constraint multipliers are slightly different: the scalars λ_a become a reaction covector only after being paired with the constraint one-forms, $\lambda_a d\Phi^a$.

A useful test is to cover the displayed equations and summarize the argument in sentences. When the prose summary breaks down, return to the model, space, principle, and interpretation before trying more algebra. Long derivations in these notes are meant to be reconstructible from those four pieces.

Laboratories

The worked examples, exercises, and computational scripts are placed at the ends of the chapters whose ideas they test. A laboratory is where a formalism proves that it can survive contact with

a specific model: a coordinate change, a constraint force, a resonance, a pressure multiplier, a finite-time ejection criterion, or a gauge holonomy.

Laboratories are checkpoints for the reading rule: state the model, name the variables, derive the diagnostic, and interpret what the diagnostic measures. For teaching, this gives three layers inside each example: local conventions and assumptions, the derivation of the diagnostic, and the interpretation. Stability, bifurcation, transport, reaction force, pressure projection, ejection, and holonomy are interpretations only after the model and diagnostic have been specified.

In laboratory sections, figures introduce the geometry before the algebra and should be checked again after the final boxed result. A figure earns its place by making a specific equation, reduction, boundary condition, or diagnostic criterion visible.

Problem-To-Demo Conversion Rules

Every computational laboratory in this repository should record the following metadata:

model + variables + parameters + units + invariants + numerical method.

For Hamiltonian demos, the minimum checks are

$\Delta H(t)$, $\Delta J(t)$ if angular momentum is conserved, step-size sensitivity.

For probabilistic demos such as asteroid ejection, the output must also state

sampling distribution, random seed, time horizon, ejection criterion.

These entries make a numerical picture into a reproducible mechanics experiment. The same rule applies in reverse when reading a demo: the code should reveal the equations it integrates, the units in which parameters are measured, the events it declares, and the invariants it monitors. A plot closes the argument after the model and diagnostics have been specified.

Chapter 2

Lagrangian Mechanics

2.1 Mechanical Data

A mechanics problem begins by naming its space of possible configurations. Once that space is named, the rest of the theory can be formulated independently of any particular coordinate chart.

A pendulum moves on a circle, not on a line. A double pendulum lives on a torus. A rigid body lives on $\text{SO}(3)$, not in a box of nine independent matrix entries. Choosing the configuration space correctly is therefore not a matter of taste; it is the first physical idealization. If the chosen space contains redundant coordinates, constraints must later repair that redundancy. If the chosen space already reflects the physical degrees of freedom, the variational principle can act directly on the geometry of the problem.

Definition 2.1 (Finite-dimensional mechanical system). *A regular finite-dimensional Lagrangian mechanical system consists of:*

1. *a smooth n -dimensional configuration manifold Q ;*
2. *a time interval $I = [t_0, t_1]$;*
3. *a Lagrangian $L : TQ \times I \rightarrow \mathbb{R}$;*
4. *a class of admissible curves $q : I \rightarrow Q$, possibly satisfying constraints.*

The action of an admissible curve is

$$S[q] = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t) dt.$$

Assumption 2.1 (Coordinate patch for derivations). *Derivations in this chapter are written in one coordinate chart $(q^1, \dots, q^n)$. The equations obtained are tensorial or intrinsic when the objects used to build them are intrinsic. Coordinates are a calculation device, not an additional physical structure.*

Roadmap. The chapter has three jobs. First, it derives the Euler-Lagrange equations from Hamilton's principle, including boundary terms, constraints, and Noether charges. Second, it treats Kepler motion and the rigid body as examples in which those ideas control a complete

calculation. Third, it prepares the boundary momentum and energy concepts that become phase-space coordinates in Hamiltonian mechanics. The variational principle separates coordinate choices from geometric structure and places forces, constraints, and conserved quantities in a common framework.

The primary object here is a path in configuration space, and the variational principle selects the physical path. The phase-space viewpoint is developed in Chapter 3. In that order, momenta and symplectic forms appear as consequences of the endpoint variation and the Legendre transform.

Derivation checkpoints. Whenever this chapter differentiates an action, check three things before following the algebra: which variables are varied, which endpoint conditions are imposed, and which boundary terms survive. Whenever it invokes a symmetry, check whether the symmetry acts on configuration space and whether the action is invariant up to an allowed boundary term. These checks are small, but they prevent the most common mistakes: using an inadmissible virtual displacement or calling a quantity conserved before verifying the symmetry hypothesis.

2.2 Variations And The Euler-Lagrange Equations

What does it mean for the action to be stationary? It means that the first derivative of $S[q_\varepsilon]$ vanishes for every admissible infinitesimal change of the path.

Remark 2.1 (Stationary, not necessarily least). *Hamilton's principle, formulated in 1834–1835, is a critical-point principle for the time integral of $L = T - V$. The older phrase “principle of least action” refers to a different functional: the action at fixed energy, due to Maupertuis (1744) and Euler. In modern Lagrangian mechanics the action need not be a minimum; it may be a saddle or a maximum for some classes of variations. The catenoid soap film is the familiar variational counterexample: a stationary surface need not be the minimizing surface once the boundary data are large enough. The word stationary is therefore the accurate word: the first variation vanishes.*

Let $q_\varepsilon(t)$ be a smooth one-parameter family of curves with $q_0(t) = q(t)$. The variation vector field along q is

$$\eta(t) = \left. \frac{\partial q_\varepsilon(t)}{\partial \varepsilon} \right|_{\varepsilon=0}, \quad \eta^i(t) = \left. \frac{\partial q_\varepsilon^i(t)}{\partial \varepsilon} \right|_{\varepsilon=0}.$$

For fixed-endpoint variations we impose

$$\eta(t_0) = 0, \quad \eta(t_1) = 0.$$

The variational calculation below has one essential idea. The path $q(t)$ is a critical point of the action when every infinitesimal deformation of the path changes the action only at second order. Differentiating the action produces a term proportional to $\dot{\eta}$; integration by parts moves the derivative onto the momentum, and the endpoint condition kills the boundary term. What is left must vanish for every interior virtual displacement η , which forces the Euler-Lagrange equations pointwise in time.

Derivation 2.1 (Euler-Lagrange equations). *The first variation of the action is*

$$\begin{aligned} \delta S &= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{t_0}^{t_1} L(q_\varepsilon, \dot{q}_\varepsilon, t) dt \\ &= \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q^i} \eta^i + \frac{\partial L}{\partial \dot{q}^i} \dot{\eta}^i \right) dt. \end{aligned}$$

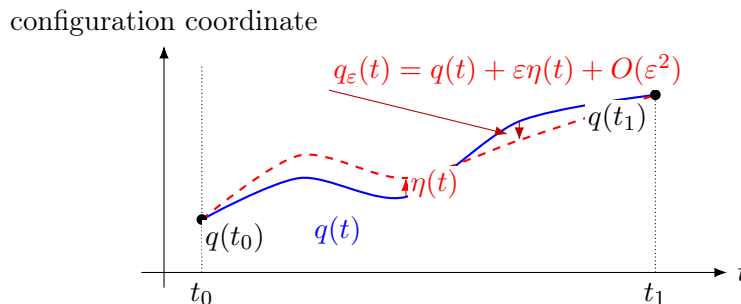


Figure 2.1: A fixed-endpoint variation changes the path in the interior while holding the initial and final configurations fixed. The infinitesimal change is the vector field $\eta(t) = \delta q(t)$ along the path.

Integrate the second term by parts:

$$\int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}^i} \dot{\eta}^i dt = \left[\frac{\partial L}{\partial \dot{q}^i} \eta^i \right]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \eta^i dt.$$

The boundary term vanishes because $\eta(t_0) = \eta(t_1) = 0$. Therefore

$$\delta S = \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q^i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \right] \eta^i dt.$$

The fundamental lemma of the calculus of variations says that if this integral vanishes for all compactly supported variations η , then the coefficient of every η^i vanishes. Hence

$$\boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = 0}.$$

Fundamental lemma used above. Under the standing smoothness assumptions on L , q , and the variation, the bracketed coefficient in the variation formula is continuous. The lemma therefore applies with f equal to that coefficient. Let $f(t)$ be continuous on (t_0, t_1) , and suppose

$$\int_{t_0}^{t_1} f(t) \eta(t) dt = 0$$

for every smooth compactly supported test function η . If $f(t_*) > 0$ at some interior point, continuity gives an interval $[a, b]$ on which $f \geq f(t_*)/2$. Choose a nonnegative smooth bump η supported in $[a, b]$ and not identically zero. Then the integral is positive, a contradiction. Applying the same argument to $-\eta$ rules out $f(t_*) < 0$. Thus $f = 0$ pointwise. Applying this to each component of the bracketed coefficient gives the Euler-Lagrange equations. $\square$

The Euler-Lagrange equation is local in time because the variations are arbitrary in the interior of the interval. A single global statement, $\delta S = 0$, becomes a pointwise differential equation because $\eta(t)$ can be chosen independently near each time. Boundary conditions and endpoint terms are part of the variational data: they decide which variations are admissible and which physical quantities appear at the endpoints.

For a regular Lagrangian, the velocity Hessian

$$W_{ij}(q, \dot{q}, t) = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$$

is nonsingular. The Euler-Lagrange equations can then be solved locally for $\ddot{q}^i$ as smooth functions of $(q, \dot{q}, t)$. Standard ODE existence and uniqueness therefore gives a unique local solution for each initial state $(q(t_0), \dot{q}(t_0))$, as long as the solution remains in the chosen chart and no singularity is reached.

2.3 Boundary Terms And Forces

Integration by parts also defines momentum. Without imposing fixed endpoints, the boundary contribution is

$$[p_i \eta^i]_{t_0}^{t_1}, \quad p_i = \frac{\partial L}{\partial \dot{q}^i}.$$

Thus p_i is the coefficient of endpoint displacement in the first variation of the action. This boundary-term definition reduces to “mass times velocity” only in special coordinates for special Lagrangians.

If nonconservative generalized forces $Q_i(q, \dot{q}, t)$ are present, the Lagrange-d'Alembert principle adds virtual work to Hamilton's principle. This is the variational form of d'Alembert's 1743 virtual-work principle, later absorbed into Lagrange's analytic mechanics. The admissible virtual displacements are the same ones used in the variational problem, and the external force does virtual work $Q_i \eta^i$. Thus

$$\delta S + \int_{t_0}^{t_1} Q_i \eta^i dt = 0,$$

which gives

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = Q_i.$$

This formula is also the bridge to continua, where the virtual work is supplied by tractions, pressure forces, or dissipative stresses.

The generalized force Q_i is a covector along the path, not necessarily a Euclidean force vector. In Cartesian coordinates for a particle the distinction is invisible, but in generalized coordinates $Q_i \eta^i$ is the virtual work done by the force under the virtual displacement η . This is why the same variational formula can describe torques, constraint reactions, pressure work, and elastic boundary tractions once the correct configuration space has been named.

2.4 Natural Lagrangians And The Metric Form

Many mechanical systems have a Lagrangian of the form

$$L(q, \dot{q}) = T(q, \dot{q}) - V(q) = \frac{1}{2} g_{ij}(q) \dot{q}^i \dot{q}^j - V(q),$$

where g_{ij} is a positive-definite mass metric on Q , and V is potential energy.

Remark 2.2 (Terminology). *Here “natural” is conventional terminology: the kinetic energy is quadratic in velocity and the remaining term is a potential energy on configuration space.*

The word “metric” here has physical content. It tells us how much kinetic energy a given velocity has, and it may depend on configuration. For a point particle in Cartesian coordinates $g_{ij} = m\delta_{ij}$. For a pendulum, double pendulum, or rigid body written in angles, the same kinetic energy becomes a metric on the configuration space. Christoffel symbols can appear for two distinct reasons that should not be confused. In flat Euclidean space written in curvilinear coordinates they are coordinate inertial terms; on a genuinely curved configuration space they also encode the curvature of the mass metric. In both cases they are not new applied forces.

Derivation 2.2 (Geodesic-plus-force equation). *For the natural Lagrangian,*

$$\frac{\partial L}{\partial \dot{q}^i} = g_{ij} \dot{q}^j,$$

so

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} = g_{ij} \ddot{q}^j + \frac{\partial g_{ij}}{\partial q^k} \dot{q}^k \dot{q}^j.$$

Also

$$\frac{\partial L}{\partial q^i} = \frac{1}{2} \frac{\partial g_{jk}}{\partial q^i} \dot{q}^j \dot{q}^k - \frac{\partial V}{\partial q^i}.$$

The Euler-Lagrange equation becomes

$$g_{ij} \ddot{q}^j + \left(\frac{\partial g_{ij}}{\partial q^k} - \frac{1}{2} \frac{\partial g_{jk}}{\partial q^i} \right) \dot{q}^k \dot{q}^j = - \frac{\partial V}{\partial q^i}.$$

After symmetrizing the velocity terms and raising an index with $g^{\ell i}$,

$$\boxed{\ddot{q}^\ell + \Gamma_{jk}^\ell \dot{q}^j \dot{q}^k = -g^{\ell i} \frac{\partial V}{\partial q^i}},$$

where

$$\Gamma_{jk}^\ell = \frac{1}{2} g^{\ell i} \left(\frac{\partial g_{ij}}{\partial q^k} + \frac{\partial g_{ik}}{\partial q^j} - \frac{\partial g_{jk}}{\partial q^i} \right)$$

are the Christoffel symbols of the mass metric.

Thus even “force-free” motion on a curved configuration space need not look like $\ddot{q} = 0$ in coordinates. The Christoffel terms are inertial terms created by the geometry of Q and the chosen coordinates. The invariant content is that the covariant acceleration equals the metric gradient of the potential:

$$\nabla_{\dot{q}} \dot{q} = -\text{grad}_g V.$$

This is the coordinate-free form of the boxed equation.

2.5 Holonomic Constraints

A holonomic constraint is a condition

$$\Phi^a(q, t) = 0, \quad a = 1, \dots, k,$$

that cuts out an admissible submanifold of configuration space. If one works in the ambient coordinates q^i , admissible variations must preserve the constraint to first order:

$$\frac{\partial \Phi^a}{\partial q^i} \eta^i = 0.$$

The Euler-Lagrange residual therefore has to vanish only on vectors tangent to the constraint surface. The multiplier rule, in the form Lagrange used in *Mecanique analytique* (1788), says that a covector vanishing on this tangent space is a linear combination of the constraint gradients. The linear-algebra statement is this: when the k one-forms $d\Phi^a$ are independent, the tangent space is

$$T_q C = \{\eta : d\Phi^a(\eta) = 0, a = 1, \dots, k\},$$

and its annihilator is

$$(T_q C)^0 = \text{span}\{d\Phi^1, \dots, d\Phi^k\}.$$

Indeed, the map $A : T_q Q \rightarrow \mathbb{R}^k$, $A(\eta) = (d\Phi^1(\eta), \dots, d\Phi^k(\eta))$, has rank k , so its dual map $A^*(\lambda) = \lambda_a d\Phi^a$ has image equal to the annihilator of $\ker A = T_q C$. This is the finite-dimensional fact behind the multiplier rule. Thus constraint regularity means the gradients of the constraints have constant rank k on the constraint surface. Under this assumption the constrained equations take the multiplier form

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = \lambda_a \frac{\partial \Phi^a}{\partial q^i}.$$

The multipliers λ_a are not arbitrary forces. They are whatever constraint reactions are needed to keep the motion on $\Phi^a = 0$. If instead one parametrizes the constraint surface directly, the λ_a disappear and the same physics is expressed with fewer coordinates.

2.6 Symmetry And Noether's Theorem

Continuous symmetries of the action produce conserved quantities. Noether's 1918 theorem is the precise statement: every one-parameter group of symmetries gives a conserved Noether charge. A symmetry is a transformation of the configuration space that leaves the action unchanged, or changes it only by an endpoint term. The infinitesimal version of that transformation is a vector field ξ_Q along which the action has zero interior first variation. The conserved quantity is the momentum paired with that vector field, corrected by the endpoint term when one is present.

Definition 2.2 (Infinitesimal generator). *Let a Lie group G act on Q . For $\xi \in \mathfrak{g}$, the infinitesimal generator is the vector field*

$$\xi_Q(q) = \left. \frac{d}{ds} \right|_{s=0} \exp(s\xi) \cdot q.$$

In coordinates, write $\xi_Q = \xi_Q^i(q) \partial / \partial q^i$.

Assumption 2.2 (Lagrangian symmetry). *The action of G is a symmetry if*

$$L(g \cdot q, T_q g \cdot \dot{q}, t) = L(q, \dot{q}, t)$$

for all $g \in G$, $q \in Q$, and $\dot{q} \in T_q Q$.

Some important symmetries preserve the action only up to an endpoint term. If the infinitesimal variation of the Lagrangian has the form

$$\delta_\xi L = \frac{d}{dt} B_\xi(q, t),$$

then the action changes by $B_\xi(t_1) - B_\xi(t_0)$, so fixed-endpoint variations still give the same equations of motion. The conserved Noether quantity is then

$$J_\xi = p_i \xi_Q^i - B_\xi.$$

Galilean boosts of a free particle are the standard example: the Lagrangian is not strictly invariant, but its change is a total time derivative, and the boost charge includes the corresponding boundary term.

Proposition 2.1 (Noether theorem). *For every $\xi \in \mathfrak{g}$, the quantity*

$$J_\xi(q, \dot{q}) = \frac{\partial L}{\partial \dot{q}^i} \xi_Q^i(q) - B_\xi(q, t) = p_i \xi_Q^i(q) - B_\xi(q, t)$$

is conserved along solutions of the Euler-Lagrange equations whenever $\delta_\xi L = dB_\xi/dt$. For a strict symmetry, $B_\xi = 0$.

Proof. Differentiate the symmetry condition along $g = \exp(s\xi)$ at $s = 0$:

$$\frac{dB_\xi}{dt} = \frac{\partial L}{\partial q^i} \xi_Q^i + \frac{\partial L}{\partial \dot{q}^i} \frac{d}{dt} \xi_Q^i(q(t)).$$

Along an Euler-Lagrange solution,

$$\frac{\partial L}{\partial q^i} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i}.$$

Substitute this into the previous equation:

$$\frac{dB_\xi}{dt} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \xi_Q^i + \frac{\partial L}{\partial \dot{q}^i} \frac{d}{dt} \xi_Q^i = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \xi_Q^i \right).$$

Therefore $d(p_i \xi_Q^i - B_\xi)/dt = 0$. □

Example 2.1 (Galilean boost as a boundary-term symmetry). *For a free particle on the line,*

$$L = \frac{1}{2} m \dot{x}^2.$$

An infinitesimal boost with parameter v changes the path by $\delta x = vt$ and $\delta \dot{x} = v$. Thus

$$\delta L = m \dot{x} v = \frac{d}{dt} (mvx).$$

Here $B = mvx$, and the Noether charge is

$$J = pvt - mvx = mv(\dot{x}t - x).$$

Removing the arbitrary boost parameter gives the conserved center-of-mass quantity $m(x - \dot{x}t)$. The Lagrangian was not invariant as a function, but the action changed only by an endpoint term.

Example 2.2 (Translation and rotation). *For a particle in $\mathbb{R}^3$ with $L = \frac{1}{2} m \|\dot{x}\|^2 - V(x)$, translation symmetry in direction a gives $J_a = m\dot{x} \cdot a$. If V is invariant under all translations, linear momentum is conserved. Rotational symmetry about axis a gives*

$$J_a = (x \times m\dot{x}) \cdot a,$$

so angular momentum is conserved when V is rotationally invariant.

2.7 Basic Example: Kepler Motion From Central Forces

Kepler motion shows how cyclic coordinates, Noether's theorem, and reduction by symmetry turn a force law into the geometry of conic sections.

Assumption 2.3 (Kepler model). *We study a particle of reduced mass μ moving in $\mathbb{R}^3 \setminus \{0\}$ under an attractive inverse-square central potential. The Lagrangian is*

$$L(r, \dot{r}) = \frac{1}{2}\mu \|\dot{r}\|^2 + \frac{k}{\|r\|}, \quad k = Gm_1m_2 > 0.$$

The coordinate $r = x_1 - x_2$ is the relative separation of the two bodies.

Rotational invariance gives conservation of angular momentum

$$J = r \times p = \mu r \times \dot{r}.$$

Since $J \cdot r = 0$ and $J \cdot \dot{r} = 0$, the motion stays in the plane perpendicular to J . In polar coordinates (ρ, θ) in that plane,

$$L = \frac{1}{2}\mu(\dot{\rho}^2 + \rho^2\dot{\theta}^2) + \frac{k}{\rho}.$$

The angle θ is cyclic, hence

$$\ell = \frac{\partial L}{\partial \dot{\theta}} = \mu\rho^2\dot{\theta}$$

is conserved. The energy is

$$E = \frac{1}{2}\mu\dot{\rho}^2 + \frac{\ell^2}{2\mu\rho^2} - \frac{k}{\rho}.$$

Thus the radial motion is one-dimensional motion in the effective potential

$$V_{\text{eff}}(\rho) = \frac{\ell^2}{2\mu\rho^2} - \frac{k}{\rho}.$$

Here ℓ is the ordinary, massful angular momentum of the relative two-body problem. Section 2.10.2 gives the bookkeeping conversion to the specific angular momentum used in the asteroid chapter.

Derivation 2.3 (Orbit equation). *Set $u = 1/\rho$. Since $\ell = \mu\rho^2\dot{\theta}$,*

$$\frac{d}{dt} = \dot{\theta} \frac{d}{d\theta} = \frac{\ell}{\mu} u^2 \frac{d}{d\theta}.$$

The radial Euler-Lagrange equation is

$$\mu(\ddot{\rho} - \rho\dot{\theta}^2) = -\frac{k}{\rho^2}.$$

In terms of $u(\theta)$ this becomes

$$\frac{d^2u}{d\theta^2} + u = \frac{\mu k}{\ell^2}.$$

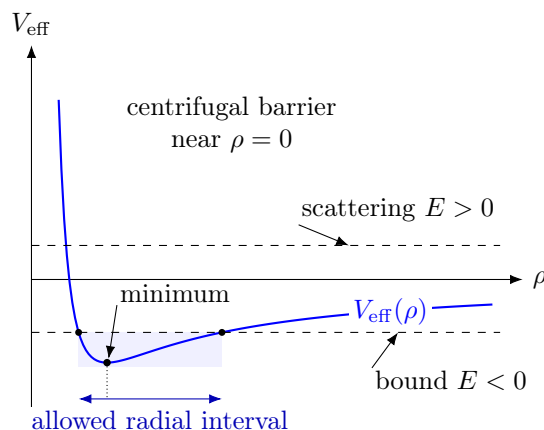


Figure 2.2: The Kepler problem reduces to one-dimensional radial motion in an effective potential. For a fixed negative energy, the radial coordinate is trapped between two turning points; for positive energy there is only a closest approach before scattering.

Therefore

$$u(\theta) = \frac{\mu k}{\ell^2} [1 + e \cos(\theta - \theta_0)],$$

equivalently

$$\rho(\theta) = \frac{\ell^2 / (\mu k)}{1 + e \cos(\theta - \theta_0)}.$$

The parameter e is the eccentricity. The conic is an ellipse, parabola, or hyperbola according as $e < 1$, $e = 1$, or $e > 1$.

The Kepler problem also has the Laplace-Runge-Lenz vector

$$A = p \times J - \mu k \frac{r}{\|r\|},$$

which is conserved and points toward periapse. The conservation is derived in Section 2.10.2; here it is introduced as the extra quantity that explains why the conic orientation does not precess in the ideal inverse-square problem.

Remark 2.3 (Bertrand's theorem and Kepler's hidden symmetry). *Bertrand's theorem (1873) gives the sharp structural statement: among central force problems, only the inverse-square potential and the harmonic oscillator make all bounded orbits closed. The Laplace-Runge-Lenz vector is the inverse-square instance of this extra symmetry; in the negative-energy Kepler problem it enlarges the rotational $SO(3)$ symmetry to the hidden $SO(4)$ symmetry of bound motion. The vector itself goes back to Laplace's celestial-mechanics work; Pauli (1926) used it in the hydrogen-atom calculation, and Fock (1935) gave the momentum-sphere interpretation that makes the $SO(4)$ explicit.*

2.8 Basic Example: Rigid Body As A Lagrangian System

The rigid body is the first example in these notes whose configuration space is a nonlinear Lie group. At the Lagrangian level, this already forces us to distinguish coordinates, frames, angular

velocity, and kinetic energy. Later chapters return to the same system as a reduced Hamiltonian flow and as an integrable model, but the essential geometric data are visible already here. The reduced equations derived below were first written by Euler in 1758.

Assumption 2.4 (Free rigid body). *A material point X in the body frame is located at*

$$x(t) = R(t)X, \quad R(t) \in \text{SO}(3),$$

with the center of mass fixed at the origin. The body is rigid, so all dynamics is in the attitude $R(t)$. No external torque acts.

The body angular velocity is defined by

$$R^{-1}\dot{R} = \widehat{\Omega}, \quad \widehat{\Omega}v = \Omega \times v.$$

The kinetic energy is

$$L(R, \dot{R}) = T = \frac{1}{2}\Omega^T I \Omega,$$

where I is the inertia tensor in the body frame. In principal axes,

$$I = \text{diag}(I_1, I_2, I_3), \quad L = \frac{1}{2}(I_1\Omega_1^2 + I_2\Omega_2^2 + I_3\Omega_3^2).$$

Derivation 2.4 (Euler equation from constrained variations). *A variation inside $\text{SO}(3)$ has the form*

$$\delta R = R\widehat{\Sigma},$$

where $\Sigma(t_0) = \Sigma(t_1) = 0$. This is the left-trivialized form of a tangent vector to $\text{SO}(3)$: multiplication by R^{-1} identifies $T_R \text{SO}(3)$ with $\mathfrak{so}(3)$. This implies

$$\delta\Omega = \dot{\Sigma} + \Omega \times \Sigma.$$

Indeed, differentiating $R^T R = I$ gives

$$R^T \delta R + (\delta R)^T R = 0,$$

so $R^T \delta R \in \mathfrak{so}(3)$, which is why $\delta R = R\widehat{\Sigma}$. Then

$$\delta(R^T \dot{R}) = \delta R^T \dot{R} + R^T \delta \dot{R} = -\widehat{\Sigma}\widehat{\Omega} + \widehat{\dot{\Sigma}} + \widehat{\Omega}\widehat{\Sigma} = \widehat{\dot{\Sigma}} + [\widehat{\Omega}, \widehat{\Sigma}],$$

and the hat map satisfies

$$[\widehat{\Omega}, \widehat{\Sigma}] = \widehat{\Omega \times \Sigma}.$$

On a test vector v , this identity is

$$[\widehat{\Omega}, \widehat{\Sigma}]v = \Omega \times (\Sigma \times v) - \Sigma \times (\Omega \times v) = (\Omega \times \Sigma) \times v,$$

which is the Jacobi identity for the cross product written as a commutator. This is the stated formula under the hat isomorphism. Let $M = \partial L / \partial \Omega = I\Omega$. Then

$$\begin{aligned} \delta S &= \int M \cdot (\dot{\Sigma} + \Omega \times \Sigma) dt \\ &= \int (-\dot{M} + M \times \Omega) \cdot \Sigma dt. \end{aligned}$$

Stationarity gives

$$\dot{M} = M \times \Omega,$$

or, with the hat convention $\hat{\Omega}v = \Omega \times v$ used throughout,

$$\dot{M} + \Omega \times M = 0.$$

In principal axes this is

$$I_1 \dot{\Omega}_1 = (I_2 - I_3) \Omega_2 \Omega_3,$$

$$I_2 \dot{\Omega}_2 = (I_3 - I_1) \Omega_3 \Omega_1,$$

$$I_3 \dot{\Omega}_3 = (I_1 - I_2) \Omega_1 \Omega_2.$$

2.9 What To Take Forward

- Hamilton's principle starts from a configuration manifold Q , paths $q(t) \in Q$, and an action $S = \int L dt$. Stationarity gives the Euler-Lagrange equations, and the endpoint term defines conjugate momentum.
- Natural Lagrangians encode kinetic energy as a mass metric. Forces, constraints, and nonconservative virtual work enter as covectors, multipliers, or potentials, depending on how they act on virtual displacements.
- Noether's theorem turns continuous symmetries into conserved quantities. Kepler motion and the rigid body show how these quantities organize exact solutions and prepare the Legendre transform to Hamiltonian phase space.

2.10 Chapter-End Laboratories And Exercises

The laboratories below test the Lagrangian ideas of this chapter: coordinate independence, symmetry reduction, central-force stability, time-dependent Lagrangians, effective potentials, and contact forces.

2.10.1 Form-Invariance Of Lagrange Equations

We begin with the coordinate-independence of the Euler-Lagrange equations. Let

$$q^i = q^i(s^1, \dots, s^n, t)$$

be a regular time-dependent point transformation. Regular means that the Jacobian $\partial q^i / \partial s^a$ has rank n . The transformed Lagrangian is

$$L'(s, \dot{s}, t) = L(q(s, t), \dot{q}(s, \dot{s}, t), t),$$

where

$$\dot{q}^i = \frac{\partial q^i}{\partial s^a} \dot{s}^a + \frac{\partial q^i}{\partial t}.$$

Two elementary identities do the work:

$$\frac{\partial \dot{q}^i}{\partial \dot{s}^a} = \frac{\partial q^i}{\partial s^a}, \quad \frac{d}{dt} \left(\frac{\partial q^i}{\partial s^a} \right) = \frac{\partial \dot{q}^i}{\partial s^a}.$$

Using the chain rule,

$$\frac{\partial L'}{\partial \dot{s}^a} = \frac{\partial L}{\partial \dot{q}^i} \frac{\partial q^i}{\partial s^a},$$

and

$$\frac{\partial L'}{\partial s^a} = \frac{\partial L}{\partial q^i} \frac{\partial q^i}{\partial s^a} + \frac{\partial L}{\partial \dot{q}^i} \frac{\partial \dot{q}^i}{\partial s^a}.$$

Therefore

$$\frac{d}{dt} \frac{\partial L'}{\partial \dot{s}^a} - \frac{\partial L'}{\partial s^a} = \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} \right) \frac{\partial q^i}{\partial s^a}.$$

If $q(t)$ satisfies the Euler-Lagrange equations, the right side vanishes; if the transformation is regular, the converse holds as well. Thus the Euler-Lagrange equation is not tied to a preferred coordinate system.

2.10.2 Central Forces From First Principles

The central-force problem is the prototype of reduction by symmetry. Rotational invariance gives conservation of angular momentum; angular momentum confines the motion to a plane; polar coordinates in that plane reduce the dynamics to a one-dimensional radial equation with an effective potential. The later Kepler and precession calculations are refinements of this same reduction.

The lesson is broader than celestial mechanics. A symmetry gives a conserved quantity; the conserved quantity can remove a variable; the reduced equation then describes motion on a smaller space. Central forces are a transparent first example because the geometry is visible in ordinary Euclidean space.

Let a particle of mass m move in a central potential $V(r)$, where

$$r = \|x\|, \quad x \in \mathbb{R}^3 \setminus \{0\}.$$

The Lagrangian is

$$L = \frac{1}{2} m \|\dot{x}\|^2 - V(r).$$

Rotational invariance gives angular momentum

$$J = x \times p = m x \times \dot{x}.$$

Since

$$\frac{dJ}{dt} = \dot{x} \times m \dot{x} + x \times m \ddot{x} = x \times \left(-V'(r) \frac{x}{r} \right) = 0,$$

the vector J is constant. Because $J \cdot x = 0$ and $J \cdot \dot{x} = 0$, the motion lies in the plane perpendicular to J . Use polar coordinates (r, θ) in this plane:

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r).$$

The angle θ is cyclic, so

$$\ell = \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

is conserved. The energy is

$$E = \frac{1}{2} m \dot{r}^2 + \frac{\ell^2}{2 m r^2} + V(r).$$

Thus the radial motion is one-dimensional motion in

$$V_{\text{eff}}(r) = V(r) + \frac{\ell^2}{2mr^2}.$$

Here ℓ is the massful angular momentum. To compare with the specific-energy Kepler convention used in Section 5.2, divide the Hamiltonian, energy, and angular momentum by the particle mass; then $\ell_{\text{spec}} = \ell/m = r^2\dot{\theta}$. The orbit equation is unchanged after this bookkeeping conversion, provided k/m is read as the gravitational parameter GM .

Binet Equation

Let

$$u(\theta) = \frac{1}{r}.$$

Since

$$\dot{\theta} = \frac{\ell}{mr^2} = \frac{\ell}{m}u^2,$$

we have

$$\frac{d}{dt} = \frac{\ell}{m}u^2 \frac{d}{d\theta}.$$

Also

$$\dot{r} = \frac{d}{dt}(u^{-1}) = -u^{-2}\dot{u} = -\frac{\ell}{m}u',$$

where a prime denotes $d/d\theta$. Differentiating once more,

$$\ddot{r} = -\frac{\ell}{m} \frac{du'}{dt} = -\frac{\ell^2}{m^2}u^2u''.$$

The radial equation is

$$m(\ddot{r} - r\dot{\theta}^2) = F(r), \quad F(r) = -V'(r).$$

Substitute the formulas above:

$$m \left(-\frac{\ell^2}{m^2}u^2u'' - \frac{1}{u} \frac{\ell^2}{m^2}u^4 \right) = -\frac{\ell^2}{m}u^2(u'' + u) = F(1/u).$$

Therefore

$$\boxed{u'' + u = -\frac{m}{\ell^2 u^2} F(1/u).}$$

This is Binet's equation. It turns a central-force problem into a differential equation for the orbit shape $r(\theta)$.

Kepler Conics And The Laplace-Runge-Lenz Vector

For the Kepler potential

$$V(r) = -\frac{k}{r}, \quad k > 0,$$

the force is

$$F(r) = -\frac{k}{r^2}.$$

Binet's equation becomes

$$u'' + u = \frac{mk}{\ell^2}.$$

The solution is

$$u(\theta) = \frac{mk}{\ell^2} [1 + e \cos(\theta - \theta_0)],$$

equivalently

$$r(\theta) = \frac{\ell^2/(mk)}{1 + e \cos(\theta - \theta_0)}.$$

This is a conic section with focus at the force center.

The extra conserved vector is

$$A = p \times J - mk \frac{x}{r}.$$

To verify conservation, write $\hat{r} = x/r$. Since J is constant,

$$\dot{A} = \dot{p} \times J - mk \dot{\hat{r}}.$$

The force law gives

$$\dot{p} = -\frac{k}{r^2} \hat{r}.$$

Use

$$J = mx \times \dot{x} = mr^2 \dot{\theta} \hat{z} = \ell \hat{z}$$

in the orbital plane. Then

$$\dot{\hat{r}} = \dot{\theta} \hat{\theta},$$

and

$$\dot{p} \times J = -\frac{k}{r^2} \hat{r} \times (\ell \hat{z}) = \frac{k\ell}{r^2} \hat{\theta} = mk \dot{\theta} \hat{\theta} = mk \dot{\hat{r}}.$$

Hence

$$\dot{A} = 0.$$

The vector A points toward periapse. Taking the dot product with x ,

$$A \cdot x = (p \times J) \cdot x - mkr.$$

By cyclic permutation,

$$(p \times J) \cdot x = J \cdot (x \times p) = J \cdot J = \ell^2.$$

Therefore

$$Ar \cos(\theta - \theta_0) = \ell^2 - mkr,$$

which rearranges to

$$r = \frac{\ell^2/(mk)}{1 + (A/(mk)) \cos(\theta - \theta_0)}.$$

Thus the eccentricity is

$$e = \frac{\|A\|}{mk}.$$

The conic equation and the hidden conservation law are the same fact written in two languages.

Small Periapse Precession From A $1/r^3$ Perturbation

A useful perturbation of the Kepler potential is

$$V(r) = -\frac{k}{r} - \frac{\epsilon}{r^3}, \quad 0 < |\epsilon| \ll 1.$$

This is a pedagogical proxy for any small correction that breaks the exact inverse-square symmetry while remaining central. The Schwarzschild laboratory below gives the relativistic source of a closely related leading precession effect. The radial force is

$$F(r) = -V'(r) = -\frac{k}{r^2} - \frac{3\epsilon}{r^4}.$$

Binet's equation gives

$$u'' + u = \frac{mk}{\ell^2} + \frac{3m\epsilon}{\ell^2}u^2.$$

For a nearly circular orbit, write

$$u(\theta) = u_0 + \eta(\theta), \quad |\eta| \ll u_0.$$

The constant u_0 solves

$$u_0 = \frac{mk}{\ell^2} + \frac{3m\epsilon}{\ell^2}u_0^2.$$

Linearizing in η ,

$$\eta'' + \eta = \frac{6m\epsilon u_0}{\ell^2}\eta,$$

where terms of order $\epsilon\eta^2$ and higher are discarded. The small quantity η/u_0 measures the radial oscillation amplitude, while ϵ measures the departure from the Kepler force law; the displayed linear equation keeps the terms needed for the leading frequency shift. Equivalently,

$$\eta'' + \kappa^2\eta = 0, \quad \kappa^2 = 1 - \frac{6m\epsilon u_0}{\ell^2}.$$

The radial oscillation has angular period

$$\Delta\theta_r = \frac{2\pi}{\kappa}.$$

The periapse advance per radial cycle is the excess over 2π :

$$\Delta\varpi = 2\pi \left(\frac{1}{\kappa} - 1 \right) \approx \frac{6\pi m\epsilon u_0}{\ell^2}.$$

Thus the $1/r^3$ correction breaks the closed Kepler ellipse and produces precession. The same algebraic structure appears in general relativity. Einstein used it in 1915 to explain Mercury's anomalous perihelion advance; the present subsection treats only a classical effective potential, but the calculation has the same shape.

2.10.3 Damped Oscillator From A Time-Dependent Lagrangian

Let

$$L(q, \dot{q}, t) = e^{\gamma t} \left(\frac{1}{2} m \dot{q}^2 - \frac{1}{2} k q^2 \right).$$

The Euler-Lagrange equation is

$$\frac{d}{dt}(e^{\gamma t} m \dot{q}) + e^{\gamma t} k q = 0,$$

equivalently

$$\boxed{m\ddot{q} + m\gamma\dot{q} + kq = 0.}$$

Although this is a dissipative equation, it comes from a time-dependent Lagrangian. The canonical energy is not conserved because L depends explicitly on time.

Set

$$s = e^{\gamma t/2} q, \quad q = e^{-\gamma t/2} s.$$

Then

$$\dot{q} = e^{-\gamma t/2} \left(\dot{s} - \frac{\gamma}{2} s \right).$$

Substituting into the action gives

$$L = \frac{1}{2} m \left(\dot{s} - \frac{\gamma}{2} s \right)^2 - \frac{1}{2} k s^2.$$

The cross term is a total derivative:

$$-\frac{1}{2} m \gamma s \dot{s} = -\frac{d}{dt} \left(\frac{1}{4} m \gamma s^2 \right).$$

Discarding that total derivative, the equivalent Lagrangian is

$$L_{\text{eff}} = \frac{1}{2} m \dot{s}^2 - \frac{1}{2} m \left(\frac{k}{m} - \frac{\gamma^2}{4} \right) s^2.$$

Thus

$$\ddot{s} + \left(\frac{k}{m} - \frac{\gamma^2}{4} \right) s = 0.$$

The transformation has moved damping into a shifted oscillator frequency.

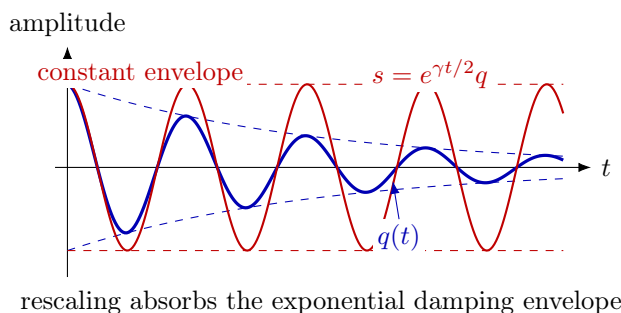


Figure 2.3: Damped-oscillator laboratory. In the original coordinate q , the motion has a decaying envelope. The rescaled coordinate $s = e^{\gamma t/2} q$ obeys an undamped oscillator equation with shifted frequency.

2.10.4 Rotating Hoop: Effective Potential And Bifurcation

A bead of mass m moves without friction on a hoop of radius a . The hoop is vertical and rotates about its vertical diameter with imposed angular speed ω . Let θ be the angle of the bead measured from the downward vertical. The bead position has speed

$$v^2 = a^2 \dot{\theta}^2 + a^2 \omega^2 \sin^2 \theta.$$

The Lagrangian is

$$L = \frac{1}{2} m a^2 \dot{\theta}^2 + \frac{1}{2} m a^2 \omega^2 \sin^2 \theta + m g a \cos \theta.$$

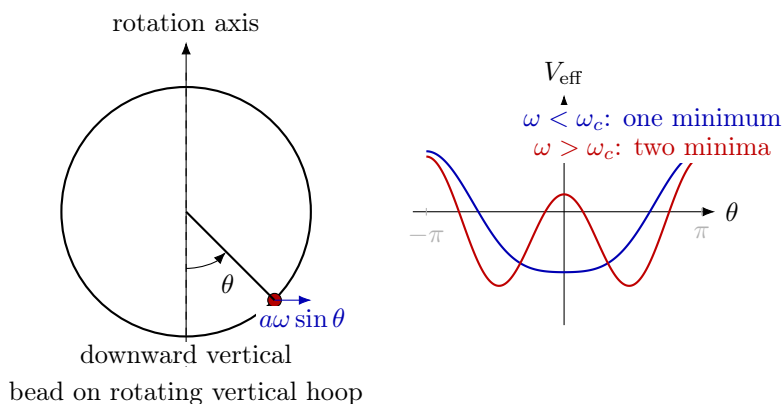


Figure 2.4: Rotating-hoop laboratory. The geometry fixes the kinetic term $a^2 \omega^2 \sin^2 \theta$, and the effective potential changes from one bottom minimum to two off-bottom minima when ω crosses $\omega_c = \sqrt{g/a}$. The potential curves are schematic and emphasize the bifurcation, not numerical scale.

This is a one-dimensional system with effective potential

$$V_{\text{eff}}(\theta) = -\frac{1}{2} m a^2 \omega^2 \sin^2 \theta - m g a \cos \theta.$$

Equilibria satisfy

$$\frac{dV_{\text{eff}}}{d\theta} = m a \sin \theta (g - a \omega^2 \cos \theta) = 0.$$

Thus either

$$\theta = 0, \pi,$$

or else

$$\cos \theta = \frac{g}{a\omega^2}.$$

The off-bottom equilibria exist only when

$$\omega \geq \omega_c = \sqrt{\frac{g}{a}}.$$

At $\omega = \omega_c$, the bottom equilibrium changes stability and two symmetric equilibria appear. This is a concrete mechanics realization of a pitchfork bifurcation in an effective potential.

2.10.5 Rolling Contact: Contact Forces As Multipliers

Rolling examples force one to distinguish coordinates, constraints, and reaction forces.

Hoop Rolling On A Fixed Cylinder

A hoop of mass m and radius r rolls without slipping on the outside of a fixed cylinder of radius R . Let θ be the angular position of the hoop center measured from the top. The center of mass moves on a circle of radius $R + r$. The rolling condition gives

$$r\dot{\phi} = (R + r)\dot{\theta},$$

where ϕ is the spin angle of the hoop. Since the hoop has

$$I_{\text{hoop}} = mr^2,$$

the kinetic energy is

$$T = \frac{1}{2}m(R + r)^2\dot{\theta}^2 + \frac{1}{2}mr^2\dot{\phi}^2 = m(R + r)^2\dot{\theta}^2.$$

Starting from rest at the top, energy gives

$$m(R + r)^2\dot{\theta}^2 = mg(R + r)(1 - \cos \theta).$$

The radial equation is

$$mg \cos \theta - N = m(R + r)\dot{\theta}^2,$$

where N is the normal force. Loss of contact occurs at $N = 0$. Using the energy relation at $N = 0$,

$$g \cos \theta = (R + r)\dot{\theta}^2 = g(1 - \cos \theta),$$

so

$$\boxed{\cos \theta = \frac{1}{2}}.$$

The hoop leaves the cylinder at $\theta = \pi/3$ from the top.

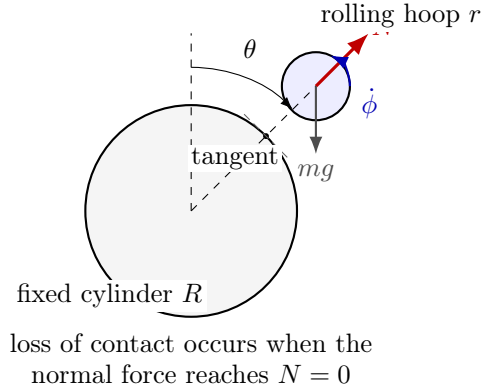


Figure 2.5: Rolling-contact laboratory. The radial force balance must be kept until the end because the contact force N is the diagnostic for leaving the surface. Eliminating the rolling constraint gives the motion, but $N = 0$ decides the loss-of-contact angle.

Solid Cylinder Inside A Hollow Cylinder

Now consider a solid cylinder of radius R rolling inside a fixed hollow cylinder of radius $2R$. The center of mass moves on a circle of radius R . With rolling imposed, the kinetic energy is

$$T = \frac{1}{2}MR^2\dot{\theta}^2 + \frac{1}{4}MR^2\dot{\theta}^2,$$

because a solid cylinder has moment of inertia $I = (1/2)MR^2$. Taking the potential as $V = -MgR\cos\theta$, the Lagrangian is

$$L = \frac{3}{4}MR^2\dot{\theta}^2 + MgR\cos\theta.$$

The equation of motion is

$$\frac{3}{2}MR^2\ddot{\theta} + MgR\sin\theta = 0,$$

equivalently

$$\frac{3}{2}\ddot{\theta} + \frac{g}{R}\sin\theta = 0.$$

For small oscillations,

$$\boxed{\omega_{\text{small}} = \sqrt{\frac{2g}{3R}}}.$$

2.10.6 Schwarzschild-Type Central-Force Stability

Consider the central Lagrangian

$$L = -m\sqrt{A(r) - \frac{\dot{r}^2}{A(r)} - r^2\dot{\theta}^2}, \quad A(r) = 1 - \frac{R}{r}, \quad r > R.$$

This is the geodesic Lagrangian for the Schwarzschild metric, derived by Schwarzschild in 1916, with horizon at $r = R$, restricted to the equatorial plane, in units $c = 1$. Here the parameter is

Schwarzschild coordinate time. The model is therefore the reduced square-root Lagrangian for spatial motion, not the reparametrization-invariant proper-time action. We use it as a classical central-force laboratory whose effective-potential analysis already yields the correct innermost stable circular-orbit radius. Define

$$B = A - \frac{\dot{r}^2}{A} - r^2 \dot{\theta}^2.$$

The conserved energy and angular momentum are

$$E = \frac{mA}{\sqrt{B}}, \quad L_z = \frac{mr^2 \dot{\theta}}{\sqrt{B}}.$$

The angular momentum follows from the cyclic coordinate θ . The energy is the autonomous-Lagrangian first integral traditionally called the Beltrami integral. It is conserved here because this reduced Lagrangian has no explicit time dependence:

$$\dot{r} \frac{\partial L}{\partial \dot{r}} + \dot{\theta} \frac{\partial L}{\partial \dot{\theta}} - L = \frac{mA}{\sqrt{B}}.$$

Equivalently, it is the time-translation Noether charge $\dot{q}^i \partial L / \partial \dot{q}^i - L$; after the Legendre transform the same expression is the Hamiltonian. Eliminating $\dot{\theta}$ and B gives the radial equation in effective potential form:

$$\dot{r}^2 = \frac{A^2}{E^2} \left[E^2 - A \left(m^2 + \frac{L_z^2}{r^2} \right) \right].$$

Thus circular orbits are critical points of

$$U_{\text{eff}}(r) = A(r) \left(m^2 + \frac{L_z^2}{r^2} \right)$$

at fixed L_z . The circular-orbit condition $U'_{\text{eff}}(r_0) = 0$ gives

$$L_z^2 = \frac{Rm^2 r_0^2}{2r_0 - 3R}.$$

Hence circular orbits require $r_0 > 3R/2$. Stability is determined by

$$U''_{\text{eff}}(r_0) = \frac{2Rm^2(r_0 - 3R)}{r_0^3(2r_0 - 3R)}.$$

Therefore

$r_0 > 3R \text{ is stable,} \quad \frac{3}{2}R < r_0 < 3R \text{ is unstable.}$
--

This is the same structural calculation used in more realistic relativistic orbit problems: circular-orbit stability is a second-derivative question for an effective potential at fixed angular momentum.

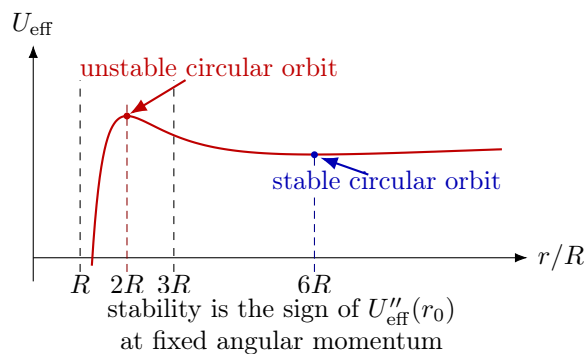


Figure 2.6: Effective-potential picture for the Schwarzschild-type laboratory. Circular orbits are critical points. The plotted curve is an affine rescaling of $-1/r + 4/r^2 - 4/r^3$, whose circular-orbit extrema occur at $r = 2R$ and $r = 6R$: the inner extremum is unstable and the outer one is stable.

2.11 Associated Demos

This chapter is mostly analytic, but two later demos are natural companions for its examples:

- `demos/python/hamiltonian_pendulum.py`, read after the Legendre transform in Chapter 3, turns effective potential thinking into a phase portrait.
- `demos/python/rigid_body_euler_top.py`, read after Chapter 4, continues the Lagrangian rigid-body derivation into reduced Euler equations and attitude reconstruction.

The same examples are first derived from the action, then revisited in phase-space or reduced variables. The computation adds a later layer to the same model.

Chapter 3

Hamiltonian Mechanics And Phase Space

Phase space is a parameter space for histories. A point $(q, p) \in T^*Q$ is one possible state at one instant; the Hamiltonian flow extends it forward and backward in time. Two histories that meet at the same phase-space point are the same history, at least for as long as the equations have a unique solution.

The Lagrangian chapter described motion as stationary paths in configuration space. Hamiltonian mechanics reorganizes the same dynamics on phase space and brings the symplectic form, Poisson brackets, canonical transformations, and phase-space volume into the foreground. These tools are the language in which integrability, action-angle variables, perturbation theory, and KAM theory are most naturally stated.

Throughout this chapter Q is a smooth configuration manifold of dimension n , with local coordinates q^i . The cotangent bundle T^*Q has induced coordinates (q^i, p_i) . When a Lagrangian is regular, the Legendre map carries $(q, \dot{q})$ to (q, p) , and the Hamiltonian H is the corresponding energy function written in these phase-space coordinates. The assumptions behind this passage are stated explicitly below, because singular Lagrangians and gauge constraints require extra reduction machinery that belongs to more advanced treatments.

Roadmap. The chapter has four layers. First, the Legendre transform explains when velocity data on TQ can be replaced by momentum data on T^*Q . Second, the canonical one-form and symplectic form derive Hamilton's equations from geometry. Third, Poisson brackets, the phase-space action, and canonical transformations identify the maps that preserve Hamiltonian structure. Fourth, the numerical lesson explains why a simulation should preserve geometry as well as approximate trajectories.

Keep one distinction in view throughout the chapter. A point of phase space is instantaneous data, while a Hamiltonian flow is the rule that moves that data. The symplectic form supplies the structure that turns the differential of H into a vector field and decides which coordinate changes are canonical.

3.1 Legendre Transform And Hamiltonian

The Legendre transform sends $(q, \dot{q}) \in TQ$ to $(q, p) \in T^*Q$ by

$$p_i = \frac{\partial L}{\partial \dot{q}^i}.$$

For a regular Lagrangian this can be inverted to express $\dot{q}^i$ as a function of (q, p, t) . Concretely, regularity means that the velocity Hessian

$$W_{ij}(q, \dot{q}, t) = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$$

is nonsingular. The inverse function theorem then says that, locally in phase space, the equations $p_i = \partial L / \partial \dot{q}^i$ can be solved for $\dot{q}^i = \dot{q}^i(q, p, t)$. The Hamiltonian is

$$H(q, p, t) = p_i \dot{q}^i(q, p, t) - L(q, \dot{q}(q, p, t), t).$$

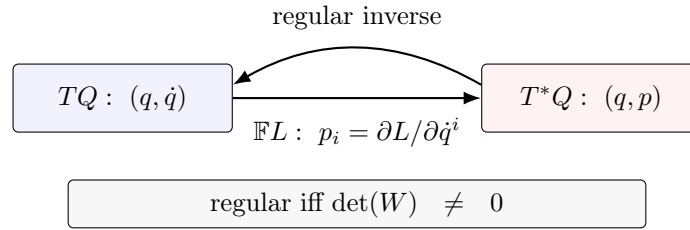


Figure 3.1: The Legendre transform from tangent-bundle data to cotangent-bundle data. Singular Lagrangians require constraints; the Hamiltonian construction in this chapter assumes the regular case.

Derivation 3.1 (Hamilton's equations from the Legendre transform). *In this calculation H is regarded as a function of (q, p, t) . The velocity $\dot{q} = \dot{q}(q, p, t)$ is the inverse image of (q, p) under the Legendre map, so its differential appears temporarily and then cancels. This is where regularity is used. Differentiate H :*

$$\begin{aligned} dH &= \dot{q}^i dp_i + p_i d\dot{q}^i - \frac{\partial L}{\partial q^i} dq^i - \frac{\partial L}{\partial \dot{q}^i} d\dot{q}^i - \frac{\partial L}{\partial t} dt \\ &= \dot{q}^i dp_i - \frac{\partial L}{\partial q^i} dq^i - \frac{\partial L}{\partial t} dt, \end{aligned}$$

because $p_i = \partial L / \partial \dot{q}^i$. Thus

$$\frac{\partial H}{\partial p_i} = \dot{q}^i, \quad \frac{\partial H}{\partial q^i} = -\frac{\partial L}{\partial q^i}.$$

Using the Euler-Lagrange equation $\dot{p}_i = \partial L / \partial q^i$, we get

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}}.$$

The cancellation of the $d\dot{q}$ terms is the algebraic heart of the Legendre transform. It says that once momentum has been defined by $p_i = \partial L / \partial \dot{q}^i$, the Hamiltonian differential contains only dq, dp, dt . Hamilton's equations are therefore a first-order system on T^*Q , with (q, p) as the instantaneous phase-space data.

The Legendre transform changes the bookkeeping of initial data. A Lagrangian state $(q, \dot{q})$ records a position and a velocity; a Hamiltonian state (q, p) records a position and the covector that measures how the action changes under endpoint displacement. The symplectic form below is built from that endpoint one-form, so Hamiltonian mechanics retains the variational origin of momentum.

Singular systems begin when the velocity Hessian has null directions. Then some velocities cannot be solved from the momenta, and some momenta become constraints. Electromagnetic gauge theory, relativistic reparametrization-invariant particles, and ideal incompressible fluids all have versions of this phenomenon. The present chapter stays in the regular case so that the geometry of T^*Q can be developed before Dirac constraints enter.

3.2 Phase Space As Initial Data And Histories

For a deterministic Hamiltonian system, phase space parametrizes the solutions. If H is smooth and the Hamiltonian vector field is complete on the time interval under study, then every point

$$z_0 = (q_0, p_0) \in T^*Q$$

determines a unique trajectory $z(t) = \Phi_H^t(z_0)$, and every trajectory is obtained this way from its value at any one time. Thus phase space is the set of all possible instantaneous states, but it is also a coordinate system on the family of all classical histories. The caveat is that this identification uses existence and uniqueness of the equations of motion; singular potentials, collisions, constraints, or finite-time blowup can invalidate the global statement even when the local Hamiltonian equations make sense. For example, the attractive Kepler Hamiltonian has regular equations away from $r = 0$, but a collision orbit reaches $r = 0$ in finite time unless one adds a separate regularization. Completeness is therefore a statement about the vector field on the chosen phase space, not a formal consequence of writing Hamilton's equations.

3.3 The Symplectic Form

The cotangent bundle T^*Q carries a canonical one-form, often called the Liouville one-form,

$$\Theta = p_i dq^i.$$

The name credits Liouville's nineteenth-century work on volume preservation and integrability in Hamiltonian systems; the geometric object is the tautological one-form on T^*Q . This formula is intrinsic under changes of coordinates on the base manifold Q , although it is written in induced cotangent coordinates. If $\tilde{q}^j = \tilde{q}^j(q)$ and the induced cotangent coordinates satisfy

$$\tilde{p}_j = \frac{\partial q^i}{\partial \tilde{q}^j} p_i,$$

then the chain rule gives

$$\tilde{p}_j d\tilde{q}^j = \frac{\partial q^i}{\partial \tilde{q}^j} p_i \frac{\partial \tilde{q}^j}{\partial q^k} dq^k = p_i dq^i.$$

Thus Θ is the same one-form in every coordinate system on Q . The canonical symplectic form is

$$\omega = -d\Theta = dq^i \wedge dp_i.$$

The sign comes from the antisymmetry of the wedge product:

$$-d(p_i dq^i) = -dp_i \wedge dq^i = dq^i \wedge dp_i.$$

This sign convention fixes the later Hamiltonian equation $\iota_{X_H}\omega = dH$. Some standard references, including Arnold's *Mathematical Methods of Classical Mechanics*, use the opposite sign for the canonical symplectic form and compensate by the corresponding convention for Hamiltonian vector fields or Poisson brackets. The geometry is unchanged; only the sign bookkeeping changes. It is closed because

$$d\omega = -d^2\Theta = 0.$$

In arbitrary local phase-space coordinates

$$x^m, \quad m = 1, \dots, 2n,$$

one can write

$$\omega = \frac{1}{2}\omega_{mn}(x) dx^m \wedge dx^n,$$

where $\omega_{mn} = -\omega_{nm}$, $\det(\omega_{mn}) \neq 0$, and

$$d\omega = 0.$$

The closedness condition is what makes ω locally equivalent to the canonical form; this is Darboux's theorem (1882). Nondegeneracy is what lets the Hamiltonian determine a unique vector field. In canonical coordinates $x = (q, p)$,

$$(\omega_{mn}) = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Darboux's proof can be outlined as a local induction. Choose a function q^1 with nonzero differential and define Y_1 by $\iota_{Y_1}\omega = -dq^1$. Nondegeneracy makes Y_1 unique, and $dq^1(Y_1) = \omega(Y_1, Y_1) = 0$, so Y_1 is tangent to the level sets of q^1 . Choose a local hypersurface transverse to Y_1 , set $p_1 = 0$ on it, and extend p_1 by requiring $Y_1 p_1 = 1$ along the Y_1 -flow. Then $\{q^1, p_1\} = 1$, with the sign convention below. The common kernel of dq^1 and dp_1 , modulo the two directions already constructed, carries a nondegenerate two-form of dimension $2n - 2$. Closedness is inherited because restriction and quotient pullbacks commute with d ; this is the point at which $d\omega = 0$ enters the induction. Repeating the construction produces a coordinate chart in which $\omega = dq^i \wedge dp_i$. The Poisson bracket can be expressed intrinsically by the bivector inverse to the symplectic form, with the sign convention fixed by Hamilton's equations below. Once ω is given, the bracket is the corresponding Lie operation on functions. In arbitrary coordinates, let π^{mn} be defined by

$$\pi^{mn}\omega_{nk} = \delta^m_k.$$

With the convention $\iota_{X_H}\omega = dH$, this gives

$$\{f, g\} = -\pi^{mn}\partial_m f \partial_n g,$$

which reduces to the displayed canonical formula in coordinates (q^i, p_i) . Indeed, for

$$\omega = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \quad \pi = \omega^{-1} = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix},$$

the extra minus sign gives $\{f, g\} = \partial_{q^i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q^i} g$.

The two defining properties of ω have different jobs. Nondegeneracy lets one solve $\iota_{X_H} \omega = dH$ for a unique vector field, just as a mass matrix lets one solve force equals mass times acceleration. Closedness is the integrability condition behind canonical coordinates, Stokes-type arguments, and the preservation of symplectic area. Hamiltonian mechanics needs both.

The Hamiltonian vector field X_H is defined by

$$\iota_{X_H} \omega = dH.$$

Here ι_X denotes interior contraction: if ω is a two-form, then $(\iota_X \omega)(Y) = \omega(X, Y)$ for every vector field Y . The coordinate-free formulation became standard in the twentieth century through Lichnerowicz's symplectic-geometry program, Kirillov's coadjoint-orbit work, and Souriau's *Structure des systemes dynamiques* (1970). The formula is not an additional physical postulate; it is Hamilton's equations written without choosing canonical coordinates. If

$$X_H = A^i \frac{\partial}{\partial q^i} + B_i \frac{\partial}{\partial p_i},$$

then

$$\iota_{X_H} \omega = A^i dp_i - B_i dq^i.$$

Equating this with

$$dH = \frac{\partial H}{\partial q^i} dq^i + \frac{\partial H}{\partial p_i} dp_i$$

gives

$$A^i = \frac{\partial H}{\partial p_i}, \quad B_i = -\frac{\partial H}{\partial q^i}.$$

Thus X_H is Hamilton's equation written geometrically.

3.4 Poisson Brackets And Time Evolution

For $f, g \in C^\infty(T^*Q)$, define

$$\{f, g\} = \frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i}.$$

If $z(t) = (q(t), p(t))$ follows Hamilton's equation, then

$$\frac{d}{dt} f(q(t), p(t), t) = \frac{\partial f}{\partial t} + \{f, H\}.$$

A time-independent observable f is conserved exactly when

$$\{f, H\} = 0.$$

Proposition 3.1 (Liouville volume preservation). *Hamiltonian flow preserves the phase-space volume form*

$$\Omega = \frac{1}{n!} \omega^n.$$

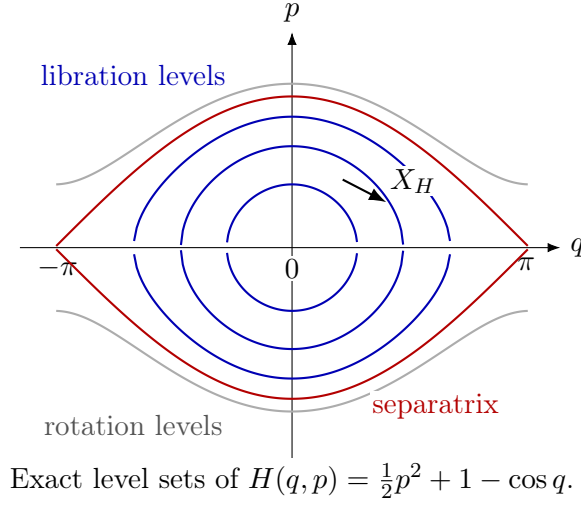


Figure 3.2: The pendulum phase plane. Libration curves have $0 < E < 2$, the separatrix has $E = 2$, and rotation curves have $E > 2$. The Hamiltonian vector field is tangent to energy level sets because $\dot{H} = \{H, H\} = 0$.

Proof. Liouville's 1838 volume-preservation theorem is a direct consequence of Cartan's formula for Lie derivatives.¹

$$\mathcal{L}_{X_H}\omega = d(\iota_{X_H}\omega) + \iota_{X_H}d\omega.$$

Since $\iota_{X_H}\omega = dH$ and $d\omega = 0$,

$$\mathcal{L}_{X_H}\omega = ddH = 0.$$

Therefore $\mathcal{L}_{X_H}\omega^n = 0$, so the associated volume form is preserved. $\square$

3.5 Phase-Space Action

Hamilton's equations can also be derived from the phase-space action

$$S_H[q, p] = \int_{t_0}^{t_1} (p_i \dot{q}^i - H(q, p, t)) dt.$$

Varying q and p independently, with $\delta q(t_0) = \delta q(t_1) = 0$, gives

$$\delta S_H = \int_{t_0}^{t_1} \left[\left(\dot{q}^i - \frac{\partial H}{\partial p_i} \right) \delta p_i - \left(\dot{p}_i + \frac{\partial H}{\partial q^i} \right) \delta q^i \right] dt.$$

Since δq and δp are arbitrary in the interior, the stationary curves are exactly Hamiltonian trajectories.

¹For any differential form α and vector field X , $\mathcal{L}_X\alpha = d(\iota_X\alpha) + \iota_Xd\alpha$. One proves it first on functions and one-forms, then uses that both sides are graded derivations of the exterior algebra.

3.6 Canonical Transformations

A diffeomorphism $\Phi : T^*Q \rightarrow T^*Q$ is canonical, or symplectic, if

$$\Phi^*\omega = \omega.$$

Canonical transformations preserve Poisson brackets:

$$\{f \circ \Phi, g \circ \Phi\} = \{f, g\} \circ \Phi.$$

Indeed, $\Phi^*\omega = \omega$ implies that the inverse bivector is also preserved, $\Phi_*\pi = \pi$. Applying the bivector formula for the bracket to $d(f \circ \Phi) = \Phi^*(df)$ and $d(g \circ \Phi) = \Phi^*(dg)$ gives the displayed identity. This is why canonical coordinates may change but Hamiltonian mechanics keeps the same form. Later, action-angle variables will be canonical coordinates adapted to integrable motion.

Derivation 3.2 (Generating function for a canonical transformation). *Let (q, p) be old canonical coordinates and (Q, P) new ones. A convenient type of generating function depends on the new momenta and old positions:*

$$F = F(P, q).$$

The starting point is preservation of the canonical two-form:

$$dq^i \wedge dp_i = dQ^i \wedge dP_i.$$

Equivalently, the one-form $p_i dq^i + Q^i dP_i$ is locally exact, because

$$d(p_i dq^i + Q^i dP_i) = -dq^i \wedge dp_i + dQ^i \wedge dP_i = 0.$$

By the Poincare lemma, a closed one-form on a contractible coordinate patch is the differential of a local function. The type-II generating function is this local potential. Impose

$$Q^i = \left. \frac{\partial F}{\partial P_i} \right|_q, \quad p_i = \left. \frac{\partial F}{\partial q^i} \right|_P.$$

Then

$$dF = Q^i dP_i + p_i dq^i.$$

Taking another exterior derivative gives

$$0 = dQ^i \wedge dP_i + dp_i \wedge dq^i.$$

Therefore

$$dQ^i \wedge dP_i = dq^i \wedge dp_i.$$

The canonical symplectic form is preserved, so the transformation is canonical. In coordinates this is the same cancellation one sees by expanding

$$dQ^i \wedge dP_i = \left(\frac{\partial^2 F}{\partial P_i \partial q^j} dq^j + \frac{\partial^2 F}{\partial P_i \partial P_j} dP_j \right) \wedge dP_i,$$

where the pure $dP_i \wedge dP_j$ term vanishes because second derivatives are symmetric while the wedge product is antisymmetric.

The generating function is useful because it replaces the condition $\Phi^*\omega = \omega$, which is a two-form equation, by the exactness of a one-form. Locally, a canonical transformation is often found by guessing the right exact differential. Globally, the same calculation must be patched across coordinate charts, which is why action-angle generating functions can be multivalued even when the symplectic form is perfectly well defined.

The type-II function used above is one member of the standard local family, systematized in Jacobi's Hamiltonian lectures of the 1840s. The four common choices are distinguished by which old and new variables are treated as independent:

type	generating function	differential convention
$F_1(q, Q)$	F_1	$p \, dq - P \, dQ = dF_1$
$F_2(q, P)$	F_2	$p \, dq + Q \, dP = dF_2$
$F_3(p, Q)$	F_3	$-q \, dp - P \, dQ = dF_3$
$F_4(p, P)$	F_4	$-q \, dp + Q \, dP = dF_4$

Each line is a different Legendre transform of the same exact one-form. The signs follow from preserving $p \, dq - P \, dQ$ up to an exact differential.

3.7 Numerical Lesson

The equations in this chapter are exact statements about smooth systems. A numerical integrator creates a discrete map

$$z_{n+1} = \Psi_{\Delta t}(z_n).$$

For long-time Hamiltonian problems, whether $\Psi_{\Delta t}$ preserves a symplectic form or a modified Hamiltonian can matter more than the local order of the method. More precisely, a symplectic method of order p often has a formal modified Hamiltonian

$$\tilde{H}_{\Delta t} = H + \Delta t^p H_p + \Delta t^{p+1} H_{p+1} + \dots$$

whose exact time- Δt flow agrees with the numerical map up to the relevant asymptotic order. The series is generally asymptotic rather than convergent, but for analytic problems it explains why energy error often stays bounded over very long times instead of drifting monotonically. The pendulum, standard map, and asteroid demos are therefore experiments in preserving structure. Symplectic integrators were developed in modern numerical form in the 1980s and 1990s. Ruth (1983) gave the first explicit schemes; Channell-Scovel and Sanz-Serna systematized them; Yoshida (1990) constructed high-order composition methods. Hairer, Lubich, and Wanner later assembled the backward-error analysis that explains why these methods preserve a modified Hamiltonian over long times.

The practical lesson is diagnostic. A good mechanics simulation reports not only a trajectory but also the quantities that the continuous problem says should be preserved or nearly preserved. Energy drift, angular-momentum drift, phase-space area, and event criteria are part of the model audit trail. Without them, a numerical plot can look convincing while solving the wrong mechanical problem.

3.8 What To Take Forward

The rest of the notes repeatedly reuse three lessons sharpened in this chapter.

- The Legendre transform is a structural step, not a change of notation: it is valid only where velocities can be recovered from momenta.
- The canonical one-form records the endpoint momentum information from the Lagrangian action; its exterior derivative is the symplectic form that controls Hamiltonian evolution.
- Hamiltonian mechanics is geometry on T^*Q : the symplectic form determines Hamilton's equations, volume preservation, Poisson brackets, and the meaning of a canonical numerical method.

3.9 Chapter-End Laboratories And Exercises

The laboratories below test when a Lagrangian can be converted into a Hamiltonian system and how the resulting phase portrait organizes motion.

3.9.1 A Nonstandard Lagrangian And Regularity

This laboratory separates two ideas that are often conflated. A Lagrangian may produce a familiar-looking equation of motion and still fail to be globally regular. Regularity is a statement about whether velocities can be solved from momenta. Without it, the Legendre transform can fold phase space and the Hamiltonian description must be handled with care.

Consider the one-dimensional Lagrangian

$$L(x, \dot{x}) = \frac{m^2 \dot{x}^4}{12} + m \dot{x}^2 V(x) - V(x)^2.$$

The momentum is

$$p = \frac{\partial L}{\partial \dot{x}} = \frac{m^2}{3} \dot{x}^3 + 2mV\dot{x}.$$

The Euler-Lagrange equation gives

$$\begin{aligned} 0 &= \frac{d}{dt} \left(\frac{m^2}{3} \dot{x}^3 + 2mV\dot{x} \right) - (m\dot{x}^2 V' - 2VV') \\ &= (m\dot{x}^2 + 2V)(m\ddot{x} + V'). \end{aligned}$$

There are two branches. The Newtonian branch is

$$m\ddot{x} = -V'(x).$$

The second branch,

$$m\dot{x}^2 + 2V(x) = 0,$$

is a singular energy-surface condition. The velocity Hessian

$$\frac{\partial^2 L}{\partial \dot{x}^2} = m(m\dot{x}^2 + 2V)$$

vanishes on this second branch. This example shows how a Lagrangian can produce familiar equations on a regular branch while being singular on another branch. Regularity is the hypothesis that lets the Legendre transform define a Hamiltonian system.

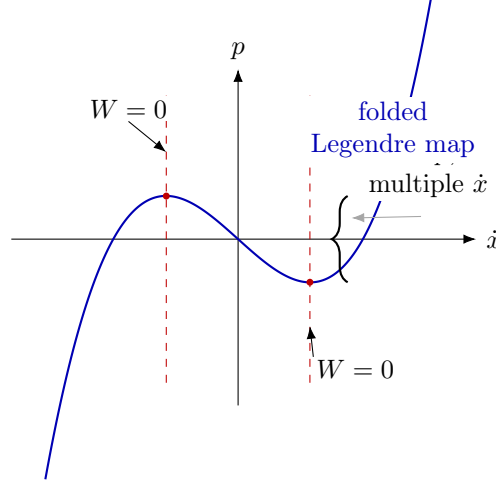


Figure 3.3: Nonstandard Lagrangian laboratory. For a representative region with $V < 0$, the momentum-velocity map can fold where the velocity Hessian $W = \partial^2 L / \partial \dot{x}^2$ vanishes. At such points the Legendre transform is not locally invertible, even though the Euler-Lagrange equation has a familiar Newtonian branch.

3.9.2 Pendulum Phase Space And The Separatrix

The simple pendulum is the canonical one-degree-of-freedom Hamiltonian example. Let θ be the angle from the downward vertical. The Lagrangian is

$$L = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell \cos \theta.$$

The momentum and Hamiltonian are

$$p_\theta = m\ell^2\dot{\theta}, \quad H(\theta, p_\theta) = \frac{p_\theta^2}{2m\ell^2} - mg\ell \cos \theta.$$

Hamilton's equations are

$$\dot{\theta} = \frac{p_\theta}{m\ell^2}, \quad \dot{p}_\theta = -mg\ell \sin \theta.$$

The equilibria are

$$(\theta, p_\theta) = (0, 0) \quad \text{and} \quad (\pi, 0)$$

modulo 2π . The downward equilibrium is elliptic; the upward equilibrium is hyperbolic.

The separatrix is the energy level through the hyperbolic point:

$$H = mg\ell.$$

On this level,

$$\frac{p_\theta^2}{2m\ell^2} = mg\ell(1 + \cos \theta) = 2mg\ell \cos^2 \frac{\theta}{2}.$$

Therefore

$$p_\theta = \pm 2m\ell\sqrt{g\ell} \cos \frac{\theta}{2}$$

for $-\pi \leq \theta \leq \pi$. The separatrix divides libration orbits, which oscillate around $\theta = 0$, from rotation orbits, which wind around the circle. This picture is the one-degree-of-freedom ancestor of separatrix splitting in perturbed Hamiltonian systems.

3.10 Associated Demos

The associated phase-space demo is `demos/python/hamiltonian_pendulum.py`. It plots pendulum trajectories and the separatrix in (θ, p_θ) . Read it as a check on this chapter's central claim: phase portraits display Hamiltonian level sets and the symplectic flow on phase space.

Chapter 4

Integrability And The Rigid Body

4.1 The Rigid Body As A Bridge To Integrability

The free rigid body is a compact model in which several central ideas meet. Its motion is non-linear; its configuration space is the Lie group $\text{SO}(3)$; its reduced description is Hamiltonian on angular-momentum space; and it is completely integrable. At the same time, small modifications such as gravity acting on a heavy top lead toward nonintegrability and chaos. The classical integrable exceptions beyond the free body are Lagrange's symmetric heavy top (1788) and Kovalevskaya's top (1888), with other special cases requiring additional hypotheses. Husson's 1906 analysis made this rarity precise for the classical heavy top under its stated assumptions. The rarity of such exceptions is part of the lesson.

The most vivid symptom is the intermediate-axis flip. Spin a book about its longest or shortest principal axis and the rotation is stable; spin it about the middle axis and the book turns over. The behavior comes from the geometry of two conserved quantities, energy and angular momentum, intersecting on the angular-momentum sphere. The rigid body is a place where geometric pictures become part of the argument: reduction, stability, Poincaré rolling, and invariant tori are all different views of the same motion.

Assumption 4.1 (Rigid body model). *A rigid body is an ideal continuum whose internal distances do not change. A material point is labelled by a body coordinate $X \in B \subset \mathbb{R}^3$. Its spatial position is*

$$x(t) = r(t) + R(t)X,$$

where $r(t) \in \mathbb{R}^3$ is the center-of-mass position and $R(t) \in \text{SO}(3)$ is the attitude matrix. The free rigid body chapter suppresses translation by working in the center-of-mass frame.

Three layers to keep separate. Rigid-body mechanics mixes three descriptions that are often compressed into one line of notation. The kinematic layer says how $R(t) \in \text{SO}(3)$ defines angular velocity. The dynamical layer says how the kinetic energy and the inertia tensor produce Euler's equations. The reduced Hamiltonian layer says why angular momentum spheres carry a Lie-Poisson flow. Each layer has its own variables and its own conserved quantities. Keeping the layers separate preserves the distinction between body components and spatial components.

Roadmap. The chapter first derives the free rigid body's kinematics, Euler equations, reduction, and stability, then abstracts the lesson into Liouville integrability and invariant tori.

4.2 The Rotation Group And Angular Velocity

The rotation group is

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} : R^T R = I, \det R = 1\}.$$

The matrix R has columns equal to the three orthonormal body axes $\hat{e}_1, \hat{e}_2, \hat{e}_3$ as seen in space:

$$R = (\hat{e}_1 \ \hat{e}_2 \ \hat{e}_3).$$

Thus a point of configuration space is not a triple of unconstrained vectors; it is an orthonormal oriented frame. The condition $R^T R = I$ gives six constraints on nine matrix entries, leaving three degrees of freedom.

The spatial angular velocity ω_{space} is defined by the fact that every body axis instantaneously rotates according to

$$\frac{d\hat{e}_a}{dt} = \omega_{\text{space}} \times \hat{e}_a,$$

or, in components,

$$\dot{R}_{ia} = \epsilon_{ijk} (\omega_{\text{space}})_j R_{ka}.$$

Differentiate $R^T R = I$:

$$\dot{R}^T R + R^T \dot{R} = 0.$$

Thus $R^T \dot{R}$ is skew-symmetric. Every skew-symmetric 3×3 matrix is uniquely of the form

$$\hat{\Omega} = \begin{pmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{pmatrix},$$

where $\hat{\Omega}v = \Omega \times v$. The body angular velocity is defined by

$$R^T \dot{R} = \hat{\Omega}.$$

The hat map is a Lie algebra isomorphism:

$$[\hat{a}, \hat{b}] = \widehat{a \times b}.$$

This identity is the bridge between matrix commutators in $\mathfrak{so}(3)$ and vector cross products in the rigid-body equations. The spatial angular velocity ω_{space} satisfies

$$\dot{R}R^T = \hat{\omega}_{\text{space}}, \quad \omega_{\text{space}} = R\Omega.$$

This space/body distinction is one of the main sources of sign errors in rigid body mechanics. The vector ω_{space} is expressed in fixed space axes, while Ω is expressed in the moving body axes. They describe the same instantaneous rotation, but derivatives of body components must also differentiate the moving frame. Euler's equation is where this bookkeeping becomes dynamical.

Derivative rules for moving frames. If $A(t)$ is a vector written in body components, then its spatial vector is $a(t) = R(t)A(t)$. Differentiating gives

$$\dot{a} = \dot{R}A + R\dot{A} = R(\widehat{\Omega}A + \dot{A}) = R(\dot{A} + \Omega \times A).$$

Thus

$$\dot{a}_{\text{space}} = R \left(\dot{A}_{\text{body}} + \Omega \times A_{\text{body}} \right).$$

Conversely, if a is fixed in space and $A = R^T a$ are its body components, then

$$\dot{A} = -\widehat{\Omega}A = -\Omega \times A.$$

This rule explains the form of Euler's equation before any variational calculation is done: a vector can be conserved in space while its body components precess because the axes themselves are moving.

4.3 Euler Angles, Metric, And The Topology Of SO(3)

One can parametrize a dense open set of SO(3) by Euler angles

$$(\phi, \theta, \psi), \quad 0 \leq \phi < 2\pi, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \psi < 2\pi.$$

For the Euler-angle convention used below, the body axes are

$$\begin{aligned} \hat{e}_1 &= \begin{pmatrix} \cos \phi \cos \psi - \sin \phi \cos \theta \sin \psi \\ \sin \phi \cos \psi + \cos \phi \cos \theta \sin \psi \\ \sin \theta \sin \psi \end{pmatrix}, \\ \hat{e}_2 &= \begin{pmatrix} -\cos \phi \sin \psi - \sin \phi \cos \theta \cos \psi \\ -\sin \phi \sin \psi + \cos \phi \cos \theta \cos \psi \\ \sin \theta \cos \psi \end{pmatrix}, \\ \hat{e}_3 &= \begin{pmatrix} \sin \phi \sin \theta \\ -\cos \phi \sin \theta \\ \cos \theta \end{pmatrix}. \end{aligned}$$

The convention matters for signs in formulas, but the lesson is geometric: Euler angles are a coordinate chart, not the space itself. The chart is singular at $\theta = 0, \pi$, even though SO(3) is a smooth homogeneous space.

A coordinate-independent line element on SO(3) is

$$ds^2 = -\frac{1}{2} \text{tr} (R^{-1} dR)^2.$$

This is the bi-invariant metric induced by minus the Killing-form normalization on $\mathfrak{so}(3)$; the minus sign makes the square norm positive because skew matrices have negative trace square. For this standard Euler-angle convention, this becomes

$$ds^2 = d\phi^2 + d\theta^2 + d\psi^2 + 2 \cos \theta d\phi d\psi.$$

Introduce

$$\xi_1 = \phi + \psi, \quad \xi_2 = \phi - \psi.$$

Then

$$ds^2 = d\theta^2 + \cos^2 \frac{\theta}{2} d\xi_1^2 + \sin^2 \frac{\theta}{2} d\xi_2^2.$$

This formula shows the topology pictorially. At one endpoint of the θ -interval the ξ_2 -circle shrinks; at the other endpoint the ξ_1 -circle shrinks. Gluing these solid tori gives S^3 , and the remaining identification is the antipodal identification. Hence

$$\mathrm{SO}(3) \simeq S^3/\mathbb{Z}_2.$$

This is why a rigid body admits local angle coordinates but no single global nondegenerate Euler-angle chart.

4.4 Kinetic Energy And The Inertia Tensor

Take the center of mass as origin in body coordinates:

$$\int_B X \rho(X) d^3X = 0,$$

where $\rho(X)$ is the mass density. A point X has velocity

$$\dot{x} = \dot{R}X = R\hat{\Omega}X = R(\Omega \times X).$$

Since R preserves Euclidean lengths,

$$\|\dot{x}\|^2 = \|\Omega \times X\|^2.$$

The kinetic energy is

$$T = \frac{1}{2} \int_B \rho(X) \|\Omega \times X\|^2 d^3X.$$

Use the vector identity

$$\|\Omega \times X\|^2 = \|\Omega\|^2 \|X\|^2 - (\Omega \cdot X)^2,$$

to write the kinetic energy as

$$T = \frac{1}{2} \Omega^T I \Omega,$$

where

$$I_{ij} = \int_B \rho(X) (\|X\|^2 \delta_{ij} - X_i X_j) d^3X.$$

The matrix I is the inertia tensor in the body frame. In principal axes,

$$I = \mathrm{diag}(I_1, I_2, I_3),$$

and

$$T = \frac{1}{2} (I_1 \Omega_1^2 + I_2 \Omega_2^2 + I_3 \Omega_3^2).$$

Equivalently, using the spatial angular velocity ω_{space} and the body axes $\hat{e}_a$,

$$L = \frac{1}{2} \sum_{a=1}^3 I_a (\hat{e}_a \cdot \omega_{\mathrm{space}})^2.$$

Here L denotes the Lagrangian. Later symbols such as L_z denote angular momentum components; the subscripted notation is always specified locally. The spatial angular momentum is

$$J_i = \frac{\partial L}{\partial (\omega_{\text{space}})_i} = \sum_{a=1}^3 I_a (\hat{e}_a)_i (\hat{e}_a \cdot \omega_{\text{space}}).$$

Its body components are

$$\tilde{J}_a = \hat{e}_a \cdot J = I_a \Omega_a,$$

and the Hamiltonian is

$$H = \sum_{a=1}^3 \frac{\tilde{J}_a^2}{2I_a}.$$

4.5 Full Phase Space: $T^* \text{SO}(3)$

The Hamiltonian phase space of a finite-dimensional mechanical system with configuration space Q is the cotangent bundle T^*Q . Therefore the unreduced phase space of a rigid body with fixed center of mass is

$$T^* \text{SO}(3).$$

If q^i , $i = 1, 2, 3$, are any nondegenerate local coordinates on $\text{SO}(3)$, the canonical momenta are

$$p_i = \frac{\partial L}{\partial \dot{q}^i}.$$

Under a change of coordinates $q^i = q^i(\tilde{q})$, the momenta transform as a one-form:

$$\tilde{p}_j = \frac{\partial q^i}{\partial \tilde{q}^j} p_i.$$

This is the coordinate calculation behind the intrinsic statement $(q, p) \in T^*Q$.

Remark 4.1 (Angular momentum is not a canonical momentum). *Let M_a be the components of body angular momentum in principal axes:*

$$M_a = I_a \Omega_a.$$

The M_a are excellent coordinates on the fiber of $T^ \text{SO}(3)$ after a choice of body frame, but they are not canonical momenta conjugate to Euler angles. Locally one has a nondegenerate linear relation*

$$p_i = A_{ia}(q) M_a,$$

where the matrix $A(q)$ depends on the chosen angle coordinates. The canonical symplectic form is still

$$\omega = dq^i \wedge dp_i,$$

in any cotangent coordinate chart, but when it is expressed in the noncanonical variables M_a , the Poisson brackets become the rigid-body Lie-Poisson brackets; the variables M_a are not canonical coordinates with $\{M_a, M_b\} = 0$. Globally, because $\text{SO}(3)$ is a Lie group, left or right translation trivializes the cotangent bundle:

$$T^* \text{SO}(3) \simeq \text{SO}(3) \times \mathfrak{so}(3)^*.$$

This global trivialization is the precise sense in which the angular-momentum components give global fiber coordinates, even though they are noncanonical.

4.6 Variations On $\text{SO}(3)$

A variation of $R(t) \in \text{SO}(3)$ must remain tangent to the group. Write

$$\delta R = R\hat{\Sigma},$$

where $\Sigma(t) \in \mathbb{R}^3$ vanishes at the time endpoints. The induced variation of body angular velocity is

$$\delta\hat{\Omega} = \delta(R^T \dot{R}) = \dot{\hat{\Sigma}} + [\hat{\Omega}, \hat{\Sigma}].$$

Under the hat isomorphism this becomes

$$\delta\Omega = \dot{\Sigma} + \Omega \times \Sigma.$$

Derivation 4.1 (Euler-Poincare equation for the free rigid body). *This derivation differs from the ordinary Euler-Lagrange calculation because Ω is not an independent velocity coordinate. It is the body-frame velocity $R^{-1}\dot{R}$, so its allowed variation must come from an allowed variation of $R \in \text{SO}(3)$. The formula $\delta\Omega = \dot{\Sigma} + \Omega \times \Sigma$ is the constraint that keeps the variation on the rotation group. Poincare introduced this reduced variational form in 1901; the modern geometric setting, where it appears as cotangent-bundle reduction by a Lie group, is the Marsden-Weinstein framework (1974). The reduced action is*

$$S[R] = \int_{t_0}^{t_1} \ell(\Omega) dt, \quad \ell(\Omega) = \frac{1}{2} \Omega^T I \Omega.$$

Let

$$M = \frac{\partial \ell}{\partial \Omega} = I \Omega.$$

Then

$$\begin{aligned} \delta S &= \int_{t_0}^{t_1} M \cdot (\dot{\Sigma} + \Omega \times \Sigma) dt \\ &= [M \cdot \Sigma]_{t_0}^{t_1} - \int_{t_0}^{t_1} \dot{M} \cdot \Sigma dt + \int_{t_0}^{t_1} M \cdot (\Omega \times \Sigma) dt. \end{aligned}$$

The boundary term vanishes. Since

$$M \cdot (\Omega \times \Sigma) = (M \times \Omega) \cdot \Sigma,$$

stationarity for arbitrary Σ gives

$$-\dot{M} + M \times \Omega = 0.$$

Equivalently,

$$\boxed{\dot{M} + \Omega \times M = 0}$$

with the hat convention $\hat{\Omega}v = \Omega \times v$ used throughout these notes. In components this is

$$I_1 \dot{\Omega}_1 = (I_2 - I_3) \Omega_2 \Omega_3,$$

and cyclic permutations.

The term $\Omega \times M$ is not an external torque. It is the correction that appears because M is expressed in the rotating body frame. In the space frame the angular momentum $J = RM$ is constant; in the body frame the components of the same vector change because the axes themselves rotate. Euler derived these equations in 1765 in *Theoria motus corporum solidorum*; the modern reduced variational form is the Euler-Poincare version of the same calculation.

4.7 Euler Equations In Principal Axes

With $M_i = I_i \Omega_i$, Euler's equations are

$$\begin{aligned} I_1 \dot{\Omega}_1 &= (I_2 - I_3) \Omega_2 \Omega_3, \\ I_2 \dot{\Omega}_2 &= (I_3 - I_1) \Omega_3 \Omega_1, \\ I_3 \dot{\Omega}_3 &= (I_1 - I_2) \Omega_1 \Omega_2. \end{aligned}$$

These equations are autonomous and quadratic. They conserve the kinetic energy

$$H = \frac{1}{2} (I_1 \Omega_1^2 + I_2 \Omega_2^2 + I_3 \Omega_3^2) = \frac{1}{2} \left(\frac{M_1^2}{I_1} + \frac{M_2^2}{I_2} + \frac{M_3^2}{I_3} \right)$$

and the squared angular momentum

$$C = \|M\|^2 = M_1^2 + M_2^2 + M_3^2.$$

Direct conservation check. Since $M = I\Omega$,

$$\dot{H} = \Omega \cdot \dot{M}.$$

Using $\dot{M} = -\Omega \times M$,

$$\dot{H} = -\Omega \cdot (\Omega \times M) = 0.$$

Similarly,

$$\dot{C} = 2M \cdot \dot{M} = -2M \cdot (\Omega \times M) = 0.$$

□

4.8 Reconstruction Of The Attitude

Euler's equations are the reduced equations: they determine the body angular momentum $M(t)$ or body angular velocity $\Omega(t) = I^{-1}M(t)$, but they do not by themselves tell us the attitude $R(t)$. Once the reduced solution is known, the attitude is recovered from the reconstruction equation

$$\boxed{\dot{R} = R\hat{\Omega}, \quad R(0) = R_0.}$$

This is a linear equation on the group along the already-computed reduced motion. In general its solution is a path-ordered exponential,

$$R(t) = R_0 \mathcal{P} \exp \left(\int_0^t \hat{\Omega}(s) ds \right),$$

where $\mathcal{P}$ denotes path ordering, because $\hat{\Omega}(t_1)$ and $\hat{\Omega}(t_2)$ need not commute. The solution remains in $\text{SO}(3)$. Indeed,

$$\frac{d}{dt} (R^T R) = (R\hat{\Omega})^T R + R^T R\hat{\Omega} = -\hat{\Omega} + \hat{\Omega} = 0,$$

so $R^T R$ stays equal to I if it is I initially; continuity keeps $\det R = 1$.

The same equation explains why spatial angular momentum is conserved. Define

$$J(t) = R(t)M(t).$$

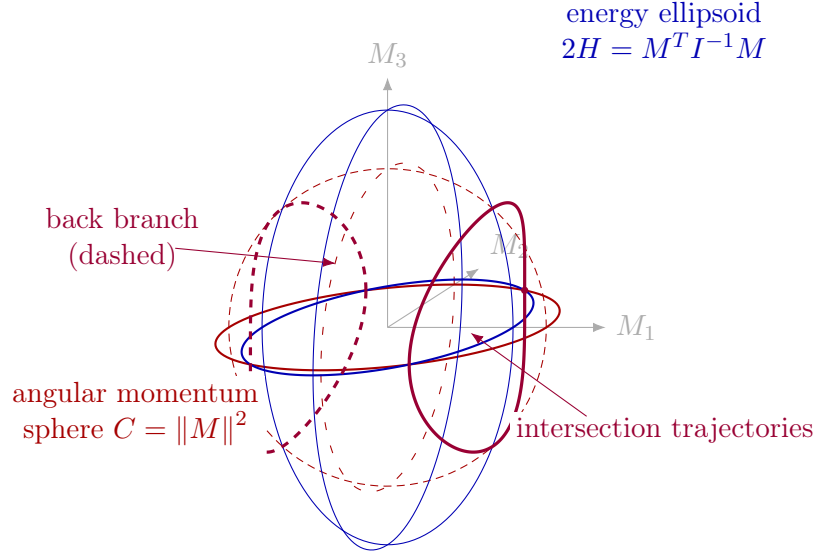


Figure 4.2: Rigid-body reduction in angular-momentum space. Conservation of $\|M\|^2$ confines the motion to a sphere, conservation of H confines it to an ellipsoid, and the reduced trajectory is their intersection. The solid and dashed purple curves are the two projected branches of the exact three-dimensional intersection for the sphere $\|M\|^2 = 4$ and the representative ellipsoid with semi-axes $\sqrt{5/2}, \sqrt{5}, \sqrt{15/2}$.

At a point Ω on this ellipsoid, the normal vector is proportional to

$$\nabla_{\Omega}(\Omega^T I \Omega) = 2I\Omega = 2M.$$

In spatial coordinates the corresponding angular velocity is

$$\omega_{\text{space}} = R\Omega,$$

and the normal is the fixed spatial angular momentum

$$J = RM.$$

Because

$$J \cdot \omega_{\text{space}} = (RM) \cdot (R\Omega) = M \cdot \Omega = 2H,$$

the tip of ω_{space} lies in the fixed plane

$$\Pi_J = \{\omega : J \cdot \omega = 2H\}.$$

This plane is perpendicular to J and has distance $2H/\|J\|$ from the origin. It is tangent to the moving inertia ellipsoid at the instantaneous angular velocity. Indeed, the tangent plane to $\Omega^T I \Omega = 2H$ has normal $M = I\Omega$; after rotation to space the normal is $J = RM$, the normal of Π_J . The point of contact is $\omega_{\text{space}} = R\Omega$, and it lies in Π_J by the displayed identity $J \cdot \omega_{\text{space}} = 2H$.

The no-slip statement is also kinematic. The instantaneous axis of the rigid motion is ω_{space} , so the point of the moving ellipsoid whose spatial position is the contact point has zero instantaneous velocity relative to the fixed plane. Thus the free rigid body can be pictured as the inertia ellipsoid rolling without slipping on the fixed invariable plane.

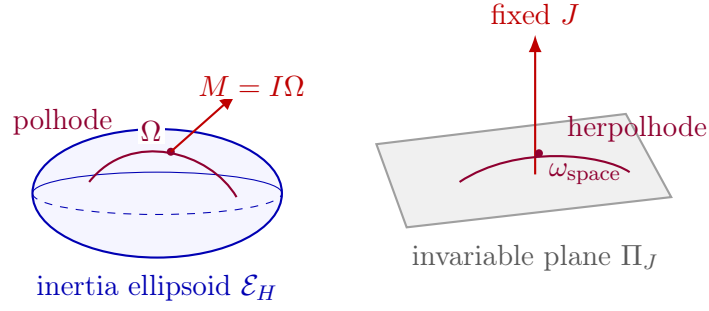


Figure 4.3: Poincaré construction. In body angular-velocity space, the motion lies on the inertia ellipsoid and traces the polhode. In space, the tip of angular velocity lies in the fixed invariant plane perpendicular to the conserved spatial angular momentum J ; its trace is the herpolhode.

Poincaré's construction is another geometric view of the same conservation of H and J , drawn in angular-velocity space. Its value is conceptual: it makes clear why the reduced trajectory is a curve, why the invariant plane is fixed in space, and why body-frame quantities can move while the spatial angular momentum does not.

4.11 Stability Of Principal-Axis Rotations

Assume

$$I_1 < I_2 < I_3.$$

A steady rotation about a principal axis is a solution. For example,

$$\Omega(t) = (\Omega_0, 0, 0)$$

is rotation about the first axis.

Derivation 4.2 (Linear stability). *Perturb rotation about the first axis:*

$$\Omega_1 = \Omega_0 + \alpha_1, \quad \Omega_2 = \alpha_2, \quad \Omega_3 = \alpha_3,$$

and keep only linear terms. Then

$$\dot{\alpha}_1 = 0,$$

while

$$I_2 \dot{\alpha}_2 = (I_3 - I_1) \Omega_0 \alpha_3, \quad I_3 \dot{\alpha}_3 = (I_1 - I_2) \Omega_0 \alpha_2.$$

Differentiate the first of these:

$$\ddot{\alpha}_2 = \frac{(I_3 - I_1)(I_1 - I_2)}{I_2 I_3} \Omega_0^2 \alpha_2.$$

Since $I_3 - I_1 > 0$ and $I_1 - I_2 < 0$, the coefficient is negative, so α_2 oscillates. Rotation about the first axis is linearly stable.

For rotation about the second axis, the analogous coefficient is

$$\frac{(I_2 - I_3)(I_1 - I_2)}{I_1 I_3} \Omega_0^2 > 0,$$

because both factors in the numerator are negative. The perturbation therefore has exponential solutions. Thus the intermediate axis is unstable. Rotation about the third axis is stable.

The geometric reason is that the energy restricted to a fixed angular momentum sphere has extrema at the smallest and largest inertia axes but a saddle at the intermediate axis. This instability is the tennis-racket theorem; it was filmed by cosmonaut Vladimir Dzhanibekov on Salyut-7 in 1985. The mathematics is already in Euler's 1765 rigid-body equations.

Derivation 4.3 (Energy-Casimir stability test). *The same conclusion can be read without solving the linearized equations. Work on the angular-momentum sphere*

$$C(M) = M_1^2 + M_2^2 + M_3^2 = c^2.$$

Near rotation about the first principal axis, write

$$M = (M_1, \eta_2, \eta_3), \quad M_1 = \sqrt{c^2 - \eta_2^2 - \eta_3^2}.$$

Expanding the Hamiltonian to second order gives

$$H - H_0 = \frac{1}{2} \left(\frac{1}{I_2} - \frac{1}{I_1} \right) \eta_2^2 + \frac{1}{2} \left(\frac{1}{I_3} - \frac{1}{I_1} \right) \eta_3^2 + O(\eta^4).$$

If $I_1 < I_2 < I_3$, both quadratic coefficients are negative. The first-axis rotation is therefore a strict local maximum of H on the sphere $C = c^2$. The third-axis rotation is similarly a strict local minimum. Either extremum prevents nearby trajectories with the same C and nearly the same H from leaving a small neighborhood.

For the intermediate axis,

$$M = (\eta_1, M_2, \eta_3),$$

and

$$H - H_0 = \frac{1}{2} \left(\frac{1}{I_1} - \frac{1}{I_2} \right) \eta_1^2 + \frac{1}{2} \left(\frac{1}{I_3} - \frac{1}{I_2} \right) \eta_3^2 + O(\eta^4).$$

The two coefficients have opposite signs, so the point is a saddle of the energy restricted to the angular-momentum sphere. This is the energy-Casimir form of the intermediate-axis instability.

4.12 Lie-Poisson Structure

The reduced rigid body is Hamiltonian on $\mathfrak{so}(3)^* \cong \mathbb{R}^3$, but not with canonical coordinates. The Lie-Poisson bracket is

$$\{F, G\}(M) = -M \cdot (\nabla F(M) \times \nabla G(M)).$$

In coordinate functions this says

$$\{M_i, M_j\} = -\epsilon_{ijk} M_k.$$

This is the minus-Lie-Poisson convention. The minus sign matches the body-frame variables used above and gives the boxed Euler equation $\dot{M} = -\Omega \times M$. Had we used spatial angular momentum components as reduced variables, the sign would be opposite. This is another place where the body/space distinction is not cosmetic: it is the $\pm$ Lie-Poisson dichotomy coming from left versus right trivialization of $T^*\mathrm{SO}(3)$. The physical motion is the same, but the reduced bracket changes sign.

For

$$H(M) = \frac{1}{2} M \cdot I^{-1} M, \quad \nabla H = I^{-1} M = \Omega,$$

the equation for any observable F is

$$\dot{F} = \{F, H\}.$$

Taking $F(M) = M_i$ yields

$$\dot{M} = -\Omega \times M.$$

Definition 4.1 (Casimir). *A Casimir is a function C satisfying*

$$\{C, F\} = 0$$

for every observable F . For the rigid body bracket,

$$C(M) = \|M\|^2$$

is a Casimir. Its level sets are the coadjoint orbits, which here are spheres.

Casimirs foliate the reduced phase space. For the rigid body, $\|M\|^2$ labels each angular-momentum sphere, and the Hamiltonian flow lives on these spheres. Each sphere carries a genuine two-dimensional symplectic structure, and the rigid body restricted to one such sphere has one degree of freedom. Here a coadjoint orbit means the orbit of a covector $M \in \mathfrak{so}(3)^*$ under the coadjoint action of $\mathrm{SO}(3)$. After identifying $\mathfrak{so}(3)^*$ with $\mathbb{R}^3$, that action is ordinary rotation, so the nonzero orbits are spheres $\|M\| = \text{constant}$. This is the rigid-body instance of Marsden-Weinstein symplectic reduction (1974): the canonical phase space $T^*\mathrm{SO}(3)$ descends to coadjoint orbits with their Kirillov-Kostant-Souriau symplectic forms. This is the reduced reason that the intersection curves in Figure 4.2 are the phase curves of the system.

4.13 Liouville Integrability And Invariant Tori

The rigid-body calculations above show a pattern: conservation laws confine motion to lower-dimensional sets, and the reduced flow on those sets can be simple even when the original coordinates look complicated. Liouville integrability is the precise Hamiltonian version of this pattern. It is not just “having many constants of motion.” The constants must be independent, they must Poisson commute, and their common level set must be regular enough to carry the commuting Hamiltonian flows.

The three hypotheses have separate meanings. Independence says the level set has the expected dimension. Involution says the Hamiltonian flows generated by the integrals commute. Compactness says the flow coordinates eventually repeat, turning straight-line motion on $\mathbb{R}^n$ into straight-line motion on a torus. Connectedness says one is discussing a single component of the common level set; otherwise the theorem applies component by component. Completeness of the commuting flows is the dynamical hypothesis that lets the $\mathbb{R}^n$ -action be followed for all flow times. Dropping any one of these assumptions changes the conclusion.

This is why the theorem should not be read backwards. Seeing a bounded-looking orbit or many conserved-looking quantities in a simulation does not by itself prove integrability. One must know that the integrals are globally defined on the phase-space region under study, independent on the level set, and in involution. The free rigid body passes these tests; a slightly forced or gravity-coupled top often fails them.

Definition 4.2 (Liouville integrability). *A Hamiltonian system on a $2n$ -dimensional symplectic manifold is Liouville integrable if there are n independent functions*

$$F_1 = H, F_2, \dots, F_n$$

such that

$$\{F_i, F_j\} = 0$$

for all i, j , and the associated Hamiltonian flows are complete on the regular level sets under discussion. The functions are said to be in involution.

Liouville proved the analytic integration-by-quadratures part in 1855: once enough commuting integrals are present, the remaining motion can be obtained by quadratures. The modern compact-level-set formulation, due to Arnold (1963), is now called the Liouville-Arnold theorem; it identifies regular compact level sets as tori carrying linear flow. Jacobi's 1839 integration of geodesics on an ellipsoid is a classical early example beyond Kepler and the rigid body: the separated coordinates are not a coordinate trick but a sign of hidden commuting integrals.

Example 4.1 (Basic integrable systems). *Any autonomous Hamiltonian system with one degree of freedom is integrable: the Hamiltonian itself supplies the one conserved quantity. The fixed-COM rigid body has three degrees of freedom. The three spatial functions*

$$H, \quad J_3, \quad J^2$$

are independent and mutually Poisson commuting on regular sets, even though the individual components J_1, J_2, J_3 do not commute with one another. This is the rigid-body example used throughout this chapter.

Assumption 4.2 (Regular compact level set). *Fix constants $f = (f_1, \dots, f_n)$ and define*

$$M_f = \{x \in P : F_i(x) = f_i, \ i = 1, \dots, n\},$$

where P is the $2n$ -dimensional phase space. We assume that the differentials dF_i are independent on M_f and that M_f is compact and connected.

The regularity assumption says that M_f is a smooth n -dimensional submanifold. Its tangent vectors $v \in T_x M_f$ are those satisfying

$$dF_i(v) = 0 \quad \text{for all } i.$$

Let X_{F_i} be the Hamiltonian vector field of F_i , defined by

$$\iota_{X_{F_i}} \omega = dF_i.$$

Then

$$dF_j(X_{F_i}) = \{F_j, F_i\} = 0.$$

Thus every X_{F_i} is tangent to M_f . The commutator of two Hamiltonian vector fields is

$$[X_{F_i}, X_{F_j}] = -X_{\{F_i, F_j\}},$$

so involution implies

$$[X_{F_i}, X_{F_j}] = 0.$$

We therefore get n independent commuting vector fields on M_f .

Derivation 4.4 (Why the regular compact level set is a torus). *The proof has a simple geometric shape. The Hamiltonian vector fields X_{F_i} give n independent directions tangent to M_f , and because the integrals Poisson commute, flowing in one direction and then another gives the same point as flowing in the opposite order. Thus M_f is locally coordinatized by “flow times.” Compactness is what turns these flow-time coordinates from $\mathbb{R}^n$ into periodic coordinates.*

Commuting vector fields have commuting flows. Starting at a point $x_0 \in M_f$, flow for times $s = (s^1, \dots, s^n)$ along the n vector fields:

$$\Phi_s(x_0) = \exp(s^1 X_{F_1}) \cdots \exp(s^n X_{F_n}) x_0.$$

Because the flows commute, this defines an action of $\mathbb{R}^n$ on M_f . Completeness is used here: without complete flows, this action would only be defined for small flow times, and the period-lattice quotient below would have no global meaning. Since M_f is compact and the vector fields are smooth, the flows on M_f are complete in the cases considered here. This is the standard escape lemma for ODEs: a maximal solution of a smooth vector field that remained defined only on a finite time interval would have compact closure, the vector field would be bounded there, and the local existence theorem would extend the solution past the endpoint. Independence of the vector fields makes the action locally free. More concretely, the constant-rank theorem applied to the map $s \mapsto \Phi_s(x_0)$ says that its derivative at $s = 0$ is an isomorphism onto $T_{x_0} M_f$; hence the flow times s^i are local coordinates. The only possible global identifications are translations

$$s \sim s + \lambda, \quad \lambda \in \Lambda,$$

where $\Lambda \subset \mathbb{R}^n$ is the period lattice of the orbit. Compactness forces Λ to have rank n . If Λ had rank $k < n$, then the structure theorem for discrete subgroups of $\mathbb{R}^n$ gives a linear basis in which Λ is generated by k independent period vectors, so

$$\mathbb{R}^n / \Lambda \simeq \mathbb{T}^k \times \mathbb{R}^{n-k},$$

which is noncompact. The action is locally free, so each orbit is open in the n -manifold M_f . Since M_f is connected, that open orbit is the whole level set. Hence

$$M_f \simeq \mathbb{R}^n / \Lambda \simeq \mathbb{T}^n.$$

These are the invariant Liouville tori.

Enough commuting flows on a compact regular level set force the level set to be a torus. The angle variables introduced below are coordinates along these flows, and the action variables are the quantities that label which torus one is on.

The symplectic form vanishes when restricted to an invariant torus. Indeed, on the tangent basis X_{F_i} ,

$$\omega(X_{F_i}, X_{F_j}) = dF_i(X_{F_j}) = \{F_i, F_j\} = 0.$$

Since these vectors span $T_x M_f$, the torus is Lagrangian:

$$\omega|_{M_f} = 0.$$

Derivation 4.5 (Coordinate proof that the invariant torus is Lagrangian). *A useful coordinate check is the following. Use canonical coordinates (q_i, p_i) , and suppose the level set is written locally as*

$$F_i(p, q) = f_i, \quad i = 1, \dots, n.$$

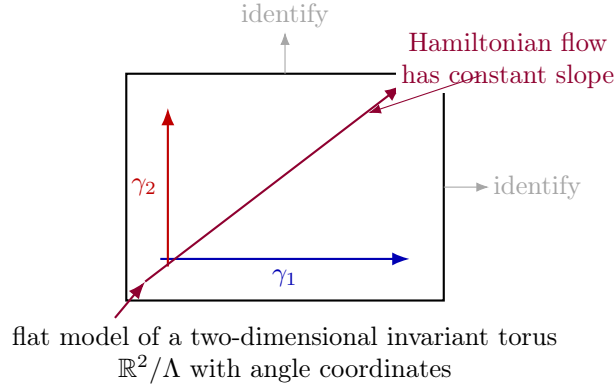


Figure 4.4: A Liouville torus is locally a space of commuting flow times with periodic identifications. Action variables label the torus; angle variables are coordinates along the fundamental cycles.

If the equations can be solved locally for

$$p_i = p_i(q, F),$$

then

$$\begin{aligned} \omega &= dq_i \wedge dp_i \\ &= dq_i \wedge \left(\frac{\partial p_i}{\partial q_j} \Big|_F dq_j + \frac{\partial p_i}{\partial F_j} \Big|_q dF_j \right). \end{aligned}$$

The key claim is

$$\frac{\partial p_i}{\partial q_j} \Big|_F = \frac{\partial p_j}{\partial q_i} \Big|_F.$$

Then the $dq_i \wedge dq_j$ contribution vanishes by antisymmetry, and

$$\omega = \frac{\partial p_i}{\partial F_j} \Big|_q dq_i \wedge dF_j.$$

On the invariant torus $F_j = f_j$, $dF_j = 0$, so

$$\omega|_{M_f} = 0.$$

This coordinate proof is local on a chart where the projection of the torus to the q -coordinates has full rank. At caustics or turning sets that projection fails, the same Lagrangian conclusion remains true by the coordinate-free argument above: at a regular point the independent Hamiltonian fields X_{F_i} still span $T_x M_f$, and their pairwise symplectic products are still $\{F_i, F_j\} = 0$. What fails at a caustic is only the choice of q as local coordinates on the torus. The displayed function $p(q, F)$ must then be replaced by another local branch or another canonical chart.

To prove the claim, differentiate the identities $F_k(p(F, q), q) = F_k$ at fixed F :

$$0 = \frac{\partial F_k}{\partial p_j} \Big|_q \frac{\partial p_j}{\partial q_i} \Big|_F + \frac{\partial F_k}{\partial q_i} \Big|_p.$$

Let

$$A_{kj} = \left. \frac{\partial F_k}{\partial p_j} \right|_q, \quad M_{ji} = \left. \frac{\partial p_j}{\partial q_i} \right|_F, \quad B_{ki} = \left. \frac{\partial F_k}{\partial q_i} \right|_p.$$

Then

$$AM + B = 0, \quad M = -A^{-1}B.$$

The involution condition gives

$$0 = \{F_i, F_j\} = B_{ik}A_{jk} - A_{ik}B_{jk},$$

or, equivalently,

$$AB^T - BA^T = 0.$$

Hence

$$A^{-1}B = B^T(A^T)^{-1} = (A^{-1}B)^T,$$

so $M = M^T$, proving the symmetry claim.

4.14 Action Variables And Angle Variables

Choose a basis of one-cycles $\gamma_1, \dots, \gamma_n$ on an invariant torus M_f . In canonical coordinates the action variables are

$$I_j = \frac{1}{2\pi} \oint_{\gamma_j} p_i dq^i.$$

Actions are therefore not arbitrary constants of integration. They are phase-space areas measured around the independent cycles of the torus. This is why they are insensitive to small deformations of the cycle on a fixed torus. They are not unique, however: changing the integer cycle basis of $H_1(\mathbb{T}^n; \mathbb{Z})$ transforms the action vector by an integral unimodular matrix, $I \mapsto AI$ with $A \in \text{GL}(n, \mathbb{Z})$ (or $\text{SL}(n, \mathbb{Z})$ after fixing orientation). Thus a local choice of action variables is part of the action-angle chart, not a canonical object independent of cycle basis. They are invariant under continuous deformations of γ_j on the torus. If γ_j and γ'_j bound a two-chain $D \subset M_f$, then by Stokes' theorem

$$\oint_{\gamma_j} p_i dq^i - \oint_{\gamma'_j} p_i dq^i = \int_D d(p_i dq^i) = \int_D \omega = 0.$$

This is also why action variables entered early quantum theory: the Bohr-Sommerfeld quantization rules, especially Sommerfeld's 1916 old-quantum conditions, imposed $\oint p dq = nh$ on precisely these periods. The modern classical use is not quantization, but the same periods are the coordinates that survive canonical perturbation theory; in one degree of freedom, the relation between action and period is also the classical seed of WKB quantization. Thus I_j depends on the torus, not on the representative loop. Under regularity assumptions, the I_j can replace the original integrals F_j as labels for nearby tori.

To find conjugate angles, construct a local generating function

$$S(I, q) = \int^q p_i(I, q') dq'^i,$$

where the integration is performed on a local branch. The compatibility condition for this definition is

$$\left. \frac{\partial p_i}{\partial q^j} \right|_I = \left. \frac{\partial p_j}{\partial q^i} \right|_I,$$

which is another form of $\omega|_{M_I} = 0$. Define

$$\theta^j = \left. \frac{\partial S}{\partial I_j} \right|_q.$$

This generating function proves that the new coordinates are canonical. Since

$$dS = p_i dq^i + \theta^j dI_j,$$

taking an exterior derivative gives

$$0 = dp_i \wedge dq^i + d\theta^j \wedge dI_j.$$

Thus

$$dq^i \wedge dp_i = d\theta^j \wedge dI_j,$$

so the symplectic form takes canonical action-angle form. When one goes around the cycle γ_k , the multivalued function S jumps by

$$\Delta_{\gamma_k} S = \oint_{\gamma_k} p_i dq^i = 2\pi I_k.$$

Therefore

$$\Delta_{\gamma_k} \theta^j = \frac{\partial}{\partial I_j} (2\pi I_k) = 2\pi \delta^j_k.$$

The θ^j are angles on the torus:

$$\theta^j \sim \theta^j + 2\pi.$$

In these action-angle coordinates the Hamiltonian is a function of the actions alone,

$$H = H(I),$$

and Hamilton's equations reduce to

$$\dot{I}_j = 0, \quad \dot{\theta}^j = \omega^j(I) = \frac{\partial H}{\partial I_j}.$$

Thus motion on each invariant torus is linear. If the frequencies are rationally related, the orbit closes; otherwise the orbit is dense on the torus.

The word “linear” refers to the angle coordinates, not to the original configuration variables. A dense quasiperiodic curve on a torus can look quite complicated after being projected to ordinary coordinates. Action-angle variables are valuable precisely because they separate the simple invariant motion on the torus from the sometimes complicated embedding of that torus in phase space.

4.15 Action Integrals In One Degree Of Freedom

The word action now appears in a second, related sense. Hamilton's principle uses the time integral $\int L dt$ as a functional on paths. Action-angle theory uses

$$I_j = \frac{1}{2\pi} \oint_{\gamma_j} p_i dq^i$$

as a coordinate labelling a closed orbit or invariant torus. The common ingredient is the canonical one-form $p_i dq^i$, whose periods are the action variables.

The abstract definition

$$I_j = \frac{1}{2\pi} \oint_{\gamma_j} p_i dq^i$$

is easiest to understand first in one degree of freedom. Suppose an autonomous system has Hamiltonian

$$H(q, p) = \frac{p^2}{2m} + V(q)$$

and a bounded orbit between turning points $q_-(E)$ and $q_+(E)$. On the energy curve $H = E$,

$$p(q, E) = \pm \sqrt{2m(E - V(q))}.$$

The closed phase-space loop has a positive-momentum branch and a negative-momentum branch, so the action is

$$I(E) = \frac{1}{2\pi} \oint p dq = \frac{1}{\pi} \int_{q_-(E)}^{q_+(E)} \sqrt{2m(E - V(q))} dq.$$

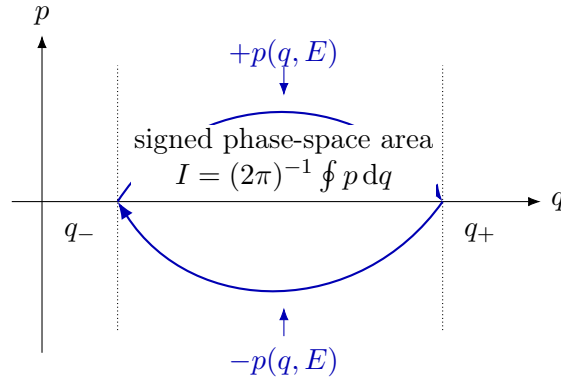


Figure 4.5: For a one-dimensional bound orbit the action is the signed phase-space area enclosed by the energy curve divided by 2π . The turning points contribute no endpoint term when differentiating with respect to energy because $p(q_{\pm}, E) = 0$.

The period follows by differentiating $I(E)$. The moving endpoint terms vanish because $p(q_{\pm}(E), E) = 0$:

$$\frac{dI}{dE} = \frac{1}{\pi} \int_{q_-}^{q_+} \frac{m dq}{\sqrt{2m(E - V(q))}} = \frac{1}{2\pi} T(E).$$

Therefore

$$\omega(E) = \frac{dH}{dI} = \left(\frac{dI}{dE} \right)^{-1}, \quad T(E) = 2\pi \frac{dI}{dE}.$$

This formula is the one-dimensional seed of action-angle theory. The same period integral reappears on higher-dimensional invariant tori by integrating the canonical one-form around a basis of cycles.

Example 4.2 (Harmonic oscillator). *For*

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega_0^2 q^2,$$

the phase curve is an ellipse:

$$\frac{p^2}{2mE} + \frac{q^2}{2E/(m\omega_0^2)} = 1.$$

The area enclosed by the ellipse is

$$\pi\sqrt{2mE} \sqrt{\frac{2E}{m\omega_0^2}} = \frac{2\pi E}{\omega_0}.$$

Thus

$$I = \frac{1}{2\pi} \text{Area} = \frac{E}{\omega_0}, \quad H(I) = \omega_0 I.$$

The conjugate angle advances uniformly:

$$\dot{\theta} = \frac{\partial H}{\partial I} = \omega_0.$$

4.16 Rigid Body Tori

The full rigid body with fixed center of mass has three degrees of freedom. On $T^*\text{SO}(3)$ the functions

$$H, \quad J_3, \quad J^2$$

are mutually Poisson commuting when J_3 is a fixed spatial component of angular momentum and $J^2 = \|J\|^2$. For regular values, their common level set is a three-torus.

The phrase “regular values” hides the familiar exceptional motions. Principal axis rotations and separatrix-level motions are not generic tori: the level set can pinch, a cycle can collapse, or the period can diverge. Liouville integrability describes the open dense region of phase space where the commuting flows are independent and compact; the exceptional sets are precisely where the qualitative changes in rigid-body motion occur.

The geometry can be seen in two stages. First, the body angular momentum $M = R^T J$ is constrained by

$$J^2 = M_1^2 + M_2^2 + M_3^2 = \text{constant}$$

and

$$H = \frac{1}{2} \left(\frac{M_1^2}{I_1} + \frac{M_2^2}{I_2} + \frac{M_3^2}{I_3} \right) = \text{constant}.$$

This puts M on the closed curve obtained by intersecting a sphere with an ellipsoid. Second, fixing J_3 constrains the orientation R : the third spatial component of $J = RM$ is fixed. For a given M on the curve, this condition leaves a circle of allowed attitudes. A circle fibered over the momentum-space circle gives a two-dimensional torus in the reduced picture, and including the remaining cyclic angle gives the full three-torus.

The reduced free rigid body has one effective degree of freedom on each angular momentum sphere. The energy level on that sphere gives closed curves except at separatrices. This is the concrete meaning of integrability here.

4.17 Rigid Body Kinematics As A Gauge Prototype

The rigid body has one orientation $R(t)$. A Cosserat body assigns an orientation to each material point:

$$R(t) \longrightarrow R(X, t).$$

The rigid-body angular velocity

$$R^T \dot{R}$$

becomes a time connection, while the spatial derivatives

$$R^T \partial_i R$$

become material connection components. The same operation that turns an attitude matrix into angular velocity therefore reappears, point by point, as the connection language of deformable bodies.

4.18 What To Take Forward

- A rigid body lives on $\text{SO}(3)$, not in a global set of Euler angles. Body and spatial angular momentum describe the same vector in different frames, and the sign of the reduced Lie-Poisson bracket depends on that choice.
- The free rigid body is integrable because energy and angular momentum confine the reduced motion to curves on angular-momentum spheres; after reconstruction these curves lift to invariant tori.
- Liouville tori are produced by commuting Hamiltonian flows, with action variables defined by period integrals of the canonical one-form. The operation $R^T dR$ used here will reappear as the connection language of deformable bodies and Cosserat continua.

4.19 Chapter-End Laboratories And Exercises

These calculations connect action variables to concrete integrable systems and show how the rigid body fits into the same torus-based language.

4.19.1 Action Variables For The Isotropic Oscillator In Polar Coordinates

A canonical test case for action-angle coordinates is the two-dimensional isotropic oscillator

$$H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + \frac{1}{2}m\omega^2 r^2.$$

The angular momentum

$$L = p_\theta$$

is conserved. The angular action is immediate:

$$I_\theta = \frac{1}{2\pi} \oint p_\theta d\theta = L.$$

The radial action is

$$I_r = \frac{1}{\pi} \int_{r_-}^{r_+} \sqrt{2mE - \frac{L^2}{r^2} - m^2\omega^2 r^2} dr.$$

Rather than evaluate this integral by brute force, use a structural argument. For the Cartesian oscillator,

$$H = \omega(I_x + I_y).$$

The angular momentum generates rotations in the plane. For nonnegative L , the circular orbit has all its action in rotation and no radial oscillation. On a circular orbit,

$$I_r = 0, \quad E = \omega L.$$

Radial oscillation has twice the angular oscillator frequency: one full radial cycle occurs while the Cartesian oscillator runs through half of its ellipse. Therefore the Hamiltonian in polar actions must be

$$\boxed{H = \omega(2I_r + |I_\theta|)}.$$

Solving for the radial action,

$$\boxed{I_r = \frac{E}{2\omega} - \frac{|L|}{2}}.$$

This formula records the cycle-basis dependence of action variables. Cartesian actions and polar actions are related by an integral linear change of torus cycles, not by an arbitrary smooth coordinate change.

4.19.2 Symmetric Free Top: Actions And Dense Motion

For a symmetric free rigid body with

$$I_1 = I_2 > I_3,$$

use Euler angles (ϕ, θ, ψ) . The conserved quantities are

$$H, \quad J^2, \quad J_z.$$

The momentum conjugate to ψ is the body-axis component of angular momentum:

$$p_\psi = M_3.$$

The momentum conjugate to ϕ is the spatial vertical component:

$$p_\phi = J_z.$$

The total angular momentum satisfies the standard Euler-angle identity

$$J^2 = p_\theta^2 + \frac{(p_\phi - p_\psi \cos \theta)^2}{\sin^2 \theta} + p_\psi^2.$$

This equation is the rigid-body analogue of the angular-momentum identity in spherical coordinates. It says that p_θ is not an independent constant; it is determined by J , $J_z = p_\phi$, $M_3 = p_\psi$, and θ .

The convention $I_1 = I_2 > I_3$ is the prolate convention: the moment about the symmetry axis is smaller than the transverse moments. An oblate symmetric top has the reversed inequality. The formulas below adapt by the same algebra with the corresponding chamber inequalities. For the symmetric top, whose classical heavy-top version was analyzed by Lagrange in 1788,

$$H = \frac{J^2 - M_3^2}{2I_1} + \frac{M_3^2}{2I_3}.$$

Solving for M_3^2 gives

$$M_3^2 = \frac{2I_1 I_3}{I_1 - I_3} \left(H - \frac{J^2}{2I_1} \right).$$

Thus one action is

$$I_\psi = M_3 = \sqrt{\frac{2I_1 I_3}{I_1 - I_3} \left(H - \frac{J^2}{2I_1} \right)},$$

with sign chosen by the orientation of the ψ -cycle. Another action is

$$I_\phi = J_z.$$

The remaining action is the cycle traced by the nutation angle θ :

$$I_\theta = \frac{1}{2\pi} \oint p_\theta d\theta = \frac{1}{\pi} \int_{\theta_-}^{\theta_+} \sqrt{J^2 - M_3^2 - \frac{(J_z - M_3 \cos \theta)^2}{\sin^2 \theta}} d\theta.$$

The endpoints $\theta_\pm$ are the roots where the square root vanishes. To see the evaluation, set

$$a = J_z, \quad b = M_3, \quad x = \cos \theta.$$

Then

$$I_\theta = \frac{1}{\pi} \int_{x_-}^{x_+} \frac{\sqrt{J^2(1-x^2) - (a-bx)^2}}{1-x^2} dx,$$

where

$$x_\pm = \frac{ab \pm \sqrt{(J^2 - a^2)(J^2 - b^2)}}{J^2}.$$

This integral is the symplectic area on the angular-momentum sphere between the two latitude circles $J_z = a$ and $M_3 = b$, divided by 2π . The cap-area formula is elementary in spherical coordinates. Write the coadjoint sphere as $M = Jn$, with $n = (\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta)$. The Kirillov form is

$$\omega_{\text{KKS}} = J \sin \vartheta d\vartheta \wedge d\varphi,$$

so the cap above $M_z = c$, where $\cos \vartheta = c/J$, has area

$$\int_0^{2\pi} \int_0^{\arccos(c/J)} J \sin \vartheta \, d\vartheta \, d\varphi = 2\pi(J - c).$$

Thus a cap of height $h = J - c$ has symplectic area $2\pi h$. The two fixed projections remove caps determined by $|a|$ and $|b|$; the remaining nutation cycle has height $J - \max\{|a|, |b|\}$. Hence

$$I_\theta = J - \max\{|J_z|, |M_3|\}.$$

Thus the simple formula

$$I_\theta = J - |J_z|$$

is the local chamber $|J_z| \geq |M_3|$. In the other chamber the limiting projection is the body-axis component, and the same cycle convention gives $I_\theta = J - |M_3|$. This chamber dependence is not a new physical conservation law; it records which projection of the angular momentum sphere sets the polar turning circle. The spin action $I_\psi = M_3$ remains a separate action measuring rotation about the body's symmetry axis. On the chamber wall $|J_z| = |M_3|$, the two polar circles touch the same turning latitude and the chosen nutation cycle changes its representative. The formula is continuous there, but the local action chart changes because the cycle basis used to describe the torus is changing.

In action variables, this means

$$J = I_\theta + \max\{|I_\phi|, |I_\psi|\},$$

and hence the free symmetric-top Hamiltonian is

$$H(I_\theta, I_\phi, I_\psi) = \frac{(I_\theta + \max\{|I_\phi|, |I_\psi|\})^2 - I_\psi^2}{2I_1} + \frac{I_\psi^2}{2I_3},$$

within a fixed chamber. The absolute values and maximum are the price paid for using Euler-angle cycles across coordinate singularities; once a chamber is chosen, the Hamiltonian is smooth in the corresponding local action variables.

Let α_j denote the angle variables conjugate to the action variables $I_j = (I_\theta, I_\phi, I_\psi)$ in the chosen chamber. The angle equations are

$$\dot{\alpha}_j = \frac{\partial H}{\partial I_j}.$$

The Hamiltonian does not depend on the inertial direction of the angular momentum. This degeneracy comes from the continuous symmetry of rotations about the spatial angular-momentum direction, not from chaotic dynamics. In the nonsymmetric free rigid body the generic invariant sets are still Liouville tori, but the action integrals involve elliptic functions in place of the elementary chamber formula above. Concretely, set

$$A = 2H, \quad B = J^2,$$

and take $I_1 < I_2 < I_3$. The two algebraic integrals

$$A = I_1\Omega_1^2 + I_2\Omega_2^2 + I_3\Omega_3^2, \quad B = I_1^2\Omega_1^2 + I_2^2\Omega_2^2 + I_3^2\Omega_3^2$$

solve for Ω_1^2 and Ω_3^2 in terms of $x = \Omega_2^2$:

$$\Omega_1^2 = \frac{B - AI_3 - I_2(I_2 - I_3)x}{I_1(I_1 - I_3)},$$

$$\Omega_3^2 = \frac{AI_1 - B - I_2(I_1 - I_2)x}{I_3(I_1 - I_3)}.$$

Substituting these expressions into $I_2\dot{\Omega}_2 = (I_3 - I_1)\Omega_3\Omega_1$ gives a quartic polynomial in Ω_2 . Let

$$x_1 = \frac{AI_3 - B}{I_2(I_3 - I_2)}, \quad x_3 = \frac{B - AI_1}{I_2(I_2 - I_1)}.$$

The root x_1 is where $\Omega_1 = 0$, and x_3 is where $\Omega_3 = 0$. In a physical chamber one must also require $\Omega_1^2, \Omega_3^2 \geq 0$. Thus, for the chamber in which the intermediate-axis component changes sign,

$$0 \leq x = \Omega_2^2 \leq a^2, \quad a^2 = \min\{x_1, x_3\}, \quad b^2 = \max\{x_1, x_3\}.$$

The equation then takes the form

$$\dot{\Omega}_2^2 = \frac{(I_3 - I_2)(I_2 - I_1)}{I_1 I_3} (a^2 - \Omega_2^2)(b^2 - \Omega_2^2).$$

Hence, after choosing the origin of time in this chamber,

$$\boxed{\Omega_2(t) = a \operatorname{sn}(\nu(t - t_0), k)}, \quad \nu = b \sqrt{\frac{(I_3 - I_2)(I_2 - I_1)}{I_1 I_3}}, \quad k^2 = \frac{a^2}{b^2}.$$

To check the sn form directly, use

$$\frac{d}{du} \operatorname{sn}(u, k) = \operatorname{cn}(u, k) \operatorname{dn}(u, k),$$

together with

$$a^2 - a^2 \operatorname{sn}^2 = a^2 \operatorname{cn}^2, \quad b^2 - a^2 \operatorname{sn}^2 = b^2 \operatorname{dn}^2.$$

Then $\dot{\Omega}_2^2$ equals the quartic above precisely when

$$\nu = b \sqrt{\frac{(I_3 - I_2)(I_2 - I_1)}{I_1 I_3}}.$$

The dn factor appears instead in the principal-axis component that does not change sign in the chosen chamber: if $a^2 = x_1$, then Ω_1 is proportional to cn and Ω_3 is proportional to dn; if $a^2 = x_3$, the roles of Ω_1 and Ω_3 are interchanged. This chamber accounting is the elliptic-function version of the intermediate-axis instability: the intermediate component naturally passes through zero, while the dominant principal-axis component keeps its sign. The elementary symmetric-top formula is therefore a degenerate limit of the elliptic-function solution.

4.20 Associated Demos

The script `demos/python/rigid_body_euler_top.py` integrates Euler's equations, reconstructs the attitude with

$$R_{n+1} = R_n \exp(\Delta t \hat{\Omega}_{n+1/2}),$$

and reports drift in H , $C = \|M\|^2$, the spatial momentum $J = RM$, and the constraints $R^T R = I$, $\det R = 1$. The Mathematica script `demos/mathematica/RigidBodyEulerTop.wl` gives a symbolic and numerical companion for the reduced equations. A useful check is to compare the numerical drift in H , C , and J for several time steps and initial conditions near the intermediate axis.

Chapter 5

Perturbation, KAM, And Breakdown Of Invariant Tori

5.1 From Integrable Motion To Transport

A massless asteroid orbits the Sun while Jupiter perturbs it. Without Jupiter, the problem is Kepler motion, one of the most integrable systems in classical mechanics. After Jupiter is included, resonances appear, invariant tori fragment, and transport channels open. The full restricted three-body formulation, including Lagrange points and ejection statistics, is developed separately in Chapter 6.

This question changed the meaning of “solving” a mechanical system. The nineteenth-century hope was that enough first integrals would reduce the motion to quadratures. Poincare’s 1890 negative theorem showed that generic perturbations of an integrable Hamiltonian need not admit additional analytic first integrals beyond the energy; his later work on the three-body problem exposed the stable-unstable manifold tangles behind this failure. Kolmogorov, Arnold, and Moser then identified the survival theorem inside the failure: sufficiently irrational invariant tori persist, while resonant regions become the sites of islands, separatrix splitting, and transport. The standard map is this chapter’s compact laboratory; the asteroid problem is the gravitational setting we revisit once the perturbative mechanisms are clear.

The precise Poincare statement is not that no special integral can ever occur. It is a generic obstruction theorem. In lecture-note form: for a nondegenerate integrable Hamiltonian $H_0(I)$ and a generic analytic perturbation $H_0 + \varepsilon H_1$, there is no additional first integral analytic in (I, θ, ε) , independent of the Hamiltonian, and reducing at $\varepsilon = 0$ to a genuinely new integral on the resonant region. The circular restricted three-body problem realizes this obstruction physically: the Jacobi constant survives, but one should not expect a second analytic integral that restores two-degree-of-freedom integrability.

Assumption 5.1 (Celestial perturbation model). *When a celestial example is needed, this chapter uses a planar restricted model:*

1. the Sun has mass $M_\odot$;
2. Jupiter has mass M_J and follows a prescribed circular orbit;
3. asteroids are massless test particles;
4. asteroid-asteroid interactions are neglected;

5. all motion is confined to one plane.

This model is a controlled Hamiltonian laboratory for perturbation, resonance, and transport. It is not intended as a high-precision Solar System ephemeris.

Roadmap. The chapter uses Kepler motion as the integrable reference, canonical perturbation theory to expose small divisors, KAM theory for torus survival, and resonance geometry for torus breakdown.

Appendix A gives the more technical proof architecture behind the KAM statement used here: analytic domains, the homological equation, twist and isoenergetic variants, small-divisor estimates, and the Cantor-measure conclusion. The present chapter keeps those estimates at the level needed to understand the mechanics.

Three levels run through the discussion. The analytic level asks whether a near-identity canonical transformation can remove angle dependence. The geometric level asks which invariant tori, islands, and separatrices organize phase space. The numerical level asks how an ensemble moves through that geometry over a finite time. The asteroid ejection experiment in the next chapter is meaningful only because these levels explain where transport channels come from.

How approximations are labelled. Perturbation theory is useful only when the bookkeeping is explicit. In this chapter H_0 denotes the exactly integrable reference problem, εH_1 denotes a controlled perturbation, $O(\varepsilon^2)$ denotes terms dropped by a first-order calculation, and resonance conditions such as $k \cdot \omega = 0$ identify where that first-order calculation is least trustworthy. Numerical experiments later in the chapter test the transport suggested by these estimates over a stated finite time horizon.

5.2 Kepler Motion As The Integrable Reference

For two bodies of masses m_1, m_2 , let

$$r = x_1 - x_2$$

be the relative position and

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

the reduced mass. The relative Hamiltonian is

$$H(r, p) = \frac{\|p\|^2}{2\mu} - \frac{Gm_1 m_2}{\|r\|}.$$

For a massless asteroid around the Sun, we divide by asteroid mass and work with the specific Hamiltonian

$$h(r, v) = \frac{1}{2} \|v\|^2 - \frac{GM_\odot}{r}, \quad r = \|r\|.$$

Derivation 5.1 (Kepler orbit equation). *The angular momentum per unit mass is*

$$\ell = r^2 \dot{\theta}.$$

This ℓ is the specific angular momentum. In Chapter 2, the two-body Kepler derivation used the massful angular momentum $\ell_{\text{mass}} = \mu r^2 \dot{\theta}$; the two symbols differ by the reduced mass because this chapter has divided the Hamiltonian by the asteroid mass. Section 2.10.2 gives the same Binet

equation with an arbitrary particle mass m ; setting $m = 1$ gives the specific convention used here. For a central potential, ℓ is conserved. The radial equation is

$$\ddot{r} - r\dot{\theta}^2 = -\frac{GM_{\odot}}{r^2}.$$

Set $u = 1/r$. Using

$$\frac{d}{dt} = \dot{\theta} \frac{d}{d\theta} = \ell u^2 \frac{d}{d\theta},$$

the radial equation loses its explicit time dependence and becomes Binet's equation for the orbit shape:

$$\frac{d^2 u}{d\theta^2} + u = \frac{GM_{\odot}}{\ell^2}.$$

Therefore

$$u(\theta) = \frac{GM_{\odot}}{\ell^2} (1 + e \cos(\theta - \varpi)),$$

equivalently

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos(\theta - \varpi)}.$$

Here a is the semimajor axis, e is eccentricity, and ϖ is the longitude of periapse.

Bound Kepler orbits have energy

$$h = -\frac{GM_{\odot}}{2a}.$$

Kepler's third law follows from the period T :

$$n^2 a^3 = GM_{\odot}, \quad n = \frac{2\pi}{T}.$$

5.3 Action-Angle View Of Kepler Motion

The Kepler problem is more than solvable; it is superintegrable. For perturbation theory, the most useful idea is that unperturbed motion can be written in action-angle form:

$$H_0 = H_0(I), \quad \dot{I} = 0, \quad \dot{\theta} = \omega(I).$$

For a near-integrable Hamiltonian,

$$H(I, \theta) = H_0(I) + \varepsilon H_1(I, \theta),$$

the actions are no longer exactly constant:

$$\dot{I} = -\varepsilon \frac{\partial H_1}{\partial \theta}, \quad \dot{\theta} = \frac{\partial H_0}{\partial I} + \varepsilon \frac{\partial H_1}{\partial I}.$$

The small parameter ε should be interpreted physically. In the Sun-Jupiter-asteroid problem, a natural scale is $M_J/M_{\odot}$, though close encounters and resonances can amplify its effect.

5.4 Resonance

Definition 5.1 (Resonance). *A resonance occurs when there is a nonzero integer vector $k \in \mathbb{Z}^n$ such that*

$$k \cdot \omega(I) = 0.$$

Near resonance, phase combinations that normally average out become slow and can accumulate over long times.

For mean-motion resonance with Jupiter, a $p : q$ resonance means

$$\frac{n}{n_J} \approx \frac{p}{q}.$$

Using Kepler's law,

$$\frac{n}{n_J} \approx \left(\frac{a_J}{a} \right)^{3/2},$$

so the nominal resonance location is

$$a_{p:q} = a_J \left(\frac{q}{p} \right)^{2/3}.$$

This formula is only the center of a resonance zone. The width of the zone depends on the perturbation strength, eccentricity, and the order of the resonance. The point of marking the nominal location is to identify where slow angle combinations can form; the later chaos discussion explains what happens inside and near those zones. The missing-asteroid bands at these resonances are the Kirkwood gaps, identified empirically by Daniel Kirkwood in 1866 before modern Hamiltonian perturbation theory supplied the phase-space explanation.

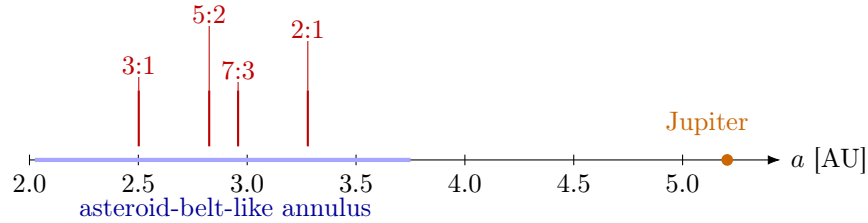


Figure 5.1: Nominal mean-motion resonance locations with Jupiter, shown across an asteroid-belt-like range. The resonance positions use $a_{p:q} = a_J(q/p)^{2/3}$; the figure now includes Jupiter's orbital radius to show that the listed resonances lie interior to Jupiter.

5.5 Perturbing An Integrable System

The central perturbation question is the following. Suppose the unperturbed Hamiltonian is integrable and has action-angle coordinates:

$$H_0 = H_0(I), \quad (I, \theta) \in U \times \mathbb{T}^N, \quad \dot{I} = 0, \quad \dot{\theta} = \omega_0(I),$$

where

$$\omega_0(I) = \frac{\partial H_0}{\partial I}.$$

Now add a small generic perturbation:

$$H(I, \theta) = H_0(I) + \varepsilon H_1(I, \theta), \quad 0 < \varepsilon \ll 1.$$

What happens to the invariant tori $I = \text{constant}$?

Two tempting answers are both incomplete. One might hope that the perturbed system remains integrable after a clever change of variables. One might also expect that a generic perturbation destroys the tori and makes a typical orbit wander ergodically through the available energy surface. KAM theory gives the actual answer: for small analytic perturbations, most sufficiently irrational tori persist, while resonant regions create islands, separatrices, and chaotic transport.

Three things often get conflated. Kolmogorov sketched the survival mechanism in 1954; the Fermi-Pasta-Ulam-Tsingou recurrence, computed in 1953–1955, was a parallel numerical surprise about near-integrable finite lattices, not a proof of KAM; the modern theorem is due to Moser's C^k theorem (1962) and Arnold's analytic proof (1963), built on Siegel's 1942 small-divisor work. Pöschel (1982) and Eliasson (1988) supplied later refinements. Zabusky and Kruskal (1965) later connected FPUT recurrence to solitons in the KdV continuum limit; that is another near-integrable mechanism, not the KAM theorem itself. The chronological point is conceptual: small perturbations can leave most invariant tori intact while still allowing chaotic regions.

5.6 Canonical Perturbation Theory And Small Divisors

We first ask whether a canonical transformation can remove the angle dependence order by order. The system is no longer exactly separable; the useful target is a near-identity generating function that makes the new Hamiltonian independent of angles up to a controlled order in ε . Use a type-II generating function

$$S(I', \theta) = I' \cdot \theta + \varepsilon S_1(I', \theta) + \varepsilon^2 S_2 + \cdots.$$

The transformation is

$$I_i = \frac{\partial S}{\partial \theta_i}, \quad \theta'_i = \frac{\partial S}{\partial I'_i}.$$

To first order,

$$I = I' + \varepsilon \frac{\partial S_1}{\partial \theta} + O(\varepsilon^2).$$

If the transformed Hamiltonian is angle-independent to order ε ,

$$H'(I') = H(I, \theta),$$

then expanding gives

$$H'(I') = H_0(I') + \varepsilon \left[\omega_0(I') \cdot \frac{\partial S_1}{\partial \theta} + H_1(I', \theta) \right] + O(\varepsilon^2).$$

Thus we need

$$\omega_0(I') \cdot \frac{\partial S_1}{\partial \theta} = f_1(I') - H_1(I', \theta),$$

where $f_1(I')$ will become the first correction to the new Hamiltonian. Since S_1 must be periodic in each angle, the average over the torus of the left side is zero. Therefore

$$f_1(I') = \langle H_1(I', \theta) \rangle_\theta.$$

Expand the perturbation in Fourier modes:

$$H_1(I, \theta) = \sum_{k \in \mathbb{Z}^N} H_{1k}(I) e^{ik \cdot \theta}.$$

The $k = 0$ mode is the average. For $k \neq 0$, the homological equation gives

$$i(k \cdot \omega_0(I')) S_{1k}(I') = -H_{1k}(I'),$$

so

$$S_{1k}(I') = \frac{iH_{1k}(I')}{k \cdot \omega_0(I')} \quad (k \neq 0).$$

The denominators $k \cdot \omega_0$ are the small divisors.

Definition 5.2 (Resonant torus). *The unperturbed torus with action I is resonant if there exists*

$$k \in \mathbb{Z}^N \setminus \{0\}$$

such that

$$k \cdot \omega_0(I) = 0.$$

On such a torus the phase $k \cdot \theta$ is constant along the unperturbed motion, so the corresponding perturbation does not average away.

Even if no divisor is exactly zero, the Fourier series for S_1 can diverge if some $k \cdot \omega_0$ are too small too often. This is the analytic heart of KAM theory.

5.7 Fourier Decay And The Diophantine Condition

Small divisors can be overcome when the numerators decay rapidly. If $H_1(I, \theta)$ is real analytic in θ , it extends to a complex strip

$$|\operatorname{Im} \theta_i| < \sigma \quad \text{for each angle component } i.$$

Shifting the Fourier integral into the complex strip gives the estimate

$$|H_{1k}(I)| \leq C(I) e^{-\sigma|k|_1}, \quad |k|_1 = |k_1| + \cdots + |k_N|.$$

Thus large Fourier modes are exponentially suppressed. The estimate is the analytic reason KAM is possible: small divisors grow at worst polynomially under a Diophantine condition, while analytic Fourier coefficients shrink exponentially. Smooth but nonanalytic perturbations require more delicate bookkeeping because the numerator may decay only as a power of $|k|$.

Example 5.1 (Fourier decay distinguishes analytic and nonanalytic functions). *One-angle examples make the difficulty visible. The series*

$$\sum_{k=1}^{\infty} \frac{\sin k\theta}{k}$$

has Fourier coefficients that decay only like $1/k$; its graph has a jump and is not real analytic. Likewise

$$\sum_{k=1}^{\infty} \frac{\cos k\theta}{k^2} = \frac{1}{4}(\theta - \pi)^2 - \frac{\pi^2}{12}, \quad |\theta| < 2\pi,$$

is periodic but not analytic at the endpoints of a period. A genuinely real-analytic example is

$$\sum_{k=-\infty}^{\infty} \cos(k\theta) e^{-k^2/2},$$

a theta-function-type series with rapidly suppressed Fourier coefficients.

The frequency vector must also avoid being too well approximated by resonances.

Definition 5.3 (Diophantine frequency). *A vector $\omega \in \mathbb{R}^N$ is Diophantine with constants $\alpha > 0$ and $\tau > N - 1$ if*

$$|k \cdot \omega| \geq \frac{\alpha}{|k|_1^\tau} \quad \text{for all } k \in \mathbb{Z}^N \setminus \{0\}.$$

For Diophantine $\omega_0(I')$,

$$|S_{1k}(I')| \leq \frac{C(I')}{\alpha} |k|_1^\tau e^{-\sigma|k|_1},$$

so the first-order generating function is well-defined. The polynomial factor from the small divisor cannot beat the exponential decay from analyticity.

Example 5.2 (How irrational is irrational?). *For $N = 2$, the Diophantine inequality can be recast as a rational approximation condition for*

$$x = \frac{\omega_1}{\omega_2}.$$

It asks that there be constants $A > 0$ and $\tau \geq 1$ such that

$$\left| x - \frac{p}{q} \right| \geq \frac{A}{q^{\tau+1}}$$

for all rational approximants p/q . For $x = \sqrt{2}$, use

$$f(x) = x^2 - 2 = 0.$$

If $p/q = x - \eta$, then

$$f(p/q) = \frac{p^2 - 2q^2}{q^2}.$$

The numerator is a nonzero integer unless $p/q = \sqrt{2}$, so its absolute value is at least one. The same estimate can be made without a formal Taylor expansion:

$$\left| \frac{p}{q} - \sqrt{2} \right| \left| \frac{p}{q} + \sqrt{2} \right| = \left| \frac{p^2 - 2q^2}{q^2} \right| \geq \frac{1}{q^2}.$$

If $|p/q - \sqrt{2}| \geq 1$, the desired lower bound is immediate after decreasing A . Otherwise $|p/q + \sqrt{2}| \leq 1 + 2\sqrt{2}$, and therefore

$$|\eta| \geq \frac{A}{q^2}$$

for some $A > 0$. Thus $\sqrt{2}$ satisfies a Diophantine-type lower bound. The sharp bound above corresponds to the constant-type exponent $\tau = 1$; the same estimate implies the weaker KAM inequalities with any $\tau > 1$ after decreasing A . This distinction matters because the measure estimate for two frequencies uses the strict condition $\tau > N - 1 = 1$. More generally, algebraic irrational numbers obey polynomial rational-approximation bounds. By contrast, numbers such as

$$x = \sum_{n=0}^{\infty} \frac{1}{2^{n!}}$$

are too well approximated by rationals and fail every fixed lower bound of this kind.

The Diophantine condition excludes a dense set of resonant hyperplanes, but the excluded set has small measure when α is small. To see the estimate, fix a bounded frequency domain $\Omega \subset \mathbb{R}^N$. For a given $k \neq 0$, the bad set is the slab

$$R_k = \left\{ \omega \in \Omega : |k \cdot \omega| < \frac{\alpha}{|k|_1^\tau} \right\}.$$

Its thickness normal to the hyperplane $k \cdot \omega = 0$ is of order

$$\frac{\alpha}{|k|_1^{\tau+1}},$$

because the normal vector has length comparable to $|k|_1$. Hence

$$\text{Vol}(R_k \cap \Omega) \leq C_\Omega \frac{\alpha}{|k|_1^{\tau+1}}.$$

Summing over $k \neq 0$ converges when $\tau + 1 > N$, i.e. $\tau > N - 1$. Thus the non-Diophantine set can be made to have arbitrarily small measure by taking α small, even though resonances are dense.

5.8 KAM Theorem: Statement And Sketch

At this point the perturbation problem has two competing facts. Analyticity makes high Fourier modes small, which helps the perturbation series. Small divisors make near-resonant modes large, which threatens the perturbation series. KAM theory is the result of balancing these facts: it restricts attention to frequencies that are irrational in a quantified way, and it asks the unperturbed Hamiltonian to twist enough that frequency can be adjusted when the torus is deformed.

The theorem is best read as a statement about labelled invariant tori, not about individual trajectories. An unperturbed action value I determines a frequency $\omega_0(I) = \partial H_0 / \partial I$. If that frequency is Diophantine with constants (α, τ) , the twist condition below holds, and ε is small enough relative to α , then the perturbed system has a nearby torus carrying quasiperiodic motion with the corresponding frequency. Points on resonant or nearly resonant tori are precisely the points for which this labelling argument breaks down.

The theorem does not give one uniform conclusion for all initial conditions. A small perturbation produces a mixed phase space: most sufficiently irrational tori survive for small ε , low-order resonances open into island chains, separatrices split, and gaps between surviving tori become possible transport channels. The rest of the chapter is organized around this mixed picture.

Assumption 5.2 (Kolmogorov nondegeneracy). *The integrable Hamiltonian $H_0(I)$ has a locally nondegenerate frequency map:*

$$\det \left(\frac{\partial^2 H_0}{\partial I_i \partial I_j} \right) \neq 0.$$

Equivalently, nearby actions give genuinely different frequency vectors.

Proposition 5.1 (KAM theorem, lecture-note form). *Let*

$$H(I, \theta) = H_0(I) + \varepsilon H_1(I, \theta)$$

be a nearly integrable Hamiltonian on $U \times \mathbb{T}^N$, analytic in the standard theorem, and assume that H_0 is Kolmogorov nondegenerate. Fix $\tau > N - 1$. For each Diophantine constant $\alpha > 0$, if $|\varepsilon|$ is small enough relative to α , every unperturbed torus whose frequency satisfies the (α, τ) Diophantine inequality persists as a nearby invariant torus of the perturbed system. For fixed α the surviving family omits the $O(\alpha)$ resonant set; choosing $\alpha = \alpha(\varepsilon) \rightarrow 0$ slowly enough as $\varepsilon \rightarrow 0$ gives a Cantor family whose complement has measure tending to zero.

The proof is not ordinary first-order perturbation theory. The first-order calculation above shows the mechanism, but higher orders produce new small divisors and require control of the coordinate change itself:

$$I = I' + \varepsilon \frac{\partial S_1}{\partial \theta} + \cdots, \quad \theta' = \theta + \varepsilon \frac{\partial S_1}{\partial I'} + \cdots.$$

The important scale warning is:

$$\frac{\partial S_1}{\partial I'}, \frac{\partial S_1}{\partial \theta} = O(\alpha^{-1}),$$

so the coordinate change is small only when ε/α is small. A common lecture-note form is therefore: for $|\varepsilon| < \delta \alpha^2$, the tori whose frequencies satisfy the Diophantine inequality with constant α persist and have a complement of $O(\alpha)$ measure in the relevant action domain.

The exponent and smallness conditions keep track of two competing estimates. If H_1 is analytic in a strip of width r , its Fourier coefficients satisfy a bound of the form

$$|H_{1,k}| \leq C e^{-r|k|}.$$

Solving the homological equation divides by $k \cdot \omega$, and the Diophantine condition gives

$$\left| \frac{H_{1,k}}{k \cdot \omega} \right| \leq \frac{C}{\alpha} |k|^\tau e^{-r|k|}.$$

The polynomial factor $|k|^\tau$ is harmless compared with exponential decay, provided one gives up a little analyticity width. This is the analytic heart of the proof: small divisors are dangerous, but not fatal, on the Diophantine set.

Here is the promised estimate in explicit form. Suppose H_1 is analytic in the strip $|\operatorname{Im} \theta| < r$ and

$$|H_{1,k}| \leq C e^{-r|k|}.$$

On a Diophantine torus,

$$|S_{1,k}| = \left| \frac{H_{1,k}}{k \cdot \omega} \right| \leq \frac{C}{\alpha} |k|^\tau e^{-r|k|}.$$

If $0 < r' < r$, then on the smaller strip $|\operatorname{Im} \theta| < r'$,

$$\sum_{k \neq 0} |S_{1,k}| e^{r'|k|} \leq \frac{C}{\alpha} \sum_{k \neq 0} |k|^\tau e^{-(r-r')|k|} < \infty.$$

The last sum converges because exponential decay beats every polynomial. Thus the first-order generating function is analytic on every slightly smaller strip. The price is the factor α^{-1} and the loss of width $r - r'$.

KAM uses an iterative correction scheme, but it should not be confused with ordinary finite-dimensional Newton's method. Solving the homological equation loses derivatives or analytic width, so the proof must shrink analytic domains, or in differentiable settings use a Nash-Moser type scheme, to compensate. At each step, one removes most of the angle dependence and makes the error quadratically smaller. Diophantine estimates keep the small divisors under control; nondegeneracy lets one adjust the action label so the desired frequency remains fixed.

The action adjustment is not cosmetic. After the oscillatory part of H_1 is removed, the averaged term changes the frequency by

$$\omega(I) \mapsto \omega(I) + \varepsilon \partial_I \langle H_1 \rangle(I) + O(\varepsilon^2).$$

To keep a torus with prescribed Diophantine frequency ω_* , one changes the action label by δI and solves, to first order,

$$D\omega_0(I_*) \delta I + \varepsilon \partial_I \langle H_1 \rangle(I_*) = 0.$$

Kolmogorov nondegeneracy is exactly the invertibility needed for this equation. If the energy rather than the full frequency is held fixed, one uses the related isoenergetic nondegeneracy condition, formulated with the bordered matrix of $D\omega_0$ and ω_0 . This note uses the frequency-labelled Kolmogorov version.

The phrase “quadratically smaller” has the same algebraic origin as Newton's method. In a one-mode model

$$H = H_0(I) + \epsilon h(I) \cos(k \cdot \theta),$$

choose S_1 so that $\{H_0, S_1\} = -h(I) \cos(k \cdot \theta)$. The transformed Hamiltonian begins

$$H \circ \Phi_{\epsilon S_1} = H_0 + \epsilon \langle H_1 \rangle + \frac{\epsilon^2}{2} \{H_1, S_1\} + O(\epsilon^3).$$

The order- ϵ oscillation has been cancelled; the remaining oscillatory part is order ϵ^2 , while the denominator $k \cdot \omega$ is controlled by the Diophantine condition.

In analytic norms the bookkeeping has the following typical form. If ϵ_n measures the remaining perturbation on a strip of width r_n and one sacrifices $\rho_n = r_n - r_{n+1}$ of analyticity at the next step, then an estimate of the schematic form

$$\epsilon_{n+1} \leq C \alpha^{-2} \rho_n^{-2\tau-2} \epsilon_n^2$$

expresses the Newton improvement. The constants and powers depend on the chosen norm, but the structure is stable: small divisors cost powers of α^{-1} and ρ_n^{-1} , while the new perturbation is quadratic in the old one. Choosing ϵ_0 sufficiently small relative to α and shrinking the domains by a summable sequence ρ_n makes the iteration converge.

In proof-outline form, the iteration has four repeating moves:

- linearize : separate resonant size from nonresonant modes,
- solve : invert $\omega \cdot \partial_\theta$ using the Diophantine bound,
- renormalize : change the action label using nondegeneracy,
- shrink : reduce the analytic domain so derivatives stay controlled.

The theorem is difficult because all four moves must be made quantitatively at once. The conceptual point is that sufficiently irrational frequencies avoid the worst resonances, while twist lets the torus move to the nearby frequency that the perturbed system actually supports. The smallness condition is often summarized schematically as $|\varepsilon| \lesssim \alpha^2$ for a fixed differentiability class and domain. The exponent is not a universal classroom constant; it records the analytic loss in the Newton step. Solving the homological equation costs a divisor $|k \cdot \omega|^{-1}$, and controlling the transformed Hamiltonian after domain shrinkage costs another small-divisor margin. The perturbation must be small enough for the quadratic improvement of the Newton scheme to beat these losses.

Read the theorem as a filtered survival statement. Start with all invariant tori of the integrable system. Remove narrow neighborhoods of the low-order resonance surfaces $k \cdot \omega = 0$. The remaining set is not an interval of actions but a Cantor-like family: it is full of holes at all orders, yet the total measure of the holes can be made small. The KAM iteration then proves that each torus in this filtered family has a nearby perturbed torus with conjugate quasiperiodic motion.

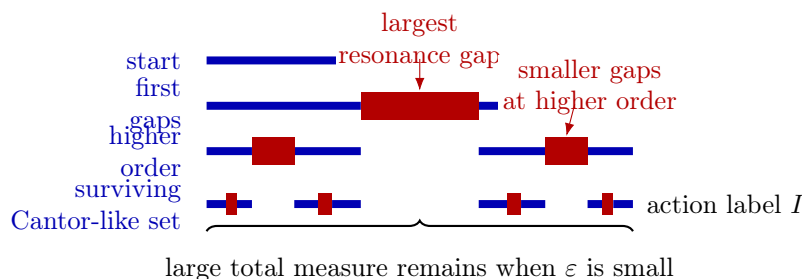


Figure 5.2: KAM survival is Cantor-like. Resonant action labels are removed in thin layers around $k \cdot \omega = 0$; the surviving Diophantine labels are nowhere all of the line, but their total measure approaches the integrable measure as $\varepsilon \rightarrow 0$.

The theorem does not say that every torus survives. Resonant tori are destroyed or reorganized into lower-dimensional structures. Nor does it say that all surviving tori are visible in a given numerical section. In systems with $N = 2$, surviving one-dimensional invariant curves on a Poincare section can block transport. In higher-dimensional systems, the geometry leaves more room for slow drift through resonance channels.

5.9 Breakdown Of Invariant Tori

KAM theory is a survival theorem, but a mechanics course also has to explain which structures lie outside its hypotheses. A surviving irrational torus is a barrier in a two-dimensional section; a resonant torus is instead a place where perturbations can accumulate coherently. The route from regular motion to chaotic transport begins by studying these resonant tori one by one, then asking when neighboring resonances overlap and open a connected transport channel.

The tori most exposed to perturbation are those whose frequencies are rationally related. On such a torus there exists a nonzero integer vector $k \in \mathbb{Z}^N$ with

$$k \cdot \omega_0(I_*) = 0,$$

where I_* is the resonant action and $\omega_0(I) = \partial H_0 / \partial I$ is the unperturbed frequency. The Fourier mode $e^{ik \cdot \theta}$ is no longer rapidly oscillating on that torus; it is phase locked. Averaging does not remove it. This is the first and most important distinction: nonresonant terms average away to leading order, while resonant terms build a new slow dynamics.

5.9.1 What It Means For A Torus To Be A Barrier

The word “breakdown” can mean several related things, so we state the geometric setting carefully.

Assumption 5.3 (Two-degree-of-freedom reduction). *When discussing transport barriers, assume either:*

1. *a Hamiltonian flow with two degrees of freedom, restricted to one fixed energy surface and viewed on a transverse Poincare section; or*
2. *an area-preserving twist map of the annulus*

$$A = \mathbb{T} \times I_{\text{ann}}, \quad (\theta, J) \in \mathbb{T} \times I_{\text{ann}},$$

where $I_{\text{ann}} \subset \mathbb{R}$ is an interval.

In either case the section is two-dimensional, and a surviving invariant torus appears as a one-dimensional invariant curve.

An invariant curve on the annulus is a set

$$\Gamma = \{(\theta, J_\Gamma(\theta)) : \theta \in \mathbb{T}\}$$

that is mapped to itself. If an orbit starts below Γ , it cannot later appear above Γ , because crossing would contradict uniqueness and invariance. Thus, in a two-dimensional Poincare section, invariant curves are literal transport barriers.

This codimension-one fact is special. For a Hamiltonian with N degrees of freedom, a KAM torus has dimension N , while a fixed energy surface has dimension $2N - 1$. The codimension inside the energy surface is therefore

$$(2N - 1) - N = N - 1.$$

For $N = 2$, this is codimension one and the torus can separate the energy surface. For $N \geq 3$, the codimension is at least two, so a single torus does not divide the energy surface into disconnected regions. This is why Arnold diffusion is possible in higher-dimensional nearly integrable systems: motion can drift through connected resonance channels even when many KAM tori survive. Arnold’s 1964 example showed this mechanism explicitly. Nekhoroshev’s 1977 theorem supplies the complementary stability statement for steep analytic Hamiltonians: for sufficiently small perturbations, action drift is typically confined for exponentially long times even though the theorem does not exclude drift on still longer scales.

5.9.2 Resonant Normal Form And Island Width

Now consider a nearly integrable Hamiltonian

$$H(I, \theta) = H_0(I) + \varepsilon H_1(I, \theta), \quad H_1(I, \theta) = \sum_{m \in \mathbb{Z}^N} H_{1m}(I) e^{im \cdot \theta}.$$

Let I_* satisfy the primitive resonance condition

$$k \cdot \omega_0(I_*) = 0, \quad k \in \mathbb{Z}^N, \quad \gcd(k_1, \dots, k_N) = 1.$$

Here primitive means that the resonance has not been written as an integer multiple of a lower-order resonance.

Near this resonance, choose a canonical linear change of angle-action variables in which

$$\psi = k \cdot \theta$$

is one of the new angles. Because k is primitive, it can be completed to the first row of an integer unimodular matrix $A \in \text{GL}(N, \mathbb{Z})$. Setting $\Theta = A\theta$ and $J = A^{-T}I$ preserves the canonical one-form:

$$I \cdot d\theta = J \cdot d\Theta.$$

The first component of Θ is ψ , and the first component of J is the action conjugate to ψ . Let $J_* = J(I_*)$. The remaining angles are fast away from secondary resonances. Averaging over those fast angles leaves, to leading order,

$$H_{\text{res}}(\Delta J, \psi) = \frac{1}{2} a_k (\Delta J)^2 + \varepsilon |V_k| \cos(\psi - \phi_k), \quad \Delta J = J - J_*.$$

All symbols in this expression are local data at the resonance:

- ψ is the slow resonant phase;
- ΔJ is the displacement of the resonant action;
- $a_k \neq 0$ is the effective twist, obtained by restricting the Hessian $\partial_I^2 H_0(I_*)$ to the resonant direction;
- V_k is the resonant Fourier coefficient of H_1 , after the same canonical change and fast-angle averaging;
- ϕ_k is the phase of V_k .

This is the Hamiltonian of a pendulum. Its equations are

$$\dot{\psi} = a_k \Delta J, \quad \dot{\Delta J} = \varepsilon |V_k| \sin(\psi - \phi_k).$$

Equilibria of this pendulum are the centers and saddles of a resonant island chain. The separatrix of the pendulum gives a first estimate of the width of the resonance.

To compute that width, shift ψ so that the stable center is at $\psi = 0$, and replace a_k by $|a_k|$ if necessary by reversing ΔJ . Up to an irrelevant additive constant, write

$$H_{\text{pend}} = \frac{1}{2} |a_k| (\Delta J)^2 - \varepsilon |V_k| \cos \psi.$$

The center has energy $-\varepsilon|V_k|$, and the saddle at $\psi = \pi$ has energy $+\varepsilon|V_k|$. On the separatrix, the largest value of $|\Delta J|$ occurs at $\psi = 0$. Therefore

$$\frac{1}{2}|a_k|(\Delta J_{\max})^2 - \varepsilon|V_k| = +\varepsilon|V_k|.$$

Solving gives the resonance half-width

$$\Delta J_{\max} = 2\sqrt{\frac{\varepsilon|V_k|}{|a_k|}}.$$

The full pendulum width in the resonant action is twice this value. A perturbation of size ε carves resonant layers of width $\sqrt{\varepsilon}$. The layer is therefore much wider than naive $O(\varepsilon)$ -counting predicts.

5.9.3 Chirikov Overlap

Suppose two resonances, labelled by k and ℓ , occur at resonant action values J_k and J_ℓ along the same local action direction. Let ΔJ_k and ΔJ_ℓ denote their half-widths, estimated by the pendulum calculation above. The Chirikov overlap criterion says that large chaotic layers should be expected when

$$\Delta J_k + \Delta J_\ell \gtrsim |J_k - J_\ell|.$$

The Chirikov criterion is a physical overlap test for the separatrix layers of neighboring resonances, introduced by Chirikov in the plasma-physics literature and systematized in his 1979 review. It is not a theorem with a sharp universal constant. Before overlap, islands may be surrounded by surviving invariant curves. After overlap, hyperbolic manifolds from different resonances intersect and produce a connected chaotic sea.

The criterion also explains why low-order resonances dominate pictures. A Fourier coefficient associated with a large vector k is exponentially small in an analytic Hamiltonian, while the resonance width depends on $\sqrt{|V_k|}$. Low-order Fourier modes tend to have the largest islands, so they create the first visible transport channels.

5.9.4 Poincare-Birkhoff Splitting Of A Rational Curve

The pendulum normal form explains the local shape of one resonant layer. The Poincare-Birkhoff theorem explains why rational invariant curves of area-preserving twist maps break into periodic orbits.

Proposition 5.2 (Poincare-Birkhoff theorem, informal form). *Let F be an area-preserving twist map of an annulus. If a rational invariant curve of rotation number p/q is destroyed by a perturbation while nearby boundary curves twist in opposite directions under F^q , then F^q has at least $2q$ fixed points in the resonant annulus, equivalently at least two q -periodic orbits of F . Generically these periodic points alternate between elliptic and hyperbolic type around the annulus.*

The hypothesis about opposite twist is the section-map version of being on opposite sides of a resonance: one can choose an intermediate annular curve C' whose images under F^q are displaced in opposite angular directions on the two sides, forcing fixed points of the q -fold map between

them. Elliptic points become island centers. Hyperbolic points have stable and unstable manifolds. If those manifolds coincide, they form a separatrix. If they split and intersect transversely, the separatrix is replaced by a homoclinic tangle. The theorem is therefore the rigid-annulus counterpart of the local pendulum normal form: the normal form predicts a resonant island, while Poincare-Birkhoff says that an area-preserving twist map must supply alternating elliptic and hyperbolic periodic points in that annulus.

This theorem gives the global fixed-point guarantee behind the island-chain picture for area-preserving twist maps. The local resonant normal form already explains the pendulum-like mechanism; Poincare-Birkhoff supplies the stronger annulus-level statement that elliptic and hyperbolic periodic points must appear under a twist perturbation of a rational invariant curve. The next section derives the local mechanism directly for a Poincare map, including the area-preservation argument that forces fixed points of the resonant iterate.

The proof of Poincare-Birkhoff uses fixed-point arguments on the annulus and is outside the elementary perturbative estimates developed here. We use it as a structural theorem about area-preserving twist maps; standard self-contained treatments appear in Poincare's 1912 "last geometric theorem," Birkhoff's 1913 proof, and modern accounts such as Moser-Zehnder.

5.9.5 Separatrix Splitting And Homoclinic Tangles

A resonant pendulum has a separatrix because the stable and unstable manifolds of the saddle coincide. This coincidence is fragile. Under a perturbation, the stable and unstable manifolds become two different curves:

$$\begin{aligned} W^s &= \text{stable manifold of the hyperbolic orbit,} \\ W^u &= \text{unstable manifold of the hyperbolic orbit.} \end{aligned}$$

If W^s and W^u cross transversely once, area preservation forces an infinite sequence of further intersections. The resulting homoclinic tangle is the geometric core of deterministic chaos: strips of phase space are stretched, folded, and reinserted near the saddle.

One first-order way to measure separatrix splitting is the Melnikov function (Melnikov, 1963). The homoclinic-tangle picture itself goes back to Poincare's *Methodes nouvelles* in the 1890s. Assume a one-degree-of-freedom separatrix Hamiltonian $H_{\text{sep}}(z)$, with separatrix orbit $z_0(t)$, is perturbed periodically:

$$H(z, t) = H_{\text{sep}}(z) + \varepsilon H_{\text{pert}}(z, t).$$

The unperturbed stable and unstable branches are assumed to coincide along this separatrix, and the Melnikov calculation below is used only when the first-order zero is simple. The phase t_0 records where the perturbation is in its period when the orbit passes near the saddle. The leading signed distance between W^u and W^s is proportional to

$$M(t_0) = \int_{-\infty}^{\infty} \{H_{\text{sep}}, H_{\text{pert}}\}(z_0(t - t_0), t) dt.$$

To see the formula, use H_{sep} itself as a transverse coordinate near the unperturbed separatrix. Along a perturbed trajectory,

$$\frac{d}{dt} H_{\text{sep}}(z(t)) = \varepsilon \{H_{\text{sep}}, H_{\text{pert}}\}(z(t), t) + O(\varepsilon^2).$$

The unstable branch accumulates this change while it leaves the saddle, and the stable branch accumulates the corresponding change while it approaches the saddle. Subtracting the two first-order energy shifts and evaluating on the unperturbed separatrix $z_0(t - t_0)$ gives the integral $M(t_0)$. A simple zero of M means the signed transverse distance changes sign, hence the two manifolds cross transversely. If $M(t_0)$ has a simple zero, then the two manifolds intersect transversely for sufficiently small nonzero ε . Thus a single computable integral can diagnose the birth of a chaotic layer near a separatrix. The final implication from transverse homoclinic intersection to symbolic dynamics uses the Smale-Birkhoff homoclinic theorem: a transverse homoclinic point of an area-preserving map implies a horseshoe on an invariant set. The Melnikov calculation supplies the transverse intersection hypothesis; the Smale-Birkhoff theorem supplies the rigorous chaos conclusion.

5.9.6 Cantori And Partial Barriers

For area-preserving maps, the strongest irrational invariant curves are often those whose rotation numbers are hardest to approximate by rationals. The golden-mean rotation number

$$\rho_g = \frac{\sqrt{5} - 1}{2}$$

is the standard example because its continued fraction is $[1; 1, 1, 1, \dots]$, making it maximally poorly approximated among irrationals in the precise sense of Hurwitz's theorem: every irrational x has infinitely many rationals p/q with

$$\left| x - \frac{p}{q} \right| < \frac{1}{\sqrt{5} q^2},$$

and the constant $\sqrt{5}$ is sharp, with equality approached by the golden ratio and its fractional-linear equivalents.

When an irrational invariant curve finally breaks, Aubry-Mather theory (Mather, 1982; Aubry, 1983) supplies the replacement object. For an area-preserving twist map and an irrational rotation number ρ , there is an invariant Aubry-Mather set with rotation number ρ . If the KAM curve survives, this set is the curve itself. If the curve has broken, the set is a Cantor-like invariant set called a cantorus. Its gaps prevent it from being a continuous barrier, but small gaps can still make transport through the associated turnstile lobes very slow. This existence-and-order statement is a separate variational theorem, not a consequence of KAM: KAM proves persistence of sufficiently nonresonant smooth curves under small perturbations, while Aubry-Mather theory constructs ordered minimizing orbits even after the smooth curve has broken. One variational model is the Frenkel-Kontorova form of an exact twist map: orbits are critical sequences of an action

$$\mathcal{S}(\{\theta_n\}) = \sum_n L(\theta_n, \theta_{n+1}),$$

with a prescribed mean rotation number. Minimizing sequences with irrational rotation number are ordered and cannot cross. When the invariant curve exists, these minimizers fill it; after breakup, the same ordered minimizers survive only on a Cantor set.

For the standard map, a famous numerical benchmark is that the last invariant curve with golden-mean rotation breaks near

$$K_c \approx 0.971635.$$

This number is obtained by numerical criteria such as Greene’s 1979 residue criterion, later sharpened by MacKay’s renormalization work. For a rational periodic orbit of period q , the residue of the q -fold derivative is

$$R = \frac{2 - \text{tr}(DT_K^q)}{4}.$$

For an elliptic periodic point $0 < R < 1$; for a hyperbolic point $R < 0$ or $R > 1$. Greene’s residue conjecture remains a conjecture: it says that an irrational invariant curve with rotation number ρ exists precisely when the residues of periodic orbits whose rotation numbers p_n/q_n approach ρ remain bounded in the appropriate limiting sense. The number $K_c \approx 0.971635$ is the value at which this criterion predicts breakdown of the golden-mean invariant curve in the standard map. Direct numerical simulations and MacKay-Stark style renormalization results support the criterion in important cases, but it should not be read as a proved theorem in the generality of its numerical use.

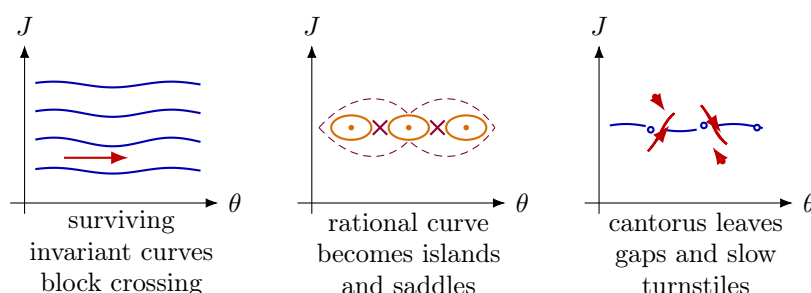


Figure 5.3: Three stages in the loss of transport barriers on a two-dimensional section. Left: invariant curves separate the annulus. Middle: a rational curve splits into an island chain with hyperbolic points. Right: an irrational curve may leave a cantorus, which permits transport only through gaps.

The accompanying Python script `demos/python/standard_map_torus_breakdown.py` uses the standard map on the cylinder. It initializes an ensemble near the golden-mean unperturbed rotation number, leaves p unwrapped, and measures finite-time momentum spreading. A large spread is evidence of transport through broken barriers; a small spread over finite time is not a proof of survival. The corresponding Makefile target is `standard-map-breakdown-demo`, and the Wolfram Language companion is `demos/mathematica/StandardMapTorusBreakdown.wl`. This distinction is essential: numerical trajectories illustrate the geometry, while KAM, Poincaré-Birkhoff theory, and separatrix-splitting criteria explain why the geometry is there.

5.10 From Resonance To Chaos: Poincaré Map Picture

A Poincaré section converts continuous Hamiltonian motion into an area-preserving map. This loses the time parametrization but keeps the geometry needed to see resonance. Near a resonant torus, some iterate of the unperturbed map is the identity; after perturbation, that near-identity map unfolds into fixed points, island chains, and separatrices. Poincaré introduced this return-map viewpoint in the *Methodes nouvelles*; it is one of the reasons that modern Hamiltonian chaos is usually drawn on a section before it is described in full phase space.

This is why maps are so useful in a chapter about flows. A continuous orbit can hide its organization in time; a section records each return to a transverse surface and turns invariant tori

into curves, resonances into island chains, and chaotic layers into visible scatter. The standard map isolates this return-map geometry in its simplest widely used form.

For two degrees of freedom, begin with an integrable Hamiltonian

$$H_0 = H_0(I_1, I_2), \quad (I_1, I_2, \theta_1, \theta_2) \in U \times \mathbb{T}^2,$$

and fix an energy surface $H_0 = E$. Choose a Poincare section S , for example $\theta_2 = 0$. Starting from $x \in S$, follow the orbit until it next intersects S . This defines the return map

$$T : S \rightarrow S.$$

The unperturbed flow has

$$\dot{\theta}_1 = \omega_1(I), \quad \dot{\theta}_2 = \omega_2(I).$$

Returning to $\theta_2 = 0$ takes time $2\pi/\omega_2$, so the return map twists the remaining angle by

$$\Delta\theta_1 = 2\pi \frac{\omega_1}{\omega_2}.$$

On a resonant torus with $\omega_1/\omega_2 = p/q$,

$$T^q = \text{id}$$

for the unperturbed return map T . A concrete period-three resonance is

$$\frac{\omega_1}{\omega_2} = \frac{1}{3}, \quad T : \theta_1 \mapsto \theta_1 + \frac{2\pi}{3}, \quad T^3 = \text{id}.$$

Let $C \subset S$ be the invariant curve obtained by intersecting this resonant torus with the section. Nearby invariant curves C_- and C_+ have slightly different rotation numbers. With the usual twist assumption, T^3 rotates points one way on C_- and the opposite way on C_+ .

Now perturb the Hamiltonian:

$$H = H_0 + \varepsilon H_1.$$

The perturbed Poincare map T_ε is still defined by first return to S , even though the invariant tori may no longer exist. Near $C \cap S$, T_ε^3 is close to the identity. At fixed energy it has the local form

$$T_\varepsilon^3(I_1, \theta_1) = (I_1 + f_\varepsilon(I_1, \theta_1), \theta_1 + g_\varepsilon(I_1, \theta_1)).$$

For small ε , the opposite twists on C_- and C_+ persist. Therefore an intermediate curve C' has

$$g_\varepsilon(I_1, \theta_1) = 0;$$

on C' , the third-return map produces no angular displacement.

The missing radial displacement is controlled by area preservation. Hamiltonian flow preserves ω , and hence the Poincare map preserves the induced area form on the section. Equivalently, for a closed curve γ bounding a disk D_γ in the section,

$$\oint_\gamma p \, dq = - \int_{D_\gamma} \omega,$$

so the signed symplectic area enclosed by γ is invariant. If f_ε were nonzero with a fixed sign everywhere on C' , the curve would be displaced monotonically inward or outward under T_ε^3 , changing the enclosed area. Thus f_ε must have at least a pair of zeros on C' . At such points

$$f_\varepsilon = 0, \quad g_\varepsilon = 0,$$

so T_ε^3 has fixed points. In the period-three example, if x is a fixed point of T_ε^3 , then $T_\varepsilon(x)$ is also a fixed point of T_ε^3 . Thus the resonant curve generically produces six fixed points: three elliptic and three hyperbolic, alternating around the resonant layer.

Elliptic points are surrounded by island chains and new local invariant curves. Hyperbolic points have stable and unstable manifolds. Once those manifolds intersect transversely, the separatrix is replaced by a chaotic layer. Farther from the resonance, sufficiently Diophantine KAM curves persist and confine motion.

This is the local mechanism behind the pictures one sees in the standard map, the driven pendulum, and the asteroid-belt resonances: KAM tori survive away from strong resonances, while resonant tori break into islands and chaotic separatrix webs. For the real asteroid belt, this local statement is only the beginning of a quantitative model. Resonance widths, eccentricity dependence, secular precession, and close encounters require the celestial variables of the restricted three-body problem rather than the single-pendulum normal form alone.

5.11 The Standard Map As A Minimal Model

The standard map is

$$\begin{aligned} p_{n+1} &= p_n + K \sin q_n, \\ q_{n+1} &= q_n + p_{n+1} \pmod{2\pi}. \end{aligned}$$

This map is often called the Chirikov standard map. It emerged from kicked rotor and resonance-overlap models in the 1960s, and Chirikov's 1979 review made it the canonical laboratory for the transition from invariant curves to global Hamiltonian chaos. This is naturally a map of the cylinder, with q taken modulo 2π and $p \in \mathbb{R}$. For phase portraits on a compact square, the Python demo reduces both q and p modulo 2π , giving the corresponding map on the two-torus. It is area-preserving because the Jacobian matrix has determinant one:

$$D\Psi = \begin{pmatrix} 1 + K \cos q_n & 1 \\ K \cos q_n & 1 \end{pmatrix}, \quad \det D\Psi = 1.$$

The parameter K controls resonance overlap. For small K , invariant curves dominate. For larger K , chaotic regions grow. The compact two-torus portrait is useful for seeing islands, but it hides whether an orbit has crossed many momentum cells on the cylinder. For transport questions one should keep p unwrapped; this is the convention used in the torus-breakdown diagnostic demo.

Example 5.3 (Period-three resonance in the standard map). *A common drift-kick convention for the standard map is*

$$T_K(\theta, p) = (\theta + p, p + K \sin(\theta + p)),$$

which is symplectically equivalent to the convention above after shifting where the kick is placed. If

$$D(\theta, p) = (\theta + p, p), \quad Kc(\theta, p) = (\theta, p + K \sin \theta),$$

then the map used above is

$$D \circ Kc,$$

whereas the drift-kick convention is

$$Kc \circ D.$$

These differ by where the stroboscopic section is placed along the kicked flow; the phase portraits and resonant normal forms are conjugate after this section shift. At $K = 0$, the invariant curve

$$C : \quad p = \frac{2\pi}{3}$$

has

$$T_0^3 = \text{id}$$

because the angle advances by 2π after three iterates.

Let

$$p = \frac{2\pi}{3} + \delta(\theta), \quad \delta = O(K).$$

To first order in K and δ ,

$$\theta_3 - \theta - 2\pi = 3\delta + 2K \sin\left(\theta + \frac{2\pi}{3}\right) + K \sin\left(\theta + \frac{4\pi}{3}\right) + O(K^2),$$

where θ_3 denotes the angle component after three iterates of T_K . The nearby curve C' on which T_K^3 fixes the angle therefore satisfies

$$\delta(\theta) = \frac{K}{2} \sin \theta - \frac{K\sqrt{3}}{6} \cos \theta + O(K^2).$$

For example, at $\theta = \pi/3$,

$$p = \frac{2\pi}{3} + \frac{K}{2\sqrt{3}} + O(K^2),$$

which is the initial condition singled out in the corresponding numerical exercise. At the next order, area preservation forces the radial displacement on C' to have alternating zeros. These become the six period-three points: three elliptic island centers and three hyperbolic points with separatrix structure.

5.12 What To Take Forward

The main logical chain of this chapter is the following. Complete integrability gives invariant tori carrying quasi-periodic motion. In action-angle variables an integrable Hamiltonian depends only on the actions, so the frequencies are gradients of $H_0(I)$. A small perturbation introduces angle-dependent terms. Canonical perturbation theory tries to remove those terms order by order, but the homological equation divides by integer frequency combinations $k \cdot \omega(I)$. Small divisors are the obstruction that distinguishes robust quasi-periodic motion from resonant dynamics.

KAM theory says that sufficiently nonresonant tori with nondegenerate frequency maps survive the small analytic perturbations considered in this chapter, though the tori deform. Resonant tori are more fragile. Near resonance, the averaged dynamics reduces to pendulum-like islands, separatrices, and stochastic layers. As perturbations grow, resonance overlap destroys transport barriers and produces large chaotic regions.

The standard map is a minimal laboratory where these stages can be seen in one picture: rotational invariant curves at small coupling, island chains around rational rotation numbers, cantori and partial barriers near the breakup threshold, and widespread diffusion when barriers are gone.

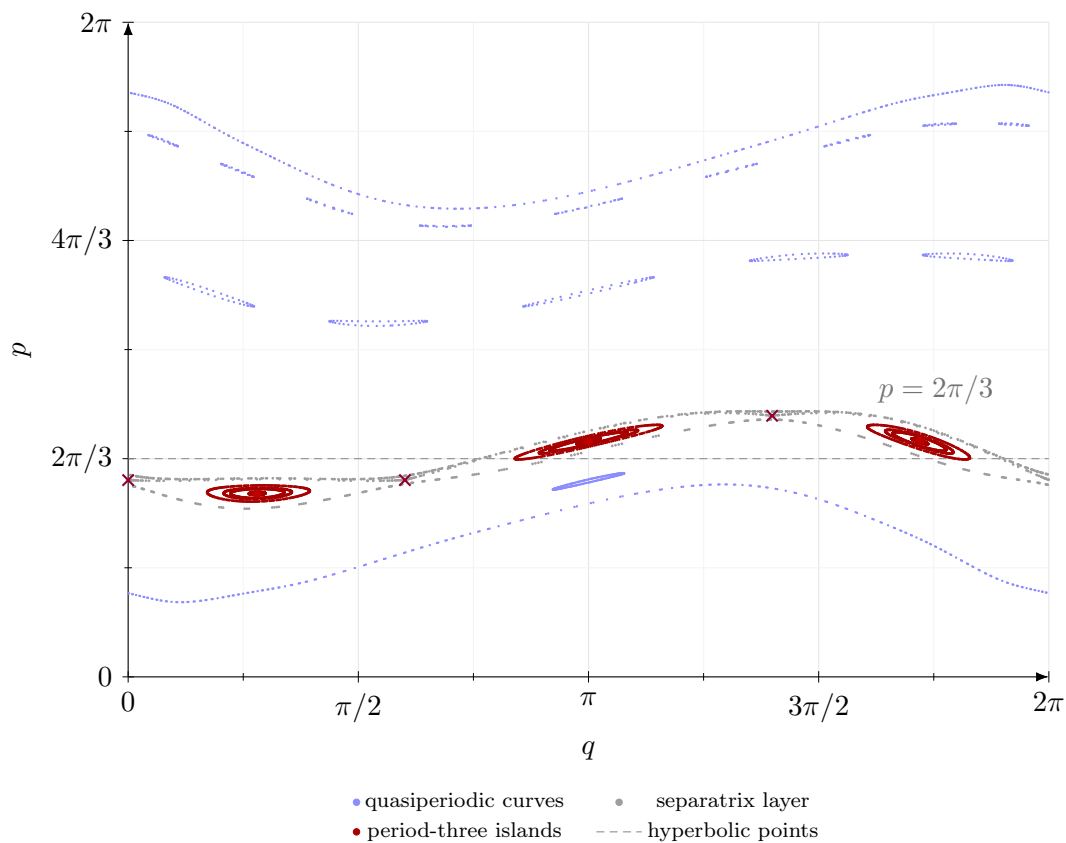


Figure 5.4: Actual iterates of the standard map on the compact two-torus at $K = 0.65$. Blue points lie on quasiperiodic invariant curves away from the resonance, red points trace orbits near the elliptic period-three island chain, gray points sample the local separatrix layer, and purple crosses mark the hyperbolic period-three orbit.

The three-body problem in Chapter 6 is a principal physical arena for these ideas. It is not included here because its Newtonian reduction, restricted limit, Lagrange points, and asteroid-belt ensembles deserve a self-contained treatment.

5.13 Chapter-End Laboratories And Exercises

The laboratories below are finite-dimensional tests of the perturbative logic: slow resonant angles, pendulum normal forms, small divisors, island chains, and the onset of chaotic regions.

5.13.1 Driven One-Angle Hamiltonian And Resonant Normal Form

Consider the one-angle time-periodic perturbation model

$$H(I, \theta, t) = \frac{1}{2}I^2 + \epsilon \cos(n\theta - \nu t).$$

The unperturbed frequency is

$$\dot{\theta} = I.$$

Resonance occurs when the phase

$$\psi = n\theta - \nu t$$

is slow, i.e.

$$nI - \nu \approx 0.$$

Thus

$$I_{\text{res}} = \frac{\nu}{n}.$$

To make the system autonomous, introduce the extended phase-space pair (t, E_t) , with Hamiltonian

$$K = E_t + \frac{1}{2}I^2 + \epsilon \cos(n\theta - \nu t).$$

Use the type-II generating function

$$F_2(\theta, t; J, \mathcal{E}) = (n\theta - \nu t)J + t\mathcal{E}.$$

It gives

$$\psi = \frac{\partial F_2}{\partial J} = n\theta - \nu t, \quad \tau = \frac{\partial F_2}{\partial \mathcal{E}} = t,$$

and

$$I = \frac{\partial F_2}{\partial \theta} = nJ, \quad E_t = \frac{\partial F_2}{\partial t} = -\nu J + \mathcal{E}.$$

Thus the autonomous Hamiltonian becomes, up to the spectator variable $\mathcal{E}$,

$$K' = \mathcal{E} - \nu J + \frac{1}{2}n^2J^2 + \epsilon \cos \psi.$$

The resonance condition is

$$\frac{\partial K'}{\partial J} = -\nu + n^2J = 0,$$

so

$$J_{\text{res}} = \frac{\nu}{n^2}, \quad I_{\text{res}} = nJ_{\text{res}} = \frac{\nu}{n}.$$

Set

$$J = J_{\text{res}} + P.$$

After dropping constants and the decoupled $\mathcal{E}$, the resonant Hamiltonian is

$$K_{\text{res}}(P, \psi) = \frac{1}{2}n^2P^2 + \epsilon \cos \psi.$$

This is a pendulum Hamiltonian with effective inverse mass n^2 . It has elliptic islands, hyperbolic fixed points, and a separatrix. This is why planetary mean-motion resonances, the standard map, and the driven pendulum share the same local geometry.

5.13.2 Henon-Heiles As The Finite-Dimensional Chaos Bridge

Henon and Heiles introduced this Hamiltonian in 1964 as a galactic-dynamics model, and it became one of the first smooth Hamiltonian systems where numerical Poincare sections made the breakdown of integrability visible. The Henon-Heiles Hamiltonian is

$$H_{\text{HH}} = H_0 + \lambda_{\text{HH}}H_1,$$

with

$$H_0 = \frac{1}{2}(p_x^2 + p_y^2 + x^2 + y^2), \quad H_1 = x^2y - \frac{1}{3}y^3.$$

For the isotropic oscillator, introduce action-angle variables

$$\begin{aligned} x &= \sqrt{2I_1} \sin \theta_1, & p_x &= \sqrt{2I_1} \cos \theta_1, \\ y &= \sqrt{2I_2} \sin \theta_2, & p_y &= \sqrt{2I_2} \cos \theta_2. \end{aligned}$$

Then

$$H_0 = I_1 + I_2, \quad \omega_0 = (1, 1).$$

The first-order homological equation for a generating function S_1 is

$$\omega_0 \cdot \partial_\theta S_1 = \langle H_1 \rangle - H_1.$$

In Fourier components,

$$i(k_1 + k_2)S_{1k} = -H_{1k}.$$

The denominator vanishes whenever

$$k_1 + k_2 = 0.$$

At first order the cubic perturbation has no average and its Fourier expansion contains no nonzero mode with $k_1 + k_2 = 0$. The first-order homological equation can therefore be solved for all modes that appear. At second order, however, the canonical transformation generates resonant Fourier modes with $k_1 + k_2 = 0$. Those modes cannot be removed by a periodic S_2 . This is the small-divisor obstruction in its most concrete classroom form: the unperturbed oscillator has equal frequencies, so resonant combinations of angles do not average away. For example, the second-order transformed Hamiltonian contains

$$\frac{\lambda_{\text{HH}}^2}{2} \{H_1, S_1\}.$$

Products of first-order Fourier modes add their wave vectors. Nonresonant modes k and k' in H_1 can therefore produce combinations such as $k + k' = (2, -2)$, which satisfy $(k + k') \cdot \omega_0 = 0$. The obstruction is not that the first-order cubic term is already resonant; it is that the near-identity transformation creates resonant terms at the next order.

The Henon-Heiles system is a useful finite-dimensional bridge from integrability to Hamiltonian chaos. At low energy, Poincare sections show deformed invariant curves. At higher energy, resonance islands and chaotic regions appear. Unlike the asteroid problem, all variables and forces remain visible in a four-dimensional phase space.

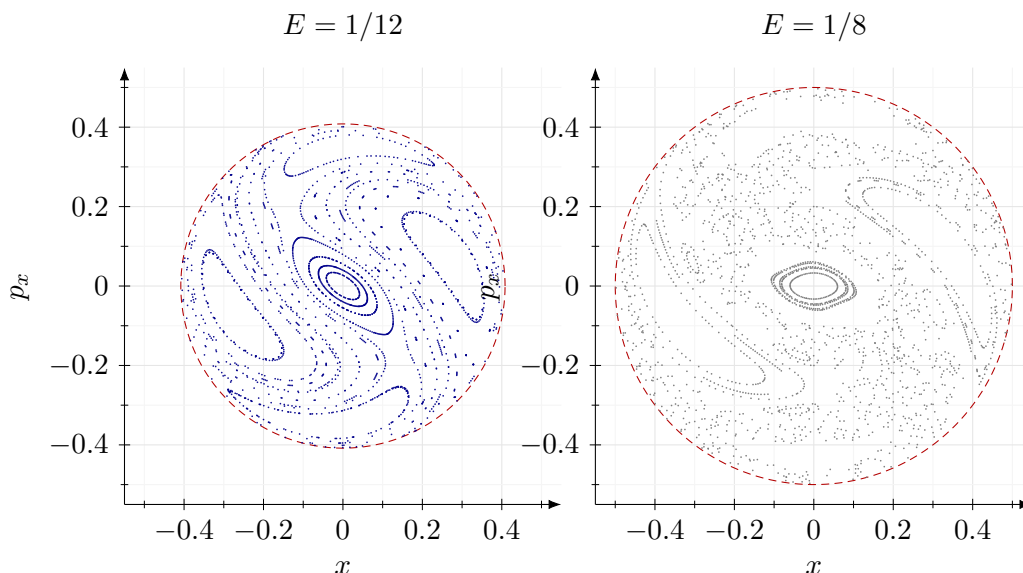


Figure 5.5: Numerical Poincare sections of the Henon-Heiles flow on $y = 0$, $p_y > 0$. Each plotted point is an upward section crossing from a velocity-Verlet integration of the Hamiltonian in the text. The dashed red circle is the kinematic boundary $x^2 + p_x^2 = 2E$ on the section, where $p_y = 0$. At $E = 1/12$ the section is dominated by deformed invariant curves; by $E = 1/8$ a chaotic sea coexists with surviving islands and smaller regular regions.

5.13.3 Questions For The Reader

1. In the homological equation $\omega(I) \cdot \partial_\theta S_1 = -(H_1 - \langle H_1 \rangle)$, identify where the Diophantine condition is used. What fails if ω is rational?
2. For a one-degree-of-freedom resonant normal form, draw the phase portrait and identify elliptic fixed points, hyperbolic fixed points, separatrices, and island width. Which part of this picture persists in an area-preserving map?
3. Use the standard-map figure in this chapter to compare three types of phase-space organization: invariant curves, resonance islands, and chaotic layers. Which of these can block global radial transport in action?
4. Explain why Liouville volume preservation is compatible with chaos. Which property of the dynamics changes when invariant tori break?

5.14 Associated Demos

The associated torus-breakdown and finite-dimensional chaos demos are:

- `demos/python/standard_map_torus_breakdown.py`
- `demos/python/henon_heiles_poincare.py`
- `demos/mathematica/StandardMapTorusBreakdown.wl`

They reproduce the qualitative transition from invariant curves to resonance islands and chaotic layers. They are intentionally small so that the phase portrait can be compared directly with the perturbative derivation.

Chapter 6

The Three-Body Problem

Two bodies under mutual gravity trace conic sections: ellipses, parabolas, hyperbolas, and rectilinear collision orbits as degenerate limiting cases. Three bodies do not admit a comparable closed-form solution, a universal one-dimensional radial reduction, or generic integrability. What survives is a rich phase-space geometry: rotating-triangle exact solutions, equilibrium points in the rotating frame, resonant islands, chaotic scattering, and long-time transport.

Sundman (1907 announcement, 1912 publication) constructed a convergent series solution away from triple collision, but the result is not a substitute for integrability. Binary collisions can be regularized by changing variables and time scale. In the planar problem, the Levi-Civita change

$$z = u^2, \quad dt = |u|^2 ds$$

turns the inverse-square collision singularity into a regular passage in the u -plane: since $|z| = |u|^2$, the singular potential $1/|z|$ becomes $1/|u|^2$, and multiplication by the Sundman time factor $dt/ds = |u|^2$ cancels the leading pole. Equivalently, in the fixed-energy Jacobi form, multiplying the Kepler expression $T - GMm/|z| - E$ by $|u|^2$ turns the singular potential term into the smooth constant $-GMm$. Sundman's construction combines this local regularization with analytic continuation and a global time reparametrization. Triple collision is the obstruction left out of the global statement, and Belorizky emphasized in 1930 that Sundman's series converges far too slowly for practical computation.

This chapter connects every phenomenon back to a controlled reduction of Newton's law. We first formulate the full Newtonian three-body problem and reduce translations using Jacobi coordinates. We then derive the circular restricted three-body problem as a controlled limit and compute its rotating-frame Hamiltonian, Jacobi constant, zero-velocity curves, and Lagrange points. Finally we connect that model to the Sun-Jupiter-asteroid demo and to ejection statistics.

Roadmap. The chapter moves through three models in a fixed order. The full three-body problem supplies the Hamiltonian, the conserved quantities, and the reason generic integrability fails. The circular restricted three-body problem supplies the rotating frame, Jacobi constant, zero-velocity curves, and Lagrange points. The heliocentric Sun-Jupiter-asteroid model supplies the finite-time ensemble experiment. Each model keeps part of the previous one and discards part of it; the text names those decisions so that numerical ejection is tied to a specified model and time scale.

Model hierarchy. The reader should keep four levels separate:

two-body Kepler	: integrable reference problem,
full three-body	: Newtonian Hamiltonian with all masses active,
CR3BP	: two primaries prescribed, one massless test body,
asteroid ensemble	: finite-time numerical statistic.

The later levels are controlled reductions chosen to isolate transport, escape channels, and resonance-driven loss in a teachable setting.

Appendix B records the numerical protocol behind the last two levels: coordinate choices, event definitions, resonance windows, uncertainty estimates, and convergence checks. The chapter introduces these choices where they first affect the mechanics; the appendix gathers them in one place for reproducibility.

Historical context. Poincare’s prize essay for King Oscar II’s 1889 competition was substantially revised after he discovered an error; Weierstrass, on the prize jury, demanded a corrected version. The revision exposed homoclinic intersections and became the foundation of *Methodes nouvelles de la mecanique celeste* (1892–1899). Birkhoff (1915) extended the program by making return maps and periodic orbits operational tools of celestial mechanics. Painleve’s 1895 singularity theorem clarified a separate issue: in the Newtonian three-body problem, every finite-time singularity is caused by a collision. Xia’s 1992 noncollision singularities for five bodies show that this is a special feature of low N , not a general safety theorem for Newtonian gravity.

6.1 The Newtonian Three-Body Problem

The restricted problem will be derived as a limit of the Newtonian three-body problem; we therefore start with the full problem. The classical conserved quantities, momentum, angular momentum, and energy, survive, but they no longer reduce the dynamics to a one-dimensional radial equation. The restricted problem is not a different force law; it is a controlled limit and coordinate choice inside the same Newtonian interaction.

Let $q_i(t) \in \mathbb{R}^3$ be the inertial position of mass $m_i > 0$, and write

$$\mathbf{r}_{ij} = q_i - q_j, \quad r_{ij} = \|\mathbf{r}_{ij}\|.$$

The Lagrangian is

$$L(q, \dot{q}) = \sum_{i=1}^3 \frac{1}{2} m_i \|\dot{q}_i\|^2 + \sum_{1 \leq i < j \leq 3} \frac{G m_i m_j}{r_{ij}},$$

because $L = T - V$ and the Newtonian potential energy is negative:

$$V(q) = - \sum_{i < j} \frac{G m_i m_j}{r_{ij}}.$$

The Euler-Lagrange equation for q_i gives

$$m_i \ddot{q}_i = \sum_{j \neq i} G m_i m_j \frac{q_j - q_i}{\|q_j - q_i\|^3}.$$

After dividing by m_i ,

$$\ddot{q}_i = \sum_{j \neq i} G m_j \frac{q_j - q_i}{\|q_j - q_i\|^3}.$$

This is a closed Hamiltonian system with canonical momenta $p_i = m_i \dot{q}_i$ and Hamiltonian

$$H(q, p) = \sum_{i=1}^3 \frac{\|p_i\|^2}{2m_i} - \sum_{i < j} \frac{G m_i m_j}{r_{ij}}.$$

The center of mass

$$R_{\text{cm}} = \frac{1}{M} \sum_i m_i q_i, \quad M = m_1 + m_2 + m_3,$$

moves uniformly because the internal gravitational forces cancel in pairs. The total momentum, angular momentum, and energy are

$$P = \sum_i p_i, \quad J = \sum_i q_i \times p_i, \quad H = \text{constant}.$$

These conserved quantities are essential, but they do not make the generic three-body problem integrable. After removing center-of-mass motion, the planar problem has four configuration degrees of freedom and an eight-dimensional phase space. Fixing angular momentum and quotienting the remaining planar rotation removes two phase dimensions; a fixed-energy reduced space is still five-dimensional. This is vastly richer than the one-degree radial problem left by the two-body reduction, leaving enough room for separatrices, homoclinic tangles, chaotic scattering, and transport.

6.1.1 Translation Reduction And Jacobi Coordinates

The center-of-mass degree of freedom is dynamically trivial. Removing it shows which variables remain in the interacting problem. Let

$$m_{12} = m_1 + m_2, \quad M = m_{12} + m_3,$$

and define Jacobi coordinates

$$R = \frac{m_1 q_1 + m_2 q_2 + m_3 q_3}{M}, \quad \rho_1 = q_2 - q_1, \quad \rho_2 = q_3 - \frac{m_1 q_1 + m_2 q_2}{m_{12}}.$$

Here ρ_1 is the relative vector inside the pair (1, 2), while ρ_2 points from the center of mass of that pair to the third body. The inverse transformation is

$$\begin{aligned} q_1 &= R - \frac{m_2}{m_{12}} \rho_1 - \frac{m_3}{M} \rho_2, \\ q_2 &= R + \frac{m_1}{m_{12}} \rho_1 - \frac{m_3}{M} \rho_2, \\ q_3 &= R + \frac{m_{12}}{M} \rho_2. \end{aligned}$$

Substituting these expressions into the kinetic energy and collecting terms gives the diagonal form

$$T = \frac{1}{2} M \left\| \dot{R} \right\|^2 + \frac{1}{2} \mu_1 \left\| \dot{\rho}_1 \right\|^2 + \frac{1}{2} \mu_2 \left\| \dot{\rho}_2 \right\|^2, \quad \mu_1 = \frac{m_1 m_2}{m_{12}}, \quad \mu_2 = \frac{m_3 m_{12}}{M}.$$

The absence of cross terms is the point of Jacobi coordinates: the mass metric has been diagonalized after translations are separated.

It is useful in celestial mechanics to write the attractive potential with a positive sign,

$$U = -V > 0.$$

With this convention the Hamiltonian is kinetic minus attraction, $H = T - U$. The attraction potential becomes

$$U(\rho_1, \rho_2) = \frac{Gm_1m_2}{\|\rho_1\|} + \frac{Gm_1m_3}{\left\|\rho_2 + \frac{m_2}{m_{12}}\rho_1\right\|} + \frac{Gm_2m_3}{\left\|\rho_2 - \frac{m_1}{m_{12}}\rho_1\right\|}.$$

In the center-of-mass frame $R = 0$, the relative Hamiltonian is therefore

$$H_{\text{rel}} = \frac{\|\pi_1\|^2}{2\mu_1} + \frac{\|\pi_2\|^2}{2\mu_2} - U(\rho_1, \rho_2),$$

where $\pi_a = \mu_a \dot{\rho}_a$. This formula shows what changed from the two-body problem: there are now two coupled relative vectors, and each inverse-square attraction depends on a different linear combination of them.

Derivation 6.1 (Lagrange-Jacobi identity). *In the center-of-mass frame define the moment of inertia*

$$I_{\text{cm}} = \sum_{i=1}^3 m_i \|q_i\|^2 = \mu_1 \|\rho_1\|^2 + \mu_2 \|\rho_2\|^2.$$

Differentiate once:

$$\dot{I}_{\text{cm}} = 2 \sum_i m_i q_i \cdot \dot{q}_i,$$

and then again:

$$\ddot{I}_{\text{cm}} = 4T + 2 \sum_i q_i \cdot m_i \ddot{q}_i.$$

Since $m_i \ddot{q}_i = \partial U / \partial q_i$ and U is homogeneous of degree -1 , Euler's theorem gives

$$\sum_i q_i \cdot \frac{\partial U}{\partial q_i} = -U.$$

Using $H = T - U$ gives the two equivalent forms

$$\boxed{\ddot{I}_{\text{cm}} = 4T - 2U = 4H + 2U.}$$

This identity is a diagnostic for the size of the configuration. For a bounded noncollisional motion, the long-time average of $\ddot{I}_{\text{cm}}$ vanishes, so time averaging gives the virial relation $2\langle T \rangle = \langle U \rangle$. For escape or near-collision, I_{cm} instead records the large-scale expansion or collapse of the configuration.

Remark 6.1 (Why the two-body reduction does not generalize). *For two bodies, the center-of-mass reduction leaves one relative vector in a central potential. Angular momentum reduces the dynamics to a one-dimensional radial problem, while the Laplace-Runge-Lenz vector gives the conic orbit because the inverse-square potential has a hidden Kepler symmetry. Central forces alone do not give that vector; generic central potentials still precess. For three bodies there is no single relative radius and no central force after reduction. Pairwise forces keep changing the effective center seen by each body, so the Kepler-only constants are no longer constants of the full motion.*

6.1.2 Central Configurations And Special Solutions

Some exact motions survive this complexity. A configuration a_1, a_2, a_3 , with center of mass at the origin, is a central configuration if there is a scalar λ such that

$$\sum_{j \neq i} Gm_j \frac{a_j - a_i}{\|a_j - a_i\|^3} = -\lambda a_i \quad (i = 1, 2, 3).$$

Then the ansatz

$$q_i(t) = R(\omega t)a_i$$

solves the equations in a plane provided $\omega^2 = \lambda$. The shape rotates rigidly even though the force law is gravitational.

Example 6.1 (Lagrange equilateral solution). *Place three bodies at the vertices of an equilateral triangle and choose the center of mass as origin. The gravitational force on each body points toward the center of mass, so the central-configuration equation holds. For three equal masses m and side length s , the force on one mass has magnitude*

$$2 \frac{Gm^2}{s^2} \cos \frac{\pi}{6} = \sqrt{3} \frac{Gm^2}{s^2}.$$

The distance from a vertex to the centroid is $s/\sqrt{3}$. Equating this to the centripetal force $m\omega^2 s/\sqrt{3}$ gives

$$\omega^2 = \frac{3Gm}{s^3}.$$

This is a genuine three-body orbit with fixed shape.

Lagrange found the equilateral family in 1772; the collinear configurations were Euler's, from 1767. Both are finite-mass ancestors of the Lagrange points in the restricted problem.

Example 6.2 (Euler collinear solutions). *There are also central configurations with the three bodies on one line. The relative separations are fixed in a rotating frame, and the line rotates uniformly. Unlike the equilateral case, the ratios of the two gaps depend on the three masses and are determined by an algebraic equation. These collinear solutions are the ancestors of L_1, L_2, L_3 in the restricted problem. For the ordering m_1, m_2, m_3 on a line, set the distance $m_2 m_3$ equal to x times the distance $m_1 m_2$. Euler's calculation gives the quintic*

$$(m_1 + m_2)x^5 + (3m_1 + 2m_2)x^4 + (3m_1 + m_2)x^3 \\ - (m_2 + 3m_3)x^2 - (2m_2 + 3m_3)x - (m_2 + m_3) = 0.$$

Its unique positive root gives the gap ratio for that ordering; the other two collinear configurations come from the other orderings of the masses.

Here is the reduction behind the polynomial. Place the bodies at

$$q_1 = 0, \quad q_2 = d, \quad q_3 = d(1 + x),$$

and subtract the center-of-mass acceleration, since a central configuration is allowed to translate uniformly before the rotating-frame description is chosen. The central-configuration equation says that the gravitational acceleration of each body differs from the common center-of-mass acceleration by $-\omega^2(q_i - q_{\text{cm}})$. Taking the difference of the equations for (q_2, q_1) and for (q_3, q_2)

eliminates the center-of-mass acceleration and leaves two expressions for the same ω^2 . Equating them, clearing the denominators d^2 , x^2d^2 , and $(1+x)^2d^2$, and collecting powers of x gives the displayed quintic. Explicitly, with the common factor G/d^2 suppressed,

$$\begin{aligned} a_2 - a_1 &= -(m_1 + m_2) + m_3 \left(\frac{1}{x^2} - \frac{1}{(1+x)^2} \right), \\ a_3 - a_2 &= m_1 \left(1 - \frac{1}{(1+x)^2} \right) - \frac{m_2 + m_3}{x^2}. \end{aligned}$$

The central-configuration condition requires $a_3 - a_2 = x(a_2 - a_1)$. Multiplying this equation by $x^2(1+x)^2$ produces the quintic above. Thus x is the only shape variable, while d sets the overall scale.

Section 6.7.1 turns the equal-mass equilateral solution into a numerical laboratory. There the exact rotating triangle is used as the comparison orbit, and small perturbations of the initial velocities show how a central configuration can organize nearby motion without making the full three-body problem integrable. Chenciner and Montgomery (2000) discovered the figure-eight periodic orbit: three equal masses chase one another along a single planar curve, with zero total angular momentum. It is not a central configuration, and like Lagrange's and Euler's solutions it does not make the generic three-body problem integrable. Special exact orbits can exist without integrability.

6.2 Circular Restricted Three-Body Problem

The circular restricted three-body problem is the standard autonomous model behind the Sun-Jupiter-asteroid story. Two massive primaries move on circular Kepler orbits about their common center of mass, and a third body has zero mass, so it feels the primaries but does not affect them. This is the setting of Jacobi's integral (1836) and of Hill's lunar restricted problem (1878), and the setting in which Birkhoff's periodic-orbit program made return maps and periodic points operational. After choosing units in which

$$G(m_1 + m_2) = 1, \quad \text{primary separation} = 1, \quad \text{angular speed} = 1,$$

define the mass parameter

$$\mu_{\text{CR3BP}} = \frac{m_2}{m_1 + m_2}, \quad 0 < \mu_{\text{CR3BP}} \leq \frac{1}{2}.$$

In the uniformly rotating barycentric frame the primaries are fixed at

$$P_1 = (-\mu_{\text{CR3BP}}, 0), \quad P_2 = (1 - \mu_{\text{CR3BP}}, 0).$$

For compactness in this section write $\mu = \mu_{\text{CR3BP}}$. If the test particle has rotating-frame position $r = (x, y)$, set

$$r_1 = \sqrt{(x + \mu)^2 + y^2}, \quad r_2 = \sqrt{(x - 1 + \mu)^2 + y^2}.$$

The effective potential is

$$\Omega_{\text{eff}}(x, y) = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}.$$

The first term is the centrifugal potential; the last two are the gravitational potentials of the fixed primaries.

Derivation 6.2 (Rotating-frame equations). Let $\rho(t) = R(t)r(t)$ be the inertial position, where $R(t)$ rotates with angular velocity $\hat{z}$. Differentiating twice gives

$$\ddot{\rho} = R(t) (\ddot{r} + 2\hat{z} \times \dot{r} + \hat{z} \times (\hat{z} \times r)).$$

In the plane, $\hat{z} \times (\hat{z} \times r) = -r$. The inertial gravitational acceleration, expressed in the rotating frame, is

$$-(1-\mu)\frac{r-P_1}{r_1^3} - \mu\frac{r-P_2}{r_2^3} = \nabla \left(\frac{1-\mu}{r_1} + \frac{\mu}{r_2} \right).$$

Therefore

$$\ddot{r} + 2\hat{z} \times \dot{r} - r = \nabla \left(\frac{1-\mu}{r_1} + \frac{\mu}{r_2} \right),$$

equivalently

$$\ddot{r} + 2\hat{z} \times \dot{r} = \nabla \Omega_{\text{eff}}(r).$$

The last step moves the centrifugal term $-r$ to the right-hand side and uses $\nabla[\frac{1}{2}(x^2 + y^2)] = r$. In components,

$$\ddot{x} - 2\dot{y} = \Omega_x, \quad \ddot{y} + 2\dot{x} = \Omega_y.$$

The Coriolis force is velocity dependent and does no work in the rotating frame, but the deeper reason a Jacobi integral exists is that the rotating-frame Hamiltonian is autonomous. The conserved quantity is rotating-frame energy, equivalently inertial energy minus ωL_z in dimensional variables.

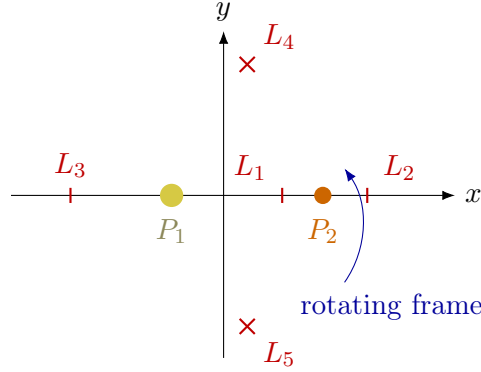


Figure 6.1: Circular restricted three-body geometry in rotating barycentric coordinates. The primaries are fixed, while the test particle moves under gravity, centrifugal force, and Coriolis deflection.

Derivation 6.3 (Jacobi integral). Dot the rotating-frame equation with $\dot{r}$:

$$\dot{r} \cdot \ddot{r} + 2\dot{r} \cdot (\hat{z} \times \dot{r}) = \dot{r} \cdot \nabla \Omega_{\text{eff}}.$$

The Coriolis term vanishes because $\dot{r} \cdot (\hat{z} \times \dot{r}) = 0$. Hence

$$\frac{d}{dt} \left(\frac{1}{2} \|\dot{r}\|^2 \right) = \frac{d}{dt} \Omega_{\text{eff}}(r(t)).$$

Therefore

$$\frac{1}{2} \|\dot{\mathbf{r}}\|^2 - \Omega_{\text{eff}} = \text{constant}.$$

It is customary to multiply by -2 and define the Jacobi constant

$$C_J = 2\Omega_{\text{eff}}(x, y) - (\dot{x}^2 + \dot{y}^2).$$

This is the energy integral of the autonomous rotating-frame problem. Equivalently, before nondimensionalizing the angular speed to one, the conserved rotating-frame Hamiltonian is the inertial energy shifted by $-\omega L_z$; the Jacobi constant is the customary sign-reversed version of that autonomous Hamiltonian.

Derivation 6.4 (Canonical Hamiltonian in the rotating frame). The rotating-frame equations also come from a genuine autonomous Hamiltonian, but the canonical momenta are not the velocity components. The Lagrangian per unit test-particle mass is

$$L_{\text{rot}} = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) + x\dot{y} - y\dot{x} + \Omega_{\text{eff}}(x, y).$$

The linear velocity term is the Coriolis term in variational form. Hence

$$p_x = \frac{\partial L_{\text{rot}}}{\partial \dot{x}} = \dot{x} - y, \quad p_y = \frac{\partial L_{\text{rot}}}{\partial \dot{y}} = \dot{y} + x.$$

Solving for velocity gives $\dot{x} = p_x + y$ and $\dot{y} = p_y - x$. The Legendre transform gives

$$\begin{aligned} H_{\text{rot}} &= p_x \dot{x} + p_y \dot{y} - L_{\text{rot}} \\ &= \frac{1}{2}(p_x^2 + p_y^2) + yp_x - xp_y - \frac{1-\mu}{r_1} - \frac{\mu}{r_2}. \end{aligned}$$

Equivalently,

$$H_{\text{rot}} = \frac{1}{2}((p_x + y)^2 + (p_y - x)^2) - \Omega_{\text{eff}}(x, y).$$

Along solutions

$$C_J = -2H_{\text{rot}}.$$

This equality is often the safest way to translate between the geometric zero-velocity picture and the canonical phase-space picture: velocities are natural for drawing allowed regions, while (x, y, p_x, p_y) are natural for Hamiltonian perturbation theory and Poincaré maps.

The Jacobi constant gives an immediate geometric constraint. Since the squared speed is nonnegative,

$$\dot{x}^2 + \dot{y}^2 = 2\Omega_{\text{eff}}(x, y) - C_J \geq 0.$$

Thus the allowed region in the rotating plane is

$$2\Omega_{\text{eff}}(x, y) \geq C_J,$$

and curves satisfying

$$2\Omega_{\text{eff}}(x, y) = C_J$$

are zero-velocity curves. They are not physical walls; they are projections of the energy surface to configuration space. A test particle cannot cross a forbidden region because that would require imaginary speed. When the zero-velocity neck near L_1 or L_2 opens, transport between regions becomes possible. This is the rotating-frame picture behind many capture, escape, and ejection mechanisms. The allowed components are often called Hill regions. Near a secondary primary, the small allowed lobe gives the Hill-sphere approximation used in celestial mechanics: it is the configuration-space shadow of the same Jacobi-energy constraint, not a literal material boundary.

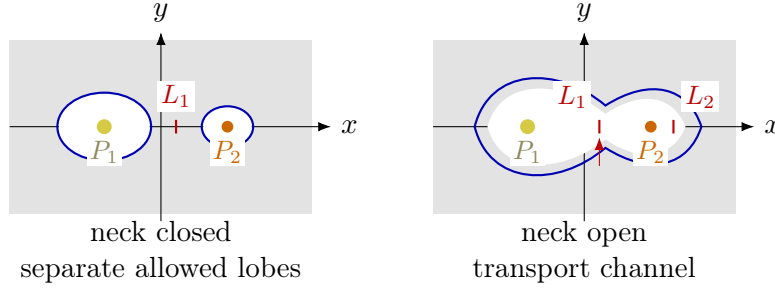


Figure 6.2: Schematic zero-velocity curves in the rotating CR3BP plane. Gray regions are forbidden by $2\Omega_{\text{eff}} < C_J$; white regions are allowed projections of the energy surface. Changing C_J can open a neck near L_1 , creating a channel between previously separated regions.

6.2.1 Lagrange Points

Equilibria in the rotating frame satisfy

$$\dot{x} = \dot{y} = \ddot{x} = \ddot{y} = 0, \quad \nabla\Omega_{\text{eff}}(x, y) = 0.$$

There are five such points. The collinear points L_1, L_2, L_3 lie on the x -axis and solve the scalar equation

$$x - (1 - \mu) \frac{x + \mu}{|x + \mu|^3} - \mu \frac{x - 1 + \mu}{|x - 1 + \mu|^3} = 0.$$

The absolute values encode which side of each primary the point lies on. There is one solution between the primaries, one to the right of P_2 , and one to the left of P_1 .

The triangular points are explicit:

$$L_4 = \left(\frac{1}{2} - \mu, \frac{\sqrt{3}}{2} \right), \quad L_5 = \left(\frac{1}{2} - \mu, -\frac{\sqrt{3}}{2} \right).$$

At either point $r_1 = r_2 = 1$. Substituting L_4 , for example, gives

$$\begin{aligned} \Omega_x &= x - (1 - \mu)(x + \mu) - \mu(x - 1 + \mu) \\ &= \left(\frac{1}{2} - \mu \right) - (1 - \mu)\frac{1}{2} - \mu\left(-\frac{1}{2} \right) = 0, \end{aligned}$$

and similarly $\Omega_y = 0$. The gravitational and centrifugal forces balance in the rotating frame.

Remark 6.2 (Linear stability of L_4, L_5). *Linearizing about L_4 or L_5 gives stable oscillations when*

$$\mu(1 - \mu) < \frac{1}{27}.$$

Here is the calculation. Write a perturbation as (ξ, η) around L_4 . At L_4 , differentiate

$$\Omega_{\text{eff}} = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}$$

twice. If $X_1 = x + \mu$, $X_2 = x - 1 + \mu$, and $m_1 = 1 - \mu$, $m_2 = \mu$, then

$$\begin{aligned}\Omega_{xx} &= 1 - \sum_{a=1}^2 \frac{m_a}{r_a^3} + 3 \sum_{a=1}^2 \frac{m_a X_a^2}{r_a^5}, \\ \Omega_{yy} &= 1 - \sum_{a=1}^2 \frac{m_a}{r_a^3} + 3 \sum_{a=1}^2 \frac{m_a y^2}{r_a^5}, \\ \Omega_{xy} &= 3 \sum_{a=1}^2 \frac{m_a X_a y}{r_a^5}.\end{aligned}$$

At L_4 , $r_1 = r_2 = 1$, $X_1 = 1/2$, $X_2 = -1/2$, and $y = \sqrt{3}/2$, so

$$\Omega_{xx} = \frac{3}{4}, \quad \Omega_{yy} = \frac{9}{4}, \quad \Omega_{xy} = \frac{3\sqrt{3}}{4}(1 - 2\mu).$$

The linearized equations

$$\ddot{\xi} - 2\dot{\eta} = \Omega_{xx}\xi + \Omega_{xy}\eta, \quad \ddot{\eta} + 2\dot{\xi} = \Omega_{xy}\xi + \Omega_{yy}\eta$$

have characteristic polynomial

$$\lambda^4 + \lambda^2 + \frac{27}{4}\mu(1 - \mu) = 0.$$

Put $s = \lambda^2$. Then

$$s^2 + s + \frac{27}{4}\mu(1 - \mu) = 0, \quad s_{\pm} = \frac{-1 \pm \sqrt{1 - 27\mu(1 - \mu)}}{2}.$$

All four eigenvalues are purely imaginary exactly when the two roots $s_{\pm}$ are real and negative. Since their sum is -1 and their product is positive for $0 < \mu < 1$, this is equivalent to the discriminant condition

$$1 - 27\mu(1 - \mu) \geq 0.$$

Equivalently, for $\mu \leq 1/2$,

$$\mu < \mu_R = \frac{1}{2} \left(1 - \sqrt{\frac{23}{27}} \right) \approx 0.03852.$$

The Sun-Jupiter value is about 9.53×10^{-4} , far below this threshold. Trojan motion near the triangular points is therefore stable in this linearized nontrivial three-body setting. The stability criterion was derived by Gascheau (1843) and Routh (1875); its Solar System relevance was confirmed in 1906, when Max Wolf discovered the first Trojan asteroid, 588 Achilles, near Jupiter's L_4 .

6.2.2 Connection To The Asteroid-Belt Model

The CR3BP is autonomous in the rotating barycentric frame and has the Jacobi constant. The ejection-probability demo uses a closely related heliocentric form in which Jupiter is prescribed on a circular orbit and the asteroid is integrated in Sun-centered coordinates. The two descriptions are related by a time-dependent rotation and translation, but they emphasize different things:

- rotating barycentric CR3BP: Jacobi constant, zero-velocity curves, and Lagrange points;
- heliocentric asteroid ensemble: Kepler elements, resonance bins, and finite-time ejection statistics.

Both are useful. The first gives the clean phase-space geometry of escape channels. The second makes direct contact with semimajor axes, mean-motion resonances, and the asteroid-belt simulations.

The CR3BP companion script computes $L_1, \dots, L_5$, plots representative zero-velocity curves, and prints the drift of the Jacobi constant along the numerical trajectory. Its time stepper is classical fourth-order Runge-Kutta: this is transparent for a first classroom calculation, but it is not symplectic and it does not preserve the Jacobi integral exactly. Long-time transport claims should therefore be checked against the printed Jacobi drift and, in research calculations, against a structure-aware integrator.

6.3 Heliocentric Sun-Jupiter-Asteroid Equations

Let $r \in \mathbb{R}^2$ be the asteroid position relative to the Sun. Jupiter is prescribed by

$$r_J(t) = a_J(\cos n_J t, \sin n_J t), \quad n_J^2 = \frac{G(M_\odot + M_J)}{a_J^3}.$$

In a heliocentric non-inertial frame, the asteroid acceleration is

$$\ddot{r} = -\frac{GM_\odot r}{\|r\|^3} - GM_J \left(\frac{r - r_J}{\|r - r_J\|^3} + \frac{r_J}{\|r_J\|^3} \right).$$

The first term is solar gravity. The first part of the Jupiter term is direct gravitational attraction to Jupiter. The final term is the indirect term caused by the acceleration of the heliocentric frame.

Equivalently, the asteroid obeys the explicitly time-dependent Hamiltonian

$$H_{\text{hel}}(r, p, t) = \frac{1}{2} \|p\|^2 - \frac{GM_\odot}{\|r\|} - \frac{GM_J}{\|r - r_J(t)\|} + GM_J \frac{r \cdot r_J(t)}{\|r_J(t)\|^3}.$$

The last term is the potential whose gradient gives the indirect acceleration. Because $r_J(t)$ is periodic, this is a periodically forced one-particle Hamiltonian. It can be made autonomous by adjoining the forcing phase $\phi = n_J t$ and its conjugate action I_ϕ . With

$$r_J(\phi) = a_J(\cos \phi, \sin \phi),$$

the extended Hamiltonian is

$$K(r, p, \phi, I_\phi) = H_{\text{hel}}(r, p, \phi) + n_J I_\phi,$$

so Hamilton's equation gives $\dot{\phi} = \partial K / \partial I_\phi = n_J$. The resulting autonomous system has more than one degree of freedom and can exhibit resonance and chaotic transport.

Derivation 6.5 (Indirect term). *Let x be the asteroid position in an inertial frame and $x_\odot$ the Sun position. The heliocentric vector is $r = x - x_\odot$. Then*

$$\ddot{r} = \ddot{x} - \ddot{x}_\odot.$$

The direct Jupiter acceleration of the asteroid is

$$-GM_J \frac{x - x_J}{\|x - x_J\|^3}.$$

The acceleration of the Sun due to Jupiter is

$$\ddot{x}_\odot = GM_J \frac{x_J - x_\odot}{\|x_J - x_\odot\|^3} = GM_J \frac{r_J}{\|r_J\|^3}.$$

Subtracting $\ddot{x}_\odot$ produces the indirect term

$$-GM_J \frac{r_J}{\|r_J\|^3}.$$

Remark 6.3 (Why the indirect term matters). *Omitting the indirect term corresponds to holding the Sun fixed while Jupiter pulls on the asteroid. That is sometimes useful as a simplified classroom model, but it is not the heliocentric form of the restricted three-body problem. Including the indirect term ensures that the coordinate origin accelerates exactly as the Sun does in the prescribed Sun-Jupiter motion.*

6.4 Numerical Ejection Probability

The highlighted simulation samples N test particles with initial semimajor axes $a_0 \in [a_{\min}, a_{\max}]$, small eccentricities, random orbital phases, and random periapee directions. Let T be the integration horizon.

Definition 6.1 (Finite-time ejection probability). *For a specified model, sampling distribution, time step, and ejection criterion, the finite-time ejection probability estimate is*

$$\hat{P}_{\text{ej}}(T) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\text{particle } i \text{ is ejected before time } T\}.$$

Remark 6.4 (Ejection criterion in the demo). *The Python demo counts a particle as ejected if it reaches a large heliocentric radius or if its heliocentric two-body energy is positive after it has moved outside Jupiter's orbital scale. In the default configuration this means*

$$r > 20 \text{ AU} \quad \text{or} \quad E_\odot > 0 \text{ and } r > a_J,$$

with $a_J = 5.2044 \text{ AU}$. The same run separately records close approaches to Jupiter, using the default threshold 0.03 AU , and collisions with the inner solar cutoff, using the default threshold 0.02 AU .

$\hat{P}_{\text{ej}}(T)$ is not a property of the three-body problem in isolation; it is a property of an experiment. Five inputs determine it:

model + initial ensemble + time horizon + numerical method + loss criterion.

The same physical question with different choices yields different numerical values.

The demo also labels each semimajor-axis bin by its nearest low-order interior mean-motion resonance with Jupiter. The nominal resonance locations are

$$a_{p:q} = a_J \left(\frac{q}{p} \right)^{2/3}, \quad (p:q) \in \{3:1, 5:2, 7:3, 2:1\}.$$

These labels are a diagnostic overlay. A semimajor-axis bin is wider and cruder than a resonance island, and the actual resonance width depends on eccentricity and perturbation strength. The overlay asks whether large loss fractions cluster near the same places where perturbation theory predicts slow resonant angles.

For a bin containing n_b sampled particles and e_b ejections, the reported fraction is

$$\hat{p}_b = \frac{e_b}{n_b}.$$

If the particles in that bin are treated as independent Bernoulli trials, a rough sampling uncertainty is

$$\text{SE}(\hat{p}_b) \approx \sqrt{\frac{\hat{p}_b(1 - \hat{p}_b)}{n_b}}.$$

This is only a Monte Carlo error bar for the chosen numerical experiment. It does not measure modeling error, timestep error, or the long-time instability of the idealized Hamiltonian system.

6.5 Integrator Used In The Demo

The demo uses a kick-drift-kick style update. If $a(r, t)$ is the acceleration, then

$$\begin{aligned} v_{n+1/2} &= v_n + \frac{\Delta t}{2} a(r_n, t_n), \\ r_{n+1} &= r_n + \Delta t v_{n+1/2}, \\ v_{n+1} &= v_{n+1/2} + \frac{\Delta t}{2} a(r_{n+1}, t_{n+1}). \end{aligned}$$

As written in physical time with prescribed Jupiter forcing, this update is not the symplectic map of the autonomous extended Hamiltonian K . It is still pedagogically preferable to an opaque black-box integrator because every force evaluation is visible.

6.6 What To Take Forward

- The full three-body problem is Newtonian and Hamiltonian, but its translation and rotation symmetries do not reduce it to a one-degree-of-freedom radial equation. The failure is structural, not a lack of clever coordinates: Bruns proved in 1887 that there are no additional integrals algebraic in the coordinates, momenta, and time, apart from the classical ones; Painleve extended the algebraic scope to velocity variables. Poincare's 1890 theorem addressed a different class, ruling out generic additional analytic integrals in the perturbative setting. The obstruction is the same small-divisor mechanism seen in Chapter 5: a formal integral must solve homological equations whose resonant denominators cannot be made to vanish generically. These theorems are not identical statements with different names: Bruns and Painleve limit algebraic first integrals of the full problem, while Poincare explains why analytic integrability fails under generic perturbation of an integrable model.

- Jacobi coordinates remove center-of-mass motion and diagonalize the mass metric. They reveal two coupled relative vectors, which is the simplest coordinate-level reason the two-body Kepler solution does not generalize.
- Central configurations are exact special motions, not generic solutions. They are valuable because they organize nearby dynamics and survive as Lagrange-point structures in the restricted problem.
- The circular restricted three-body problem supplies an autonomous rotating-frame model with Lagrange points, zero-velocity curves, a canonical Hamiltonian, and the Jacobi constant. These structures are the standard geometric language for capture and escape channels.
- The heliocentric Sun-Jupiter-asteroid model is a periodically forced Kepler problem. Its indirect term is part of the coordinate transformation, not an optional correction.
- Numerical ejection probabilities are finite-time, model-dependent observables. They become meaningful only when the ensemble, integration time, timestep, invariants, and loss criterion are stated.

6.7 Chapter-End Laboratories And Exercises

This laboratory is the finite-mass counterpart of the restricted Sun-Jupiter-asteroid simulations: derive Newton's equations, choose controlled initial data, and monitor invariants before interpreting instability.

6.7.1 Equal-Mass Three-Body Laboratory

For an equal-mass three-body laboratory, take three masses in the plane with pair potential

$$V = -k \left(\frac{1}{r_{12}} + \frac{1}{r_{23}} + \frac{1}{r_{13}} \right),$$

where

$$r_{ij} = \|r_i - r_j\|.$$

The Lagrangian is

$$L = \sum_{i=1}^3 \frac{1}{2} m \|\dot{r}_i\|^2 + k \left(\frac{1}{r_{12}} + \frac{1}{r_{23}} + \frac{1}{r_{13}} \right).$$

The equations are

$$m\ddot{r}_i = -k \sum_{j \neq i} \frac{r_i - r_j}{\|r_i - r_j\|^3}.$$

The symmetric initial positions are

$$r_1(0) = (1, 0), \quad r_2(0) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad r_3(0) = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2} \right).$$

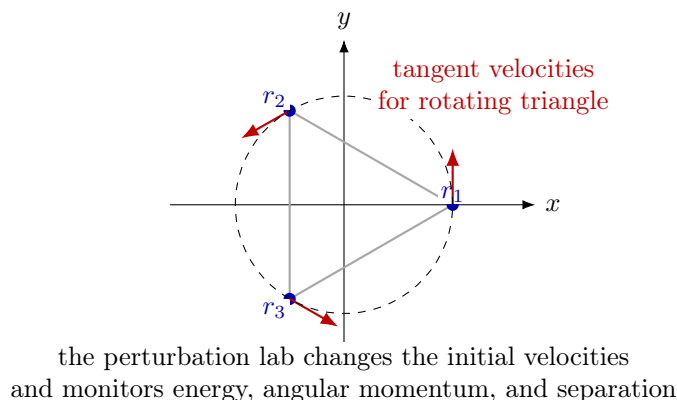


Figure 6.3: Equal-mass three-body laboratory. The unperturbed comparison solution starts from an equilateral triangle with tangential velocities; perturbing this initial data is a controlled way to watch the loss of symmetric motion.

For the rotating equilateral solution, velocities are tangent to the circumcircle. A representative perturbed initial condition uses a speed parameter $v = 0.8$ and perturbation $\epsilon = 0.05$, with initial velocities such as

$$\dot{x}_1(0) = \epsilon, \quad \dot{x}_2(0) = -\frac{\sqrt{3}}{2}v - \epsilon, \quad \dot{x}_3(0) = \frac{\sqrt{3}}{2}v.$$

The workflow matters more than the particular number 0.8: derive Newton's equations from a Lagrangian, integrate the exact pairwise forces, monitor energy and angular momentum, and compare unperturbed symmetric motion with perturbed motion. This is the finite-mass cousin of the Sun-Jupiter-asteroid ejection simulation.

6.7.2 Questions For The Reader

1. Starting from the inertial three-body Lagrangian, derive the center-of-mass equation and explain why internal gravitational forces cancel in pairs.
2. Verify the Jacobi-coordinate kinetic energy formula by substituting the inverse transformation for q_1, q_2, q_3 . Which mass factors would be wrong if the center of mass of bodies 1 and 2 were not used in ρ_2 ?
3. Use the homogeneity of U to rederive the Lagrange-Jacobi identity. What does it imply for a bounded noncollisional orbit?
4. For the Sun-Jupiter value of μ_{CR3BP} , where are L_1, L_2, L_3 relative to Jupiter?
5. For a chosen Jacobi constant, which zero-velocity necks are open?
6. Show explicitly that $C_J = -2H_{\text{rot}}$ after converting canonical momenta to rotating-frame velocities.
7. How does $\hat{P}_{\text{ej}}(T)$ change with T ?
8. Which semimajor-axis bins have the largest ejection fractions?
9. What changes when M_J is scaled upward?

10. Which losses are close encounters, and which are clean ejections?
11. How sensitive are the results to Δt ?
12. Are apparent resonance structures stable under a different random seed?

6.8 Associated Demos

- `demos/python/circular_restricted_three_body.py`
- `demos/python/asteroid_ejection_probability.py`
- `demos/mathematica/CircularRestrictedThreeBody.wl`
- `demos/mathematica/AsteroidEjectionProbability.wl`

Use the CR3BP demo first: it checks the Jacobi constant and draws the zero-velocity geometry. Use the ejection-probability demo second: it turns the same dynamical ideas into an ensemble statistic, so the meaningful outputs are not only trajectories but also loss criteria, timestep sensitivity, resonance bins, and finite-time uncertainty.

Chapter 7

Elastic Bodies

7.1 Elastic Solids As Material Continua

Pull a rubber band and let go. It snaps back. Push a finger through honey and the honey flows around the finger. The rubber band stores energy in shear; the honey responds through a rate-dependent stress. That distinction separates an elastic solid from a simple fluid.

Elasticity is the continuum setting closest to the finite-dimensional mechanics developed earlier because every material point keeps a persistent label. This lets the configuration-space logic of particles and rigid bodies pass into field theory with minimal translation. The basic unknown is the material map

$$\varphi : B \times I \rightarrow \mathbb{R}^3, \quad x = \varphi(X, t),$$

where $X \in B$ is the label of a material point and x is its current spatial position. The central questions are local and material: how the deformation gradient measures stretch, which strain fields can come from a single-valued displacement, how stored energy produces stress, and how boundary-value problems reduce the field equations to measurable stiffness laws. Chapter 8 keeps the material body but separates overall rigid placement from internal shape. Chapter 9 changes the constitutive question: material labels can still be used in derivations, but the state of a simple fluid is organized most effectively by Eulerian fields and rate-dependent stress.

Roadmap. The chapter has one main narrative. Deformation gradients measure local stretching, compatibility distinguishes true displacement fields from incompatible strain data, and a hyperelastic action gives elastodynamic equations and stress. The finite-strain theory is kept first because it makes objectivity and stress measures unambiguous; the linear theory is then obtained as a controlled small-displacement limit.

The guiding distinction is between intrinsic deformation and frame description. Quantities such as $C = F^T F$ and the Green strain do not change under a rigid rotation of the observer. Stress measures such as P and σ live in different configurations and are related by the deformation gradient. Each stress or strain tensor is introduced together with the space on which it acts and the boundary term that defines its mechanical meaning.

Appendix C returns to these choices in a more technical register, including weak forms, boundary terms, incompressibility as a constraint, and the functional-analytic caveats behind existence and regularity statements.

Assumption 7.1 (Regular deformation map). *Unless stated otherwise, $\varphi(\cdot, t)$ is a smooth orientation-preserving embedding of the reference body into space:*

$$J(X, t) = \det F(X, t) > 0.$$

The inequality $J > 0$ rules out local interpenetration and orientation reversal. Global injectivity is stronger than $J > 0$. We assume it when contact or self-intersection matters; otherwise the local stress derivations use only nonsingularity of F .

Definition 7.1 (Reference versus spatial densities). *The reference density $\rho_0(X)$ is mass per reference volume. The spatial density $\rho(x, t)$ is mass per current volume. Conservation of mass for a material volume gives*

$$\rho(\varphi(X, t), t)J(X, t) = \rho_0(X).$$

Thus J converts volume elements and ρ_0 converts kinetic energy, momentum balance, and stress work back to the fixed material domain B .

7.2 Deformation Gradient And Strain

Elasticity begins by comparing nearby material points before and after deformation. Local strain is measured by the derivative of the displacement map: two deformations that differ by a rigid translation have the same local strain. This is why the deformation gradient F , and the metric-like tensor $C = F^T F$, are introduced before stresses and constitutive laws.

The deformation gradient is the local linearization of the deformation map:

$$F^i{}_A = \frac{\partial \varphi^i}{\partial X^A}.$$

If δX is a small material vector, then its spatial image is

$$\delta x^i = F^i{}_A \delta X^A.$$

The right Cauchy-Green tensor measures squared lengths after deformation:

$$|\delta x|^2 = C_{AB} \delta X^A \delta X^B, \quad C_{AB} = F^i{}_A F^i{}_B.$$

The Green strain is

$$E_{AB} = \frac{1}{2}(C_{AB} - \delta_{AB}).$$

The names are part of the logic: Cauchy's finite-deformation tensors and Green's strain organize deformation by comparing metrics, not by subtracting points. Truesdell and Noll later made the objectivity requirements systematic in modern continuum mechanics. The Green strain vanishes for every rigid motion

$$\varphi(X) = a + QX, \quad Q \in \text{SO}(3),$$

because $F = Q$ and $Q^T Q = I$. This invariance is why E , not $F - I$, is a geometric strain measure for finite deformations. It is not the only such measure. The Almansi strain, Biot strain, Hencky logarithmic strain, and the Seth-Hill family are also objective finite-strain measures; they agree to first order in small strain and differ in how they organize finite stretch.

For small displacements, the finite strain reduces to the symmetric gradient of the displacement. To see this, write

$$\varphi(X, t) = X + \xi(X, t),$$

one has

$$F^i_A = \delta^i_A + \partial_A \xi^i.$$

Keeping only first-order displacement gradients,

$$E_{AB} = \frac{1}{2}(\partial_A \xi_B + \partial_B \xi_A) + O(|\nabla \xi|^2).$$

The infinitesimal strain is therefore

$$e_{AB} = \frac{1}{2}(\partial_A \xi_B + \partial_B \xi_A).$$

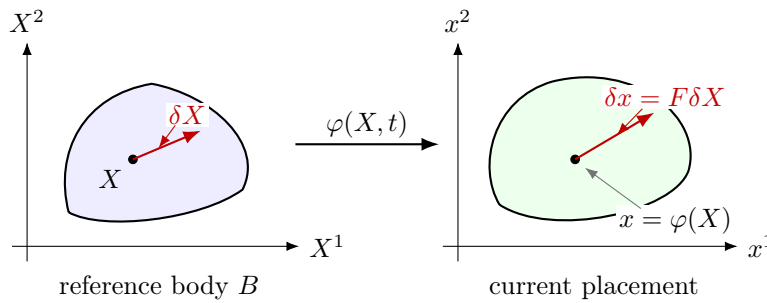


Figure 7.1: The deformation gradient $F = T_X \varphi$ is the tangent map of the deformation. It sends an infinitesimal material vector δX to the spatial vector $\delta x = F \delta X$. Strain is obtained by comparing the lengths and angles of these vectors, not by comparing absolute positions.

7.3 Polar Decomposition And Objectivity

The deformation gradient contains both rigid rotation and intrinsic stretch. For $J = \det F > 0$, polar decomposition gives

$$F = R_p U = V R_p,$$

where $R_p \in \text{SO}(3)$, and U, V are symmetric positive-definite stretch tensors. The right stretch U acts on material vectors before the polar rotation; the left stretch V acts on spatial vectors after the rotation. The right Cauchy-Green tensor removes the rigid rotation:

$$C = F^T F = U^2.$$

Similarly,

$$B = F F^T = V^2.$$

Derivation 7.1 (Constructing the polar factors). *Since F is nonsingular, $C = F^T F$ is symmetric positive definite. It has a unique symmetric positive square root*

$$U = C^{1/2}.$$

The square root is constructed by the spectral theorem. Write

$$C = O\Lambda O^T, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_3), \quad \lambda_i > 0,$$

with O orthogonal. Define

$$C^{1/2} = O \text{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \sqrt{\lambda_3}) O^T.$$

This matrix is symmetric positive definite and squares to C . If S is another symmetric positive square root, then S commutes with $C = S^2$ and preserves the eigenspaces of C ; on each eigenspace positivity selects the positive square root. Hence $S = C^{1/2}$. Define

$$R_p = FU^{-1}.$$

Then

$$R_p^T R_p = U^{-1} F^T F U^{-1} = U^{-1} U^2 U^{-1} = I.$$

Because $\det F > 0$ and $\det U > 0$, $\det R_p = 1$. Thus $R_p \in \text{SO}(3)$. Finally, setting

$$V = R_p U R_p^T$$

gives $F = R_p U = V R_p$. The construction also proves uniqueness: the positive square root $U = C^{1/2}$ is unique, so $R_p = FU^{-1}$ and $V = R_p U R_p^T$ are unique for every F with $\det F > 0$.

The eigenvalues of U are the principal stretches. They are intrinsic measurements of local length change: a unit material vector in a principal direction is stretched by $\lambda_i > 0$ before the rigid rotation R_p is applied. The polar rotation is not the same thing as material spin; it is the rotation factor in the instantaneous deformation gradient.

This distinction is the doorway to objectivity. A constitutive law for an ordinary elastic solid may depend on stretch invariants such as those of C , and it must be invariant under a superposed rigid rotation of the observer. Polar decomposition makes the separation visible: R_p tells how the local material frame is carried into space, while U and C record the intrinsic stretching that can store elastic energy.

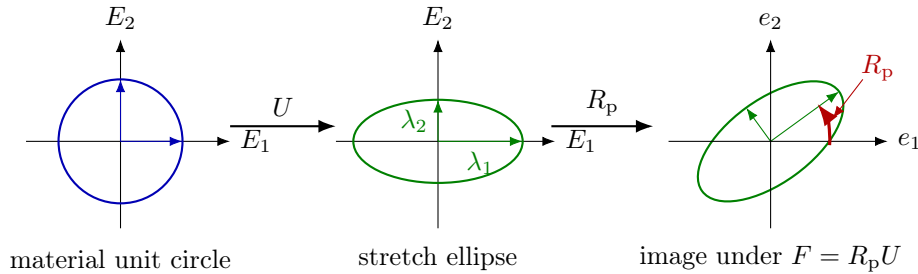


Figure 7.2: Polar decomposition separates stretch from rigid rotation. In a principal two-dimensional slice, U sends the material unit circle to an ellipse with semi-axes equal to principal stretches; R_p then rotates that stretched figure without changing its lengths.

This decomposition clarifies frame indifference. If an observer superposes a rigid rotation $Q \in \text{SO}(3)$, the deformation gradient becomes QF , but

$$(QF)^T(QF) = F^T F = C.$$

Thus, for an ordinary hyperelastic solid, frame indifference is equivalent to the statement that the stored energy factors through C :

$$W(F, X) = \widehat{W}(C, X).$$

The qualifier “ordinary” matters: if the material carries additional internal directors or micro-rotations, as in Cosserat or liquid-crystal models, those variables must be transformed along with F , and frame indifference is a statement about all of them together. The converse is immediate: if $W(F, X) = \widehat{W}(F^T F, X)$, then

$$W(QF, X) = \widehat{W}(F^T Q^T Q F, X) = \widehat{W}(F^T F, X) = W(F, X).$$

For orientation-preserving deformations with $J > 0$, polar decomposition gives the other direction: $F = R_p U$, so frame indifference removes R_p and leaves dependence only on $U^2 = C$. For an isotropic material, the material has no preferred direction in the reference body, so $\widehat{W}$ may depend on C only through its principal invariants:

$$I_1 = \operatorname{tr} C, \quad I_2 = \frac{1}{2} ((\operatorname{tr} C)^2 - \operatorname{tr}(C^2)), \quad I_3 = \det C = J^2.$$

The distinction is important. Frame indifference is invariance under rotations of the spatial observer. Isotropy is invariance under rotations of the material reference frame. A crystal can be frame indifferent but not isotropic.

Example 7.1 (Simple shear is not a rigid rotation). *For the simple shear*

$$\varphi(X^1, X^2, X^3) = (X^1 + \gamma X^2, X^2, X^3),$$

the deformation gradient is

$$F = \begin{pmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Although $\det F = 1$, the body is not undeformed:

$$C = F^T F = \begin{pmatrix} 1 & \gamma & 0 \\ \gamma & 1 + \gamma^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_{12} = \frac{\gamma}{2}.$$

The volume is preserved, but right angles in the reference square are changed. This is the basic solid-mechanics distinction between volume change and shear strain.

7.4 Compatibility

Not every matrix field is a deformation gradient. If

$$F^i{}_A = \partial_A \varphi^i$$

for a smooth deformation, then mixed partial derivatives commute:

$$\boxed{\partial_A F^i{}_B - \partial_B F^i{}_A = 0.}$$

On a simply connected body, this curl-free condition is also enough to recover φ from F , up to a rigid translation.

For small strain, the compatibility question is subtler because e is only the symmetric part of $\nabla\xi$. Given a symmetric tensor field e_{ij} , we ask whether there exists a displacement ξ such that

$$e_{ij} = \frac{1}{2}(\partial_i \xi_j + \partial_j \xi_i).$$

The answer is controlled by the Saint-Venant tensor:

$$\boxed{\epsilon_{ik\ell}\epsilon_{jmn}\partial_k\partial_m e_{\ell n} = 0.}$$

Saint-Venant formulated the compatibility condition in 1864; Beltrami later recast it in the double-curl form in 1886. Its geometric content is that incompatibility records defects, residual stress, or a non-Euclidean reference geometry inside the strain field. This is often written

$$\text{curl curl } e = 0.$$

When the condition fails, the local strain field cannot be assembled into a single-valued displacement without cuts, defects, residual stress, or a non-Euclidean reference geometry.

Derivation 7.2 (Saint-Venant compatibility as linearized curvature). *A small strain field defines the metric*

$$g_{ij} = \delta_{ij} + 2e_{ij}.$$

To first order in e , the Christoffel symbols are

$$\Gamma^i_{jk} = \partial_j e_{ik} + \partial_k e_{ij} - \partial_i e_{jk},$$

where indices are raised and lowered with δ_{ij} . The linearized Riemann tensor is therefore

$$R_{iljm} = \partial_j \Gamma_{ilm} - \partial_m \Gamma_{ilj} = \partial_j \partial_\ell e_{im} + \partial_m \partial_i e_{\ell j} - \partial_j \partial_i e_{\ell m} - \partial_m \partial_\ell e_{ij}.$$

If $e_{ij} = \frac{1}{2}(\partial_i \xi_j + \partial_j \xi_i)$, this curvature vanishes because mixed derivatives commute. In three dimensions, the Saint-Venant tensor is the double dual of this linearized curvature:

$$(\text{curl curl } e)_{ij} = \epsilon_{ik\ell}\epsilon_{jmn}\partial_k\partial_m e_{\ell n}.$$

Thus the displayed compatibility condition is exactly the linearized flatness condition.

On a simply connected domain the condition is also sufficient. If $\text{curl curl } e = 0$, the linearized curvature of $g = \delta + 2e$ vanishes. Locally there are vector fields ξ solving

$$\partial_i \xi_j + \partial_j \xi_i = 2e_{ij}.$$

The obstruction to patching these local solutions is a rigid motion on each overlap, because the difference of two solutions has zero symmetric gradient. On a simply connected body these rigid-motion cocycles can be chosen consistently, giving a single-valued displacement ξ unique up to a global rigid motion. This is the sufficiency part of Saint-Venant compatibility; on a multiply connected body, periods around noncontractible loops must also vanish.

Here is a more constructive way to read the last sentence. Choose a base point X_0 , prescribe $\xi(X_0)$ and the antisymmetric part $\omega_{ij}(X_0) = -\omega_{ji}(X_0)$, and write

$$\partial_i \xi_j = e_{ij} + \omega_{ij}.$$

Differentiating and eliminating ξ gives first-order equations for ω . The integrability conditions for this first-order system are precisely $\text{curl curl } e = 0$. Once those equations are satisfied, (ξ, ω) may be transported along any path from X_0 . Path independence on a simply connected body follows from the same flatness condition. Equivalently, the displacement can be reconstructed by the Volterra-Cesaro path integral

$$\xi_i(X) = \xi_i(X_0) + \int_{X_0}^X (e_{ij} + \omega_{ij}) dX^j,$$

where ω_{ij} is transported along the same path by the first-order system implied by compatibility. Vanishing linearized curvature makes the integral path independent on a simply connected body. Changing the initial $\xi(X_0)$ adds a translation, and changing $\omega(X_0)$ adds an infinitesimal rotation. Thus the only freedom left by the compatible strain is the expected rigid motion.

7.5 Hyperelastic Action And Piola Stress

A hyperelastic material has stored energy density $W(F, X)$ per reference volume. Let $\rho_0(X)$ be reference mass density. The action is

$$S[\varphi] = \int_{t_0}^{t_1} \int_B \left(\frac{1}{2} \rho_0 |\partial_t \varphi|^2 - W(F, X) \right) d^3 X dt.$$

Assumption 7.2 (Hyperelastic material). *The stress is derived from stored energy:*

$$P^A{}_i = \frac{\partial W}{\partial F^i{}_A}.$$

This assumption excludes hypoelastic, Cauchy-elastic but non-potential, viscous, plastic, fracture, and history-dependent materials unless additional internal variables or dissipation laws are added.

Derivation 7.3 (Elastodynamic equation). *Let $\delta\varphi = \eta$, with $\eta = 0$ at the time endpoints. Since*

$$\delta F^i{}_A = \partial_A \eta^i,$$

the first variation is

$$\begin{aligned} \delta S &= \int_{t_0}^{t_1} \int_B (\rho_0 \dot{\varphi}_i \dot{\eta}^i - P^A{}_i \partial_A \eta^i) d^3 X dt \\ &= \int_{t_0}^{t_1} \int_B (-\rho_0 \ddot{\varphi}_i + \partial_A P^A{}_i) \eta^i d^3 X dt - \int_{t_0}^{t_1} \int_{\partial B} P^A{}_i N_A \eta^i dS dt. \end{aligned}$$

For arbitrary interior variations,

$$\boxed{\rho_0 \ddot{\varphi}_i = \partial_A P^A{}_i.}$$

This is the material form of momentum balance with no body force. A body force per unit mass b_i would add $+\rho_0 b_i$ to the right-hand side. If an external boundary traction t_i per reference area is included through virtual work, the natural boundary condition is

$$P^A{}_i N_A = t_i.$$

The derivation also explains why P , not σ , is the stress that appears naturally in a reference-domain action: variations are differentiated with respect to material coordinates X^A , so the boundary term is a force per reference area.

7.6 Spatial Stress And Frame Indifference

The first Piola-Kirchhoff stress P measures force per reference area. The Cauchy stress σ measures force per current spatial area. The two stress tensors represent the same physical traction, but on different surface elements.

Two additional stress measures are often useful in finite strain and will reappear in Appendix C. If the stored energy is written as $W(F) = \widehat{W}(C)$, with $C = F^T F$, the second Piola-Kirchhoff stress is the material tensor

$$S = 2 \frac{\partial \widehat{W}}{\partial C}, \quad P = F S.$$

It is work-conjugate to the Green strain $E = (C - I)/2$, because

$$P : \dot{F} = S : \dot{E}.$$

The Kirchhoff stress is the spatial tensor

$$\tau = J\sigma = P F^T.$$

Thus σ measures current traction per current area, P converts reference surface elements to spatial force, S pulls both slots back to the material description, and τ is the volume-weighted spatial stress. These are not four different forces; they are four representations of the same stress power and traction balance.

Throughout this section, capital indices and normals are material/reference objects, while lower-case indices and normals are spatial/current objects. In linearized elasticity the two descriptions coincide to leading order, but the distinction is essential before linearization. Cauchy's 1822 tetrahedron argument is the classical reason a traction law linear in the surface normal must be represented by a stress tensor.

Proposition 7.1 (Cauchy's traction theorem). *At a fixed point and time, suppose the traction $t(n)$ on a plane with unit normal n is continuous in n , local, and satisfies force balance on shrinking material volumes. Then there is a unique tensor σ such that*

$$t(n) = \sigma n.$$

Proof. Use Cauchy's tetrahedron. Let the oblique face have area A and outward normal n , and let the three coordinate faces have outward normals $-e_i$ and areas $A_i = n_i A$, first for $n_i \geq 0$. Force balance on the tetrahedron gives

$$t(n)A + \sum_i t(-e_i)A_i + O(V) = 0,$$

where V is the tetrahedron volume. Shrink the tetrahedron while keeping its shape fixed. Traction terms scale like area and the inertial and body-force terms scale like volume, so $O(V)/A \rightarrow 0$. More explicitly, locality and continuity imply that the tractions on the shrinking faces differ from

their values at the limiting point by $o(1)$ uniformly for the finitely many face normals used in the tetrahedron; after multiplying by areas this gives an $o(A)$ correction. The volume force is $O(h^3)$ for a tetrahedron of size h , while $A = O(h^2)$. Dividing by A and taking $h \rightarrow 0$ removes all non-surface terms. Hence

$$t(n) = - \sum_i n_i t(-e_i).$$

Applying the same argument to a thin pillbox gives $t(-e_i) = -t(e_i)$. Therefore $t(n) = \sum_i n_i t(e_i)$ for normals in the chosen octant. Repeating the tetrahedron in each octant gives the same formula whenever the signs of the components of n are fixed. Continuity of $t(n)$ on the unit sphere then extends the formula across the coordinate great circles where one component vanishes. Define σ by taking its i -th column to be $t(e_i)$. Then $t(n) = \sigma n$ for every unit normal n , and uniqueness follows by evaluating the formula on the basis normals e_i . $\square$

Derivation 7.4 (Piola transform from traction balance). *Let a material surface element have reference unit normal N and reference area dS . Its spatial image has unit normal n and area da . Nanson's formula states*

$$n_i da = J(F^{-T})_{Ai} N_A dS.$$

The force across the same cut can be written in material variables as

$$df_i = P^A_i N_A dS$$

or in spatial variables as

$$df_i = \sigma_{ij} n_j da.$$

Substituting Nanson's formula gives

$$P^A_i N_A dS = \sigma_{ij} J(F^{-T})_{Aj} N_A dS.$$

Since N_A is arbitrary,

$$P^A_i = J \sigma_{ij} (F^{-T})_{Aj}.$$

Multiplying by F^k_A and using

$$(F^{-T})_{Aj} F^k_A = \delta^k_j$$

gives

$$\boxed{\sigma_{ij} = \frac{1}{J} P^A_i F^j_A.}$$

A stored energy is frame indifferent if a superposed rigid rotation does not change it:

$$W(QF, X) = W(F, X) \quad \text{for all } Q \in \text{SO}(3).$$

Let $Q(\varepsilon) = I + \varepsilon A + O(\varepsilon^2)$, with $A^T = -A$. Differentiating frame indifference at $\varepsilon = 0$ gives

$$0 = P^A_i A^i_j F^j_A.$$

Since A is any skew matrix, the tensor PF^T must be symmetric:

$$P^A_i F^j_A = P^A_j F^i_A.$$

Equivalently, if $B = PF^T - (PF^T)^T$, then in three dimensions $B_{ij} = \epsilon_{ijk}b_k$ for some axial vector b . The identity $B : A = 0$ for every skew A forces $b = 0$, hence $B = 0$. Therefore $\sigma = \sigma^T$. Thus symmetry of stress follows from rotational invariance of stored energy, matching angular-momentum balance.

When $W(F, X) = \widehat{W}(C, X)$, this symmetry can also be seen directly. Since

$$\delta C = F^T \delta F + \delta F^T F,$$

the variation of W is

$$\delta W = \frac{\partial \widehat{W}}{\partial C} : \delta C = 2 \left(F \frac{\partial \widehat{W}}{\partial C} \right) : \delta F,$$

where $\partial \widehat{W} / \partial C$ is symmetric. Comparing with $\delta W = P : \delta F$ gives

$$P = 2F \frac{\partial \widehat{W}}{\partial C}.$$

Because $\partial \widehat{W} / \partial C$ is symmetric,

$$PF^T = 2F \frac{\partial \widehat{W}}{\partial C} F^T$$

is symmetric, and therefore so is $\sigma = J^{-1}PF^T$.

7.7 Linear Isotropic Elasticity

For small strain in an isotropic homogeneous solid, the quadratic energy density is

$$W(e) = \frac{\lambda}{2}(\text{tr } e)^2 + \mu e_{ij}e_{ij},$$

where λ, μ are Lamé moduli. In this linearized section, the reference and spatial Euclidean frames have been identified to first order, so lower-case indices i, j are used for components of the displacement gradient and strain. The stress is

$$\sigma_{ij} = \frac{\partial W}{\partial e_{ij}} = \lambda(\text{tr } e)\delta_{ij} + 2\mu e_{ij}.$$

In a crystalline solid, one microscopic route to such an energy is the Cauchy-Born assumption: atoms in a unit cell are assumed to follow the macroscopic deformation gradient F affinely, so the lattice energy becomes a function $W(F)$. The assumption is useful but not automatic; defects, multi-lattices, phase transitions, and short-wavelength instabilities can make the continuum energy depart from the affine lattice prediction. The linearized equation of motion is

$$\rho_0 \partial_t^2 \xi_i = \partial_j \sigma_{ij} + \rho_0 b_i.$$

Substituting the stress-strain relation,

$$\begin{aligned} \partial_j \sigma_{ij} &= \lambda \partial_i (\partial_k \xi_k) + \mu \partial_j (\partial_i \xi_j + \partial_j \xi_i) \\ &= (\lambda + \mu) \partial_i (\nabla \cdot \xi) + \mu \Delta \xi_i. \end{aligned}$$

Hence

$$\rho_0 \partial_t^2 \xi = (\lambda + \mu) \nabla(\nabla \cdot \xi) + \mu \Delta \xi + \rho_0 b.$$

Using

$$\Delta \xi = \nabla(\nabla \cdot \xi) - \nabla \times \nabla \times \xi,$$

one sees two wave families:

$$c_P = \sqrt{\frac{\lambda + 2\mu}{\rho_0}}, \quad c_S = \sqrt{\frac{\mu}{\rho_0}}.$$

Longitudinal waves change volume; shear waves change shape at nearly fixed volume.

Derivation 7.5 (Plane-wave split into P and S modes). *Set $b = 0$ and try a plane wave*

$$\xi(X, t) = a e^{i(k \cdot X - \omega t)},$$

where a is a constant polarization vector and k is the wave vector. Then

$$\nabla(\nabla \cdot \xi) = -k(k \cdot a) e^{i(k \cdot X - \omega t)}, \quad \Delta \xi = -|k|^2 a e^{i(k \cdot X - \omega t)}.$$

Substituting into the linear elastodynamic equation gives

$$\rho_0 \omega^2 a = (\lambda + \mu) k(k \cdot a) + \mu |k|^2 a.$$

If $a \parallel k$, then $k(k \cdot a) = |k|^2 a$, so

$$\omega^2 = c_P^2 |k|^2, \quad c_P^2 = \frac{\lambda + 2\mu}{\rho_0}.$$

If $a \perp k$, then $k \cdot a = 0$, so

$$\omega^2 = c_S^2 |k|^2, \quad c_S^2 = \frac{\mu}{\rho_0}.$$

Thus the eigenvectors of the acoustic tensor split into one longitudinal polarization and two transverse polarizations in an isotropic solid.

Wave-speed positivity is strictly weaker than energy convexity. Propagating plane waves require $c_P^2 > 0$ and $c_S^2 > 0$, namely $\mu > 0$ and $\lambda + 2\mu > 0$. Positive definiteness of the quadratic elastic energy requires the stronger-looking but different conditions $\mu > 0$ and $K_{\text{bulk}} = \lambda + \frac{2}{3}\mu > 0$. The distinction foreshadows the hierarchy in finite elasticity: strong ellipticity tests the second variation on rank-one perturbations $a \otimes n$, rank-one convexity and quasiconvexity are stronger global conditions, and polyconvexity is a checkable sufficient condition often used in existence theory. These are not cosmetic terms: a stored energy can be frame indifferent and support real small-amplitude waves while still failing to be globally convex as a function of all entries of F . The Lamé constants entered elasticity through nineteenth-century work on isotropic solids; the P - and S -wave names are seismological terminology for the same longitudinal/transverse split. Oldham (1906) used the S -wave shadow zone to infer that Earth has a fluid outer core.

7.8 Elastic Constants And Classical Boundary-Value Problems

Stiffness laws used in the lab,

$$F = \frac{E_Y A_{cs}}{L} \Delta L, \quad T = \mu J_t \Theta', \quad E_Y I_{area} w'''' = q,$$

are reductions of the same isotropic equations. This section derives the standard bar, shaft, and beam formulas and tracks which modulus each experiment measures. The notation E_Y , ν_P , λ , μ , and K_{bulk} is therefore not interchangeable; each symbol names a different response.

Saint-Venant's principle gives the modelling justification behind many of the one-dimensional reductions below. Details of how a load is applied at an end are expected to decay over distances comparable to the cross-section size, so a slender body can have a robust interior bar, shaft, or beam solution even when the end fixtures are complicated. This is an asymptotic modelling principle, not an exact theorem for every geometry.

7.8.1 Moduli, Positivity, And Nearly Incompressible Solids

The quadratic energy

$$W(e) = \frac{\lambda}{2} (\text{tr } e)^2 + \mu e : e$$

becomes transparent after splitting strain into volume-changing and shape-changing parts. Define

$$e_0 = e - \frac{1}{3} (\text{tr } e) I, \quad K_{bulk} = \lambda + \frac{2}{3} \mu.$$

Since

$$e : e = e_0 : e_0 + \frac{1}{3} (\text{tr } e)^2,$$

the energy becomes

$$W(e) = \mu e_0 : e_0 + \frac{1}{2} K_{bulk} (\text{tr } e)^2.$$

The shear modulus μ penalizes distortion at fixed volume. The bulk modulus K_{bulk} penalizes volume change. For a stable isotropic linear elastic solid one requires

$$\mu > 0, \quad K_{bulk} > 0.$$

These inequalities have two related meanings in the linear theory. They make the quadratic energy positive on pure shear strains and pure volumetric strains, after rigid translations and rotations have been removed. They also make the acoustic wave speeds real, because the shear and longitudinal wave speeds below contain μ and $\lambda + 2\mu = K_{bulk} + \frac{4}{3}\mu$. For finite-strain elasticity, however, stability is subtler: strong ellipticity, rank-one convexity, quasiconvexity, and polyconvexity are distinct conditions. The linear isotropic model collapses these distinctions to the simple positive-definiteness test above, which is why the small-strain theory is so transparent.

Engineers often use Young's modulus E_Y and Poisson's ratio ν_P instead of λ, μ . The conversion is

$$E_Y = \mu \frac{3\lambda + 2\mu}{\lambda + \mu}, \quad \nu_P = \frac{\lambda}{2(\lambda + \mu)}.$$

Equivalently,

$$\mu = \frac{E_Y}{2(1 + \nu_P)}, \quad \lambda = \frac{E_Y \nu_P}{(1 + \nu_P)(1 - 2\nu_P)}, \quad K_{\text{bulk}} = \frac{E_Y}{3(1 - 2\nu_P)}.$$

Positive energy is equivalent to

$$E_Y > 0, \quad -1 < \nu_P < \frac{1}{2}.$$

Most ordinary solids have $0 < \nu_P < 1/2$: they contract laterally when stretched. The limit $\nu_P \rightarrow 1/2$ is the nearly incompressible limit. In that limit λ and K_{bulk} grow large while μ can remain finite, and the equations begin to resemble incompressible fluid mechanics: pressure-like stresses enforce a volume constraint.

7.8.2 Static Navier-Cauchy Equation And Boundary Data

In static linear elasticity, inertia is absent and the displacement $\xi(X)$ satisfies

$$(\lambda + \mu)\nabla(\nabla \cdot \xi) + \mu\Delta\xi + \rho_0 b = 0.$$

Equivalently,

$$\partial_j \sigma_{ij} + \rho_0 b_i = 0, \quad \sigma_{ij} = \lambda(\partial_k \xi_k) \delta_{ij} + \mu(\partial_i \xi_j + \partial_j \xi_i).$$

This is the static Navier-Cauchy equation, due to Navier (1821) in a molecular formulation and Cauchy (1822) in the stress-tensor formulation. Boundary conditions are part of the problem statement. On a displacement boundary ∂B_D , one prescribes

$$\xi = \xi_D.$$

On a traction boundary ∂B_N , one prescribes

$$\sigma n = t_N,$$

where n is the outward unit normal to the reference boundary, to linear order. The pure traction problem determines displacement only up to rigid translations and rotations; these rigid modes must be fixed by constraints or by imposing zero mean displacement and zero mean rotation.

The weak form follows from virtual work. For any test displacement η vanishing on ∂B_D ,

$$\int_B \sigma(\xi) : e(\eta) \, d^3 X = \int_B \rho_0 b \cdot \eta \, d^3 X + \int_{\partial B_N} t_N \cdot \eta \, dS.$$

This is the elastic analogue of deriving Cauchy's equation from balance laws: internal stress work equals external virtual work.

The weak form also gives uniqueness in the standard coercive case. If the body has a nonempty displacement boundary, or if rigid translations and rotations are otherwise fixed, the quadratic elastic energy is strictly convex on the admissible displacement space. Korn's 1909 inequality is the missing analytic bridge:

$$\|\xi\|_{H^1(B)} \leq C (\|e(\xi)\|_{L^2(B)} + \|\xi\|_{L^2(B)}),$$

which is often called Korn's second inequality. If $\mathcal{R}$ denotes the finite-dimensional space of infinitesimal rigid motions $r(X) = a + \Omega X$, $\Omega^T = -\Omega$, the sharper rigid-motion form is

$$\inf_{r \in \mathcal{R}} \|\xi - r\|_{H^1(B)} \leq C \|e(\xi)\|_{L^2(B)}.$$

After fixing the rigid part by displacement boundary conditions, zero mean translation and rotation, or another side condition, this becomes $\|\xi\|_{H^1} \leq C \|e(\xi)\|_{L^2}$. The inequality is an integration-by-parts statement that the skew part of $\nabla \xi$ is controlled by the symmetric part once rigid rotations are excluded; modern proofs are standard in the Korn-Friedrichs theory of elasticity. With Korn's inequality, the bilinear elastic energy is coercive, and the Lax-Milgram theorem gives a unique weak solution for compatible loads. In a pure traction problem, uniqueness holds only modulo rigid motions, which must be removed by side constraints.

7.8.3 Uniaxial Extension

Consider a slender bar of length L , cross-sectional area A_{cs} , and Young modulus E_Y . A tensile force F_{end} is applied at the end, and lateral surfaces are traction-free. The axial stress is

$$\sigma_{11} = \frac{F_{end}}{A_{cs}}, \quad \sigma_{22} = \sigma_{33} = 0.$$

Hooke's law in the E_Y, ν_P form gives

$$e_{11} = \frac{\sigma_{11}}{E_Y}, \quad e_{22} = e_{33} = -\nu_P \frac{\sigma_{11}}{E_Y}.$$

Thus the axial elongation is

$$\Delta L = \frac{F_{end} L}{E_Y A_{cs}}.$$

The negative lateral strain is the definition of Poisson's ratio:

$$\nu_P = -\frac{\text{lateral strain}}{\text{axial strain}}.$$

This formula identifies the experimental meaning of Young's modulus: it is stress divided by axial strain in a uniaxial test with free lateral contraction. The appearance of Poisson's ratio is also a historical caution: early "rari-constant" elasticity in the Cauchy-Poisson tradition overconstrained the moduli. Modern isotropic linear elasticity needs two independent elastic constants.

7.8.4 Torsion Of A Circular Shaft

Let a circular shaft of radius a and length L be twisted by torque T_{end} . In the Saint-Venant torsion solution, cross-sections rotate by an angle $\Theta(z)$ about the shaft axis. For a circular cross section, there is no warping correction, and the twist per length is constant:

$$\Theta'(z) = \frac{T_{end}}{\mu J_{polar}}, \quad J_{polar} = \frac{\pi a^4}{2}.$$

The total twist angle is

$$\Theta(L) - \Theta(0) = \frac{T_{end} L}{\mu J_{polar}}.$$

The shear stress grows linearly with radius:

$$\tau(r) = \frac{T_{\text{end}} r}{J_{\text{polar}}}, \quad 0 \leq r \leq a.$$

The maximum shear stress occurs at the boundary $r = a$. This is the torsional counterpart of the axial bar formula: stiffness is the elastic modulus times a geometric moment of the cross-section. Circular shafts are special because warping vanishes. Saint-Venant's general torsion theory allows non-circular sections, but then the cross-section warps and the simple polar-moment formula must be replaced by a torsion function. For a non-circular cross-section A , take the shaft axis to be z and let $\tau = \Theta'(z)$ be the twist per length. A Saint-Venant ansatz is

$$u_x = -\tau z y, \quad u_y = \tau z x, \quad u_z = \tau \chi(x, y),$$

where χ is the warping function. The only nonzero shear stresses are

$$\sigma_{xz} = \mu \tau (\partial_x \chi - y), \quad \sigma_{yz} = \mu \tau (\partial_y \chi + x).$$

Equilibrium $\partial_x \sigma_{xz} + \partial_y \sigma_{yz} = 0$ gives

$$\Delta \chi = 0 \quad \text{in } A,$$

and the traction-free lateral boundary condition $(\sigma_{xz}, \sigma_{yz}) \cdot n = 0$ gives

$$\frac{\partial \chi}{\partial n} = y n_x - x n_y \quad \text{on } \partial A.$$

The torque is $T_{\text{end}} = \mu \tau J_t$, where the torsional rigidity is computed from the shear-energy integral

$$J_t = \int_A [(\partial_x \chi - y)^2 + (\partial_y \chi + x)^2] \, dA.$$

For a circular section, $\chi = 0$ satisfies the boundary condition and $J_t = J_{\text{polar}}$. For a non-circular section, warping changes the rigidity and the stress distribution.

7.8.5 Euler-Bernoulli Beam Bending

The model is due to Daniel Bernoulli's elastic-energy viewpoint, around 1742, and Euler's 1744 *De curvis elasticis*. Its assumptions are deliberate: later Timoshenko theory restores shear deformation and rotary inertia when the beam is not sufficiently slender. As an asymptotic reduction from three-dimensional elasticity, it assumes a slenderness parameter $h/L \ll 1$, where h is a cross-section diameter. Keeping the leading bending energy gives strains of order $h w''$; transverse shear strains and rotary inertia enter at higher order in h/L . This is why the model is excellent for long thin beams and unreliable for short thick ones.

For a slender beam, let $x \in [0, L]$ be arclength along the undeformed neutral axis and let $w(x)$ be a small transverse deflection. The Euler-Bernoulli assumptions are:

1. slopes are small, so curvature is $w''(x)$ to leading order;
2. plane cross-sections remain plane and normal to the neutral axis;
3. shear deformation and rotary inertia are neglected.

Under these assumptions the axial displacement at height y from the neutral axis is $u_x(x, y) = -y w'(x)$ to leading order. Hence the dominant axial strain is

$$e_{xx} = \partial_x u_x = -y w''(x).$$

The three-dimensional elastic energy density $\frac{1}{2} E_Y e_{xx}^2$ therefore reduces to

$$\frac{1}{2} E_Y (w'')^2 \int_A y^2 dA = \frac{1}{2} E_Y I_{\text{area}} (w'')^2,$$

where $I_{\text{area}} = \int_A y^2 dA$. This is the leading three-dimensional origin of the one-dimensional bending energy. The bending energy is

$$U_{\text{bend}} = \frac{1}{2} \int_0^L E_Y I_{\text{area}} (w'')^2 dx,$$

where I_{area} is the second moment of area of the cross-section. With transverse distributed load $q(x)$, the potential energy is

$$\Pi[w] = \frac{1}{2} \int_0^L E_Y I_{\text{area}} (w'')^2 dx - \int_0^L q(x) w(x) dx.$$

Varying w gives

$$\begin{aligned} \delta \Pi &= \int_0^L E_Y I_{\text{area}} w'' \eta'' dx - \int_0^L q \eta dx \\ &= \int_0^L (E_Y I_{\text{area}} w'''' - q) \eta dx + [E_Y I_{\text{area}} w'' \eta' - E_Y I_{\text{area}} w''' \eta]_0^L. \end{aligned}$$

The beam equation is therefore

$$\boxed{E_Y I_{\text{area}} w'''' = q.}$$

The natural boundary quantities are the bending moment

$$M = E_Y I_{\text{area}} w''$$

and shear force

$$V_{\text{shear}} = -E_Y I_{\text{area}} w'''.$$

For a cantilever clamped at $x = 0$ and loaded by a downward end force P_{end} at $x = L$, the boundary conditions are

$$w(0) = 0, \quad w'(0) = 0, \quad M(L) = 0, \quad V_{\text{shear}}(L) = P_{\text{end}}.$$

With $q = 0$ in the interior, the solution is

$$\boxed{w(x) = \frac{P_{\text{end}} x^2 (3L - x)}{6 E_Y I_{\text{area}}}.}$$

The tip deflection is

$$\boxed{w(L) = \frac{P_{\text{end}} L^3}{3 E_Y I_{\text{area}}}.}$$

For a simply supported beam under uniform load q_0 , the deflection is

$$\boxed{w(x) = \frac{q_0 x (L^3 - 2Lx^2 + x^3)}{24 E_Y I_{\text{area}}}.}$$

It satisfies $w(0) = w(L) = 0$, zero bending moment at both supports, and $E_Y I_{\text{area}} w'''' = q_0$.

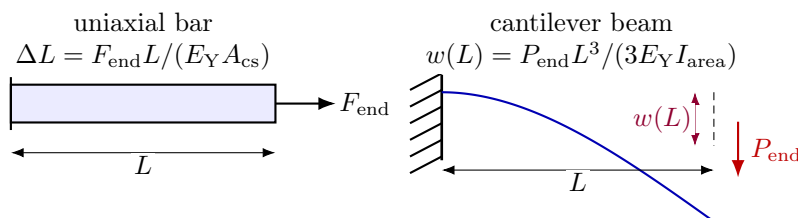


Figure 7.3: Two classical reductions of three-dimensional linear elasticity. A slender bar under axial load measures Young's modulus. A slender beam under transverse load converts elastic energy in curvature into the fourth-order Euler-Bernoulli equation.

7.9 What To Take Forward

An elastic body is described by a deformation map $\varphi(X, t)$ from a material reference body B into physical space. The deformation gradient $F^i_A = \partial\varphi^i/\partial X^A$ measures local stretch and rotation, while objective strain tensors remove the arbitrary rigid rotation. Compatibility says that not every field of strain is the strain of a single-valued displacement; it must arise from a deformation with commuting mixed derivatives, or equivalently satisfy the Saint-Venant conditions in the linear theory.

The variational derivation is parallel to finite-dimensional Lagrangian mechanics but with fields. A hyperelastic stored energy $W(F, X)$ defines the first Piola-Kirchhoff stress $P^i_A = \partial W/\partial F^i_A$, and stationarity of the action gives material momentum balance $\rho_0 \ddot{\varphi}^i = \partial_A P^i_A + \rho_0 b^i$ together with the correct traction boundary term. Frame indifference forces the stored energy to depend on deformation through rotationally invariant combinations such as $C = F^T F$, and it implies symmetry of the spatial Cauchy stress.

Linear isotropic elasticity is the small-strain limit. Its two Lamé constants encode the quadratic energy, the Navier-Cauchy equation describes displacement waves and static equilibrium, and the classical bar, torsion, and beam problems show how continuum equations reduce to experimentally measurable stiffness laws. The same material description also has a frame-theoretic extension: once each material point is allowed its own local orientation, elastic kinematics becomes the material side of a gauge theory of deforming bodies.

7.10 Chapter-End Laboratories And Exercises

The chapter-end work is organized as an audit of the elastic modelling chain: objectivity, variational stress, linearization, and classical reductions. The questions are short, but each one points to a place where continuum mechanics often becomes confusing if one forgets which configuration a tensor belongs to.

1. Starting from $C = F^T F$, show explicitly that superposing a rigid spatial rotation on φ leaves C unchanged. Which strain tensors would fail this objectivity test?
2. In the hyperelastic action, identify the boundary term whose coefficient is the first Piola-Kirchhoff traction. How does that term transform into a spatial Cauchy traction?
3. Derive the Navier-Cauchy equation from the quadratic isotropic energy and identify where the two Lamé constants enter longitudinal and transverse waves.

4. For the cantilever beam formula, check the four boundary conditions at the clamped and loaded ends. Which condition represents the applied point load?
5. Compare the bar and beam reductions in Figure 7.3. Which geometric small parameter justifies replacing a three-dimensional elastic body by a one-dimensional model?

7.10.1 Boundary-Value Laboratory: Bar Versus Beam

Use Figure 7.3 as a modelling checklist. The bar problem assumes that the dominant strain is axial and approximately uniform over each cross-section, giving the stiffness law $F = EA \Delta L/L$. The beam problem assumes that cross-sections remain approximately plane and that curvature, not axial stretch, carries the leading elastic energy, giving

$$EI w''''(x) = q(x).$$

Both are one-dimensional reductions of the same three-dimensional elastic theory. The useful comparison is therefore not “bar formula versus beam formula,” but which geometric small parameter justifies throwing away the other strain components. A good solution should state the slenderness assumption, the boundary data, the stress or moment being measured, and the energy term that survives the reduction.

7.11 Associated Demos

The associated elastic-boundary-value demos are:

- `demos/python/linear_elasticity.py`
- `demos/mathematica/LinearElasticity.wl`

They solve elementary elastic boundary-value problems and visualize displacement, strain, and stress fields. They are designed to be read beside this chapter’s derivations of the Piola stress, Cauchy stress, and linear elastic equations.

Chapter 8

Gauge Theory Of Deforming Bodies

A falling cat can reorient in mid-air without any external torque. Its angular momentum can remain zero while its orientation changes, because the body cycles through internal shapes. Related geometric-phase mechanisms appear in low-Reynolds swimming, astronaut reorientation, and the rotor laboratory below: closed loops in shape space produce rigid motions in physical space, although the constraints and shape variables differ from one system to another.

Frank Wilczek’s gauge-theoretic viewpoint, developed with Shapere in the context of swimming and self-propulsion,¹ treats the decomposition into overall rigid motion and internal deformation as a choice of gauge. The connection tells us which rigid motion accompanies an infinitesimal change of shape under a specified momentum constraint, and its curvature and holonomy describe the observable geometric phase.

The central phenomenon is geometric phase. A body can execute a closed loop in shape space and return to the same internal shape while acquiring a net rotation or translation in physical space. This is the mechanical analogue of parallel transport in a gauge field. The curvature of the connection measures the leading displacement produced by small loops, while the full holonomy records the finite motion produced by large cycles.

Roadmap. The chapter starts with the finite-dimensional Shapere-Wilczek split between rigid placement g and shape s . It then derives the mechanical connection from zero momentum and explains how this connection transforms under changes of body frame. The two-rotor model is a minimal laboratory for noncommuting connection components. The last third returns to continuum mechanics: local frames, Cosserat rods, curvature, torsion, and defects show how the same gauge language becomes a field theory over a material body.

Reading rule. There are two meanings of “gauge” in this chapter. One is finite-dimensional: choosing body axes over shape space. The other is local and continuum: choosing a frame at each material point. Both are frame choices, and both have curvatures, but they live over different base spaces. Whenever a symbol $\mathcal{A}$, A , ω , or R appears, first identify whether the base variable is shape s , time t , or material position X .

Appendix D systematizes this distinction after the examples have been developed. It separates mechanical connections, Shapere-Wilczek swimming connections, Cosserat frame fields,

¹A. Shapere and F. Wilczek, “Gauge kinematics of deformable bodies,” *American Journal of Physics* 57, 514–518 (1989), [doi:10.1119/1.15986](https://doi.org/10.1119/1.15986). Their closely related low-Reynolds-number swimming formulation appears in [doi:10.1017/S002211208900025X](https://doi.org/10.1017/S002211208900025X).

defect geometry, and fluid relabeling symmetry so that the same word “gauge” does not blur different mechanisms.

8.1 Gauge Kinematics Of Deformable Bodies

Elasticity describes deformation relative to a reference body. Gauge kinematics adds another question: how is the same changing shape placed and oriented in space? A choice of body axes at each shape gives a local description, while holonomy around a closed loop records the observable rigid motion produced by the shape cycle.

The Wilczek-Shapere point of view begins with a finite deformable body moving in space. Separate its configuration into:

$$g \in G, \quad s \in \mathcal{S},$$

where G is the rigid-motion group, such as $\text{SE}(3)$ or $\text{SO}(3)$, and $\mathcal{S}$ is shape space. A point of the full configuration space is a rigid placement g together with an internal shape s .

The subtlety is that the split into g and s is not unique. To say what the body’s orientation is at a given shape, one must choose body axes for that shape. A shape-dependent change of axes

$$h : \mathcal{S} \rightarrow G$$

changes the representative g . At this stage this is best described as the analogue of a gauge choice: the full gauge-theoretic content appears once the connection, curvature, and holonomy transformation laws are introduced below.

Geometrically, the full configuration space is locally a product

$$Q_{\text{full}} \simeq G \times \mathcal{S},$$

but no preferred product splitting is supplied by bare kinematics. A kinetic energy metric on Q_{full} , together with a momentum constraint, does pick out a distinguished splitting; that distinguished splitting is the mechanical connection. The projection

$$Q_{\text{full}} \rightarrow \mathcal{S}$$

forgets the rigid placement and remembers only shape. A connection is a rule for deciding which infinitesimal motions in Q_{full} count as “pure shape change” and which contain rigid motion. In mechanics this rule is not arbitrary: the mechanical connection is selected by the kinetic energy and the momentum constraint.

The observable content is therefore not the connection coefficient by itself. A different convention for body axes changes the coefficient, just as a different electromagnetic gauge changes the vector potential. What survives is the reconstructed rigid motion, especially the holonomy around a closed shape cycle and the curvature that controls small cycles.

8.1.1 Principal-Bundle Form Of The Split

The geometric content of the split is a principal bundle. Let Q_{full} be the full configuration space of the deforming body, after any constraints such as fixed center of mass have been imposed. The rigid-motion group G acts on Q_{full} by changing the overall placement of the same internal shape. The shape space is the quotient

$$\mathcal{S} = Q_{\text{full}}/G, \quad \pi : Q_{\text{full}} \rightarrow \mathcal{S}.$$

The point $s = \pi(q)$ remembers the shape and forgets the rigid placement.

Assumption 8.1 (Regular shape bundle). *In this chapter the G -action is assumed to be free and proper on the configurations under discussion, so that $\mathcal{S}$ is a smooth manifold and $Q_{\text{full}} \rightarrow \mathcal{S}$ is locally a product. Shapes with extra symmetry, such as a perfectly spherical body, have stabilizers and require minor orbifold or stratified-space modifications. The formulas below describe the generic regular stratum.*

A local choice of body frame is a local section

$$\sigma : \mathcal{U} \subset \mathcal{S} \rightarrow Q_{\text{full}}.$$

Every nearby configuration can then be written as

$$q = g \sigma(s), \quad g \in G, \quad s \in \mathcal{U}.$$

This expression is not unique because the section is not unique. If the internal representative is changed by a shape-dependent rigid motion

$$\sigma'(s) = h(s)^{-1} \sigma(s), \quad h : \mathcal{U} \rightarrow G,$$

then the same physical configuration is represented by

$$q = g \sigma(s) = g' \sigma'(s), \quad g' = g h(s).$$

Thus a change of section changes the coordinate g even though the physical configuration q has not changed. This is the precise analogue of a gauge transformation.

A connection is a G -equivariant rule that splits a tangent vector $\dot{q} \in T_q Q_{\text{full}}$ into a vertical rigid part and a horizontal shape-changing part. In the local coordinates (g, s) , write the body-frame group velocity as

$$\xi = g^{-1} \dot{g} \in \mathfrak{g}.$$

The horizontal condition has the form

$$\xi + \mathcal{A}_\alpha(s) \dot{s}^\alpha = 0.$$

Therefore a horizontal lift of a shape path obeys the reconstruction equation

$$g^{-1} \dot{g} = -\mathcal{A}_\alpha(s) \dot{s}^\alpha.$$

The local coefficient $\mathcal{A} = \mathcal{A}_\alpha ds^\alpha$ is the gauge potential. Under the change of section above it transforms as

$$\mathcal{A}' = \text{Ad}_{h^{-1}} \mathcal{A} - h^{-1} dh,$$

which is exactly the inhomogeneous transformation law expected of a connection. A connection supplies a chosen horizontal subspace, so the phrase “pure shape change” has a specified meaning. The mechanical connection is the particular connection selected by the kinetic-energy metric and the momentum value.

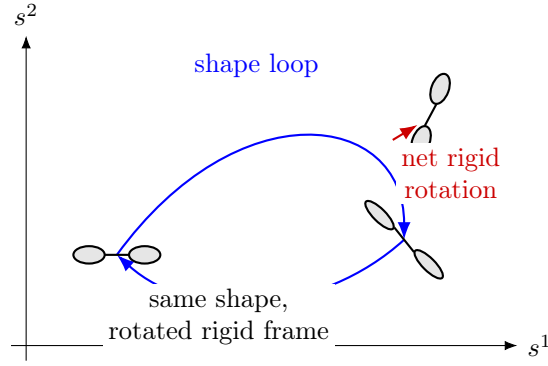


Figure 8.1: Gauge kinematics: a closed loop in shape space can produce a net rigid motion. The start and end shape can be the same while the rigid placement has changed; the effect is the holonomy of a connection.

8.1.2 Mechanical Connection From Zero Momentum

The mechanical connection is obtained by solving a constraint, not by adding a new force law. If the body has no total rigid momentum, then any internal shape velocity that would otherwise create angular or linear momentum must be cancelled by an accompanying rigid velocity. The connection is the linear rule that performs this cancellation at each shape.

The zero-momentum assumption is a modelling choice, not a mathematical necessity. It is appropriate for problems such as a freely falling cat or an isolated rotor device initialized with no angular momentum. With nonzero momentum, the same kinetic metric still defines a connection, but the reconstruction equation contains an additional momentum-driven drift. The zero-momentum case isolates geometric phase because the entire rigid motion is produced by shape change.

Let the body velocity in the moving frame be

$$\xi = g^{-1}\dot{g} \in \mathfrak{g},$$

where $\mathfrak{g}$ is the Lie algebra of G . Suppose the kinetic energy has the block form

$$T = \frac{1}{2} \begin{pmatrix} \xi \\ \dot{s} \end{pmatrix}^T \begin{pmatrix} M(s) & A(s) \\ A(s)^T & m(s) \end{pmatrix} \begin{pmatrix} \xi \\ \dot{s} \end{pmatrix}.$$

This is the quadratic form of kinetic energy written in the variables $(\xi, \dot{s})$, not a linearization of the equations of motion. The block $M(s)$ is the locked inertia operator: it measures the rigid momentum produced by a rigid velocity when the shape is held fixed, as if the internal shape coordinates were locked at that instant. The block $m(s)$ is the shape-space metric for pure deformation in the chosen frame. The off-diagonal block $A(s)$ is the crucial one: it measures the rigid momentum generated by changing shape while the chosen body frame is held fixed.

Assumption 8.2 (Regular locked inertia). *The zero-momentum reconstruction formula below assumes that $M(s)$ is invertible on the rigid motions being reconstructed. If a shape has an exact symmetry that makes some rigid velocity physically invisible, then $M(s)$ has a null direction and the regular principal-bundle description must be restricted or quotient further.*

The momentum conjugate to rigid motion is

$$\Pi = \frac{\partial T}{\partial \xi} = M(s)\xi + A(s)\dot{s}.$$

The symbol Π is the total rigid-motion momentum written in body variables: for $G = \text{SE}(3)$ it packages total linear and angular momentum. If the total angular and linear momentum vanish,

$$\Pi = 0,$$

as they do for an isolated body prepared with zero net external impulse, then Π remains zero by conservation and

$$\xi = -M(s)^{-1}A(s)\dot{s}.$$

Writing $\dot{s} = \dot{s}^\alpha \partial_\alpha$, define the connection

$$\boxed{\mathcal{A}_\alpha(s) = M(s)^{-1}A_\alpha(s).}$$

The rigid velocity is then

$$\boxed{g^{-1}\dot{g} = -\mathcal{A}_\alpha(s)\dot{s}^\alpha.}$$

This is the direct mechanical derivation of the gauge field: it is the rule that tells a zero-momentum deforming body how much rigid motion is forced by an infinitesimal shape change.

The minus sign is part of the reconstruction convention. A positive connection component means that the chosen shape change would create rigid momentum in the body frame; the actual zero-momentum motion supplies the opposite rigid velocity to cancel it. This is the mechanical meaning behind the gauge-theory formula.

8.1.3 Explicit Inertia-Operator Form

A concrete $\text{SO}(3)$ version of the same construction is useful because it turns the abstract mechanical connection into a linear algebra problem. Let

$$A = R^{-1}\dot{R} \in \mathfrak{so}(3)$$

be the angular velocity matrix of the body frame. Let $Q = Q^T$ encode the instantaneous shape through the second moment tensor

$$Q_{ij} = \int_B \rho(X) X_i X_j d^3X$$

in the chosen internal frame. The corresponding inertia tensor is

$$\tilde{I}_{ij} = \delta_{ij} \sum_{k=1}^3 Q_{kk} - Q_{ij}.$$

Let $Y(X, s(t))$ be the position of a material point in the chosen internal frame, with the center of mass at the origin. The spatial position is $x = RY$. In the body frame,

$$R^{-1}\dot{x} = AY + \dot{Y}.$$

The first term is the rigid velocity required by the frame motion; the second term is internal deformation. The body angular momentum is therefore

$$M_{\text{tot}} = \int_B \rho Y \times (AY + \dot{Y}) d^3X.$$

With the hat convention used for the rigid body, $\widehat{v}w = v \times w$, or

$$(\widehat{v})_{ij} = -\epsilon_{ijk}v_k,$$

the rigid part obeys the matrix identity

$$\widehat{I\Omega} = AQ + QA, \quad A = \widehat{\Omega}.$$

The internal part is

$$J_{\text{int}} = \int_B \rho Y \times \dot{Y} d^3X.$$

For zero total angular momentum,

$$\widehat{I\Omega} + J_{\text{int}} = 0.$$

Define the deformation-generated momentum that the rigid motion must cancel by

$$\widetilde{J} = -J_{\text{int}}.$$

Then the zero-momentum condition takes the linear matrix form

$$\boxed{AQ + QA = \widetilde{J}}.$$

Solving this equation is the explicit finite-dimensional gauge connection: the angular velocity A is determined by the shape velocity through $\widetilde{J}$.

Derivation 8.1 (Solving the finite-dimensional gauge equation). *Write*

$$A = \widehat{\Omega}, \quad A_{ij} = -\epsilon_{ijk}\Omega_k.$$

To verify the identity $AQ + QA = \widehat{I\Omega}$, diagonalize the symmetric tensor Q at the instant under consideration:

$$Q = \text{diag}(q_1, q_2, q_3).$$

Then

$$(AQ + QA)_{12} = -(q_1 + q_2)\Omega_3, \quad (AQ + QA)_{23} = -(q_2 + q_3)\Omega_1, \quad (AQ + QA)_{31} = -(q_3 + q_1)\Omega_2.$$

But the inertia tensor has principal moments

$$\widetilde{I}_1 = q_2 + q_3, \quad \widetilde{I}_2 = q_3 + q_1, \quad \widetilde{I}_3 = q_1 + q_2,$$

so these entries are exactly those of $\widehat{I\Omega}$. Since both sides transform tensorially under orthogonal changes of the internal frame, the identity holds in any frame. At zero total angular momentum,

the rigid angular momentum must equal $\tilde{J}$, the negative of the internal contribution. The linear map from angular velocity Ω to rigid angular momentum is precisely the inertia tensor:

$$\tilde{J}_i = \tilde{I}_{ij}\Omega_j.$$

Equivalently,

$$\Omega_i = (\tilde{I}^{-1})_{ij}\tilde{J}_j.$$

Therefore

$$A_{ij} = -\epsilon_{ijk}(\tilde{I}^{-1})_{kl}\tilde{J}_l.$$

This is the concrete $\text{SO}(3)$ deforming-body gauge formula. In the abstract notation above, $M(s)$ is the inertia operator $\tilde{I}(s)$, while the cross term $A(s)\dot{s}$ is the internally generated angular momentum $\tilde{J}(s, \dot{s})$. The helicopter rotor laboratory in Section 8.8.2 is the finite-dimensional special case where the shape variables are two rotor angles and this abstract inertia-operator equation reduces to a pair of scalar connection components.

8.1.4 Gauge Transformation And Holonomy

Under a shape-dependent change of body frame $h(s)$, the connection transforms as

$$\mathcal{A}' = \text{Ad}_{h^{-1}} \mathcal{A} - h^{-1}dh.$$

The sign of the inhomogeneous term follows from the reconstruction convention $g^{-1}\dot{g} = -\mathcal{A}(\dot{s})$. With the opposite convention, the same geometric connection is written with the more familiar plus sign. The base space is shape space and the structure group is the group of rigid motions.

For a path $s(t)$, the rigid motion is obtained by integrating

$$g^{-1}\dot{g} = -\mathcal{A}_\alpha(s)\dot{s}^\alpha.$$

For a closed shape cycle Γ , the net displacement is

$$g(T) = g(0) \mathcal{P} \exp \left(- \oint_{\Gamma} \mathcal{A} \right),$$

where $\mathcal{P}$ denotes path ordering. The result can be nontrivial even though the shape returns to its starting point. The path-ordered exponential is the limit of multiplying many small reconstruction steps. If $0 = t_0 < t_1 < \dots < t_N = T$, then

$$g(t_{j+1}) \approx g(t_j) \exp \left(-\mathcal{A}_\alpha(s(t_j))\dot{s}^\alpha(t_j)\Delta t_j \right),$$

and the factors are appended chronologically on the right of the current configuration. Thus the finite product for $g(T)$ is $\exp X_0 \exp X_1 \dots \exp X_{N-1}$, with earlier steps written farther to the left in this reconstruction convention. The factors must remain in this order because elements of G need not commute. Taking the mesh size to zero gives the displayed $\mathcal{P} \exp$ formula.

With the reconstruction convention $g^{-1}\dot{g} = -\mathcal{A}(\dot{s})$, the finite-dimensional curvature used for shape holonomy is

$$\mathcal{F} = d\mathcal{A} - \mathcal{A} \wedge \mathcal{A},$$

or in components,

$$\mathcal{F}_{\alpha\beta} = \partial_\alpha \mathcal{A}_\beta - \partial_\beta \mathcal{A}_\alpha - [\mathcal{A}_\alpha, \mathcal{A}_\beta].$$

This sign is tied to the minus sign in the reconstruction equation. The local frame connection used later for Cosserat and defect geometry follows the standard Maurer-Cartan convention $F = dA + A \wedge A$.

Derivation 8.2 (Small-loop holonomy). *Take shape coordinates (s^1, s^2) and traverse a tiny rectangle based at s_0 : first $+\Delta s^1$, then $+\Delta s^2$, then $-\Delta s^1$, then $-\Delta s^2$. Along each side the reconstruction equation*

$$g^{-1}\dot{g} = -\mathcal{A}_\alpha \dot{s}^\alpha$$

produces an infinitesimal group exponential. If $\mathcal{A}$ were constant, the product would be the group commutator

$$e^{-\mathcal{A}_1 \Delta s^1} e^{-\mathcal{A}_2 \Delta s^2} e^{\mathcal{A}_1 \Delta s^1} e^{\mathcal{A}_2 \Delta s^2} = \exp([\mathcal{A}_1, \mathcal{A}_2] \Delta s^1 \Delta s^2 + O(|\Delta s|^3)),$$

with the sign following the reconstruction convention. If $\mathcal{A}$ varies over the rectangle, the two s^1 -edges differ by $\partial_2 \mathcal{A}_1 \Delta s^2$, and the two s^2 -edges differ by $\partial_1 \mathcal{A}_2 \Delta s^1$. Combining the derivative terms with the commutator gives

$$g_{\text{loop}} = \exp(-\mathcal{F}_{12} \Delta s^1 \Delta s^2 + O(|\Delta s|^3)).$$

This is the finite-dimensional version of the usual curvature-as-holonomy calculation for a gauge connection.

For a small loop enclosing area element $\Delta S^{\alpha\beta}$,

$$\Delta g \approx \exp(-\mathcal{F}_{\alpha\beta} \Delta S^{\alpha\beta}).$$

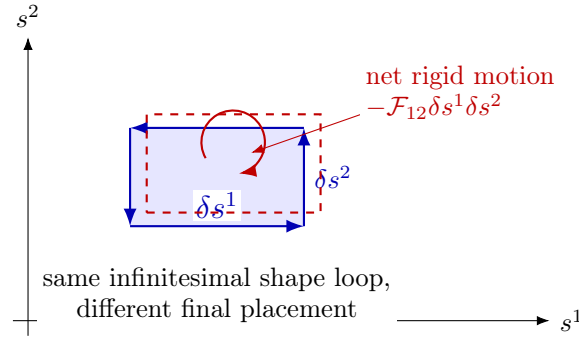


Figure 8.2: Curvature is the local density of holonomy. Traversing a small rectangle in shape space returns to the same shape but can leave a residual rigid displacement proportional to the curvature through the enclosed area.

Thus curvature is the local density of geometric phase. Berry phase, Hannay angle, and falling-cat reorientation are all holonomies of connections over parameter or shape spaces, but the constructions differ. Berry (1984) introduced the quantum $U(1)$ geometric phase; Hannay (1985) gave the classical angle holonomy on invariant tori; Shapere and Wilczek (1989) developed the gauge-kinematic program for deformable bodies; and Montgomery (1993) gave the gauge-theoretic treatment of the falling cat. The common name is geometric phase; the underlying mechanisms differ as described above.

Flat connections can still produce frame changes along open paths, but a small closed loop detects curvature. The curvature is the local observable of the gauge description. Different choices of body frame change $\mathcal{A}$, while the holonomy around a loop changes by conjugation and therefore represents the same physical displacement.

8.2 A Two-Rotor Deforming-Body Model

A minimal finite-dimensional example is a rigid carrier with two internal rotors. The rotor angles are shape variables

$$s = (\alpha, \beta) \in S^1 \times S^1.$$

This is the compact version of the helicopter-rotor laboratory in Section 8.8.2: here we strip the device down to its two noncommuting shape directions so the curvature calculation can be seen in one page. Let the carrier have isotropic inertia $I_0 \mathbf{1}$. Let one rotor spin about the body z -axis with inertia I_1 , and another spin about the body y -axis with inertia I_2 . If Ω is the carrier's body angular velocity, the rotational kinetic energy is

$$T = \frac{1}{2} I_0 \|\Omega\|^2 + \frac{1}{2} I_1 (\Omega_z + \dot{\alpha})^2 + \frac{1}{2} I_2 (\Omega_y + \dot{\beta})^2.$$

The body angular momentum conjugate to Ω is

$$\Pi = I_0 \Omega + I_1 (\Omega_z + \dot{\alpha}) e_z + I_2 (\Omega_y + \dot{\beta}) e_y.$$

Imposing zero total angular momentum gives

$$\Omega = -a \dot{\alpha} e_z - b \dot{\beta} e_y, \quad a = \frac{I_1}{I_0 + I_1}, \quad b = \frac{I_2}{I_0 + I_2}.$$

Thus

$$\mathcal{A}_\alpha = a e_z, \quad \mathcal{A}_\beta = b e_y.$$

The connection coefficients are constant, so the derivative part of curvature vanishes. But $\text{SO}(3)$ is nonabelian:

$$[e_z, e_y] = e_z \times e_y = -e_x.$$

Therefore

$$\mathcal{F}_{\alpha\beta} = -[\mathcal{A}_\alpha, \mathcal{A}_\beta] = ab e_x.$$

A small rectangular stroke with sides $\Delta\alpha, \Delta\beta$, traversed first in the α direction and then in the β direction, gives

$$\begin{aligned} g_{\text{loop}} &= \exp(-\mathcal{A}_\alpha \Delta\alpha) \exp(-\mathcal{A}_\beta \Delta\beta) \exp(\mathcal{A}_\alpha \Delta\alpha) \exp(\mathcal{A}_\beta \Delta\beta) \\ &= \exp([\mathcal{A}_\alpha, \mathcal{A}_\beta] \Delta\alpha \Delta\beta + O(\Delta^3)), \end{aligned}$$

by the Baker-Campbell-Hausdorff formula. With the conventions above the leading rotation vector is

$$-\mathcal{F}_{\alpha\beta} \Delta\alpha \Delta\beta = -ab \Delta\alpha \Delta\beta e_x;$$

reversing the traversal reverses the sign. The essential content is the holonomy: a closed loop in internal shape variables produces a rigid rotation even though the rotors return to their starting angles. Like the falling cat, this is a nonabelian $\text{SO}(3)$ holonomy over shape space, although the physical constraints and shape coordinates are different.

8.3 Swimming At Low Reynolds Number

Shapere and Wilczek's low-Reynolds-number swimming problem gives a particularly physical gauge theory. At very small Reynolds number, inertia drops out of Navier-Stokes and the surrounding fluid satisfies Stokes equations:

$$-\nabla p + \mu \Delta u = 0, \quad \nabla \cdot u = 0.$$

Boundary velocity is prescribed by the swimmer's rigid motion plus deformation rate. Because the Stokes problem is linear, the force and torque on the body depend linearly on:

$$\xi = g^{-1} \dot{g} \quad \text{and} \quad \dot{s}.$$

A freely swimming body has zero net force and torque, so the same algebra as above gives

$$\xi = -\mathcal{A}_\alpha(s) \dot{s}^\alpha.$$

The connection is now computed from a hydrodynamic boundary-value problem. More explicitly, solving the Stokes problem with unit rigid velocity gives a resistance matrix $R(s)$, and solving it with unit shape velocity gives a shape-force matrix $B(s)$. The zero-force and zero-torque condition has the linear form

$$R(s)\xi + B_\alpha(s)\dot{s}^\alpha = 0,$$

so

$$\xi = -R(s)^{-1} B_\alpha(s) \dot{s}^\alpha.$$

Thus $\mathcal{A}_\alpha = R^{-1} B_\alpha$. The phrase "solving the Stokes problem" includes the no-slip boundary condition on the swimmer surface, decay or wall conditions in the surrounding fluid, and the force/torque integrals of the hydrodynamic stress over the surface. Thus the connection is not inserted by hand; it is the finite-dimensional summary of a linear elliptic boundary-value problem. The Lorentz reciprocal theorem gives the symmetry of the Stokes resistance operator under the usual boundary conditions, which is why the rigid-motion resistance matrix is positive definite after quotienting out impossible rigid motions.

A reciprocal one-parameter stroke encloses no area in shape space and produces no net motion: this is Purcell's 1977 scallop theorem in gauge language. In one shape coordinate s , the reconstruction is

$$g^{-1} \dot{g} = -\mathcal{A}(s) \dot{s}.$$

Traversing a path from s_0 to s_1 and then retracing it reverses $\dot{s} dt = ds$, so the ordered product of rigid motions is inverted on the return leg. At zero Reynolds number there is no inertial memory to distinguish the outward stroke from the reversed stroke, hence the net displacement is trivial. A two-parameter stroke can enclose area and swim by curvature. This is the same geometric-mechanics language developed in the Marsden-Weinstein reduction tradition; later geometric swimmer models, including work by Kanso, Marsden, and Rowley, make the connection computation explicit for finite-dimensional shape families.

8.4 Local Frames In Elastic Continua

The finite-dimensional gauge picture becomes a field theory when every material point carries a local frame. A Cosserat body has, in addition to $\varphi(X, t)$, a rotation field

$$R(X, t) \in \text{SO}(3).$$

The directors are

$$d_a(X, t) = R(X, t)E_a,$$

where E_a is a fixed reference basis. The rotational connection is

$$A_A = R^{-1}\partial_A R \in \mathfrak{so}(3), \quad A_t = R^{-1}\partial_t R.$$

The translational strain measured in the local frame is

$$\Gamma_A = R^{-1}\partial_A \varphi.$$

For a rigid body, R depends only on time. For a Cosserat continuum, R depends on X , so neighboring material points can rotate relative to one another.

This is the field-theoretic version of the rigid-body frame issue. In the rigid body there is one moving frame; in a Cosserat body there is a moving frame at every material point. The connection A_A compares nearby frames, while A_t records how a given local frame evolves in time.

Definition 8.1 (Local frame gauge transformation). *Let $G(X, t) \in \text{SO}(3)$ be a change of local material frame:*

$$R'(X, t) = R(X, t)G(X, t).$$

Then

$$A'_A = G^{-1}A_AG + G^{-1}\partial_AG, \quad \Gamma'_A = G^{-1}\Gamma_A.$$

Thus A_A is a gauge connection and Γ_A is a vector-valued matter field in the local frame.

The simplest elastic energy for a Cosserat continuum penalizes deviations of these gauge-covariant strains from preferred values:

$$W_{\text{Cosserat}} = \frac{1}{2}C_{ab}^{AB}(\Gamma_A^a - \Gamma_{0A}^a)(\Gamma_B^b - \Gamma_{0B}^b) + \frac{1}{2}D_{ab}^{AB}(K_A^a - K_{0A}^a)(K_B^b - K_{0B}^b).$$

Here $K_A \in \mathbb{R}^3$ is the axial-vector form of the skew matrix A_A :

$$(A_A)^a_b = -\epsilon^a_{bc}K_A^c.$$

The tensors C and D encode stretching/shearing and bending/twisting stiffnesses. The preferred fields Γ_0, K_0 represent intrinsic strain and intrinsic curvature. This formula is schematic but important: it shows how ordinary elastic energy becomes an energy on translational strain and gauge connection data.

Remark 8.1 (Rigid body as the zero-gradient special case). *If $R(X, t) = R(t)$ and*

$$\varphi(X, t) = r(t) + R(t)X,$$

then

$$A_A = R^{-1}\partial_A R = 0, \quad \Gamma_A = R^{-1}\partial_A \varphi = E_A.$$

Thus a rigid body is the special Cosserat field with no spatial frame connection and no translational strain beyond the reference directors. A deformable Cosserat body is obtained by allowing R and Γ_A to vary with X .

8.5 Curvature, Torsion, And Defects

If a connection is globally of the form

$$A_A = R^{-1} \partial_A R,$$

then it is pure gauge and satisfies the Maurer-Cartan identity:

$$F_{AB} = \partial_A A_B - \partial_B A_A + [A_A, A_B] = 0.$$

Conversely, on a simply connected material patch, $F = 0$ implies locally that the connection is pure gauge: parallel transport is path independent, so one can define $R(X)$ by transporting a chosen initial frame from a base point to X , and this R satisfies $A = R^{-1} dR$. Path independence follows from applying Stokes' theorem to small spanning surfaces and using $F = 0$, then subdividing a homotopy between two paths into such small surfaces. Equivalently, the parallel transport around a small coordinate rectangle is

$$\exp(F_{AB} \Delta X^A \Delta X^B + O(\Delta^3)),$$

so vanishing curvature removes the infinitesimal obstruction; simple connectedness then removes the global path-dependence obstruction. Nonzero curvature means that local frames cannot be integrated into a single smooth orientation field without singularities or defects. In defect theory, curvature is associated with disclinations, or rotational defects. More precisely, in three dimensions the disclination-density tensor is the Hodge dual of the curvature two-form, just as the dislocation-density tensor below is the Hodge dual of torsion. Classical continuum defect theory was developed in the 1950s and 1960s by Kondo, Bilby, Kroner, and others.

To describe translational defects, introduce a local coframe θ^a , which measures material line elements in a local frame. The connection one-forms ω^a_b are the components of the same $\mathfrak{so}(3)$ -valued connection written above: if the matrix entries of A_A are $(A_A)^a_b$, then

$$\omega^a_b = (A_A)^a_b dX^A.$$

With this notation, the torsion two-form is

$$T^a = d\theta^a + \omega^a_b \wedge \theta^b.$$

Torsion measures closure failure of infinitesimal parallelograms and is associated with dislocation density. In three spatial dimensions the vector defect-density fields often used in materials science are Hodge duals of these two-forms; for example $(*F^a_b)_i = \frac{1}{2} \epsilon_i^{jk} F^a_{bjk}$ and $(*T^a)_i = \frac{1}{2} \epsilon_i^{jk} T^a_{jk}$. In this language:

$$\begin{aligned} \text{curvature} &\longleftrightarrow \text{rotational mismatch,} \\ \text{torsion} &\longleftrightarrow \text{translational mismatch,} \\ \text{incompatible strain} &\longleftrightarrow \text{nonintegrable local data.} \end{aligned}$$

The compatibility identities are the differential bookkeeping rules for these defect densities. With covariant exterior derivative D ,

$$DF = 0, \quad DT^a = F^a_b \wedge \theta^b.$$

They follow directly from the definitions. For curvature $F = d\omega + \omega \wedge \omega$,

$$DF = dF + \omega \wedge F - F \wedge \omega = 0,$$

because

$$\begin{aligned} dF &= d\omega \wedge \omega - \omega \wedge d\omega, \\ \omega \wedge F - F \wedge \omega &= \omega \wedge d\omega - d\omega \wedge \omega + \omega \wedge \omega \wedge \omega - \omega \wedge \omega \wedge \omega, \end{aligned}$$

so all terms cancel. For torsion,

$$DT^a = D(d\theta^a + \omega^a_b \wedge \theta^b) = F^a_b \wedge \theta^b.$$

The first identity says that disclination density satisfies a Bianchi identity. The second says that dislocation density is not independent of rotational curvature when the local frame itself has curvature. These identities are the continuum-defect analogue of $\nabla \cdot (\nabla \times A) = 0$: they tell which defect fields can be prescribed consistently.

The gauge theory records which descriptions depend on arbitrary local choices of frame and which quantities are invariant.

8.6 Cosserat Rod As A One-Dimensional Gauge Theory

A planar rod has material coordinate s , centerline $r(s) \in \mathbb{R}^2$, and orientation angle $\theta(s)$. Its director is

$$d_1 = (\cos \theta, \sin \theta).$$

Eugene and Francois Cosserat (1909) introduced the idea that a continuum point can carry an independent local frame, not only a position. Let

$$d_2 = (-\sin \theta, \cos \theta).$$

Then the moving-frame equations are

$$d'_1 = \theta' d_2, \quad d'_2 = -\theta' d_1.$$

Thus curvature is the connection coefficient for the rotation of the director frame along the rod:

$$\kappa = \theta'.$$

For an inextensible unshearable rod,

$$r'(s) = d_1(s).$$

Given κ ,

$$\theta(s) = \theta(0) + \int_0^s \kappa(\sigma) d\sigma,$$

and

$$r(s) = r(0) + \int_0^s (\cos \theta(\sigma), \sin \theta(\sigma)) d\sigma.$$

This is the reconstruction problem: a gauge field κds determines shape after choosing initial position and frame.

The bending energy

$$E_{\text{bend}} = \frac{1}{2} \int_0^L B(\kappa - \kappa_0)^2 ds$$

compares actual curvature with preferred curvature. Here B is bending stiffness and κ_0 is intrinsic curvature. In gauge language, the energy penalizes mismatch between a connection and a preferred connection.

The Euler-Lagrange equation is short in this one-dimensional case. Since $\kappa = \theta'$,

$$\delta E_{\text{bend}} = \int_0^L B(\kappa - \kappa_0) \delta \theta' \, ds = [M \delta \theta]_0^L - \int_0^L M' \delta \theta \, ds,$$

where

$$M = B(\kappa - \kappa_0)$$

is the bending moment. With fixed end angles and no distributed applied couple, stationarity gives

$$M' = 0.$$

If endpoint angles are free, the boundary term supplies the natural moment condition $M = 0$ at the free end. With external forces and the constraint $r' = d_1$, force balance adds the usual Cosserat equations

$$n'(s) + f(s) = 0, \quad M'(s) + d_1(s) \times n(s) + \ell(s) = 0,$$

where n is internal force, f is applied force per length, and ℓ is applied couple per length. These equations are the one-dimensional balance laws behind the reconstruction formulas above. In this planar notation $d_1 \times n$ denotes the scalar out-of-plane moment component; in a spatial Cosserat rod the same balance is vector-valued. They follow from the constrained variational functional

$$\int_0^L \left[\frac{1}{2} B(\theta' - \kappa_0)^2 + n \cdot (r' - d_1) - f \cdot r - \ell \theta \right] ds.$$

Variation with respect to r gives $n' + f = 0$. Variation with respect to θ gives the moment equation, because $\delta d_1 = \delta \theta d_2$ and

$$\delta [n \cdot (r' - d_1)] = -n \cdot d_2 \delta \theta = -(d_1 \times n) \delta \theta.$$

The θ' -variation contributes $-M' \delta \theta$ after integration by parts, while the applied couple contributes $-\ell \delta \theta$. The coefficient of $\delta \theta$ is therefore

$$-M' - d_1 \times n - \ell,$$

which gives $M' + d_1 \times n + \ell = 0$.

8.7 What To Take Forward

The progression from rigid body to the gauge theory of deforming bodies is:

$$\begin{aligned} \text{rigid body:} & \quad R(t), \quad R^{-1} \dot{R}, \\ \text{elastic body:} & \quad \varphi(X, t), \quad F = \partial_X \varphi, \quad W(F), \\ \text{deformable body:} & \quad g(t), \quad s(t), \quad g^{-1} \dot{g} = -\mathcal{A}(s) \dot{s}, \quad \mathcal{F} = d\mathcal{A} - \mathcal{A} \wedge \mathcal{A}, \\ \text{Cosserat continuum:} & \quad R(X, t), \quad R^{-1} \partial_A R, \\ \text{defect theory:} & \quad A, \quad F = dA + A \wedge A, \quad T = d\theta + \omega \wedge \theta. \end{aligned}$$

The first line is ordinary rigid-body mechanics. The second is classical elasticity: deformation gradients, stored energy, stress, and elastic waves. The third is Shapere-Wilczek gauge kinematics over shape space. The fourth attaches a local rigid body to every material point. The fifth allows the local frame and translation data to be incompatible, producing curvature and torsion as physical defect densities.

8.8 Chapter-End Laboratories And Exercises

These laboratories make the gauge language operational: a small shape loop measures curvature, and a finite-dimensional rotor model shows how noncommuting connection components produce body rotation.

8.8.1 Mechanical Connection: Curvature From A Small Shape Loop

Let s^1, s^2 be two shape coordinates and let the zero-momentum reconstruction equation be

$$g^{-1}\dot{g} = -\mathcal{A}_\alpha(s)\dot{s}^\alpha.$$

For a tiny rectangular loop based at s_0 , first move by Δs^1 , then Δs^2 , then back by $-\Delta s^1$, then back by $-\Delta s^2$. To second order in the side lengths, the net group element is

$$g_{\text{loop}} = \exp(-\mathcal{A}_1\Delta s^1) \exp(-\mathcal{A}_2\Delta s^2) \exp(\mathcal{A}_1\Delta s^1) \exp(\mathcal{A}_2\Delta s^2)$$

plus the correction from the variation of $\mathcal{A}_\alpha(s)$ across the loop. The Baker-Campbell-Hausdorff formula gives the commutator contribution, while the spatial variation gives derivative terms.² The result is

$$g_{\text{loop}} = \exp\left(-\mathcal{F}_{12}\Delta s^1\Delta s^2 + O(|\Delta s|^3)\right),$$

where

$$\boxed{\mathcal{F}_{12} = \partial_1\mathcal{A}_2 - \partial_2\mathcal{A}_1 - [\mathcal{A}_1, \mathcal{A}_2].}$$

This is why curvature is the local density of geometric phase. Even if the connection components are constant, a nonabelian structure group can give nonzero holonomy through the commutator term. The helicopter rotor laboratory below is a concrete instance of this constant-connection case.

8.8.2 Helicopter Rotor Model As A Gauge Connection

A finite-dimensional helicopter-rotor model makes the Shapere-Wilczek connection explicit. This subsection does not introduce a second theory. It takes the connection coefficients already derived in Section 8.2 and uses them as a worked laboratory: the shape coordinates are the two rotor angles (α, β) , the curvature is a commutator, and the rectangular stroke can be compared directly with the small-loop holonomy formula above.

The calculation has one main checkpoint. The rotor angles return to their initial values after a closed stroke, so any final rotation of the body cannot be attributed to a changed shape. It must come from the noncommuting connection components. This is the finite-dimensional version of geometric phase.

As derived in Section 8.2, the zero-momentum connection for this laboratory model is

$$\mathcal{A}_\alpha = \frac{I_1}{I_0 + I_1}e_z, \quad \mathcal{A}_\beta = \frac{I_2}{I_0 + I_2}e_y.$$

²We use only the leading commutator term $\exp X \exp Y \exp(-X) \exp(-Y) = \exp([X, Y] + O(\|(X, Y)\|^3))$. Higher terms are higher nested commutators. In the displayed loop $X = -\mathcal{A}_1\Delta s^1$ and $Y = -\mathcal{A}_2\Delta s^2$, so $[X, Y] = [\mathcal{A}_1, \mathcal{A}_2]\Delta s^1\Delta s^2$.

The components are constant, so the only curvature comes from noncommutativity. With the reconstruction convention of this chapter,

$$\mathcal{F}_{\alpha\beta} = -[\mathcal{A}_\alpha, \mathcal{A}_\beta] = \frac{I_1 I_2}{(I_0 + I_1)(I_0 + I_2)} e_x.$$

For a rectangular stroke of signed area $\Delta\alpha \Delta\beta$, the leading rigid rotation is

$$\Delta g \approx \exp(-\mathcal{F}_{\alpha\beta} \Delta\alpha \Delta\beta).$$

This is the point of the exercise: because the rotor angles return to their initial values, any final rotation of the body is holonomy, not a changed shape.

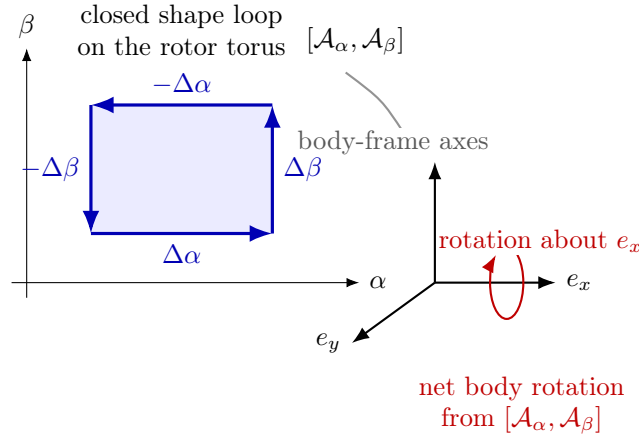


Figure 8.3: Helicopter-rotor gauge laboratory. The connection components are constant, but they point in noncommuting body directions. A closed rectangle in shape space therefore has curvature from the commutator term and can produce a net body rotation.

8.9 Associated Demos

- `demos/python/cosserat_rod_demo.py`, a planar frame reconstruction from curvature.
- `demos/python/deforming_body_gauge.py`: finite-dimensional two-rotor gauge holonomy.
- `demos/mathematica/GaugeDeformingBody.wl`, a finite-dimensional gauge-connection laboratory.
- Suggested extension: a low-Reynolds swimmer whose connection is computed from a Stokes boundary-value problem.

Chapter 9

Fluid Mechanics: Balance Laws, Navier-Stokes, And Stability

9.1 What Is A Fluid?

A Newtonian fluid does not assign stress to a static shear strain relative to a preferred reference shape. Push a finger through honey: the displaced material moves, stretches, and eventually fills in behind the finger, but no permanent shear strain is stored. Pull the finger out and the honey does not snap back. This is a constitutive idealization, not the definition of every material called a fluid: viscoelastic fluids, Bingham fluids, granular media, and superfluids require different constitutive structures.

This is why the Eulerian description is convenient for many fluid problems. We observe fields at fixed points in space: velocity, density, pressure, and stress. The moving-particle viewpoint returns through the material derivative and transport theorems, and remains essential for ideal-fluid variational principles, vortex-line transport, mixing, and free-surface motion. The historical names are literal here: Euler wrote the inviscid fluid equations in Eulerian form in the 1750s, while the Lagrangian description follows labelled material particles. The preceding elastic and deforming-body chapters kept the material reference description in the foreground; this chapter shifts the emphasis to Eulerian fields without treating them as the only natural description.

The basic fields are functions on physical space:

$$\rho(x, t), \quad u(x, t), \quad p(x, t),$$

where ρ is mass density, u is velocity, and p is pressure. A fluid particle can still be labelled by a material coordinate X , with

$$x = \varphi(X, t), \quad u(\varphi(X, t), t) = \partial_t \varphi(X, t).$$

The Eulerian view keeps x fixed and watches fluid pass through; the Lagrangian view follows the particle labelled by X .

Assumption 9.1 (Continuum fluid). *The molecular scale is much smaller than the length scales being modelled, so density, velocity, pressure, and stress are treated as smooth fields. Across shocks, contact discontinuities, or interfaces, the differential equations must be replaced by weak forms and jump conditions.*

Roadmap. The derivation is staged in the order of the modelling choices. First we learn how to differentiate quantities attached to moving fluid parcels. Then we convert integral balance laws into local partial differential equations for mass and momentum. Only after those kinematic and balance statements are in place do we introduce the Newtonian constitutive law that turns Cauchy's momentum equation into the Navier-Stokes equation. The later sections use that equation in three ways: exact laminar solutions, stability analysis, and Hamiltonian point-vortex reduction.

Elastic bodies use a material map as the primary unknown; deforming bodies add frame choices and connections to material shape. A simple fluid still permits material labels in derivations, but its closed equations are most useful as equations for Eulerian fields: density, velocity, pressure, and stress at points in space. The common continuum thread is stress and balance; the constitutive question shifts from stored deformation energy to rate-dependent stress.

Appendix C compares the elastic and fluid versions of these continuum assumptions, including weak formulations, pressure as a constraint multiplier, and the difference between smooth derivations and weak solution theory.

Derivation audit trail. The derivation has four ingredients: kinematics, mass balance, momentum balance, and constitutive law. Each section below pairs an equation with the conservation statement that produced it. Navier-Stokes appears only after the Newtonian stress law is imposed, so $\rho Du/Dt = -\nabla p + \mu \Delta u$ records a sequence of modelling choices, not a single axiom.

9.2 Material Derivative

If $f(x, t)$ is a scalar or vector field, its value along a particle path $x(t) = \varphi(X, t)$ changes as

$$\frac{d}{dt}f(x(t), t) = \partial_t f(x(t), t) + \dot{x}(t) \cdot \nabla f(x(t), t).$$

Since $\dot{x} = u(x(t), t)$, the material derivative is

$$\boxed{\frac{Df}{Dt} = \partial_t f + u \cdot \nabla f.}$$

For the velocity field itself,

$$\frac{Du}{Dt} = \partial_t u + (u \cdot \nabla)u$$

is the particle acceleration. The nonlinear term is kinematic: it appears even before any force law is chosen, because the velocity field is sampled along a moving trajectory.

9.3 Reynolds Transport Theorem

Let $V(t)$ be a material volume: it moves with the fluid, so its boundary velocity is u . For a scalar field f , we want the time derivative of

$$I(t) = \int_{V(t)} f(x, t) d^3x.$$

There are two sources of change. The value of f can change along each particle path, and the volume element occupied by neighboring particles can expand or contract. The Reynolds

transport theorem is the formula that keeps both effects. Reynolds gave the theorem in this systematic continuum-mechanics form in his 1903 treatise, but the pullback idea is already present in Euler's Lagrangian description of fluids.

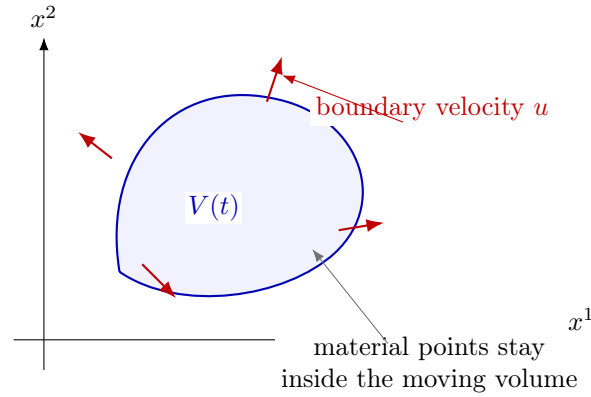


Figure 9.1: A material volume is carried by the fluid velocity. Its boundary velocity is u , so the time derivative of an integral over $V(t)$ must include both the material change of the integrand and the volume change measured by $\nabla \cdot u$.

Derivation 9.1 (Transport through a moving volume). Write $V(t) = \varphi(U, t)$ for a fixed material region U . Let

$$F^i_A = \frac{\partial \varphi^i}{\partial X^A}, \quad J = \det F.$$

Changing variables gives

$$I(t) = \int_U f(\varphi(X, t), t) J(X, t) d^3 X.$$

Differentiate:

$$\dot{I} = \int_U \left(\frac{Df}{Dt} J + f \dot{J} \right) d^3 X.$$

The Jacobian identity is

$$\dot{J} = J \nabla \cdot u.$$

For any smooth path $F(t) \in \text{GL}(n)$, Jacobi's formula is

$$\frac{d}{dt} \det F = \det F \operatorname{tr}(F^{-1} \dot{F}).$$

Indeed,

$$\det(F + \delta F) = \det F \det(I + F^{-1} \delta F) = \det F (1 + \operatorname{tr}(F^{-1} \delta F) + O(\delta F^2)).$$

Applying this formula to $J = \det F$ gives

$$\dot{J} = J \operatorname{tr}(F^{-1} \dot{F}),$$

and $\dot{F}^i_A = \partial_A u^i(\varphi, t)$, so $\operatorname{tr}(F^{-1} \dot{F}) = \partial_i u^i$. Returning to spatial variables,

$$\boxed{\frac{d}{dt} \int_{V(t)} f d^3 x = \int_{V(t)} \left(\frac{Df}{Dt} + f \nabla \cdot u \right) d^3 x = \int_{V(t)} (\partial_t f + \nabla \cdot (fu)) d^3 x.}$$

The theorem separates two effects: the field f changes along particles, and the material volume expands or contracts.

9.4 Conservation Of Mass

Mass in a material volume is constant:

$$\frac{d}{dt} \int_{V(t)} \rho d^3x = 0.$$

Using the transport theorem,

$$\int_{V(t)} (\partial_t \rho + \nabla \cdot (\rho u)) d^3x = 0.$$

Because this holds for every material volume,

$$\boxed{\partial_t \rho + \nabla \cdot (\rho u) = 0.}$$

Equivalently,

$$\frac{D\rho}{Dt} + \rho \nabla \cdot u = 0.$$

For constant-density incompressible flow,

$$\boxed{\nabla \cdot u = 0.}$$

Two statements are being separated here. Incompressibility means that material volumes are preserved, so $\nabla \cdot u = 0$. Constant density is then consistent with mass conservation, but it is not a separate equation of motion. This distinction matters in stratified or variable-density flows, where the velocity may be divergence-free in an approximation while density remains an advected scalar.

9.5 Cauchy Momentum Balance

Cauchy's 1822–1823 stress principle is the common ancestor of the elastic and fluid balance laws used in these notes. The same traction theorem that produced stress in Chapter 7 now produces the stress tensor of a moving fluid.

Newton's second law for a material volume $V(t)$ is

$$\frac{d}{dt} \int_{V(t)} \rho u d^3x = \int_{V(t)} \rho b d^3x + \int_{\partial V(t)} t(n) dS,$$

where b is body force per unit mass and $t(n)$ is surface traction on a surface whose outward unit normal is n .

Proposition 9.1 (Cauchy's traction theorem). *Under the same locality and continuity hypotheses used in the tetrahedron argument of Section 7.6, there is a tensor $\sigma(x, t)$ such that*

$$t(n) = \sigma n,$$

where σ is the Cauchy stress tensor.

Balance of angular momentum gives the symmetry of this tensor in the absence of couple stresses. For a material volume $V(t)$, moment balance about the origin is

$$\frac{d}{dt} \int_{V(t)} \rho x \times u \, d^3x = \int_{V(t)} \rho x \times b \, d^3x + \int_{\partial V(t)} x \times (\sigma n) \, dS.$$

Using the divergence theorem on the surface term gives the volume integral of $x \times (\nabla \cdot \sigma)$ plus the axial vector of $\sigma - \sigma^T$. Comparing with $x \times$ the linear momentum balance leaves only this axial term. Since the volume is arbitrary,

$$\sigma = \sigma^T.$$

In indices, the extra term is

$$\epsilon_{ijk} \sigma_{kj} e_i,$$

which is twice the axial vector of the skew tensor $(\sigma - \sigma^T)/2$. Angular-momentum balance without internal body couples forces this vector to vanish pointwise.

Derivation 9.2 (Local momentum equation). *Apply the transport theorem to $f = \rho u$. Using mass conservation, the left-hand side becomes*

$$\begin{aligned} \frac{d}{dt} \int_{V(t)} \rho u \, d^3x &= \int_{V(t)} (\partial_t(\rho u) + \nabla \cdot (\rho u \otimes u)) \, d^3x \\ &= \int_{V(t)} \rho (\partial_t u + u \cdot \nabla u) \, d^3x. \end{aligned}$$

The traction term is converted by the divergence theorem:

$$\int_{\partial V(t)} \sigma n \, dS = \int_{V(t)} \nabla \cdot \sigma \, d^3x.$$

Therefore

$$\int_{V(t)} \left[\rho \frac{Du}{Dt} - \nabla \cdot \sigma - \rho b \right] d^3x = 0.$$

Since $V(t)$ is arbitrary,

$$\boxed{\rho \frac{Du}{Dt} = \nabla \cdot \sigma + \rho b.}$$

The localization step is the continuum analogue of the fundamental lemma of the calculus of variations: if the integral of a smooth density vanishes for every sufficiently small material volume, then the density itself must vanish pointwise. This equation is not yet a closed fluid equation. It is a balance law. Closure comes from a constitutive relation for σ .

The same balance law will reappear in the elasticity chapter with a different constitutive relation and a different choice of variables. That repetition is intentional. Momentum balance is universal; the material model enters when one decides how stress depends on deformation, rate of deformation, pressure, temperature, or internal variables.

9.6 Newtonian Stress And The Navier-Stokes Equation

The name Navier-Stokes compresses four contributions. Navier (1822) gave a molecular model for viscosity; Cauchy's stress work from 1822–1827 supplied the continuum framework; Saint-Venant (1843) added continuum viscous stresses; and Stokes (1845) gave the formulation that became standard. Truesdell and Noll later systematized objectivity and constitutive response in the 1960s.

The balance laws so far are universal for continua, but they are not yet a closed fluid model: the stress tensor σ is still unknown. For a simple Newtonian fluid the closure is

$$\sigma = -pI + 2\mu D + \lambda(\nabla \cdot u)I.$$

Here p is pressure, $\mu \geq 0$ is shear viscosity, λ is the second viscosity coefficient, and

$$D_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$$

is the symmetric strain-rate tensor.

The assumptions behind the formula are local, linear, objective, and isotropic. The stress may depend on the instantaneous rate of deformation, not on the accumulated shear history; this is the point at which fluids separate from elastic solids. To see why only D appears, split the velocity gradient into symmetric and antisymmetric parts:

$$\nabla u = D + W, \quad D_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i), \quad W_{ij} = \frac{1}{2}(\partial_i u_j - \partial_j u_i).$$

The antisymmetric part W represents local rigid rotation. It should not produce viscous stress in an isotropic simple fluid. The constitutive assumptions can be summarized as:

- locality: $\sigma(x, t)$ depends on u through $\nabla u(x, t)$,
- objectivity: superposed rigid rotations do not create stress,
- isotropy: there is no preferred material direction.

Objectivity removes W . Isotropy says that a linear map from the symmetric tensor D to a symmetric stress can only use the deviatoric part of D and its trace. The symmetric strain-rate tensor decomposes as

$$D = D_0 + \frac{1}{3}(\text{tr } D)I, \quad \text{tr } D_0 = 0.$$

The representation statement used above is elementary in the linear case. A linear isotropic map L on symmetric tensors must commute with every rotation:

$$L(QDQ^T) = QL(D)Q^T.$$

The one-dimensional subspace spanned by I is invariant, and the five-dimensional space of traceless symmetric tensors is irreducible under rotations. Schur's lemma, in this elementary real representation-theoretic setting, says that L acts by a scalar on each irreducible piece. Hence $L(D) = a(\text{tr } D)I + bD_0$, which is the displayed law after renaming constants.

For constant μ, λ ,

$$\begin{aligned} (\nabla \cdot \sigma)_i &= -\partial_i p + \mu \partial_j (\partial_i u_j + \partial_j u_i) + \lambda \partial_i (\partial_k u_k) \\ &= -\partial_i p + \mu \Delta u_i + (\lambda + \mu) \partial_i (\nabla \cdot u). \end{aligned}$$

Thus the compressible Navier-Stokes momentum equation is

$$\rho(\partial_t u + u \cdot \nabla u) = -\nabla p + \mu \Delta u + (\lambda + \mu) \nabla(\nabla \cdot u) + \rho b.$$

Together with mass conservation, an equation of state, and an energy or temperature equation when needed, this describes a Newtonian compressible fluid. The displayed equation combines three independent ingredients: Cauchy momentum balance, the Newtonian stress law, and the constant-viscosity simplification. If viscosity varies with temperature or composition, the divergence of $2\mu D + \lambda(\nabla \cdot u)I$ must be kept in product-rule form.

For incompressible flow with constant ρ ,

$$\nabla \cdot u = 0,$$

and pressure plays the role, in this incompressible model, of the Lagrange multiplier enforcing this constraint. In compressible flow pressure is instead a thermodynamic variable closed by an equation of state. The incompressible equations reduce to

$$\begin{aligned} \rho(\partial_t u + u \cdot \nabla u) &= -\nabla p + \mu \Delta u + \rho b, \\ \nabla \cdot u &= 0. \end{aligned}$$

Read the incompressible equations term by term. The material acceleration $\partial_t u + u \cdot \nabla u$ is the acceleration of a moving fluid particle. The pressure gradient enforces the volume-preserving constraint and transmits normal stress. The Laplacian $\nu \Delta u$ diffuses momentum. The body force adds external acceleration. This interpretation prevents a common mistake: pressure is not specified independently in incompressible flow; it is solved simultaneously with u so that $\nabla \cdot u = 0$ remains true. Taking the divergence of the incompressible momentum equation makes this constraint role explicit:

$$-\Delta p = \rho \nabla \cdot (u \cdot \nabla u) - \rho \nabla \cdot b,$$

with boundary conditions determined by the normal component of the momentum equation at the wall. The pressure is therefore a nonlocal field: it adjusts instantaneously within the incompressible model, the formal $c_s \rightarrow \infty$ limit, to project the acceleration back onto divergence-free velocity fields. Real fluids transmit pressure disturbances at the speed of sound; incompressibility is the low-Mach, long-wavelength approximation in which that propagation time is neglected.

9.7 Initial And Boundary Data For Navier-Stokes

A Navier-Stokes problem is not specified by the differential equation alone. One must also state the initial velocity, the domain, and the boundary conditions. For incompressible flow in a fixed domain Ω , a typical initial condition is

$$u(x, 0) = u_0(x), \quad \nabla \cdot u_0 = 0.$$

The divergence-free condition is not optional: it is the compatibility condition required by incompressibility. Boundary data must also be compatible with mass conservation. If $\partial\Omega$ is an impermeable wall moving with velocity U_{wall} , then the normal velocity must satisfy

$$u \cdot n = U_{\text{wall}} \cdot n.$$

The most common viscous wall condition is no slip:

$$\boxed{u = U_{\text{wall}} \quad \text{on } \partial\Omega.}$$

It says that the fluid velocity equals the velocity of the solid boundary, both normal and tangential components included. No slip is a molecular-scale constitutive fact about many ordinary fluids at ordinary solid walls; it is not implied by incompressibility alone. By contrast, inviscid Euler flow uses only no penetration,

$$u \cdot n = U_{\text{wall}} \cdot n,$$

because tangential slip is not controlled by a viscous boundary layer in the ideal equation. Prandtl's 1904 boundary-layer theory explains how no slip at a wall can coexist with high-Reynolds-number flow that is nearly inviscid away from the wall: the tangential adjustment is confined to a thin viscous layer.

Other boundary conditions model different physical surfaces. A free-slip or stress-free boundary has

$$u \cdot n = 0, \quad (I - n \otimes n)\sigma n = 0,$$

meaning there is no normal flow and no tangential viscous traction. Periodic boundary conditions identify opposite faces of a box and are useful for studying homogeneous turbulence and exact periodic solutions such as the Taylor-Green vortex below. For three-dimensional finite-energy flows, Leray's 1934 weak solutions provide the standard global framework. They satisfy the equations in integral form and obey an energy inequality, while global smooth regularity in three dimensions remains an open problem.

Pressure boundary conditions deserve special care. For incompressible flow, pressure is not prescribed independently at a no-slip wall. Instead, taking the normal component of

$$\rho(\partial_t u + u \cdot \nabla u) = -\nabla p + \mu \Delta u + \rho b$$

gives the wall relation

$$\partial_n p = n \cdot (\mu \Delta u + \rho b - \rho(\partial_t u + u \cdot \nabla u)).$$

Together with the pressure Poisson equation, this determines pressure up to an additive constant in a closed incompressible domain. This is why numerical projection methods first advance a tentative velocity and then solve a Poisson problem for the pressure correction that restores $\nabla \cdot u = 0$.

9.8 Nondimensionalization And Reynolds Number

Let U be a typical speed, L a typical length, and define

$$x = Lx', \quad t = \frac{L}{U}t', \quad u = Uu', \quad p = \rho U^2 p'.$$

Dropping primes, the incompressible Navier-Stokes equation without body force becomes

$$\partial_t u + u \cdot \nabla u = -\nabla p + \frac{1}{\text{Re}} \Delta u, \quad \nabla \cdot u = 0,$$

where

$$\boxed{\text{Re} = \frac{UL}{\nu}}, \quad \nu = \frac{\mu}{\rho}.$$

The Reynolds number compares inertial transport to viscous diffusion. Small Re flow is viscously dominated and often reversible in time after the velocity is reversed. Large Re flow can carry disturbances long distances before viscosity damps them.

9.9 Energy Identity And Viscous Dissipation

For a smooth incompressible solution in a fixed domain Ω , assume either periodic boundary conditions or no-slip walls $u = 0$ on $\partial\Omega$. Dot the Navier-Stokes equation with u and integrate:

$$\rho \int_{\Omega} u \cdot \partial_t u \, d^3x + \rho \int_{\Omega} u \cdot (u \cdot \nabla u) \, d^3x = - \int_{\Omega} u \cdot \nabla p \, d^3x + \mu \int_{\Omega} u \cdot \Delta u \, d^3x.$$

The nonlinear term is a divergence:

$$u \cdot (u \cdot \nabla u) = u \cdot \nabla \left(\frac{1}{2} \|u\|^2 \right) = \nabla \cdot \left(\frac{1}{2} \|u\|^2 u \right),$$

because $\nabla \cdot u = 0$. The pressure term is also a divergence:

$$u \cdot \nabla p = \nabla \cdot (pu).$$

The boundary conditions remove both boundary fluxes. Integration by parts gives

$$\int_{\Omega} u \cdot \Delta u \, d^3x = - \int_{\Omega} |\nabla u|^2 \, d^3x.$$

Hence

$$\boxed{\frac{d}{dt} \int_{\Omega} \frac{1}{2} \rho \|u\|^2 \, d^3x = -\mu \int_{\Omega} |\nabla u|^2 \, d^3x \leq 0.}$$

Viscosity irreversibly converts organized kinetic energy into heat. This identity is also the first stability estimate for laminar flows: disturbances must draw energy from mean shear fast enough to beat viscous dissipation.

9.10 Vorticity And Circulation

Helmholtz (1858) proved the vortex theorems; Kelvin (1869) proved circulation conservation. Both survive in the inviscid limit and explain which parts of vorticity motion persist without viscosity.

The vorticity is

$$\omega = \nabla \times u.$$

Taking the curl of the incompressible Navier-Stokes equation gives

$$\boxed{\partial_t \omega + u \cdot \nabla \omega = \omega \cdot \nabla u + \nu \Delta \omega.}$$

The identity behind this equation is

$$\nabla \times (u \cdot \nabla u) = u \cdot \nabla \omega - \omega \cdot \nabla u + \omega \nabla \cdot u,$$

and the last term vanishes for incompressible flow. The term $\omega \cdot \nabla u$ stretches and tilts vortex lines in three dimensions. In two dimensions, vorticity is perpendicular to the plane and the stretching term vanishes:

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega.$$

This contrast is central. Two-dimensional inviscid vorticity is transported like a scalar dye, which creates many conserved quantities. Three-dimensional vorticity can be stretched by the flow itself, allowing vortex intensification and making the analysis of Navier-Stokes much harder. For a compressible inviscid barotropic fluid the corresponding identity is

$$\frac{D\omega}{Dt} = \omega \cdot \nabla u - \omega \nabla \cdot u.$$

Together with mass conservation,

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot u,$$

this gives

$$\boxed{\frac{D}{Dt} \left(\frac{\omega}{\rho} \right) = \left(\frac{\omega}{\rho} \right) \cdot \nabla u.}$$

This is the precise stretching law behind Helmholtz's vortex-line statement. For $\nu = 0$, vortex lines are transported by the flow. In two-dimensional incompressible Euler flow this reduces to scalar advection of vorticity; in three dimensions transport is accompanied by the stretching term.

Proposition 9.2 (Kelvin circulation theorem). *Let $\gamma(t)$ be a closed material loop moving with an inviscid barotropic fluid under conservative body forces. Then*

$$\frac{d}{dt} \oint_{\gamma(t)} u \cdot dx = 0.$$

Proof. Parametrize the loop by $s \mapsto x(s, t)$, with

$$\partial_t x(s, t) = u(x(s, t), t).$$

Then

$$\frac{d}{dt} \oint u \cdot \partial_s x \, ds = \oint \left(\frac{Du}{Dt} \cdot \partial_s x + u \cdot \partial_s u \right) ds.$$

For inviscid barotropic flow with conservative potential Φ ,

$$\frac{Du}{Dt} = -\nabla h - \nabla \Phi,$$

where h is specific enthalpy. The integrand becomes

$$\partial_s \left(-h - \Phi + \frac{1}{2} \|u\|^2 \right).$$

The integral of a single-valued total derivative around a closed loop is zero. $\square$

Proposition 9.3 (Helmholtz vortex theorems, inviscid barotropic form). *For a smooth inviscid barotropic fluid with conservative body force:*

1. *vortex lines are transported by the fluid motion;*
2. *the circulation around a small material loop moving with a vortex tube is constant, so the vortex-tube strength is constant;*
3. *vortex lines cannot begin or end in the interior of the fluid.*

Proof. Kelvin's theorem gives conservation of circulation around every material loop. Apply it to a small loop encircling a vortex tube. By Stokes' theorem,

$$\oint_{\partial S(t)} u \cdot dx = \int_{S(t)} \omega \cdot n \, dS,$$

so the vorticity flux through a material cross-section of the tube is constant. If a material curve is initially tangent to ω , then the inviscid vorticity equation for a barotropic fluid,

$$\frac{D}{Dt} \left(\frac{\omega}{\rho} \right) = \left(\frac{\omega}{\rho} \right) \cdot \nabla u,$$

has the same stretching form as the evolution equation for an infinitesimal material line element; hence vortex lines move with the fluid. In the incompressible case ρ is constant and this is the usual $D\omega/Dt = \omega \cdot \nabla u$. Finally, $\nabla \cdot \omega = \nabla \cdot (\nabla \times u) = 0$, so vortex flux has no interior sources or sinks. Vortex lines either close, meet a boundary, or continue to infinity. $\square$

9.11 Bernoulli's Theorem As An Inviscid First Integral

Bernoulli's *Hydrodynamica* (1738) predates Navier-Stokes by nearly a century, yet its central invariant follows cleanly from Euler's equation. For inviscid barotropic flow with conservative body force $-\nabla\Phi$,

$$\frac{Du}{Dt} = -\nabla h - \nabla\Phi,$$

where h is specific enthalpy, defined by $dh = \rho^{-1}dp$. Assume the flow is steady. The vector identity

$$(u \cdot \nabla)u = \nabla \left(\frac{1}{2}|u|^2 \right) - u \times \omega, \quad \omega = \nabla \times u,$$

gives

$$\nabla \left(\frac{1}{2}|u|^2 + h + \Phi \right) = u \times \omega.$$

Dotting with u removes the right-hand side:

$$u \cdot \nabla \left(\frac{1}{2}|u|^2 + h + \Phi \right) = 0.$$

Thus

$$\frac{1}{2}|u|^2 + h + \Phi = \text{constant along each streamline.}$$

If the steady flow is also irrotational on a connected region, then $u \times \omega = 0$, so the same Bernoulli function is constant throughout that region. The distinction matters: streamline-wise Bernoulli and global Bernoulli are different statements with different hypotheses.

9.12 How Exact Navier-Stokes Solutions Are Found

Exact Navier-Stokes solutions all exploit one of three mechanisms: a velocity field that kills $u \cdot \nabla u$ by directionality, a fully developed geometry that makes the same cancellation transverse, or a pressure field that absorbs the nonlinear term as a gradient. The examples below isolate these mechanisms.

1. *Unidirectional shear flow.* If

$$u = (V(y, t), 0, 0),$$

then $\nabla \cdot u = 0$ and

$$u \cdot \nabla u = V(y, t) \partial_x (V(y, t), 0, 0) = 0.$$

The nonlinear term vanishes because the velocity points in the x -direction but varies only in the transverse y -direction.

2. *Fully developed pipe or channel flow.* If the velocity is axial and depends only on the transverse coordinates, the same cancellation occurs. The Navier-Stokes equation reduces to an ordinary differential equation in the cross-section for steady flow, or to a heat equation for unsteady flow.
3. *Nonlinearity absorbed into pressure.* In some periodic flows, $(u \cdot \nabla)u$ is the gradient of a scalar. Since incompressible pressure is a Lagrange multiplier, that gradient can be absorbed into p . The Taylor-Green vortex is the standard example.

For a unidirectional channel flow $u = (V(y, t), 0, 0)$ with pressure $p = p(x, t)$, incompressible Navier-Stokes becomes

$$\rho \partial_t V = -\partial_x p + \mu \partial_y^2 V.$$

This single equation is the parent of Couette flow, Poiseuille flow, Stokes' first problem, and the oscillatory Stokes layer. The boundary conditions decide which solution it represents.

9.13 Laminar Flow Solutions

Laminar solutions are exact, organized solutions of Navier-Stokes. Stability theory asks which of these organized solutions persist under disturbance. Each solution follows from three choices: geometry, boundary conditions, and pressure gradient or wall motion. Changing any one of these choices changes the profile and the instability problem.

9.13.1 The Couette-Poiseuille Family

Consider two infinite plates at $y = 0$ and $y = h$. The lower plate moves with speed U_0 in the x -direction, the upper plate moves with speed U_h , and a constant pressure gradient

$$-\frac{dp}{dx} = G$$

is imposed. We seek a steady parallel solution

$$u = (V(y), 0, 0), \quad p(x) = p_0 - Gx.$$

The ansatz satisfies incompressibility. It also kills the nonlinear term:

$$(u \cdot \nabla)u = V(y)\partial_x(V(y), 0, 0) = 0.$$

The y - and z -momentum equations say that p is independent of y, z . The x -momentum equation is

$$0 = -\frac{dp}{dx} + \mu V''(y) = G + \mu V''(y).$$

Thus

$$V''(y) = -\frac{G}{\mu}.$$

Integrating twice gives

$$V(y) = -\frac{G}{2\mu}y^2 + C_1y + C_0.$$

No slip gives

$$V(0) = U_0, \quad V(h) = U_h.$$

Solving for C_0, C_1 yields the whole Couette-Poiseuille family:

$$V(y) = U_0 + \frac{U_h - U_0}{h}y + \frac{G}{2\mu}y(h - y).$$

The first two terms are wall-driven shear. The last term is pressure-driven flow. The volume flux per unit span is

$$Q = \int_0^h V(y) dy = \frac{h}{2}(U_0 + U_h) + \frac{Gh^3}{12\mu}.$$

The wall shear stresses are

$$\tau(0) = \mu V'(0) = \mu \frac{U_h - U_0}{h} + \frac{Gh}{2}, \quad \tau(h) = \mu V'(h) = \mu \frac{U_h - U_0}{h} - \frac{Gh}{2}.$$

These formulas are often the easiest way to check signs: pressure forcing increases shear at one wall and decreases it at the other.

9.13.2 Plane Couette Flow

Two infinite plates are separated by distance h . The lower plate is fixed; the upper plate moves with velocity Ue_x . Seek a steady parallel flow

$$u = (V(y), 0, 0), \quad p = \text{constant}.$$

The incompressibility condition is automatic. The x -momentum equation is

$$0 = \mu V''(y),$$

with boundary conditions $V(0) = 0, V(h) = U$. Thus

$$V(y) = U \frac{y}{h}.$$

This is the special case of the Couette-Poiseuille family with

$$U_0 = 0, \quad U_h = U, \quad G = 0.$$

The shear stress is constant:

$$\sigma_{xy} = \mu V'(y) = \mu \frac{U}{h}.$$

9.13.3 Plane Poiseuille Flow

Now both plates are fixed and a pressure gradient drives the flow. Let

$$u = (V(y), 0, 0), \quad -\frac{dp}{dx} = G > 0, \quad 0 < y < h.$$

The steady x -momentum equation is

$$0 = -\frac{dp}{dx} + \mu V''(y) = G + \mu V''(y),$$

so

$$V''(y) = -\frac{G}{\mu}.$$

With no-slip conditions $V(0) = V(h) = 0$,

$$V(y) = \frac{G}{2\mu}y(h-y).$$

This is the special case $U_0 = U_h = 0$ of the Couette-Poiseuille family. The maximum speed occurs at $y = h/2$:

$$V_{\max} = \frac{Gh^2}{8\mu}.$$

The volume flux per unit span is

$$Q = \int_0^h V(y) dy = \frac{Gh^3}{12\mu}.$$

9.13.4 Hagen-Poiseuille Pipe Flow

Hagen (1839) and Poiseuille (1840) measured this pipe-flow law experimentally; the Navier-Stokes derivation came later. The chronology is a reminder that some continuum laws entered mechanics through measurement, not from constitutive postulates.

For a circular pipe of radius a , use cylindrical radius r and seek

$$u = V(r)e_z, \quad -\frac{dp}{dz} = G > 0.$$

The axial equation is

$$0 = G + \mu \frac{1}{r} \frac{d}{dr} \left(r \frac{dV}{dr} \right).$$

Multiply by r and integrate:

$$\frac{d}{dr} (rV'(r)) = -\frac{G}{\mu}r, \quad rV'(r) = -\frac{G}{2\mu}r^2 + C_1.$$

Regularity at the centerline requires $rV'(r) \rightarrow 0$ as $r \rightarrow 0$. If $C_1 \neq 0$, then $V'(r) \sim C_1/r$ and the next integration would produce a logarithmic singularity in V . Hence $C_1 = 0$. A second integration gives

$$V(r) = -\frac{G}{4\mu}r^2 + C_2.$$

No slip at $r = a$ fixes $C_2 = Ga^2/(4\mu)$. Therefore

$$V(r) = \frac{G}{4\mu}(a^2 - r^2).$$

The flux is

$$Q = 2\pi \int_0^a V(r)r \, dr = \frac{\pi a^4 G}{8\mu}.$$

The fourth power in a is one of the most important practical consequences of viscous laminar flow.

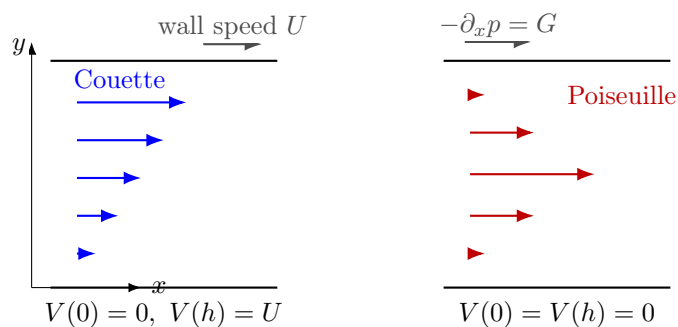


Figure 9.2: Couette flow is linear because it is driven by wall motion. Poiseuille flow is parabolic because it balances pressure forcing against viscous diffusion of momentum. The arrows are spatial velocity vectors, not a graph with velocity on the vertical axis.

9.14 Unsteady Exact Solutions

Steady laminar solutions show how pressure and wall motion balance viscosity. Unsteady exact solutions answer a different physical question: how fast does a boundary or vortex imprint its motion on the surrounding fluid? The answer is not a sound-crossing time in the incompressible model. It is the diffusive viscous time y^2/ν , and the associated length scale is $\sqrt{\nu t}$. The examples below are therefore more than formulae. They show how momentum diffuses, how oscillatory forcing produces a penetration depth, and why exact solutions are useful tests for numerical codes that claim to solve the full Navier-Stokes equation.

9.14.1 Stokes' First Problem

Stokes' 1851 first problem is the canonical similarity solution for viscous diffusion of momentum from an impulsively moved wall.

Let a viscous incompressible fluid occupy the half-space $y > 0$. For $t < 0$, both fluid and wall are at rest. At $t = 0$, the wall at $y = 0$ is instantaneously set into motion with velocity Ue_x . Seek

$$u = (V(y, t), 0, 0), \quad p = \text{constant}.$$

As before, incompressibility is automatic and $u \cdot \nabla u = 0$. The Navier-Stokes equation reduces to the heat equation

$$\partial_t V = \nu \partial_y^2 V.$$

The data are

$$V(y, 0) = 0 \quad (y > 0), \quad V(0, t) = U, \quad V(y, t) \rightarrow 0 \quad (y \rightarrow \infty).$$

The only length scale that can be made from ν and t is $\sqrt{\nu t}$, so introduce

$$\eta = \frac{y}{2\sqrt{\nu t}}, \quad V(y, t) = UF(\eta).$$

Using

$$\partial_t \eta = -\frac{\eta}{2t}, \quad \partial_y \eta = \frac{1}{2\sqrt{\nu t}},$$

the heat equation becomes

$$-\frac{U\eta}{2t}F'(\eta) = \nu U \frac{1}{4\nu t}F''(\eta), \quad F'' + 2\eta F' = 0.$$

The boundary conditions are $F(0) = 1$ and $F(\infty) = 0$. Hence

$$V(y, t) = U \operatorname{erfc}\left(\frac{y}{2\sqrt{\nu t}}\right).$$

The thickness of the region significantly affected by the wall grows like $\sqrt{\nu t}$. The wall shear stress has magnitude

$$|\tau_w(t)| = \mu |\partial_y V(0, t)| = \mu \frac{U}{\sqrt{\pi \nu t}},$$

which is singular at $t = 0^+$ because the idealized wall velocity jumps instantaneously.

9.14.2 Oscillatory Stokes Layer

Now let the wall at $y = 0$ oscillate tangentially:

$$V(0, t) = U \cos(\Omega_s t), \quad V(y, t) \rightarrow 0 \quad (y \rightarrow \infty).$$

Again V solves $\partial_t V = \nu V_{yy}$. Use a complex solution and take the real part at the end:

$$V(y, t) = \operatorname{Re} \left\{ U e^{i\Omega_s t - \lambda y} \right\}.$$

Substitution gives

$$i\Omega_s = \nu \lambda^2.$$

The root with positive real part is

$$\lambda = \frac{1+i}{\delta_s}, \quad \delta_s = \sqrt{\frac{2\nu}{\Omega_s}}.$$

Therefore

$$V(y, t) = U e^{-y/\delta_s} \cos\left(\Omega_s t - \frac{y}{\delta_s}\right).$$

The length δ_s is the oscillatory Stokes-layer thickness. Faster oscillation gives a thinner layer; larger viscosity gives a thicker layer. The phase lag y/δ_s records the time delay caused by viscous diffusion of momentum away from the moving wall.

9.14.3 Taylor-Green Vortex

The Taylor-Green vortex is an exact periodic solution of the two-dimensional incompressible Navier-Stokes equation. On a periodic square, set

$$\begin{aligned} u_x(x, y, t) &= A(t) \sin(kx) \cos(ky), \\ u_y(x, y, t) &= -A(t) \cos(kx) \sin(ky), \\ A(t) &= U_0 e^{-2\nu k^2 t}. \end{aligned}$$

The divergence vanishes because

$$\partial_x u_x + \partial_y u_y = Ak \cos(kx) \cos(ky) - Ak \cos(kx) \cos(ky) = 0.$$

The Laplacian satisfies

$$\Delta u = -2k^2 u,$$

so $\partial_t u = \nu \Delta u$. The remaining nonlinear term is not zero, but it is a gradient:

$$(u \cdot \nabla)u = \frac{A(t)^2 k}{2} (\sin(2kx), \sin(2ky)).$$

It is cancelled by the pressure

$$p(x, y, t) = p_0 + \frac{\rho A(t)^2}{4} (\cos(2kx) + \cos(2ky)).$$

Indeed,

$$-\frac{1}{\rho} \nabla p = \frac{A(t)^2 k}{2} (\sin(2kx), \sin(2ky)) = (u \cdot \nabla)u.$$

Thus the full nonlinear equation

$$\partial_t u + u \cdot \nabla u = -\frac{1}{\rho} \nabla p + \nu \Delta u$$

is satisfied exactly. The vorticity is

$$\omega_z = \partial_x u_y - \partial_y u_x = 2kA(t) \sin(kx) \sin(ky),$$

and the cell-averaged kinetic energy density is

$$\left\langle \frac{1}{2} \rho |u|^2 \right\rangle = \frac{1}{4} \rho A(t)^2 = \frac{1}{4} \rho U_0^2 e^{-4\nu k^2 t}.$$

9.15 Linear Stability Of Laminar Shear Flow

Exact laminar solutions become physically informative when their nearby motions are understood. Stability analysis supplies this local question. We linearize Navier-Stokes around a known base flow and ask whether infinitesimal disturbances grow or decay. This gives a mathematically controlled diagnostic that can be compared with experiments and simulations.

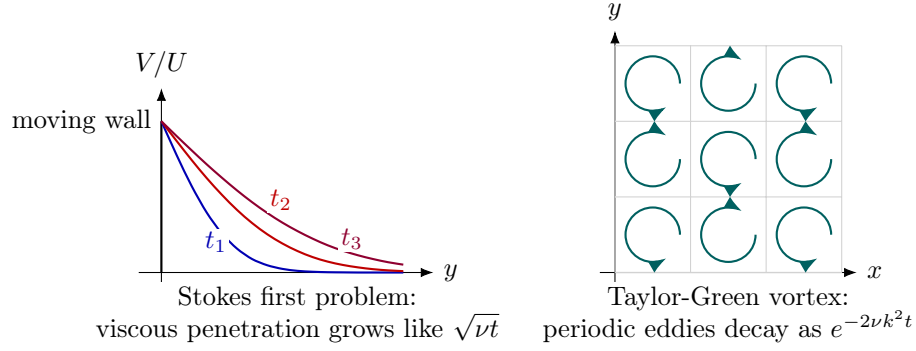


Figure 9.3: Two unsteady exact solutions. Left: an impulsively moved wall creates the profile $V/U = \operatorname{erfc}(y/(2\sqrt{\nu t}))$, sampled at three times. Right: the curved arrows show the alternating vorticity cells of the Taylor-Green velocity $(A \sin kx \cos ky, -A \cos kx \sin ky)$; advection is absorbed by pressure while viscosity damps the amplitude.

Let a steady base flow be

$$U(y)e_x.$$

Write a small perturbation as

$$u = U(y)e_x + \tilde{u}, \quad p = P(x) + \tilde{p},$$

and retain only terms linear in $\tilde{u}$. In nondimensional variables,

$$\partial_t \tilde{u} + U \partial_x \tilde{u} + \tilde{v} U'(y) e_x = -\nabla \tilde{p} + \frac{1}{\operatorname{Re}} \Delta \tilde{u}, \quad \nabla \cdot \tilde{u} = 0.$$

The term $\tilde{v} U' e_x$ is the mechanism by which cross-stream velocity tilts fluid into faster or slower parts of the mean shear and extracts energy from the base flow.

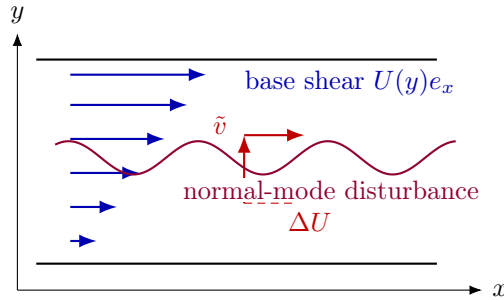


Figure 9.4: Linear shear-flow stability. A wall-normal perturbation velocity $\tilde{v}$ moves fluid across the base shear; the term $\tilde{v} U'(y) e_x$ is the linear energy-exchange channel between the base flow and the disturbance.

9.15.1 Orr-Sommerfeld Equation

The Orr-Sommerfeld operator is fourth-order and non-self-adjoint. Orr and Sommerfeld derived the normal-mode equation independently in 1907–1908. The equation is linear, but the physical

transition problem is not exhausted by linear eigenvalues: Reynolds' 1883 dye-injection experiment had already shown that pipe flow can transition to turbulence through finite disturbances.

For two-dimensional perturbations, introduce a streamfunction

$$\tilde{u} = \partial_y \psi, \quad \tilde{v} = -\partial_x \psi.$$

Squire's theorem justifies this restriction for parallel shear flows: if a three-dimensional normal mode is unstable at Reynolds number Re , then there is a two-dimensional mode unstable at a no larger effective Reynolds number. Thus the least stable linear mode can be found in the two-dimensional Orr-Sommerfeld problem. The proof idea is a change of wave-vector variables. A three-dimensional disturbance proportional to

$$e^{i(\alpha x + \beta z) + \sigma t}$$

has total horizontal wavenumber $k = (\alpha^2 + \beta^2)^{1/2}$. Squire's transformation maps it to a two-dimensional disturbance with streamwise wavenumber k , the same phase speed, and effective Reynolds number

$$\text{Re}_{2D} = \frac{|\alpha|}{k} \text{Re} \leq \text{Re}.$$

In the transformed variables, the wall-normal velocity equation keeps the Orr-Sommerfeld form with α replaced by k . The remaining three-dimensional wall-normal vorticity variable satisfies the Squire equation and is forced by the two-dimensional velocity mode. Thus any growing three-dimensional normal mode implies a growing two-dimensional mode at no larger Reynolds number. The statement is linear and applies to parallel shear flows; it is not a claim about finite-amplitude transition. Seek normal modes

$$\psi(x, y, t) = \phi(y) e^{i\alpha x + \sigma t}.$$

Taking the curl of the linearized equation removes pressure and gives the linear vorticity equation. Since the perturbation vorticity is

$$\tilde{\omega}_z = \partial_x \tilde{v} - \partial_y \tilde{u} = -(\partial_y^2 - \alpha^2) \phi e^{i\alpha x + \sigma t},$$

substitution gives a fourth-order eigenvalue equation for the wall-normal shape $\phi(y)$:

$$\boxed{(\sigma + i\alpha U)(D^2 - \alpha^2)\phi - i\alpha U''\phi = \frac{1}{\text{Re}}(D^2 - \alpha^2)^2\phi,}$$

where $D = d/dy$. No-slip and no-penetration at walls give

$$\phi = 0, \quad D\phi = 0$$

at the walls. This fourth-order eigenvalue problem is the Orr-Sommerfeld equation. A base flow is linearly unstable if some eigenvalue has

$$\text{Re } \sigma > 0.$$

For viscous flow on a bounded channel, the no-slip boundary conditions make the Orr-Sommerfeld operator a fourth-order non-self-adjoint boundary-value problem with a discrete spectrum. The eigenfunctions are not orthogonal in the way self-adjoint Sturm-Liouville modes are, and the

inviscid Rayleigh limit is singular: continuous-spectrum effects and critical layers can appear as $\text{Re} \rightarrow \infty$. This is why inviscid inflection-point criteria and viscous Orr-Sommerfeld thresholds should be compared with care rather than identified term by term. Many fluid-mechanics texts write the same normal mode as $e^{i\alpha(x-ct)}$. With the convention used here, the complex phase speed c and growth rate σ are related by

$$\sigma = -i\alpha c,$$

so $\text{Re } \sigma = \alpha \text{ Im } c$ for real α .

Remark 9.1 (Classical facts). *Plane Poiseuille flow has a linear instability above a critical Reynolds number about 5772, using centerline speed and half-channel width; Thomas (1953) established the numerical value, and Orszag (1971) refined it. Plane Couette flow is linearly stable for all Reynolds numbers in the classical Orr-Sommerfeld analysis, yet it can transition through finite-amplitude disturbances. Reynolds' 1883 dye-injection observation is the founding empirical demonstration. Stability is therefore not a single concept: linear eigenvalue stability, energy stability, transient growth, and nonlinear transition answer different questions.*

9.15.2 Energy Stability Estimate

For perturbations of a parallel flow, the kinetic energy of the disturbance satisfies

$$\frac{1}{2} \frac{d}{dt} \int_{\Omega} |\tilde{u}|^2 d^3x = - \int_{\Omega} \tilde{u} \tilde{v} U'(y) d^3x - \frac{1}{\text{Re}} \int_{\Omega} |\nabla \tilde{u}|^2 d^3x.$$

The second term is viscous dissipation. The first is production by mean shear. If dissipation dominates production for all admissible perturbations, the base flow is energy stable. If not, the estimate does not prove instability, but it reveals the only possible energy source for perturbation growth. This identity is the Reynolds-Orr energy equation for perturbations of a shear flow.

9.16 Point Vortices As A Hamiltonian Fluid Model

In two dimensions, an idealized vorticity distribution can be written as

$$\omega(x, y) = \sum_{i=1}^N \Gamma_i \delta(x - x_i, y - y_i),$$

where Γ_i is circulation. Helmholtz introduced point-vortex idealizations alongside the 1858 vortex theorems; Kirchhoff cast them as a Hamiltonian system in his 1876 lectures. Here they serve as a finite-dimensional Hamiltonian reduction of ideal-fluid motion. The point vortex equations are

$$\begin{aligned} \Gamma_i \dot{x}_i &= \frac{\partial H}{\partial y_i}, \\ \Gamma_i \dot{y}_i &= -\frac{\partial H}{\partial x_i}, \end{aligned}$$

with Hamiltonian

$$H = -\frac{1}{4\pi} \sum_{i < j} \Gamma_i \Gamma_j \log((x_i - x_j)^2 + (y_i - y_j)^2).$$

This is Hamiltonian mechanics, but the symplectic form is weighted:

$$\omega_{\text{vortex}} = \sum_i \Gamma_i dx_i \wedge dy_i.$$

When the circulations have mixed signs this form is not positive in any metric sense. This is consistent with symplectic mechanics: symplectic structure requires nondegeneracy, not positive definiteness. The weight Γ_i is the circulation carried by the i -th vortex. One can derive this finite-dimensional symplectic form by reducing the ideal-fluid Hamiltonian bracket to delta-function vorticity; in this section we use the resulting point-vortex bracket as the model.

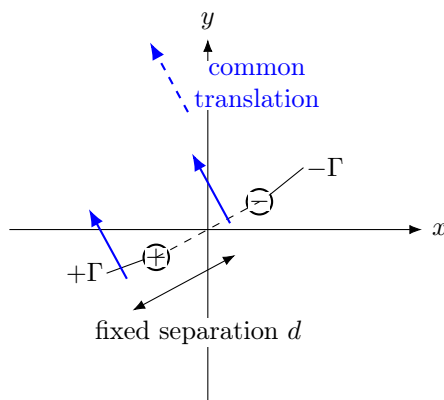


Figure 9.5: A vortex dipole is the simplest point-vortex motion: the two vortices translate together with fixed separation. More complicated point-vortex systems give finite-dimensional Hamiltonian models of ideal-fluid motion.

9.17 What To Take Forward

- The material derivative is the bridge between following particles and describing fields at fixed spatial points.
- Conservation of mass and momentum are balance laws before they are differential equations; the local PDEs come from applying the divergence theorem to arbitrary moving volumes.
- Navier-Stokes is Cauchy momentum balance plus the Newtonian stress law. Changing the constitutive law changes the continuum model.
- Exact Navier-Stokes solutions become tractable when symmetry removes the nonlinear term, when the flow is fully developed, or when advection can be absorbed into pressure.
- The Reynolds number compares inertial transport with viscous diffusion of momentum. It is the organizing scale behind laminar solutions, instabilities, and transition.
- Point vortices show that Hamiltonian structure survives inside fluid mechanics after a strong but explicit reduction.

9.18 Chapter-End Laboratories And Exercises

The following derivations isolate two recurring fluid-mechanics diagnostics: pressure as the multiplier that enforces incompressibility, and the Orr-Sommerfeld eigenvalue problem as the linear stability test for parallel shear flows.

9.18.1 Pressure As A Lagrange Multiplier In Incompressible Flow

For incompressible Navier-Stokes,

$$\partial_t u + u \cdot \nabla u = -\frac{1}{\rho} \nabla p + \nu \Delta u, \quad \nabla \cdot u = 0.$$

The pressure is not determined by a separate thermodynamic equation in the incompressible model. It is the Lagrange multiplier enforcing $\nabla \cdot u = 0$.

Take the divergence of the momentum equation. Since

$$\nabla \cdot \partial_t u = \partial_t (\nabla \cdot u) = 0 \quad \text{and} \quad \nabla \cdot \Delta u = \Delta (\nabla \cdot u) = 0,$$

this gives the pressure Poisson equation

$$\boxed{-\frac{1}{\rho} \Delta p = \nabla \cdot (u \cdot \nabla u) = \partial_i u_j \partial_j u_i.}$$

The last equality follows by expanding:

$$\partial_i (u_j \partial_j u_i) = (\partial_i u_j) (\partial_j u_i) + u_j \partial_j (\partial_i u_i),$$

and the second term vanishes by incompressibility. Thus p is whatever spatial field is needed to project the nonlinear acceleration back onto the space of divergence-free vector fields.

This is the Helmholtz projection in PDE form. On a smooth bounded or periodic domain, with the appropriate boundary conditions, a vector field a can be decomposed as

$$a = a_{\text{df}} + \nabla \phi, \quad \nabla \cdot a_{\text{df}} = 0.$$

Taking the divergence gives a Poisson equation

$$\Delta \phi = \nabla \cdot a,$$

with boundary conditions determined by the normal component. In incompressible Navier-Stokes, the pressure supplies exactly this gradient part of the acceleration.

9.18.2 Orr-Sommerfeld Boundary Conditions And Spectrum

The body of the chapter derived the Orr-Sommerfeld equation. The laboratory here records the boundary and spectral data needed to turn that differential equation into a well-posed eigenvalue problem. No penetration means

$$\tilde{v} = -\partial_x \psi = 0 \quad \Rightarrow \quad \phi = 0$$

at a wall. No slip also requires

$$\tilde{u} = \partial_y \psi = 0 \quad \Rightarrow \quad D\phi = 0.$$

Thus the Orr-Sommerfeld equation is fourth order and needs the four wall conditions $\phi = D\phi = 0$ at the two boundaries. For a channel $y \in [-1, 1]$, the eigenvalue problem may be written as

$$\mathcal{L}_{OS}\phi = \sigma \mathcal{M}\phi, \quad \mathcal{M} = D^2 - \alpha^2,$$

where

$$\mathcal{L}_{OS} = -i\alpha U(D^2 - \alpha^2) + i\alpha U'' + \frac{1}{\text{Re}}(D^2 - \alpha^2)^2.$$

The no-slip boundary conditions make this a discrete non-self-adjoint spectral problem for finite Re . Numerically, one must monitor convergence with respect to the wall-normal discretization because non-normal operators can produce spurious eigenvalues and large transient amplification even when every eigenvalue is stable.

9.19 Associated Demos

- `demos/python/navier_stokes_solutions.py`. Exact Couette-Poiseuille, pipe, Stokes-layer, and Taylor-Green solutions.
- `demos/python/fluids_vorticity.py`, a point-vortex Hamiltonian model.
- `demos/python/hamiltonian_pendulum.py`, useful for comparing separatrix geometry with shear-flow stability.
- Suggested extension: a numerical Orr-Sommerfeld spectrum for Couette and Poiseuille flow.

Appendix A

KAM Proof Architecture

This appendix gives a more explicit view of the proof mechanism behind the KAM statement in Chapter 5. It is not a substitute for a full monograph proof. Its purpose is to record the analytic estimates, the role of the twist condition, and the reason that the theorem is a Newton method on Hamiltonians rather than an ordinary first-order perturbation calculation.

A.1 Local Analytic Model

Let n be the number of degrees of freedom. The unperturbed phase space is

$$\mathcal{D} \times \mathbb{T}^n, \quad (I, \theta) = (I_1, \dots, I_n, \theta^1, \dots, \theta^n),$$

where $\mathcal{D} \subset \mathbb{R}^n$ is an open action domain and $\mathbb{T}^n = \mathbb{R}^n / (2\pi\mathbb{Z})^n$. The symplectic form is

$$\omega = \sum_{a=1}^n dI_a \wedge d\theta^a,$$

and the Hamiltonian is

$$H(I, \theta) = h(I) + \varepsilon f(I, \theta).$$

Here h is the integrable Hamiltonian, f is a perturbation, and ε is a small dimensionless parameter. The frequency map of the unperturbed motion is

$$\omega(I) = \nabla_I h(I).$$

If $\varepsilon = 0$, the torus $I = I_0$ is invariant and the angle motion is

$$\theta(t) = \theta(0) + \omega(I_0)t.$$

Assumption A.1 (Analyticity and complex strips). *For the proof outline in this appendix, assume that h and f extend holomorphically to a complex neighborhood*

$$\mathcal{D}_\rho \times \mathbb{T}_\sigma^n = \{I \in \mathbb{C}^n : \text{dist}(I, \mathcal{D}) < \rho\} \times \{\theta \in \mathbb{C}^n / (2\pi\mathbb{Z})^n : |\text{Im } \theta| < \sigma\}.$$

The constants ρ and σ are analytic radii. Estimates lose part of these radii at each coordinate change; that loss is the analytic price paid for controlling derivatives and Fourier tails.

For an analytic function

$$F(I, \theta) = \sum_{k \in \mathbb{Z}^n} F_k(I) e^{ik \cdot \theta},$$

a convenient norm is

$$\|F\|_{\rho, \sigma} = \sum_{k \in \mathbb{Z}^n} \sup_{I \in \mathcal{D}_\rho} |F_k(I)| e^{|k| \sigma}, \quad |k| = |k_1| + \cdots + |k_n|.$$

The exponential weight records the analytic decay of Fourier coefficients. The KAM proof repeatedly trades this decay against small denominators $k \cdot \omega$.

A.2 The Homological Equation

The first calculation is the one canonical perturbation theory already suggests. Fix an action I_0 and write

$$N(I) = h(I), \quad P(I, \theta) = \varepsilon f(I, \theta).$$

We seek a near-identity canonical transformation generated by a Hamiltonian $\chi(I, \theta)$. If Φ_χ^1 denotes the time-one map of the Hamiltonian flow of χ , then Lie expansion gives

$$H \circ \Phi_\chi^1 = H + \{H, \chi\} + \frac{1}{2} \{\{H, \chi\}, \chi\} + \cdots.$$

To first order in the perturbation, one tries to choose χ so that the angle-dependent part of P is removed:

$$\{N, \chi\} + P - \langle P \rangle = 0, \quad \langle P \rangle(I) = \frac{1}{(2\pi)^n} \int_{\mathbb{T}^n} P(I, \theta) d^n \theta.$$

Since $N = h(I)$,

$$\{N, \chi\} = \nabla_I h(I) \cdot \partial_\theta \chi = \omega(I) \cdot \partial_\theta \chi.$$

At $I = I_0$, the Fourier coefficients of the equation are

$$i(k \cdot \omega_0) \chi_k(I_0) + P_k(I_0) = 0, \quad k \neq 0, \quad \omega_0 = \omega(I_0),$$

so that

$$\chi_k(I_0) = \frac{i P_k(I_0)}{k \cdot \omega_0}.$$

This formula is the source of the small-divisor problem. The numerator P_k decays exponentially in $|k|$ if the perturbation is analytic, but the denominator can be very small when $k \cdot \omega_0$ nearly vanishes.

Definition A.1 (Diophantine frequency). *A vector $\omega \in \mathbb{R}^n$ is Diophantine with constants $\gamma > 0$ and $\tau > n - 1$ if*

$$|k \cdot \omega| \geq \frac{\gamma}{|k|^\tau} \quad \text{for all } 0 \neq k \in \mathbb{Z}^n.$$

The set of such vectors is denoted $\text{DC}(\gamma, \tau)$.

If $\omega_0 \in \text{DC}(\gamma, \tau)$, then

$$|\chi_k(I_0)| \leq \gamma^{-1} |k|^\tau |P_k(I_0)|.$$

The factor $|k|^\tau$ is polynomial. The analytic decay of P_k can absorb it if one gives up a smaller angle strip. More explicitly, for $0 < \sigma' < \sigma$,

$$|k|^\tau e^{-|k|\sigma} = |k|^\tau e^{-|k|(\sigma-\sigma')} e^{-|k|\sigma'} \leq C_{\tau,n} (\sigma - \sigma')^{-(\tau+n)} e^{-|k|\sigma'}.$$

Thus the generator satisfies an estimate of the form

$$\|\chi\|_{\rho,\sigma'} \leq C \gamma^{-1} (\sigma - \sigma')^{-(\tau+n)} \|P\|_{\rho,\sigma}.$$

The constants are not the conceptual point. The point is the structure: small divisors are controlled by Diophantine arithmetic, and derivatives are controlled by sacrificing analytic width.

A.3 Twist, Frequencies, And What Is Preserved

The KAM theorem does not say that every original torus $I = I_0$ remains exactly at the same action. A perturbation changes the average Hamiltonian, and the frequency map moves. To track a torus with a prescribed frequency ω_* , one needs a nondegenerate relation between actions and frequencies.

Assumption A.2 (Kolmogorov nondegeneracy). *On the action domain under consideration,*

$$\det \partial_I^2 h(I) \neq 0.$$

Equivalently, the frequency map $I \mapsto \omega(I)$ is locally invertible.

After the first averaging step the new integrable part is

$$h_+(I) = h(I) + \langle P \rangle(I) + \text{higher order terms}.$$

The action whose frequency equals ω_* generally shifts from I_0 to $I_0 + \Delta I$. Linearizing,

$$\omega_* = \nabla h_+(I_0 + \Delta I) = \nabla h(I_0) + \partial_I^2 h(I_0) \Delta I + \nabla \langle P \rangle(I_0) + \dots.$$

If $\nabla h(I_0) = \omega_*$, then

$$\Delta I = -(\partial_I^2 h(I_0))^{-1} \nabla \langle P \rangle(I_0) + \dots.$$

The twist condition is the assumption that this adjustment can be solved for uniquely. Without it, KAM theorems still exist in modified forms, but some other nondegeneracy condition must replace it.

A.4 Isoenergetic And Map Versions

The nondegeneracy condition depends on which object is being preserved. The Kolmogorov theorem fixes a frequency vector and constructs a nearby invariant torus with that frequency. In applications one often fixes an energy surface instead. Then frequencies are meaningful only up to a common rescaling of time, because the orbit on a torus with frequency ω and the orbit with frequency $c\omega$, $c > 0$, have the same unparametrized winding direction.

Definition A.2 (Isoenergetic nondegeneracy). *The integrable Hamiltonian $h(I)$ satisfies the isoenergetic nondegeneracy condition at I if*

$$\det \begin{pmatrix} \partial_I^2 h(I) & \nabla h(I) \\ \nabla h(I)^T & 0 \end{pmatrix} \neq 0.$$

This condition says that, after restricting to an energy surface, the frequency direction still changes nontrivially with the actions. It is the natural condition when the geometry of a fixed energy hypersurface is the main object.

For a two-dimensional area-preserving twist map with coordinates (θ, I) and unperturbed form

$$I_{m+1} = I_m, \quad \theta_{m+1} = \theta_m + \Omega(I_m),$$

the corresponding nondegeneracy condition is

$$\frac{d\Omega}{dI} \neq 0.$$

This is the map version of twist. It is what makes the standard map a useful laboratory: for $K = 0$,

$$p_{m+1} = p_m, \quad q_{m+1} = q_m + p_m,$$

so the rotation number changes with p . KAM curves of the perturbed map are organized by their irrational rotation numbers.

setting	preserved object	nondegeneracy condition
Hamiltonian flow, fixed frequency	torus with frequency ω	$\det \partial_I^2 h \neq 0$
Hamiltonian flow, fixed energy	frequency direction on an energy surface	bordered determinant above is nonzero
area-preserving twist map	invariant curve with rotation number ρ	$d\Omega/dI \neq 0$

A.5 One KAM Step

The proof is organized as an iteration. At the m -th stage the Hamiltonian has the form

$$H_m(I, \theta) = h_m(I) + P_m(I, \theta)$$

on a complex domain $\mathcal{D}_{\rho_m} \times \mathbb{T}_{\sigma_m}^n$, with $\|P_m\|$ small. The step has four parts.

1. *Truncate Fourier modes.* Write P_m as low modes $|k| \leq K_m$ plus a high-mode tail. Analyticity makes the tail exponentially small:

$$\left\| P_m^{>|K_m|} \right\|_{\sigma_m - \delta_m} \leq e^{-K_m \delta_m} \|P_m\|_{\sigma_m}.$$

2. *Solve the homological equation.* Remove the low-mode angle-dependent part by solving

$$\omega_m(I_*) \cdot \partial_\theta \chi_m + P_m^{\leq K_m}(I_*, \theta) - \langle P_m^{\leq K_m} \rangle(I_*) = 0.$$

3. *Apply the canonical transformation.* Use the time-one flow $\Phi_{\chi_m}^1$. The transformed Hamiltonian is

$$H_{m+1} = H_m \circ \Phi_{\chi_m}^1 = h_{m+1} + P_{m+1}.$$

4. *Recentre the action.* Use the twist condition to keep the frequency of the selected torus equal to the desired Diophantine vector.

The new perturbation contains a quadratic part, a Fourier-tail part, and domain-loss errors:

$$\|P_{m+1}\| \lesssim C \frac{\|P_m\|^2}{\gamma^2 \delta_m^{2(\tau+n)+1}} + C e^{-K_m \delta_m} \|P_m\|.$$

The parameters K_m and δ_m are chosen so that the right side is much smaller than $\|P_m\|$. This is the Newton character of the method: after the resonant obstructions are excluded and the domain losses are budgeted, the perturbation decreases roughly quadratically.

The displayed power $2(\tau+n)+1$ is a bookkeeping exponent for the analytic norm used in this appendix, not a canonical invariant of KAM theory. Different proofs distribute the same losses among Fourier summation, Cauchy estimates, and the Lie-series remainder in slightly different ways. What matters for this lecture-note proof architecture is the structure of the estimate: two small-divisor losses, a domain-loss factor, a Fourier tail, and a quadratic improvement in $\|P_m\|$.

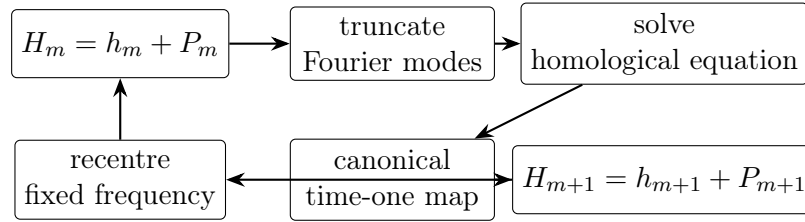


Figure A.1: A KAM step is a controlled averaging step followed by a recentring of the action. The loop contracts because the remaining perturbation is quadratic up to an exponentially small Fourier tail.

A.6 Cantor Set And Measure Estimate

The theorem keeps only those frequencies that remain safely away from resonance. For a bounded frequency domain $\Omega \subset \mathbb{R}^n$, define

$$\Omega_{\gamma,\tau} = \left\{ \omega \in \Omega : |k \cdot \omega| \geq \frac{\gamma}{|k|^\tau} \text{ for every } 0 \neq k \in \mathbb{Z}^n \right\}.$$

The complement is a union of resonance strips

$$\mathcal{R}_k = \left\{ \omega \in \Omega : |k \cdot \omega| < \frac{\gamma}{|k|^\tau} \right\}.$$

For a fixed k , this strip has thickness of order $\gamma|k|^{-(\tau+1)}$ in the direction normal to the hyperplane $k \cdot \omega = 0$. Summing over $k \neq 0$,

$$\text{meas}(\Omega \setminus \Omega_{\gamma,\tau}) \leq C_\Omega \gamma \sum_{0 \neq k \in \mathbb{Z}^n} |k|^{-(\tau+1)}.$$

The sum converges when $\tau > n - 1$. Therefore the excluded set has measure $O(\gamma)$. The surviving set is large in measure but is not open: it is a Cantor-like set obtained by removing resonance neighborhoods at all integer orders.

A.7 Precise Form Of The Conclusion

One common form of the theorem can now be read without hiding the main ingredients.

Proposition A.1 (KAM theorem, schematic analytic version). *Let*

$$H(I, \theta) = h(I) + \varepsilon f(I, \theta)$$

be real analytic on $\mathcal{D}_\rho \times \mathbb{T}_\sigma^n$. Assume that $\det \partial_I^2 h(I) \neq 0$ on $\mathcal{D}$. Fix $\tau > n - 1$ and $\gamma > 0$. If $\varepsilon \|f\|_{\rho, \sigma}$ is sufficiently small compared with a constant determined by h, ρ, σ, γ , and τ , then for each frequency in a slightly smaller Diophantine set there is an invariant Lagrangian torus of the perturbed Hamiltonian. On that torus the motion is analytically conjugate to the linear flow $\dot{\theta} = \omega$. The union of destroyed tori in the corresponding action region has measure tending to zero with ε .

The theorem is deliberately a persistence theorem, not a global solution of the perturbed dynamics. It says that sufficiently irrational invariant tori survive. It does not say that resonant tori survive, that all trajectories are regular, or that transport is absent through the gaps between tori. Those gaps are where resonant islands, separatrix splitting, cantori, and chaotic transport enter the main text.

A.8 Checklist For Applying KAM

When using KAM reasoning in examples, check the following points explicitly.

1. Identify the integrable Hamiltonian $h(I)$, the perturbation $\varepsilon f(I, \theta)$, and the action domain.
2. Verify the nondegeneracy condition being used. In Kolmogorov's theorem this is the twist condition $\det \partial_I^2 h \neq 0$.
3. State the arithmetic condition on the frequency vector and the constants γ, τ .
4. Explain which tori are excluded because they are resonant or nearly resonant.
5. Distinguish persistence of a torus from confinement of nearby orbits. In two-dimensional area-preserving maps, invariant curves are barriers. In higher-dimensional Hamiltonian flows, surviving tori need not divide an energy surface.
6. Separate mathematical smallness from physical smallness. A physically small perturbation can still have important resonant effects if the orbit is near a low-order resonance.

This checklist is the bridge from the proof architecture to the standard map, the asteroid-belt resonances, and the breakdown mechanisms discussed in Chapter 5.

Appendix B

Three-Body Numerics And Ejection Statistics

This appendix records the numerical protocol behind the three-body demonstrations. The purpose is pedagogical reproducibility. A simulation in this repository should specify its equations, units, random sampling rule, time-step rule, diagnostic invariants, and statistical estimator. Without these ingredients, an ejection probability is only a number; with them, it is a repeatable experiment whose limitations can be discussed.

B.1 Hierarchy Of Models

The notes use three related models. Their variables and invariants are not identical, so the diagnostic printed by a code must match the model being integrated.

model	degrees of freedom	main diagnostics
full Newtonian three-body problem	three massive bodies in inertial space, reduced by translation when desired	energy, total angular momentum, center-of-mass drift
circular restricted three-body problem	two primaries on a prescribed circular orbit, massless particle in a rotating frame	Jacobi integral, zero-velocity curves, Lagrange point locations
heliocentric Sun-Jupiter-asteroid ensemble	Sun and prescribed Jupiter, many noninteracting massless test particles	loss classification, binned ejection probability, resonance labels

The first model is autonomous and Hamiltonian. The circular restricted model is autonomous only after passing to the rotating frame. The heliocentric ensemble used for the asteroid lab is a time-dependent restricted model because Jupiter is prescribed as a moving source.

B.2 Units And Equations

The asteroid scripts use astronomical units, years, and solar masses:

$$G = 4\pi^2, \quad M_{\odot} = 1.$$

For a test particle with heliocentric position $r_i(t) \in \mathbb{R}^2$, velocity $v_i(t)$, and prescribed Jupiter position $r_J(t)$, the equation is

$$\ddot{r}_i = -GM_\odot \frac{r_i}{|r_i|^3} - GM_J \frac{r_i - r_J}{|r_i - r_J|^3} - GM_J \frac{r_J}{|r_J|^3}.$$

The last term is the indirect acceleration that appears because the origin is attached to the Sun rather than to an inertial barycenter. Omitting it changes the physical model as well as the numerical method, so it is exposed as an explicit code option.

The CR3BP scripts use nondimensional rotating-frame units. The total primary mass is 1, the primary separation is 1, and the rotating frame has angular velocity 1. If μ is the smaller-primary mass fraction, the primaries sit at

$$r_1 = (-\mu, 0), \quad r_2 = (1 - \mu, 0).$$

The rotating-frame particle state is $(x, y, \dot{x}, \dot{y})$, and

$$\ddot{x} - 2\dot{y} = \Omega_x, \quad \ddot{y} + 2\dot{x} = \Omega_y,$$

where

$$\Omega(x, y) = \frac{x^2 + y^2}{2} + \frac{1 - \mu}{r_1(x, y)} + \frac{\mu}{r_2(x, y)}.$$

The Jacobi integral is

$$C = 2\Omega(x, y) - \dot{x}^2 - \dot{y}^2.$$

The printed value of $\max_t |C(t) - C(0)|$ is a numerical error diagnostic.

B.3 Initial Ensembles

For the asteroid ejection laboratory, the ensemble is defined by independent draws

$$a_0 \sim U(a_{\min}, a_{\max}), \quad e_0 \sim U(0, e_{\max}), \quad M_0 \sim U(0, 2\pi), \quad \varpi_0 \sim U(0, 2\pi).$$

Here a_0 is the initial semimajor axis, e_0 is eccentricity, M_0 is mean anomaly, and ϖ_0 is periape angle. The eccentric anomaly E solves Kepler's equation

$$M_0 = E - e_0 \sin E.$$

The true anomaly f is then

$$f = 2 \arctan \left(\frac{\sqrt{1 + e_0} \sin(E/2)}{\sqrt{1 - e_0} \cos(E/2)} \right),$$

and the heliocentric Kepler data are

$$p = a_0(1 - e_0^2), \quad r = \frac{p}{1 + e_0 \cos f},$$

$$(x_{\text{orb}}, y_{\text{orb}}) = (r \cos f, r \sin f), \quad (v_{x,\text{orb}}, v_{y,\text{orb}}) = \sqrt{\frac{GM_\odot}{p}} (-\sin f, e_0 + \cos f).$$

A rotation by ϖ_0 places the orbit in the reference plane. The random seed is part of the experiment. Two reported probabilities are comparable only when the sampling law, seed policy, time horizon, and loss thresholds are stated.

B.4 Loss Classification

At each diagnostic time the particle is classified as alive or lost. The current demo uses three absorbing loss classes:

$$\begin{aligned} \text{ejected} & : |r| > R_{\text{ej}} \quad \text{or} \quad E_{\text{Kep}} > 0 \text{ and } |r| > a_J, \\ \text{solar collision} & : |r| < R_{\odot}, \\ \text{Jupiter close encounter} & : |r - r_J| < R_J^{\text{close}}. \end{aligned}$$

The Kepler energy used in the first line is

$$E_{\text{Kep}} = \frac{1}{2}|v|^2 - \frac{GM_{\odot}}{|r|}.$$

This energy is not an invariant of the perturbed problem. It is used only as a practical escape indicator. The ejection radius and close-encounter radius therefore must be printed with the output.

B.5 Time Integration And Invariant Drift

For a separable autonomous Hamiltonian $H(p, q) = T(p) + V(q)$, the velocity Verlet method

$$\begin{aligned} p_{m+1/2} &= p_m - \frac{\Delta t}{2} \nabla V(q_m), \\ q_{m+1} &= q_m + \Delta t \partial_p T(p_{m+1/2}), \\ p_{m+1} &= p_{m+1/2} - \frac{\Delta t}{2} \nabla V(q_{m+1}) \end{aligned}$$

is symplectic and time reversible. The asteroid demo uses the same kick-drift-kick idea in a time-dependent restricted field. Since the prescribed Jupiter field depends on time, the map is best understood as a transparent second-order classroom integrator rather than as a final high-precision celestial mechanics method.

The CR3BP demo uses classical fourth-order Runge-Kutta. RK4 is useful because the formula is easy to inspect, but it is not symplectic and does not preserve the Jacobi integral exactly. For long-time transport claims, a structure-aware method and a convergence study are mandatory.

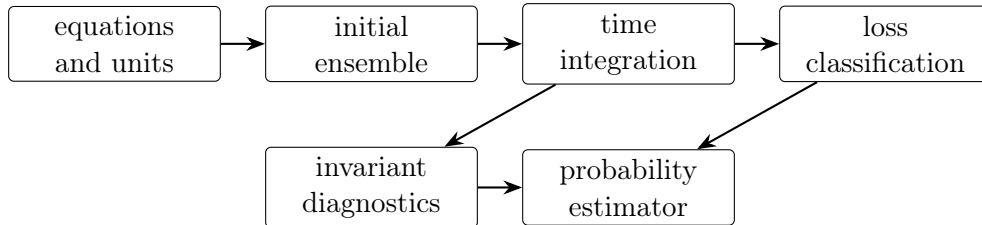


Figure B.1: A reproducible three-body numerical experiment reports model, sampling law, integrator, diagnostics, loss rules, and statistical estimator.

B.6 Ejection Probability As A Statistical Estimator

Let X_i be the indicator that asteroid i is ejected by time T :

$$X_i = \begin{cases} 1, & \text{particle } i \text{ is ejected by time } T, \\ 0, & \text{otherwise.} \end{cases}$$

The Monte Carlo estimator is

$$\hat{p}_{\text{ej}}(T) = \frac{1}{N} \sum_{i=1}^N X_i.$$

For independent samples from the specified initial ensemble,

$$\mathbb{E}\hat{p}_{\text{ej}} = p_{\text{ej}}, \quad \text{Var}(\hat{p}_{\text{ej}}) = \frac{p_{\text{ej}}(1 - p_{\text{ej}})}{N}.$$

The plug-in standard error is

$$\widehat{\text{se}} = \sqrt{\frac{\hat{p}_{\text{ej}}(1 - \hat{p}_{\text{ej}})}{N}}.$$

The same formula applies to a semimajor-axis bin after replacing N by the number of particles in that bin. Bins with few particles have large sampling error and should not be overinterpreted.

There are three different limits:

$$N \rightarrow \infty, \quad \Delta t \rightarrow 0, \quad T \rightarrow \infty.$$

Increasing the number of particles reduces sampling noise. Decreasing the time step reduces discretization error. Increasing the horizon changes the event being estimated. These limits answer different questions and should not be conflated.

B.7 Bootstrap Intervals And Resonance Windows

The binomial standard error is appropriate when one reports a single bin with independently sampled initial conditions. For plots with many bins, a bootstrap interval gives a more flexible classroom diagnostic. Given the sample

$$(a_i, X_i), \quad i = 1, \dots, N,$$

where a_i is the initial semimajor axis and X_i is an ejection indicator, form bootstrap samples by drawing N pairs with replacement from the observed list. For each resample compute the same binned ejection fractions. The empirical 16th and 84th percentiles give a one-sigma style interval; the 2.5th and 97.5th percentiles give a rough 95 percent interval. This estimates sampling variability of the chosen finite-time experiment. It does not estimate model error, timestep error, or the uncertainty caused by using a planar restricted model.

For a $p : q$ interior mean-motion resonance with Jupiter, the nominal semimajor axis is

$$a_{p:q} = a_J \left(\frac{q}{p} \right)^{2/3}.$$

A resonance-window comparison chooses a width Δa and compares three sets:

$$|a_0 - a_{p:q}| < \Delta a, \quad \Delta a \leq |a_0 - a_{p:q}| < 2\Delta a, \quad |a_0 - a_{p:q}| \geq 2\Delta a.$$

For a more dynamical diagnostic one also tracks a resonant angle. In a planar restricted problem, a typical resonant angle for an interior asteroid resonance is

$$\phi_{p:q} = p\lambda_J - q\lambda - (p - q)\varpi,$$

where λ_J is Jupiter’s mean longitude, λ is the asteroid’s mean longitude, and ϖ is the asteroid periapee longitude. Libration of $\phi_{p:q}$ is stronger evidence of resonant behavior than proximity in initial semimajor axis alone. The present demo labels bins by nominal locations; an upgraded project version can add resonant-angle time series.

B.8 Convergence Table Template

A serious numerical report should include a small convergence table. The columns should separate the three main effects:

N	Δt	T	$\hat{p}_{\text{ej}}(T)$	diagnostic
N_0	Δt_0	T_0	$\dots$	baseline
$2N_0$	Δt_0	T_0	$\dots$	sampling change
N_0	$\Delta t_0/2$	T_0	$\dots$	timestep change
N_0	Δt_0	$2T_0$	$\dots$	horizon change

Only the second row tests Monte Carlo variance. Only the third tests discretization. The fourth row changes the physical event from “ejected by T_0 ” to “ejected by $2T_0$.” This simple table prevents a common misreading of the asteroid lab: a longer run is not a more accurate estimate of the shorter-run probability.

B.9 Recommended Reporting Standard

A publication-quality classroom run should report:

1. the exact command line or configuration file;
2. the model equations, including whether the indirect term is included;
3. N , T , Δt , random seed, and initial sampling law;
4. all loss thresholds;
5. global loss counts and binned counts;
6. binomial standard errors for probabilities;
7. an invariant-drift or convergence diagnostic appropriate to the model;
8. plots with resonance markers, bin widths, and axis units.

The flagship asteroid script now follows this standard closely enough that its text, CSV, and JSON outputs can be used as lab records. A more advanced version could add symplectic extended-phase integration, adaptive encounter handling, and comparison against a high-accuracy reference integrator.

Appendix C

Continuum Mechanics Caveats

The continuum chapters use compact notation because the main text already derives the equations in detail. This appendix gathers the assumptions that are easiest to blur: what is kinematics, what is balance law, what is constitutive law, which stress measure is being used, and where idealizations enter. The goal is to prevent correct formulas from being read with the wrong meaning.

C.1 Three Layers: Kinematics, Balance, Constitutive Law

A continuum theory is built in three logically distinct layers.

1. *Kinematics* specifies the variables. For a material body this may be a deformation map $\varphi(X, t)$, deformation gradient $F = \partial\varphi/\partial X$, velocity $v(x, t)$, or director field.
2. *Balance laws* express conservation of mass, momentum, and angular momentum. They hold for broad classes of materials before a particular stress-strain relation has been chosen.
3. *Constitutive laws* specify the material response, such as $\sigma = -pI + 2\mu D$ for a Newtonian incompressible fluid or $P = \partial W/\partial F$ for a hyperelastic solid.

Most mistakes in continuum mechanics come from mixing these layers. For example, Cauchy's momentum equation

$$\rho \frac{Dv}{Dt} = \operatorname{div} \sigma + \rho b$$

is a balance law. The Navier-Stokes equation appears only after inserting a Newtonian constitutive law and, for the incompressible form, the constraint $\operatorname{div} v = 0$.

C.2 Reference And Spatial Descriptions

Let $\mathcal{B}$ be a reference body with material point X , and let $\varphi_t : \mathcal{B} \rightarrow \mathbb{R}^3$ be the placement at time t . The spatial point is

$$x = \varphi(X, t).$$

The deformation gradient is

$$F^i{}_A = \frac{\partial \varphi^i}{\partial X^A}.$$

Here Latin indices $i, j, \dots$ refer to spatial components, while capital indices $A, B, \dots$ refer to material components. The Jacobian

$$J = \det F$$

measures local volume change. If $\rho_0(X)$ is the reference mass density and $\rho(x, t)$ is the spatial density, conservation of mass gives

$$\rho(\varphi(X, t), t)J(X, t) = \rho_0(X).$$

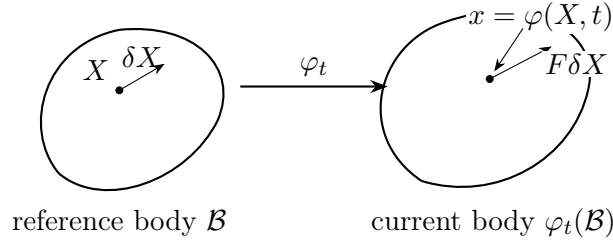


Figure C.1: Material and spatial descriptions are connected by the placement φ_t . The deformation gradient F maps material tangent vectors to spatial tangent vectors.

C.3 Stress Measures

Different stress tensors answer different questions.

symbol	name	meaning
σ	Cauchy stress	spatial traction per current area: $t = \sigma n$
P	first Piola-Kirchhoff stress	spatial force per reference area; appears in $\text{Div } P + \rho_0 b = \rho_0 \ddot{\varphi}$
S	second Piola-Kirchhoff stress	material stress work-conjugate to $E = (C - I)/2$, where $C = F^T F$
$\tau = J\sigma$	Kirchhoff stress	volume-weighted Cauchy stress, useful in finite strain

The standard Piola transform is

$$P = J\sigma F^{-T}.$$

If the material is hyperelastic with stored energy density $W(F)$ per reference volume, then

$$P = \frac{\partial W}{\partial F}.$$

Objectivity requires

$$W(RF) = W(F) \quad \text{for all } R \in \text{SO}(3),$$

so that the energy does not change under a superposed rigid rotation.

C.4 Incompressibility And Pressure

Incompressibility means different things in material and spatial language. In finite elasticity it is

$$J = \det F = 1.$$

In Eulerian fluid mechanics it is

$$\operatorname{div} v = 0.$$

The second condition follows from differentiating the first along a material trajectory:

$$\dot{J} = J \operatorname{div} v.$$

In both settings, pressure is a Lagrange multiplier for a constraint. For an incompressible hyperelastic solid one writes

$$P = \frac{\partial W}{\partial F} - p J F^{-T}.$$

For an incompressible Newtonian fluid one writes

$$\sigma = -pI + 2\mu D, \quad D = \frac{1}{2}(\nabla v + \nabla v^T).$$

The pressure is determined together with the velocity by the constraint and the boundary conditions; it is not supplied by an equation of state in the incompressible model.

C.5 Elastic Solids Versus Fluids

The notes place elastic bodies before deforming bodies and fluids because elasticity makes the material reference configuration explicit. Fluids can also be described materially, but their standard equations are Eulerian and their constitutive law does not remember a fixed rest shape.

feature	elastic solid	Newtonian fluid
primary memory	deformation from reference body	instantaneous rate of strain
typical constitutive variable	F , $C = F^T F$, or strain E	$D = (\nabla v + \nabla v^T)/2$
stress response	elastic stress from stored energy	viscous stress plus pressure
equilibrium equation	boundary-value problem for displacement	steady or unsteady velocity-pressure problem
constraint pressure	enforces $J = 1$ when incompressible	enforces $\operatorname{div} v = 0$ when incompressible

The distinction is physical. A solid can sustain shear stress at rest. A Newtonian fluid at rest has stress $-pI$; shear stress is proportional to rate of deformation.

C.6 Boundary Conditions Are Part Of The Model

The same differential equation can describe different physical problems under different boundary conditions. For elasticity, common boundary conditions are prescribed displacement

$$u = u_0 \quad \text{on } \Gamma_D$$

and prescribed traction

$$PN = T_0 \quad \text{on } \Gamma_N.$$

For a viscous fluid at a fixed wall, the usual no-slip condition is

$$v = v_{\text{wall}}.$$

For inviscid flow, only impermeability

$$v \cdot n = v_{\text{wall}} \cdot n$$

is normally imposed. Applying a no-slip condition to the Euler equation or an inviscid slip condition to a viscous wall problem changes the physical model.

C.7 Weak Forms In Elasticity

For linear elasticity on a reference domain $\Omega \subset \mathbb{R}^3$, write the displacement as $u : \Omega \rightarrow \mathbb{R}^3$ and the infinitesimal strain as

$$\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T).$$

The strong static equation is

$$-\operatorname{div} \sigma = f, \quad \sigma = \mathbb{C}\varepsilon(u),$$

with prescribed displacement on Γ_D and traction $\sigma n = t$ on Γ_N . Let v be a test displacement vanishing on Γ_D . Multiplying by v , integrating, and integrating by parts gives

$$\int_{\Omega} \sigma : \varepsilon(v) \, dx = \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} t \cdot v \, dS.$$

This is the weak form. The natural energy space is a Sobolev space

$$u \in H^1(\Omega; \mathbb{R}^3), \quad v \in H_0^1(\Omega; \mathbb{R}^3; \Gamma_D).$$

The coercivity mechanism is Korn's inequality: modulo rigid motions, the H^1 -size of u is controlled by the L^2 -size of $\varepsilon(u)$. This is why strain, although it is only the symmetric part of ∇u , is enough to control displacement after boundary conditions remove free rigid motions.

C.8 Finite Elasticity And Convexity Caveats

In finite elasticity the stored energy $W(F)$ must be frame indifferent:

$$W(RF) = W(F) \quad R \in \operatorname{SO}(3).$$

This symmetry is incompatible with ordinary convexity in F for realistic elastic energies. The direct method in the calculus of variations therefore uses weaker notions.

$$\begin{aligned} \text{polyconvexity} &\implies \text{quasiconvexity} \\ &\implies \text{rank-one convexity} \\ &\implies \text{Legendre-Hadamard ellipticity.} \end{aligned}$$

Polyconvexity means that W can be written as a convex function of

$$F, \quad \text{cof } F, \quad \det F.$$

Quasiconvexity is the condition tied directly to weak lower semicontinuity of the elastic energy. Rank-one convexity tests convexity along lines $F + s a \otimes b$, the infinitesimal directions relevant to localized gradients. The main text uses finite-strain formulas classically; existence theory for minimizers requires these additional hypotheses plus coercivity and orientation preservation.

C.9 Navier-Stokes Assumptions

The incompressible Navier-Stokes equation

$$\rho(\partial_t v + v \cdot \nabla v) = -\nabla p + \mu \Delta v + \rho b, \quad \text{div } v = 0,$$

uses the following assumptions:

1. continuum approximation: fields are smooth enough on the length scales of interest;
2. constant density ρ and viscosity μ ;
3. Newtonian, isotropic, frame-indifferent viscous stress;
4. incompressibility;
5. boundary and initial data compatible with $\text{div } v = 0$.

The equation is local and deterministic. Turbulence does not invalidate it as a model, but turbulent prediction may require statistical descriptions, resolution studies, or closure assumptions if one averages the equations.

C.10 Weak Form Of Incompressible Navier-Stokes

Let v be divergence-free and let w be a smooth divergence-free test field satisfying the homogeneous velocity boundary condition. Multiplying Navier-Stokes by w and integrating gives

$$\rho \int_{\Omega} \partial_t v \cdot w \, dx + \rho \int_{\Omega} (v \cdot \nabla) v \cdot w \, dx + 2\mu \int_{\Omega} D(v) : D(w) \, dx = \int_{\Omega} \rho b \cdot w \, dx.$$

The pressure term is absent because

$$\int_{\Omega} \nabla p \cdot w \, dx = \int_{\partial\Omega} p w \cdot n \, dS - \int_{\Omega} p \, \text{div } w \, dx = 0$$

for the usual no-penetration or no-slip test spaces. A Leray-Hopf weak solution in three dimensions satisfies the equation in this weak sense and the energy inequality

$$\frac{1}{2} \rho \|v(t)\|_{L^2}^2 + 2\mu \int_0^t \|D(v(s))\|_{L^2}^2 \, ds \leq \frac{1}{2} \rho \|v(0)\|_{L^2}^2 + \int_0^t \int_{\Omega} \rho b \cdot v \, dx \, ds.$$

Weak existence is not the same as smooth uniqueness. The three-dimensional global regularity problem asks whether smooth data always produce smooth solutions for all time; this remains outside the scope of these notes.

C.11 Energy Identities And Regularity

For smooth incompressible flow on a domain with boundary conditions that remove boundary power, multiplying Navier-Stokes by v and integrating gives

$$\frac{d}{dt} \int \frac{1}{2} \rho |v|^2 dx = - \int 2\mu D : D dx + \int \rho b \cdot v dx.$$

The pressure term drops out because

$$\int v \cdot \nabla p dx = \int \operatorname{div}(pv) dx - \int p \operatorname{div} v dx.$$

The second term is zero by incompressibility, and the first is a boundary flux.

For weak solutions, the same calculation may become an inequality rather than an equality. In three dimensions, global smoothness for arbitrary smooth data is a famous open problem. The lecture-note derivations are therefore classical derivations: they are correct under the stated smoothness assumptions, while the functional-analytic theory requires additional care.

Appendix D

Gauge Theory Taxonomy For Deforming Bodies

The deforming-body chapter uses gauge language because a body can change shape while a symmetry group moves it through space. This appendix separates the main connection concepts that appear in rigid-body reconstruction, swimming, elastic frames, and Cosserat media. The common pattern is a split between “where the body is” and “what shape the body has.”

D.1 Principal-Bundle Kinematics

Let Q be the configuration space of a body and let a Lie group G act freely on Q . For a rigid body in space, $G = \text{SE}(3)$ or $G = \text{SO}(3)$. For a planar swimmer, $G = \text{SE}(2)$. The quotient

$$S = Q/G$$

is the shape space. Locally one may write a configuration as

$$q = (g, s), \quad g \in G, \quad s \in S.$$

The body velocity is

$$\xi = g^{-1}\dot{g} \in \mathfrak{g}$$

for a left-trivialized convention. A connection is a rule that assigns a group velocity to a shape velocity:

$$\xi + A(s)\dot{s} = 0$$

for zero-momentum or force-free motion. The sign is conventional; what matters is that A is a $\mathfrak{g}$ -valued one-form on shape space.

D.2 Mechanical Connection

Suppose the kinetic energy defines a metric on Q and the G -action is a symmetry. The momentum map $J : TQ \rightarrow \mathfrak{g}^*$ measures motion along the group directions. The mechanical connection is the projection that declares a velocity horizontal when its group momentum vanishes:

$$J(q, \dot{q}) = 0.$$

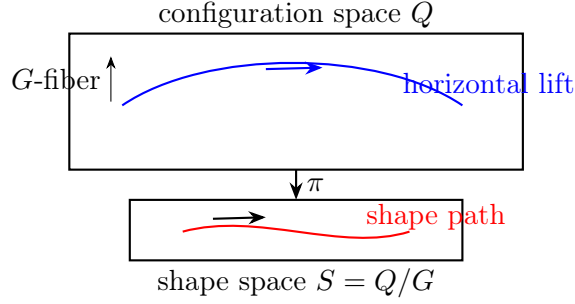


Figure D.1: A connection chooses horizontal lifts of shape-space motion. The failure of a lifted loop to close in Q is the geometric phase or holonomy.

In local variables (g, s) , the kinetic energy has the schematic form

$$T = \frac{1}{2} \begin{pmatrix} \xi \\ \dot{s} \end{pmatrix}^T \begin{pmatrix} \mathcal{I}(s) & \mathcal{K}(s) \\ \mathcal{K}(s)^T & \mathcal{M}(s) \end{pmatrix} \begin{pmatrix} \xi \\ \dot{s} \end{pmatrix}.$$

The group momentum is

$$\Pi = \frac{\partial T}{\partial \xi} = \mathcal{I}(s)\xi + \mathcal{K}(s)\dot{s}.$$

At zero momentum,

$$\xi = -\mathcal{I}(s)^{-1}\mathcal{K}(s)\dot{s}.$$

Thus

$$A(s) = \mathcal{I}(s)^{-1}\mathcal{K}(s).$$

This is the inertial connection used for falling cats, deforming rigid assemblies, and the finite-dimensional examples in the main text.

D.3 Local Connection For Swimming

At low Reynolds number, inertia is negligible. The governing equations are force and torque balance with a linear relation between body velocity and hydrodynamic force. In local coordinates one again obtains

$$\xi = -A_{\text{Stokes}}(s)\dot{s},$$

but the origin of A_{Stokes} is not kinetic energy. It is the resistance problem for the surrounding fluid. The same gauge geometry appears because group motion is reconstructed from shape motion, but the physical inner product and the equations that define horizontality are different.

D.4 Gauge Transformation

A local trivialization $q = (g, s)$ is a choice of body frame over shape space. Changing the frame by a smooth map $h : S \rightarrow G$ gives

$$g' = gh(s).$$

For the left-trivialized velocity $\xi = g^{-1}\dot{g}$, the connection one-form transforms as

$$A' = \text{Ad}_{h^{-1}} A - h^{-1}dh.$$

The minus sign follows from the reconstruction convention $\xi + A(s)\dot{s} = 0$. If one instead defines the standard connection $B = -A$, then $B' = \text{Ad}_{h^{-1}} B + h^{-1}dh$. This formula explains the word “gauge.” The local coefficients of connection depend on the chosen frame, but geometric predictions do not. Holonomy around a closed shape loop changes by conjugation, and conjugacy invariants are frame-independent.

D.5 Curvature And Small Loops

For the reconstruction convention used in the finite-dimensional deforming-body chapter, the curvature is the $\mathfrak{g}$ -valued two-form

$$F_A = dA - \frac{1}{2}[A \wedge A].$$

For a small oriented loop Σ in shape space, the leading geometric phase is

$$g_{\text{hol}} = \exp \left(- \int_{\Sigma} F_A + O(\text{area}^{3/2}) \right)$$

when noncommutativity is weak over the loop. In two shape variables (α, β) , if

$$A = A_{\alpha}(\alpha, \beta) d\alpha + A_{\beta}(\alpha, \beta) d\beta,$$

then

$$F_{\alpha\beta} = \partial_{\alpha} A_{\beta} - \partial_{\beta} A_{\alpha} - [A_{\alpha}, A_{\beta}].$$

The bracket term is the non-Abelian correction in this reconstruction sign convention. It vanishes for a translation group $\mathbb{R}^m$, but not for rotations or rigid motions.

D.6 Connection Types In These Notes

connection	horizontality condition	typical use
mechanical connection	zero group momentum $J = 0$	falling cat, internal rotors, deforming inertial bodies
Stokes or local swimming connection	zero net force and torque in viscous flow	low-Reynolds swimming and crawling
material frame connection	comparison of frames attached to a continuum	elastic directors, rods, shells
spin connection	covariant differentiation of orthonormal frames	curved surfaces and continuum frame fields
Cosserat connection	director rotation along a material coordinate	rods and micropolar continua
defect connection	incompatible local frames or coframes	disclinations, dislocations, torsion, curvature

These objects share gauge transformation laws, but they are not the same physical connection. The defining horizontal condition must always be stated.

D.7 Relation To Elasticity And Fluids

Elasticity uses a deformation gradient F and may also use local directors. Gauge language enters when one compares material frames across the body or separates a rigid group motion from internal shape. A Cosserat rod, for example, has a centerline $r(s)$ and an orthonormal frame $(d_1(s), d_2(s), d_3(s))$. The strain vector $\kappa(s)$ is defined by

$$\partial_s d_a = \kappa \times d_a.$$

This is a one-dimensional connection along the rod.

Fluids use the diffeomorphism group more directly. In ideal fluid mechanics, particle relabeling symmetry leads to Kelvin circulation and the Euler equation. In viscous fluid mechanics, dissipation breaks the purely Hamiltonian picture, but geometric structures remain useful for vorticity, transport, and reduction. The gauge-theory chapter focuses on deforming bodies because the finite-dimensional connection picture can be displayed there with the least analytic overhead.

D.8 Fluid Relabeling Versus Shape-Space Gauge Theory

Fluid mechanics has a gauge-like symmetry that is easy to confuse with the Shapere-Wilczek mechanism. In a material description, a fluid motion is a time-dependent diffeomorphism

$$\eta_t : \mathcal{B} \rightarrow \Omega.$$

Relabeling particles by a volume-preserving diffeomorphism $\psi : \mathcal{B} \rightarrow \mathcal{B}$ changes η_t to $\eta_t \circ \psi$ without changing the Eulerian velocity field

$$v(x, t) = \dot{\eta}_t(\eta_t^{-1}(x)).$$

The quotient by particle relabeling is therefore the passage from material labels to Eulerian fields. Kelvin's circulation theorem can be understood as a consequence of this relabeling symmetry in the ideal-fluid variational principle.

For a deforming swimmer or falling cat, the quotient has a different meaning. The base variables are shape variables s , the fiber is a physical rigid motion $g \in G$, and the connection reconstructs $g(t)$ from a path in shape space. The holonomy is an observable displacement or rotation. Fluid relabeling removes unobservable labels; shape-space gauge theory reconstructs observable group motion from internal deformation. Both use quotient geometry, but they answer different physical questions.

D.9 Common Pitfalls

1. Do not identify gauge dependence with physical ambiguity. Coordinates and frames are choices; holonomy and reconstructed motion are physical after a frame convention is fixed.
2. Do not use the mechanical connection for a low-Reynolds swimmer unless the kinetic-energy metric is actually the relevant physics. The Stokes connection comes from resistance, not inertia.

3. Do not infer zero phase from zero instantaneous group velocity in one frame. A closed shape loop can produce net motion through curvature.
4. Do not ignore noncommutativity for rotations. For $G = \text{SO}(3)$ or $\text{SE}(3)$, the bracket term in F_A and the ordering of finite motions matter.
5. Do not call every internal coordinate a gauge variable. The gauge group is a redundancy or symmetry acting along fibers; shape variables label the base.

The conceptual payoff is that the rigid body, deforming body, elastic director field, and swimmer are not isolated examples. They are different realizations of a common reconstruction problem: shape motion plus a connection determines motion in a symmetry group.

Further Directions

Natural continuations include Dirac constraints and symplectic reduction, Delaunay variables for celestial mechanics, resonance normal forms, Arnold diffusion, heavy-top dynamics, numerical Poincare-section construction, and structure-preserving integrators for fluids and deforming bodies. Each addition should follow the same standard used here: state the model, define the symbols, derive the equations, draw the geometry, and connect the mathematics to a reproducible computation.

Suggested Further Reading

For Lagrangian and Hamiltonian mechanics, Arnold's *Mathematical Methods of Classical Mechanics* and Goldstein, Poole, and Safko's *Classical Mechanics* are natural companions. For symplectic geometry, reduction, and momentum maps, see Abraham and Marsden's *Foundations of Mechanics* and Marsden and Ratiu's *Introduction to Mechanics and Symmetry*. For perturbation theory and KAM context, Arnold, Kozlov, and Neishtadt's *Mathematical Aspects of Classical and Celestial Mechanics* and Holmes's *Introduction to Perturbation Methods* are useful next steps. For fluids and elasticity, standard references include Batchelor's *An Introduction to Fluid Dynamics*, Drazin and Reid's *Hydrodynamic Stability*, and Marsden and Hughes's *Mathematical Foundations of Elasticity*. For geometric phases and deforming bodies, read Shapere and Wilczek's original gauge-kinematics papers alongside modern discussions of mechanical connections.